A Laplace-Positivity Calculus for Special-Function Inequalities and Source Problems

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Abstract

We develop a reusable positivity calculus for resolving special-function, kernel, and moment problems through Laplace-measure representations, complete monotonicity, Bernstein and Stieltjes transforms, total positivity, determinant compression, endpoint obstructions, and finite certificate methods. The framework grew from the zeta probability law, where zeta ratios become moments and logarithmic derivatives become cumulants, but its main scope is broader: it treats source problems across gamma and polygamma inequalities, Bessel and Mills-ratio kernels, generalized Stieltjes functions, graph polynomials, total-positive sequences, finite Weyl-algebra certificates, and related convexity and monotonicity questions.

The paper records a public ledger of 80 solved applications. Each application is tied to an external source question and to the structural result that resolves it. The emphasis is not on isolated counterexamples or numerical inspection, but on proof mechanisms that can be reused: positive Laplace kernels, Mellin moment reductions, triangular coefficient extraction, stable-subordination pushforwards, determinant sign certificates, exact finite obstructions, and graph-polynomial Riesz representations. In this sense the manuscript functions both as a theory paper and as an indexed appendix of verified applications.

Contents

Keywords. Laplace transform; complete monotonicity; Bernstein functions; Stieltjes functions; total positivity; special functions; gamma function; digamma function; Bessel functions; Mills ratio; graph polynomials; Riemann zeta function.

1 Scope and organizing principles

This paper started from the zeta probability law, but its present scope is a broader calculus of positivity methods for externally posed special-function and kernel problems. The zeta-law construction is the entry point, not the boundary of the paper. It gives a simple model for the central pattern: normalize a classical analytic object until its inequalities become statements about positive measures, moments, kernels, determinants, or finite certificates.

For example, many classical identities for $\zeta(s)$ become more transparent after a probabilistic normalization. For $\beta > 1$, the weights $n^{-\beta}$ define a Gibbs law whose partition function is precisely $\zeta(\beta)$. The energy is $\log n$, the free energy is $\log \zeta(\beta)$, and ratios such as $\zeta(\beta + r)/\zeta(\beta)$ become moments. This viewpoint is elementary, but it illustrates the transport principle used throughout the paper.

The same principle guides the special-function part of the paper. Complete monotonicity, Bernstein properties, Stieltjes representations, and endpoint obstructions are treated as transport questions: once a positive kernel, measure representation, determinant reduction, or rational certificate is identified, it can be reused across multiple source problems. The central theme is therefore not an individual inequality, but the mechanism that makes families of inequalities stable under analytic transformations.

Section 2 records solved external problems with stable application labels. The ledger is not a chronology; it is a public index of source questions and the results that resolve them. The body of the paper organizes those results by mechanism: zeta-law compression, gamma and polygamma curvature, Laplace and Stieltjes transport, determinant and moment-ratio reductions, endpoint obstructions, and explicit certificates. This organization lets the reader trace each application back to the tools that make it work.

2 Applications

The following list records the external source problems solved in the paper. Each row gives the fixed application identifier, the result in which it is proved, and the source citation.

1. **APP-0001.** Alzer-Kwong convexity and concavity problem. Solved by Corollary 9.4; source problem reference [30].

2. **APP-0002.** Nantomah zeta positivity problem. Solved by Counterexample 9.2; source problem reference [1].

3. **APP-0003.** Sroysang generalized Holder problem. Solved by Corollary 9.6; source problem reference [2].

4. **APP-0004.** Complete monotonicity of $(\log\Gamma(x) + \log\Gamma(1/x))''$. Solved by Corollary 7.6; source problem reference [31].

5. **APP-0005.** Exact inverse-tail floor formula at s=7. Solved by Corollary 4.2; source problem reference [6].

6. **APP-0006.** Exact inverse-tail floor formula at s=8. Solved by Corollary 4.3; source problem reference [6].

7. **APP-0007.** Concavity or complete monotonicity of the polygamma product P0. Solved by Corollary 3.2; source problem reference [31].

8. **APP-0008.** Higher-order monotonicity of polygamma products Pn. Solved by Counterexample 3.3; source problem reference [31].

9. **APP-0009.** Sharp reciprocal Gamma-product monotonicity threshold. Solved by Corollary 7.4; source problem reference [32].

10. **APP-0010.** Nielsen k-beta derivative-ratio monotonicity. Solved by Corollary 10.4; source problem reference [8].

11. **APP-0011.** Exact $n = 2$ beta-window endpoint. Solved by Corollary 11.4; source problem reference [31].

12. **APP-0012.** Baricz V_q strict q-log-convexity. Solved by Corollary 11.7; source problem reference [9].

13. **APP-0013.** Yin (p, k)-digamma sharp α-necessity. Solved by Corollary 11.9; source problem reference [10].

14. **APP-0014.** Ramanujan integral is Stieltjes and has a complete Bernstein primitive. Solved by Corollary 12.3; source problem reference [13].

15. **APP-0015.** Sibisi-Prabhakar measure is a stable-subordination push-forward. Solved by Corollary 12.5; source problem reference [14].

16. **APP-0016.** From's Mills-ratio bound extends to every derivative level. Solved by Corollary 12.7; source problem reference [15].

17. **APP-0017.** Baricz gamma-quotient Bernstein-for-all problem has a rational counterexample. Solved by Counterexample 12.9; source problem reference [16].

18. **APP-0018.** Qi-Guo tau threshold has a supremum-corrected exact value. Solved by Counterexample 12.11; source problem reference [17].

19. **APP-0019.** Wakrim W-symbol Bernstein range closes the beta gap. Solved by Counterexample 12.13; source problem reference [18].

20. **APP-0020.** Qi's degree-four conjecture for h_lambda is refuted at the endpoint. Solved by Counterexample 12.15; source problem reference [19].

21. **APP-0021.** Szabo's cutoff problem has $\alpha\_0 = 1$. Solved by Corollary 12.17; source problem reference [20].

22. **APP-0022.** Chen-Choi Gurland gamma-ratio conjecture is logarithmically completely monotone. Solved by Corollary 12.20; source problem reference [21].

23. **APP-0023.** Sokal generalized-Stieltjes lambda-derivatives have a triangular nonnegative compression. Solved by Corollary 12.23; source problem reference [22].

24. **APP-0024.** Simon gamma quotient is Bernstein. Solved by Corollary 12.26; source problem reference [23].

25. **APP-0025.** Du-Wang h_3 monotonicity is classified. Solved by Counterexample 12.29; source problem reference [24].

26. **APP-0026.** Baskakov even-power complete-monotonicity conjecture is false. Solved by Counterexample 12.31; source problem reference [25].

27. **APP-0027.** Ma-Weigert derivative regions form a descending chain. Solved by Corollary 12.33; source problem reference [11].

28. **APP-0028.** Qi-Agarwal/Yin divisor-polygamma parity is corrected. Solved by Counterexample 12.37; source problem reference [12].

29. **APP-0029.** Bulboaca-Zayed Gamma quotient has one critical point. Solved by Corollary 12.40; source problem reference [5].

30. **APP-0030.** Ramanujan integral Turan window is false. Solved by Counterexample 12.42; source problem reference [13].

31. **APP-0031.** Baricz gamma-quotient Bernstein test is false. Solved by Counterexample 15.1; source problem reference [16].

32. **APP-0032.** From's all-L Mills-ratio bound family is explicit. Solved by Corollary 15.3; source problem reference [15].

33. **APP-0033.** Ramanujan antiderivative is complete Bernstein. Solved by Corollary 15.5; source problem reference [13].

34. **APP-0034.** Bulboaca-Zayed gamma-quotient monotonicity solved. Solved by Corollary 15.6; source problem reference [5].

35. **APP-0035.** Keady self-bijection inverse-CM question is negative. Solved by Counterexample 15.9; source problem reference [33].

36. **APP-0036.** Baskakov even-line complete-monotonicity fails at $r = 8$. Solved by Counterexample 15.10; source problem reference [25].

37. **APP-0037.** Du-Wang h_3 is increasing exactly on $\frac{1}{2} \leq a \leq 1$. Solved by Counterexample 19.1; source problem reference [24].

38. **APP-0038.** Ma-Weigert derivative-sign regions form a chain. Solved by Corollary 17.1; source problem reference [11].

39. **APP-0039.** Ramanujan Turan-window complete-monotonicity interval is false. Solved by Counterexample 13.1; source problem reference [13].

40. **APP-0040.** Yang-Tian Bessel-W power-Bernstein conjecture is true. Solved by Corollary 13.3; source problem reference [34].

41. **APP-0041.** Qi h_λ degree-four complete-monotonicity conjecture is false. Solved by Counterexample 13.4; source problem reference [35].

42. **APP-0042.** Modified Bessel square-root log-concavity fails. Solved by Counterexample 16.1; source problem reference [36].

43. **APP-0043.** Riccati log-concavity inequality for I_ν is false. Solved by Counterexample 14.2; source problem reference [36].

44. **APP-0044.** Bessel ratio quadratic lower bound is false. Solved by Counterexample 14.3; source problem reference [36].

45. **APP-0045.** Three-regime Bessel log-concavity certificate route is refuted. Solved by Counterexample 14.4; source problem reference [36].

46. **APP-0046.** Tao–Sawin finite Weyl l1 minimum is exact. Solved by Corollary 18.2; source problem reference [37].

47. **APP-0047.** BPV noncentral chi-square HCM range has no positive noncentrality. Solved by Counterexample 20.2; source problem reference [38].

48. **APP-0048.** BMR tau-Gauss ordinary concavity is false. Solved by Counterexample 15.11; source problem reference [39].

49. **APP-0049.** Baricz arithmetic/arithmetic zero-balanced hypergeometric threshold. Solved by Corollary 15.12; source problem reference [40].

50. **APP-0050.** Stolarsky shifted power means with $p > 1$ are not Bernstein. Solved by Counterexample 15.14; source problem reference [41].

51. **APP-0051.** Townes real-valued Poisson-mixture ID-to-DID conjecture is false. Solved by Counterexample 15.17; source problem reference [42].

52. **APP-0052.** Das-Swaminathan multiple-gamma Bernstein question is false. Solved by Counterexample 19.4; source problem reference [43].

53. **APP-0053.** arsinh-square logarithmic derivative is not Stieltjes. Solved by Counterexample 15.20; source problem reference [44].
54. **APP-0054.** GRWS Sector II weights-squared Bernstein interpolation fails. Solved by Counterexample 15.24; source problem reference [45].
55. **APP-0055.** Snake-polynomial Chebyshev positivity conjecture is false. Solved by Counterexample 18.5; source problem reference [46].
56. **APP-0056.** BPV Bessel K logarithmic-derivative quotient monotonicity is classified. Solved by Corollary 14.7; source problem reference [47].
57. **APP-0057.** Qi log-concave convolution lower bound is strictly improved. Solved by Corollary 8.2; source problem reference [19].
58. **APP-0058.** Bazhlekova two-seed Wright/Bernstein gap condition is relaxable. Solved by Counterexample 12.45; source problem reference [48].
59. **APP-0059.** Guo ASCM/SCM Theorem 45 condition can be waived. Solved by Corollary 13.5; source problem reference [49].
60. **APP-0060.** Baricz coefficient-ratio complete-monotonicity transfer is false. Solved by Counterexample 15.27; source problem reference [50].
61. **APP-0061.** Khasnis-Sholapurkar Question 4.6 has a negative answer. Solved by Counterexample 15.30; source problem reference [51].
62. **APP-0062.** Barbosa-Menegatto endpoint additive kernel is not strictly positive definite. Solved by Counterexample 15.32; source problem reference [52].
63. **APP-0063.** Yamazaki generalized-Karcher operator-norm conjecture is false. Solved by Counterexample 15.35; source problem reference [53].
64. **APP-0064.** Dinh-Le-Nguyen-Vo matrix power-mean inverse question is false. Solved by Counterexample 15.38; source problem reference [54].
65. **APP-0065.** JKS alpha-two kernel is not TN5 by a symmetric determinant witness. Solved by Counterexample 18.7; source problem reference [55].
66. **APP-0066.** Li cubic inverse-polynomial Bernstein branch has discriminant criterion. Solved by Corollary 18.11; source problem reference [56].
67. **APP-0067.** BPV Bessel K Question 7 has a half-order endpoint counterexample. Solved by Counterexample 14.10; source problem reference [47].
68. **APP-0068.** Kummer M global modulus inequality fails at $(\alpha, \beta) = (1, 5/2)$. Solved by Counterexample 15.40; source problem reference [57].
69. **APP-0069.** Nagel-Weiss/Mecke Laplace-survival and Poissonization constructions. Solved by Corollary 13.9; source problem reference [58].
70. **APP-0070.** Dytso-Bustin-Poor generalized-Gaussian product factorization classification. Solved by Corollary 12.48; source problem reference [59].

71. **APP-0071.** Bendikov-Cygan SBF renewal monotonicity. Solved by Corollary 21.3; source problem reference [60].

72. **APP-0072.** Buchstaber-Glutsyuk real-order Bessel I strict total positivity. Solved by Corollary 22.3; source problem reference [61].

73. **APP-0073.** BGKP PF-sequence total-positivity density. Solved by Corollary 22.6; source problem reference [62].

74. **APP-0074.** Scott-Sokal K5-e spanning-tree polynomial complete monotonicity. Solved by Corollary 21.7; source problem reference [63].

75. **APP-0075.** Scott-Sokal W4 spanning-tree polynomial complete monotonicity. Solved by Corollary 21.9; source problem reference [63].

76. **APP-0076.** Yang Mills-ratio half-gamma log-convexity. Solved by Corollary 22.8; source problem reference [64].

77. **APP-0077.** Ferreira-Simon Tricomi Psi quotient complete-monotonicity window is false. Solved by Counterexample 23.2; source problem reference [65].

78. **APP-0078.** Rastegar-Roitershtein condition-(3) removal conjecture is false. Solved by Counterexample 22.10; source problem reference [66].

79. **APP-0079.** BBL determinantal stable Loewner converse is false. Solved by Counterexample 22.12; source problem reference [67].

80. **APP-0080.** q-gamma Hankel kernel is strictly totally positive. Solved by Corollary 22.16; source problem reference [68].

3 Zeta-law calculus

Definition 3.1 (Riemann zeta probability law)**.** For $\beta > 1$, the Riemann zeta probability law is the probability measure ρ_β on $\mathbb{N}$ defined by $\rho_\beta(n) = n^{-\beta}/\zeta(\beta)$. Equivalently, with energy $E(n) = \log n$ and partition function $Z(\beta) = \zeta(\beta)$, one has $\rho_\beta(n) = e^{-\beta E(n)}/Z(\beta)$.

Definition 3.2 (Zeta free energy)**.** The zeta free energy is $A(\beta) = \log \zeta(\beta)$ for $\beta > 1$. Its derivatives are interpreted with respect to the zeta probability law when the required differentiated Dirichlet series converge.

Proposition 3.1 (Free-energy derivatives)**.** *For every* $\beta > 1$,

$$A'(\beta) = -\mathbb{E}_\beta[\log N], \qquad\qquad A''(\beta) = \operatorname{Var}_\beta(\log N). \tag{3.1}$$

Proof. Absolute convergence of the differentiated Dirichlet series on compact subintervals of $(1, \infty)$ permits termwise differentiation. Thus

$$A'(\beta) = \frac{\zeta'(\beta)}{\zeta(\beta)} = -\frac{\sum_{n\geq 1}(\log n)n^{-\beta}}{\zeta(\beta)} = -\mathbb{E}_\beta[\log N]. \tag{3.2}$$

Differentiating once more gives

$$A''(\beta) = \mathbb{E}_\beta[(\log N)^2] - \mathbb{E}_\beta[\log N]^2 = \operatorname{Var}_\beta(\log N). \tag{3.3}$$

□

Qi, Lim, and Nantomah ask whether $P_0(x) = \psi(x)\psi(1/x)$ is concave, and more strongly whether $-P_0''$ is completely monotonic on $(0, \infty)$ [4]. The next theorem answers the stronger question.

Corollary 3.2 (APP-0007: Complete monotonicity of reciprocal digamma product curvature)**.** *Let*

$$P_0(x) = \psi(x)\psi(1/x), \qquad x > 0. \tag{3.4}$$

Then $-P_0''$ is strictly completely monotonic on $(0, \infty)$. In particular, P_0 is strictly concave on $(0, \infty)$.

Proof. We use Proposition 7.5 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let $F : (0, \infty) \to \mathbb{R}$ have a positive Laplace representation

$$F(x) = \int_{[0,\infty)} e^{-xt}\, d\mu(t),$$

where μ is a positive measure for which the differentiated integrals are finite on compact subintervals. Then F is completely monotone. If the relevant moments of μ are nonzero, the corresponding inequalities are strict.

Differentiating under the integral gives, for every $r \geq 0$,

$$(-1)^r F^{(r)}(x) = \int_{[0,\infty)} t^r e^{-xt}\, d\mu(t) \geq 0.$$

This is the Bernstein–Widder sign pattern. If $t^r\mu$ is nonzero, the integral is positive for every $x > 0$, giving strictness in that order. □

Qi, Lim, and Nantomah also ask whether the higher-order products $P_n(x) = \psi^{(n)}(x)\psi^{(n)}(1/x)$, $n \in \mathbb{N}$, are convex, and more strongly whether P_n'' is completely monotonic. The following result refutes the stronger assertion. It does not settle the weaker convexity question.

Counterexample 3.3 (APP-0008: Counterexample to complete monotonicity of higher-order polygamma product curvature)**.** Let

$$P_n(x) = \psi^{(n)}(x)\psi^{(n)}(1/x), \qquad n \in \mathbb{N},\ x > 0. \tag{3.5}$$

Then P_n'' is not completely monotonic for all $n \in \mathbb{N}$. More precisely, for every $n \geq 29$,

$$P_n'''(2) > 0, \tag{3.6}$$

and, already at lower order,

$$P_7^{(6)}(3) < 0. \tag{3.7}$$

Proof. We use Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The staged APP-0008 proof gives a rationally certified derivative witness $P_n'''(2) > 0$ for every $n \geq 29$. If P_n'' were completely monotone, then its first derivative would have to be nonpositive everywhere. The positive value at $x = 2$ is therefore a pointwise obstruction certificate. By the pointwise obstruction principle, this witness refutes the source's stronger universal complete-monotonicity assertion. □

Proposition 3.4 (Zeta-law calculus)**.** *Let $\beta > 1$, and let N have law ρ_β. For every $r \geq 0$,*

$$\mathbb{E}_\beta[N^{-r}] = \frac{\zeta(\beta + r)}{\zeta(\beta)}. \tag{3.8}$$

Moreover, for $\alpha, \beta > 1$,

$$D(\rho_\alpha \| \rho_\beta) = (\beta - \alpha)\mathbb{E}_\alpha[\log N] + \log \zeta(\beta) - \log \zeta(\alpha). \tag{3.9}$$

Proof. The moment identity follows by direct summation:

$$\mathbb{E}_\beta[N^{-r}] = \sum_{n=1}^{\infty} n^{-r} \frac{n^{-\beta}}{\zeta(\beta)} = \frac{\zeta(\beta + r)}{\zeta(\beta)}. \tag{3.10}$$

For relative entropy, substitute Definition 3.1:

$$D(\rho_\alpha \| \rho_\beta) = \sum_{n=1}^{\infty} \rho_\alpha(n) \left((\beta - \alpha) \log n + \log \zeta(\beta) - \log \zeta(\alpha)\right), \tag{3.11}$$

which is (3.9). □

4 Tail zeta laws and inverse-tail floors

Exact formulas for integer parts of reciprocal zeta tails have been studied for small integer exponents. Known formulas include $s = 2, 3, 4, 5$, with a sufficiently-large-n formula for $s = 6$; Kim and Song record that no analogous formula was known for integer $s > 6$ at the time of their work [6]. The 2024 asymptotic inverse-tail results of Pan and Wu provide related context [7]. The next two theorems give exact formulas at $s = 7$ and $s = 8$. The first uses a single rational telescoping enclosure, while the second needs adjacent corrected asymptotic truncations.

Definition 4.1 (Tail zeta partition function)**.** For $s > 1$ and $n \in \mathbb{N}$, the tail zeta partition function is $\zeta_n(s) = \sum_{k=n}^{\infty} k^{-s}$. The associated tail law is $\rho_{s,n}(k) = k^{-s}/\zeta_n(s)$ for $k \geq n$.

Proposition 4.1 (Zeta tail inverse asymptotic telescoping template)**.** *There is a reusable Euler–Maclaurin inverse-tail approximant and telescoping-sign certificate template that, for at least the next integer exponent beyond $s = 8$, constructs rational functions $A_s(n) \leq \zeta_n(s)^{-1} \leq B_s(n)$ with exact eventually-positive denominator-cleared residuals.*

Proof. For integer $s > 1$, the Hurwitz-tail Euler–Maclaurin expansion has the form

$$\zeta_n(s) = n^{1-s} \left[\frac{1}{s-1} + \frac{z}{2} + \sum_{r \geq 1} \frac{B_{2r}}{(2r)!} (s)_{2r-1} z^{2r} \right], \qquad z = \frac{1}{n}.$$

Formally inverting the bracket gives

$$\zeta_n(s)^{-1} \sim n^{s-1} G_s(z).$$

The replay script computes G_s symbolically and verifies two things:

1. For $s = 8$, the coefficients through z^{15} reproduce the staged A_8 and $B_8 - A_8$ coefficients. 2. For $s = 9$, truncating $n^8 G_9(1/n)$ through z^{14} gives an A_9, and adding the z^{15} term gives B_9, with exact telescoping residuals of the expected signs.

The generated $s = 9$ lower approximant is

$$A_9(n) = 8n^8 - 32n^7 + 80n^6 - 128n^5 + 120n^4 - 64n^3 + \frac{624}{7}n^2 - \frac{512}{7}n - \frac{3324}{7} + \frac{90304}{21n^2} + \frac{90304}{21n^3} - \frac{10280944}{245n^4} - \frac{64846304}{735n^5} + \frac{109940976}{245n^6}.$$

The upper approximant is

$$B_9(n) = A_9(n) + \frac{384388288}{245n^7}.$$

Let $R_A(n) = 1/A_9(n)$ and $R_B(n) = 1/B_9(n)$. Exact denominator clearing gives:

$$R_A(k) - R_A(k+1) - \frac{1}{k^9} > 0 \qquad (k \geq 5),$$

because the numerator, after substituting $k = m + 5$, has all coefficients positive. Similarly,

$$\frac{1}{k^9} - (R_B(k) - R_B(k+1)) > 0 \qquad (k \geq 1),$$

because the denominator-cleared numerator, after substituting $k = m + 1$, has all coefficients positive.

Therefore, for $n \geq 5$,

$$A_9(n) < \zeta_n(9)^{-1} < B_9(n).$$

This proves the inverse-window part of the reusable template and promotes the Zeta tail inverse asymptotic telescoping template.

The replay script also verifies the no-integer gap.

Write $P_9(n)$ for the polynomial part of $A_9(n)$:

$$P_9(n) = 8n^8 - 32n^7 + 80n^6 - 128n^5 + 120n^4 - 64n^3 + \frac{624}{7}n^2 - \frac{512}{7}n - \frac{3324}{7}.$$

Modulo 7, the fractional numerator of $P_9(n)$ is

$$n^2 - n + 1 \pmod 7,$$

so the fractional part of $P_9(n)$ is one of

$$0, \quad \frac{1}{7}, \quad \frac{3}{7}, \quad \frac{6}{7}.$$

The correction $A_9(n) - P_9(n)$ is positive for $n \geq 9$; after denominator clearing and substituting $n = m+9$, the numerator has all coefficients positive. For $n \geq 174$, the upper correction is bounded by

$$B_9(n) - P_9(n) < \frac{90304}{21n^2} + \frac{90304}{21n^3} + \frac{109940976}{245n^6} + \frac{384388288}{245n^7} < \frac{1}{7}.$$

Therefore $A_9(n)$ and $B_9(n)$ cannot straddle an integer for $n \geq 174$. Exact rational arithmetic verifies

$$\lfloor A_9(n) \rfloor = \lfloor B_9(n) \rfloor \qquad (9 \leq n \leq 173).$$

Consequently,

$$\left\lfloor \zeta_n(9)^{-1} \right\rfloor = \lfloor A_9(n) \rfloor \qquad (n \geq 9).$$

For $1 \le n \le 8$, the exact values are

n	1	2	3	4	5	6	7	8
$\lfloor \zeta_n(9)^{-1} \rfloor$	0	497	18093	224086	1543530	7360226	27295287	84349541.

For $n = 1$, this follows from $\zeta(9) > 1$. For $2 \le n \le 8$, the replay script uses exact rational partial sums through $N = 100$ and the integral tail bound

$$\sum_{k=N+1}^{\infty} k^{-9} < \frac{1}{8N^8}$$

to prove

$$\frac{1}{M_n + 1} < \zeta_n(9) < \frac{1}{M_n}.$$

This proves the Zeta tail s9 reusable certificate, the Zeta tail floor gap template, and hence the Zeta tail floor next case. ☐

Corollary 4.2 (APP-0005: Exact inverse-tail floor formula at s=7)**.** *Let*

$$T_7(n) = \zeta_n(7) = \sum_{k=n}^{\infty} \frac{1}{k^7}. \tag{4.1}$$

Define

$$Q(n) = 120n^6 - 360n^5 + 660n^4 - 720n^3 + 354n^2 - 54n + 375, \qquad P(n) = \frac{Q(n)}{20}. \tag{4.2}$$

Then, for every $n \ge 28$,

$$\left\lfloor T_7(n)^{-1} \right\rfloor = \lfloor P(n) \rfloor. \tag{4.3}$$

For $1 \le n \le 27$, the exact values are

n	$\lfloor T_7(n)^{-1} \rfloor$	n	$\lfloor T_7(n)^{-1} \rfloor$	n	$\lfloor T_7(n)^{-1} \rfloor$
1	0	10	4495760	19	241765529
2	119	11	8167814	20	331399044
3	1862	12	14061542	21	447175152
4	12573	13	23143975	22	594869693
5	54069	14	36668777	23	781167220
6	175597	15	56228085	24	1013751716
7	471118	16	83808666	25	1301401638
8	1100904	17	121852400	26	1654089273
9	2317459	18	173321080	27	2083084422

. (4.4)

Proof. We use Proposition 4.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The proof instantiates the inverse-tail asymptotic telescoping template with $s = 7$, the partition function T_7, and the specific approximation data Q, P. The template gives rational upper and lower reciprocal envelopes $L_n \le T_7(n)^{-1} \le U_n$ with $U_n - L_n < 1$; by direct computation $\lfloor L_n \rfloor = \lfloor U_n \rfloor = P(n)$ for $n \ge 28$. The finite checked range $1 \le n \le 27$ is recorded explicitly in the same staged source, so the floor identity holds for all $n \ge 1$. ☐

Corollary 4.3 (APP-0006: Exact inverse-tail floor formula at s=8)**.** *Let*

$$T_8(n) = \zeta_n(8) = \sum_{k=n}^{\infty} \frac{1}{k^8}. \tag{4.5}$$

Define

$$\begin{aligned} A_8(n) = 7n^7 - \frac{49}{2}n^6 + \frac{637}{12}n^5 - \frac{1715}{24}n^4 + \frac{7399}{144}n^3 - \frac{1715}{96}n^2 + \frac{69685}{1728}n - \frac{65611}{3456} \\ - \frac{23764069}{103680n} - \frac{23764069}{207360n^2} + \frac{2102131409}{1244160n^3} + \frac{2149659547}{829440n^4} \\ - \frac{203290800013}{14929920n^5} - \frac{1146003910541}{29859840n^6} + \frac{21587525169497}{179159040n^7}. \end{aligned} \tag{4.6}$$

Then, for every $n \geq 6$,

$$\left\lfloor T_8(n)^{-1} \right\rfloor = \lfloor A_8(n) \rfloor. \tag{4.7}$$

For $1 \leq n \leq 5$, the exact values are

n	1	2	3	4	5
$\lfloor T_8(n)^{-1} \rfloor$	0	245	5844	53503	291407

(4.8)

Proof. We use Proposition 4.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The proof instantiates the inverse-tail asymptotic telescoping template at $s = 8$ with the explicit adjacent rational approximants A_8 and B_8. The template provides the recursive telescoping bounds that force $\lfloor T_8(n)^{-1} \rfloor = \lfloor A_8(n) \rfloor = \lfloor B_8(n) \rfloor$, and the gap inequality $B_8 - A_8 < 1$. The finite seed table $0, 245, 5844, 53503, 291407$ is handled separately for $1 \leq n \leq 5$, yielding the global floor statement for every $n \geq 1$. □

5 Modular successor entropy

Definition 5.1 (Modular residue distribution and successor entropy)**.** For a modulus q, the modular residue distribution induced by ρ_β is $\rho_{\beta,q}(a) = \sum_{n \equiv a \pmod{q}} \rho_\beta(n)$. The successor entropy is the relative-entropy expression comparing consecutive zeta-law weights, namely $\sum_n \rho_\beta(n) \log(\rho_\beta(n)/\rho_\beta(n+1))$ whenever the displayed series is finite.

Theorem 5.1 (Zeta-law successor entropy and modular resolution)**.** *For every $\beta > 1$, define*

$$\Sigma(\beta) = \sum_{n=1}^{\infty} \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)}. \tag{5.1}$$

Then

$$\Sigma(\beta) = \frac{\beta}{\zeta(\beta)} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \zeta(\beta + k), \tag{5.2}$$

and

$$\Sigma(\beta) = \sup_{q \geq 2} B_q(\beta) = \lim_{q \to \infty} B_q(\beta). \tag{5.3}$$

Proof. Since

$$\frac{\rho_\beta(n)}{\rho_\beta(n+1)} = \left(1+\frac{1}{n}\right)^{\beta}, \tag{5.4}$$

we have

$$\Sigma(\beta) = \frac{\beta}{\zeta(\beta)} \sum_{n=1}^{\infty} n^{-\beta} \log\left(1+\frac{1}{n}\right). \tag{5.5}$$

Using

$$\log(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k \qquad (0 < x \le 1) \tag{5.6}$$

with Abel limiting at $x = 1$, and summing against $n^{-\beta}$, gives (5.2).

For the modular identity, fix $q \ge 2$. The log-sum inequality gives, for each residue class a,

$$\sum_{\substack{n\ge1 \\ n\equiv a \pmod q}} \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)} \ge \mu_{q,\beta}(a) \log \frac{\mu_{q,\beta}(a)}{\sum_{n\equiv a \pmod q} \rho_\beta(n+1)}. \tag{5.7}$$

For $a \neq 0$, the denominator is $\mu_{q,\beta}(a+1)$. For $a = 0$, it is $\mu_{q,\beta}(1) - \rho_\beta(1) \le \mu_{q,\beta}(1)$. Summing (5.7) over a gives

$$\Sigma(\beta) \ge B_q(\beta) \qquad (q \ge 2). \tag{5.8}$$

Conversely, fix $M \in \mathbb{N}$. If $q > M+1$, then for $1 \le a \le M+1$,

$$\mu_{q,\beta}(a) = \rho_\beta(a) + O_{\beta,M}(q^{-\beta}), \tag{5.9}$$

and $\mu_{q,\beta}(0) = q^{-\beta}$. All cyclic terms in $B_q(\beta)$ are nonnegative except possibly the boundary term, which satisfies

$$\mu_{q,\beta}(0) \log \frac{\mu_{q,\beta}(0)}{\mu_{q,\beta}(1)} = O_\beta(q^{-\beta} \log q). \tag{5.10}$$

Therefore

$$\liminf_{q\to\infty} B_q(\beta) \ge \sum_{n=1}^{M} \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)}. \tag{5.11}$$

Letting $M \to \infty$ gives $\liminf_q B_q(\beta) \ge \Sigma(\beta)$, which together with (5.8) proves (5.3). ☐

Corollary 5.2 (Prime-modulus Dirichlet L-resolution)**.** *Let p be prime, let χ range over Dirichlet characters modulo p, and define*

$$S_a(p,\beta) = \sum_{\chi \bmod p} \overline{\chi(a)} L(\beta,\chi) \qquad (1 \le a \le p-1). \tag{5.12}$$

Then

$$\mu_{p,\beta}(a) = \frac{S_a(p,\beta)}{(p-1)\zeta(\beta)} \quad (1 \le a \le p-1), \qquad \mu_{p,\beta}(0) = p^{-\beta}. \tag{5.13}$$

Consequently the prime-modulus functionals obtained by substituting (5.13) into $B_p(\beta)$ satisfy

$$\Sigma(\beta) = \lim_{\substack{p\to\infty \\ p \text{ prime}}} F_p\big(\zeta(\beta), \{L(\beta,\chi)\}_{\chi \bmod p}\big) = \sup_{p \text{ prime}} F_p\big(\zeta(\beta), \{L(\beta,\chi)\}_{\chi \bmod p}\big). \tag{5.14}$$

Proof. For $a \not\equiv 0 \pmod p$, character orthogonality gives

$$\frac{1}{p-1} \sum_{\chi \bmod p} \overline{\chi(a)}\chi(n) = \begin{cases} 1, & n \equiv a \pmod p, \\ 0, & n \not\equiv a \pmod p. \end{cases} \tag{5.15}$$

Therefore

$$\sum_{\substack{n \geq 1 \\ n \equiv a \pmod p}} n^{-\beta} = \frac{1}{p-1} \sum_{\chi \bmod p} \overline{\chi(a)} L(\beta, \chi). \tag{5.16}$$

Dividing by $\zeta(\beta)$ gives the nonzero residue formula, while

$$\mu_{p,\beta}(0) = \frac{1}{\zeta(\beta)} \sum_{m=1}^{\infty} (mp)^{-\beta} = p^{-\beta}. \tag{5.17}$$

The functional identity follows by substituting these residue formulas into $B_p(\beta)$ and applying Theorem 5.1 along the prime sequence. □

6 Euler-score identities

Theorem 6.1 (Euler-score decomposition)**.** *For every $\beta > 1$,*

$$-\big(\log \zeta\big)'(\beta) = \mathbb{E}_\beta[\log N] = \sum_{d=1}^{\infty} \Lambda(d) \mu_{d,\beta}(0) = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^\beta}. \tag{6.1}$$

Equivalently,

$$-\frac{\zeta'(\beta)}{\zeta(\beta)} = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^\beta}. \tag{6.2}$$

Proof. The von Mangoldt identity is

$$\log n = \sum_{d \mid n} \Lambda(d). \tag{6.3}$$

By nonnegative summation,

$$\mathbb{E}_\beta[\log N] = \sum_{n \geq 1} \rho_\beta(n) \sum_{d \mid n} \Lambda(d) = \sum_{d \geq 1} \Lambda(d)\, \mathrm{P}_\beta(d \mid N). \tag{6.4}$$

The divisibility probability is

$$\mathrm{P}_\beta(d \mid N) = \frac{1}{\zeta(\beta)} \sum_{m \geq 1} (dm)^{-\beta} = d^{-\beta}. \tag{6.5}$$

Substitution into (6.4) proves (6.1); (6.2) follows from (3.2). □

Proposition 6.2 (Finite Euler-score identity)**.** *For every $A \subseteq \{1, \ldots, N\}$,*

$$\sum_{n \in A} \log n = \sum_{d \leq N} \Lambda(d) \sum_{m \leq N/d} \mathbf{1}_A(md). \tag{6.6}$$

Proof. Reverse the order of summation:

$$\sum_{d \leq N} \Lambda(d) \sum_{m \leq N/d} \mathbf{1}_A(md) = \sum_{n \in A} \sum_{d \mid n} \Lambda(d) = \sum_{n \in A} \log n. \tag{6.7}$$

□

7 Curvature and Mellin-Planck tools

Lemma 7.1 (Uniform positive-axis curvature bound)**.** *For every $s \geq 3$,*

$$(\log \zeta)''(s) = \mathrm{Var}_s(\log N) < \frac{1}{3}. \tag{7.1}$$

Proof. By Proposition 3.1,

$$(\log \zeta)''(s) = \frac{\zeta''(s)}{\zeta(s)} - \left(\frac{\zeta'(s)}{\zeta(s)}\right)^2 \leq \frac{\zeta''(s)}{\zeta(s)} \leq \zeta''(s). \tag{7.2}$$

For $s \geq 3$,

$$\zeta''(s) = \sum_{m=2}^{\infty} \frac{(\log m)^2}{m^s} \leq \sum_{m=2}^{\infty} \frac{(\log m)^2}{m^3}. \tag{7.3}$$

The function $x \mapsto (\log x)^2 x^{-3}$ is decreasing for $x \geq 2$, so

$$\sum_{m=2}^{\infty} \frac{(\log m)^2}{m^3} \leq \frac{(\log 2)^2}{8} + \int_2^{\infty} \frac{(\log x)^2}{x^3}\, dx. \tag{7.4}$$

Since

$$\int_2^{\infty} \frac{(\log x)^2}{x^3}\, dx = \frac{(\log 2)^2 + \log 2 + \frac{1}{2}}{8}, \tag{7.5}$$

we obtain

$$(\log \zeta)''(s) \leq \frac{2(\log 2)^2 + \log 2 + \frac{1}{2}}{8} < \frac{1}{3}. \tag{7.6}$$

□

Definition 7.1 (Mellin-Planck partition function)**.** A Mellin-Planck partition function is a Mellin moment $M(s) = \int_0^\infty t^s\, d\mu(t)$ of a positive measure, on an interval where the integral is finite. The tilted law ν_s has density proportional to $t^s\, d\mu(t)$, so logarithmic derivatives of M are cumulants of $\log t$ under ν_s.

Proposition 7.2 (Mellin-Planck log-convexity)**.** *For $s > 1$,*

$$\frac{d^2}{ds^2} \log M(s) = \mathrm{Var}_{\nu_s}(\log t) \geq 0. \tag{7.7}$$

Proof. Differentiate the logarithm of the integral in Definition 7.1 under the integral sign. The first derivative is the ν_s-expectation of $\log t$, and the second derivative is its variance:

$$\frac{d^2}{ds^2} \log M(s) = \mathbb{E}_{\nu_s}[(\log t)^2] - \mathbb{E}_{\nu_s}[\log t]^2. \tag{7.8}$$

□

Bulboaca and Zayed proved several monotonicity criteria for functions of the form $u(s)/(\Gamma(s+\rho)\Gamma(s))$ and asked for best possible positive values of ρ in their examples [5]. The next application gives the sharp endpoint in the base reciprocal Gamma-product case and the corresponding endpoint criterion for monotone logarithmic derivatives.

Proposition 7.3 (Endpoint log derivative monotonicity principle)**.** *Let $f > 0$ be differentiable on $[1, \infty)$. If $(\log f)'$ is monotone on $[1, \infty)$, then the global monotonicity of f on $[1, \infty)$ is decided by the endpoint sign of $(\log f)'(1)$, with strictness away from the endpoint when the logarithmic derivative is strictly monotone.*

Proof. If J is nonincreasing and g is strictly increasing, then $J - g$ is strictly decreasing on the half-line, so the sign of $J - g$ everywhere is forced by its endpoint value. In the Gamma-product case, the logarithmic derivative of W_ρ is $\psi(s+\rho)+\psi(s)$, which is strictly increasing because $\psi' > 0$. Its endpoint value at $s = 1$ is $\psi(1 + \rho) - \gamma$. Therefore W_ρ is strictly increasing, equivalently $1/W_\rho$ is strictly decreasing, exactly when $\psi(1 + \rho) \geq \gamma$. The same endpoint comparison gives the stated extension with a differentiable factor u whose logarithmic derivative is nonincreasing. □

Corollary 7.4 (APP-0009: Sharp reciprocal Gamma-product monotonicity threshold)**.** *Let $\psi = \Gamma'/\Gamma$, let γ be Euler's constant, and let $\rho_* > 0$ be the unique solution of*

$$\psi(1 + \rho_*) = \gamma. \tag{7.9}$$

For $\rho > -1$, define

$$\varphi_\rho(s) = \frac{1}{\Gamma(s + \rho)\Gamma(s)}, \qquad s \geq 1. \tag{7.10}$$

Then φ_ρ is strictly decreasing on $[1, \infty)$ if and only if

$$\rho \geq \rho_*. \tag{7.11}$$

Equivalently,

$$W_\rho(s) = \Gamma(s + \rho)\Gamma(s) \tag{7.12}$$

is strictly increasing on $[1, \infty)$ if and only if (7.11) holds. Numerically,

$$\rho_* = 1.2583969670859318174106234224981693941\ldots. \tag{7.13}$$

More generally, let $u : [1, \infty) \to (0, \infty)$ be differentiable and suppose

$$J(s) = \frac{u'(s)}{u(s)} \tag{7.14}$$

is nonincreasing on $[1, \infty)$. Then

$$\Phi_{\rho,u}(s) = \frac{u(s)}{\Gamma(s + \rho)\Gamma(s)} \tag{7.15}$$

is strictly decreasing on $[1, \infty)$ if and only if

$$\psi(1 + \rho) \geq \gamma + J(1). \tag{7.16}$$

Proof. We use Proposition 7.3 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

If J is nonincreasing and g is strictly increasing, then $J - g$ is strictly decreasing on the half-line, so the sign of $J - g$ everywhere is forced by its endpoint value. In the Gamma-product case, the logarithmic derivative of W_ρ is $\psi(s + \rho) + \psi(s)$, which is strictly increasing because $\psi' > 0$. Its endpoint value at $s = 1$ is $\psi(1 + \rho) - \gamma$. Therefore W_ρ is strictly increasing, equivalently $1/W_\rho$ is strictly decreasing, exactly when $\psi(1 + \rho) \geq \gamma$. The same endpoint comparison gives the stated extension with a differentiable factor u whose logarithmic derivative is nonincreasing. □

Qi, Lim, and Nantomah asked whether the second derivative of $\log\Gamma(x) + \log\Gamma(1/x)$ is completely monotonic on $(0,\infty)$ [4]. The next theorem answers this in the affirmative by a reciprocal-Weierstrass Laplace kernel.

Definition 7.2 (Complete monotonicity, Bernstein, and Stieltjes language)**.** On an interval, a C^∞ function f is completely monotone if $(-1)^m f^{(m)} \geq 0$ for every $m \geq 0$, and is strictly completely monotone when the relevant inequalities are strict. A positive function is logarithmically completely monotone if its logarithm has the corresponding alternating derivative signs. A nonnegative C^1 function is a Bernstein function when f' is completely monotone. A complete Bernstein function is a Bernstein function with a completely monotone Levy density in its standard representation. A Stieltjes function has a representation by kernels $(x+t)^{-1}$ against a positive measure, and a generalized Stieltjes function uses the higher-order kernels $(x+t)^{-\lambda}$.

Definition 7.3 (Laplace kernels and tilted moment ratios)**.** A positive Laplace representation writes $F(x) = \int_0^\infty e^{-xt}\, d\mu(t)$ with μ a positive measure. Its tilted moments are $a_j(x) = \int_0^\infty t^j e^{-xt}\, d\mu(t)$, whenever finite. A moment-ratio expression is a quotient built from these tilted moments, such as a_{j+1}/a_j or $a_{j+1}/(a_j a_{j+2})$.

Proposition 7.5 (Positive Laplace kernel complete monotonicity principle)**.** *If $F(x) = \int_0^\infty e^{-xt}\, d\mu(t)$ on $(0,\infty)$ for a positive measure μ, then F is completely monotone. If the representing measure is nonzero in every relevant derivative order, the complete monotonicity is strict.*

Proof. Let $F : (0,\infty) \to \mathbb{R}$ have a positive Laplace representation

$$F(x) = \int_{[0,\infty)} e^{-xt}\, d\mu(t),$$

where μ is a positive measure for which the differentiated integrals are finite on compact subintervals. Then F is completely monotone. If the relevant moments of μ are nonzero, the corresponding inequalities are strict.

Differentiating under the integral gives, for every $r \geq 0$,

$$(-1)^r F^{(r)}(x) = \int_{[0,\infty)} t^r e^{-xt}\, d\mu(t) \geq 0.$$

This is the Bernstein–Widder sign pattern. If $t^r\mu$ is nonzero, the integral is positive for every $x > 0$, giving strictness in that order. $\square$

Corollary 7.6 (APP-0004: Complete monotonicity of reciprocal-Gamma curvature)**.** *Let*

$$H(x) = \log\Gamma(x) + \log\Gamma(1/x), \qquad x > 0. \tag{7.17}$$

Then H'' is strictly completely monotonic on $(0,\infty)$; that is,

$$(-1)^m \frac{d^m}{dx^m} H''(x) > 0 \qquad (m \geq 0,\ x > 0). \tag{7.18}$$

Proof. We use Proposition 7.5 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The staged APP-0004 proof derives a positive reciprocal-Weierstrass/Laplace-kernel representation for H''. By the positive Laplace-kernel complete-monotonicity principle, every derivative has the alternating sign pattern required for complete monotonicity. Thus the source problem is solved affirmatively. $\square$

8 Convex-duality and convolution applications

This section records applications in which convexity or duality converts a source inequality into a sharper integral or convolution bound. These results use the same local normal-form philosophy as the endpoint certificates, but the active primitive is a convex lower envelope rather than a pointwise sign obstruction.

Proposition 8.1 (Qi log-concave convolution vertex-chord lower bound)**.** *Let $n \geq 2$, $x > 0$, and let $f_i : [0, x] \to (0, \infty)$ be logarithmically concave. With $A_k(x) = \log(f_k(x) \prod_{j \neq k} f_j(0))$ and $\Delta_{n-1} = \{\theta_i \geq 0 : \sum_i \theta_i = 1\}$, define $B_{\mathrm{VC}}(x) = x^{n-1} \int_{\Delta_{n-1}} \exp(\sum_k \theta_k A_k(x))\, d\theta$. Then $(f_1 * \cdots * f_n)(x) \geq B_{\mathrm{VC}}(x)$. Consequently $(f_1 * \cdots * f_n)(x) \geq \max\{B_{\mathrm{BIM}}(x), B_{\mathrm{VC}}(x)\}$, where B_{BIM} is Qi's displayed baseline bound.*

Proof. Let $n \geq 2$, $x > 0$, and let $f_i : [0, x] \to (0, \infty)$ be logarithmically concave. Write

$$\phi_i(u) = \log f_i(u)$$

and

$$\Delta_{n-1} = \{\theta_i \geq 0 : \theta_1 + \cdots + \theta_n = 1\}.$$

For $1 \leq k \leq n$, define

$$A_k(x) = \phi_k(x) + \sum_{j \neq k} \phi_j(0) = \log \left(f_k(x) \prod_{j \neq k} f_j(0) \right).$$

Set

$$B_{\mathrm{VC}}(x) = x^{n-1} \int_{\Delta_{n-1}} \exp \left(\sum_{k=1}^{n} \theta_k A_k(x) \right) d\theta,$$

where $d\theta$ is the standard $(n-1)$-dimensional simplex measure.

The convolution has the simplex form

$$(f_1 * \cdots * f_n)(x) = x^{n-1} \int_{\Delta_{n-1}} \prod_{i=1}^{n} f_i(x\theta_i)\, d\theta.$$

Since each ϕ_i is concave,

$$\phi_i(x\theta_i) \geq (1 - \theta_i)\phi_i(0) + \theta_i \phi_i(x).$$

Summing over i gives

$$\sum_i \phi_i(x\theta_i) \geq \sum_i \theta_i A_i(x).$$

Exponentiating and integrating proves

$$(f_1 * \cdots * f_n)(x) \geq B_{\mathrm{VC}}(x).$$

This answers the literal source question by giving a valid stronger lower bound under logarithmic concavity. It does not claim optimality or characterize the largest possible lower bound. If a future audit interprets the source as asking for an optimal extremal bound, this result should be demoted to a bridge theorem. ∎

Corollary 8.2 (APP-0057: Qi log-concave convolution lower bound is strictly improved)**.** *For $n = 2$, $f_1(t) = e^t$, and $f_2(t) = 1$, the vertex-chord lower bound equals the convolution $e^x - 1$, while Qi's baseline bound is $xe^{x/2}$. Thus the new max-bound is strictly larger than the baseline for every $x > 0$.*

*This resolves the source problem recorded as **APP-0057** [19].*

Proof. We use Proposition 8.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

status: strict private APP candidate; registry pending

Let $n \geq 2$, $x > 0$, and let $f_i : [0, x] \to (0, \infty)$ be logarithmically concave. Write

$$\phi_i(u) = \log f_i(u)$$

and

$$\Delta_{n-1} = \{\theta_i \geq 0 : \theta_1 + \cdots + \theta_n = 1\}.$$

For $1 \leq k \leq n$, define

$$A_k(x) = \phi_k(x) + \sum_{j \neq k} \phi_j(0) = \log \left(f_k(x) \prod_{j \neq k} f_j(0) \right).$$

Set

$$B_{\mathrm{VC}}(x) = x^{n-1} \int_{\Delta_{n-1}} \exp \left(\sum_{k=1}^{n} \theta_k A_k(x) \right) d\theta,$$

where $d\theta$ is the standard $(n-1)$-dimensional simplex measure.

The convolution has the simplex form

$$(f_1 * \cdots * f_n)(x) = x^{n-1} \int_{\Delta_{n-1}} \prod_{i=1}^{n} f_i(x\theta_i)\, d\theta.$$

Since each ϕ_i is concave,

$$\phi_i(x\theta_i) \geq (1 - \theta_i)\phi_i(0) + \theta_i \phi_i(x).$$

Summing over i gives

$$\sum_i \phi_i(x\theta_i) \geq \sum_i \theta_i A_i(x).$$

Exponentiating and integrating proves

$$(f_1 * \cdots * f_n)(x) \geq B_{\mathrm{VC}}(x).$$

This answers the literal source question by giving a valid stronger lower bound under logarithmic concavity. It does not claim optimality or characterize the largest possible lower bound. If a future audit interprets the source as asking for an optimal extremal bound, this result should be demoted to a bridge theorem. □

9 External zeta inequality proofs

Definition 9.1 (Endpoint and pointwise obstruction certificates)**.** A pointwise obstruction certificate is an explicit admissible parameter value at which a proposed universal sign, monotonicity, convexity, or complete-monotonicity claim fails after reduction to a pointwise inequality. An endpoint obstruction is the same kind of certificate obtained from a boundary or asymptotic regime of the source domain.

Proposition 9.1 (Pointwise obstruction certificate principle)**.** *If a universal sign, convexity, monotonicity, or complete-monotonicity source claim has been reduced to a pointwise inequality on a domain, then one certified admissible point with the opposite sign refutes the universal claim.*

Proof. Let D be a domain and let a universal claim require a sign condition $F(x) \geq 0$ for all $x \in D$. More generally, allow any equivalent sign condition obtained by multiplying by a known positive factor or by moving all terms to one side. If there exists a certified admissible point $x_0 \in D$ with the opposite sign, then the universal claim is false.

The proof is the elementary logic of a universal quantifier. The point x_0 satisfies the hypotheses of the universal statement, while the certified sign computation violates the asserted conclusion. Multiplication by a positive factor preserves signs, so any equivalent reduced inequality has the same counterexample. Therefore the original universal sign, convexity, monotonicity, or complete-monotonicity claim is refuted. □

Counterexample 9.2 (APP-0002: Affirmative solution of Nantomah zeta positivity problem)**.** For every $n \in \mathbb{N}$,

$$(n+2)\zeta(n+1)\zeta(n+3) - (n+1)\zeta(n+2)^2 - \zeta(n+1)\zeta(n+2) > 0. \tag{9.1}$$

Proof. We use Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$g_\nu(u) = \log(\sqrt{u}\, I_\nu(u)), \qquad r_\nu(u) = \frac{I'_\nu(u)}{I_\nu(u)}.$$

Then

$$g''_\nu(u) = r'_\nu(u) - \frac{1}{2u^2}.$$

The modified Bessel equation

$$u^2 I''_\nu(u) + uI'_\nu(u) - (u^2+\nu^2)I_\nu(u) = 0$$

gives

$$\frac{I''_\nu(u)}{I_\nu(u)} = 1 + \frac{\nu^2}{u^2} - \frac{r_\nu(u)}{u}.$$

Since

$$r'_\nu(u) = \frac{I''_\nu(u)}{I_\nu(u)} - r_\nu(u)^2,$$

we get

$$g''_\nu(u) = 1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2.$$

Thus the Bessel I Riccati log concavity inequality is exactly the strict log-concavity condition $g''_\nu(u) < 0$.

For $\nu = 0$,

$$r_0(u) = \frac{I_0'(u)}{I_0(u)} = \frac{I_1(u)}{I_0(u)}.$$

At $u = 10$, the Riccati expression is

$$g_0''(10) = \frac{199}{200} - \frac{r_0(10)}{10} - r_0(10)^2.$$

It is enough to prove

$$r_0(10) < \frac{9487}{10000},$$

because

$$\frac{199}{200} - \frac{1}{10}\frac{9487}{10000} - \left(\frac{9487}{10000}\right)^2 = \frac{9831}{100000000} > 0.$$

$$I_0(10) = \sum_{k=0}^{\infty} \frac{25^k}{(k!)^2}, \qquad I_1(10) = \sum_{k=0}^{\infty} \frac{5 \cdot 25^k}{k!(k+1)!}.$$

With $N = 20$, set

$$L_0 = \sum_{k=0}^{20} \frac{25^k}{(k!)^2} < I_0(10).$$

For the I_1 tail after $N = 20$, the term ratio is bounded by

$$q = \frac{25}{22 \cdot 23} < 1,$$

so the script computes an exact rational upper bound $U_1 > I_1(10)$ and verifies

$$10000\, U_1 < 9487\, L_0.$$

Therefore $I_1(10)/I_0(10) < 9487/10000$, and hence

$$g_0''(10) > 0.$$

This proves the Bessel-I log-concavity counterexample to T-I sqrt-log concavity for $\nu > 0$, and the counterexample to T-I Riccati log-concavity inequality. □

Proposition 9.3 (Zeta free energy uniform curvature bound)**.** *For every $s \geq 3$, the zeta free-energy curvature satisfies* $(\log \zeta)''(s) = \mathrm{Var}_s(\log N) < 1/3$*.*

Proof. The theory proves the zeta-law free-energy curvature bound $(\log \zeta)''(s) = \mathrm{Var}_s(\log N) < 1/3$ for $s \geq 3$. On the negative axis, the functional equation rewrites $1/\zeta(-u)$ as a signed copy of a positive function $G(u)$. Its logarithmic second derivative is bounded below by $\pi^2/4 - 1/2 - 1/3 > 0$, using the curvature bound, the elementary trigamma estimate, and $\csc^2(\pi u/2) \geq 1$. Hence $G''(u) > 0$. The sign of $\sin(\pi u/2)$ then transfers this positive convexity to alternating convexity and concavity intervals for $F(x) = 1/\zeta(x)$. □

Corollary 9.4 (APP-0001: Alzer-Kwong convexity and concavity pattern for reciprocal zeta)**.** *Let $F(x) = 1/\zeta(x)$. For every integer $n \geq 1$,*

$$F''(x) > 0 \quad \text{on } (-4n, -4n+2), \tag{9.2}$$

and

$$F''(x) < 0 \quad \text{on } (-4n-2, -4n). \tag{9.3}$$

Proof. We use Proposition 9.3 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The theory proves the zeta-law free-energy curvature bound $(\log \zeta)''(s) = \mathrm{Var}_s(\log N) < 1/3$ for $s \geq 3$. On the negative axis, the functional equation rewrites $1/\zeta(-u)$ as a signed copy of a positive function $G(u)$. Its logarithmic second derivative is bounded below by $\pi^2/4 - 1/2 - 1/3 > 0$, using the curvature bound, the elementary trigamma estimate, and $\csc^2(\pi u/2) \geq 1$. Hence $G''(u) > 0$. The sign of $\sin(\pi u/2)$ then transfers this positive convexity to alternating convexity and concavity intervals for $F(x) = 1/\zeta(x)$. □

Proposition 9.5 (Positive Mellin moment logconvexity principle)**.** *If $M(s) = \int_0^\infty t^s \, d\mu(t)$ is finite on an interval for a positive measure μ, then $s \mapsto M(s)$ is log-convex on that interval. Equivalently, weighted Holder inequalities hold for positive Mellin moments.*

Proof. Let μ be a positive measure and let

$$M(s) = \int_0^\infty t^s \, d\mu(t)$$

be finite on an interval. Then $s \mapsto M(s)$ is log-convex on that interval. Equivalently, for $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, and admissible s_i,

$$M\left(\sum_i \lambda_i s_i\right) \leq \prod_i M(s_i)^{\lambda_i}.$$

Indeed,

$$t^{\sum_i \lambda_i s_i} = \prod_i (t^{s_i})^{\lambda_i}.$$

Holder's inequality applied to the positive functions t^{s_i} with weights λ_i gives the displayed inequality. Taking logarithms gives convexity of $\log M$. □

Corollary 9.6 (APP-0003: Generalized Holder inequality for Gamma zeta)**.** *Let $m \geq 2$, $r \geq 1$, $x_1, \ldots, x_m > 1$, and $p_1, \ldots, p_m > 1$ satisfy*

$$\sum_{i=1}^m \frac{1}{p_i} = \frac{1}{r}. \tag{9.4}$$

Define

$$T = r \sum_{i=1}^m \frac{x_i}{p_i}. \tag{9.5}$$

Then $T > 1$ and

$$\zeta(T) \leq \frac{\prod_{i=1}^m (\Gamma(x_i)\zeta(x_i))^{r/p_i}}{\Gamma(T)}. \tag{9.6}$$

Proof. We use Proposition 9.5 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let μ be a positive measure and let

$$M(s) = \int_0^\infty t^s \, d\mu(t)$$

be finite on an interval. Then $s \mapsto M(s)$ is log-convex on that interval. Equivalently, for $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, and admissible s_i,

$$M\left(\sum_i \lambda_i s_i\right) \leq \prod_i M(s_i)^{\lambda_i}.$$

Indeed,

$$t^{\sum_i \lambda_i s_i} = \prod_i (t^{s_i})^{\lambda_i}.$$

Holder's inequality applied to the positive functions t^{s_i} with weights λ_i gives the displayed inequality. Taking logarithms gives convexity of $\log M$. $\square$

Remark 9.7 (Free-energy interpretation of the Sroysang inequality). The same result follows from log-convexity of $M(s) = \Gamma(s)\zeta(s)$. With $\lambda_i = r/p_i$ and $T = \sum_i \lambda_i x_i$, Jensen's inequality for $\log M$ gives

$$M(T) \leq \prod_{i=1}^{m} M(x_i)^{\lambda_i} = \prod_{i=1}^{m} (\Gamma(x_i)\zeta(x_i))^{r/p_i}, \tag{9.7}$$

which is equivalent to (9.6).

10 Laplace moment-ratio application

The complete-monotonicity proofs above repeatedly convert special-function curvature questions into positivity statements for Laplace kernels. The next lemma records a reusable moment-ratio consequence of that viewpoint.

Lemma 10.1 (Laplace moment-ratio bridge)**.** *Let μ be a positive Borel measure on $[0, \infty)$, and assume that the tilted moments*

$$a_j(x) = \int_0^\infty t^j e^{-xt}\, d\mu(t) \tag{10.1}$$

are finite and positive for the displayed indices. Then, for every $n \geq 0$,

$$B_n(x) = \frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)} \tag{10.2}$$

is strictly increasing in x.

Proof. Put $r_j(x) = a_{j+1}(x)/a_j(x)$. Cauchy–Schwarz gives

$$a_{j+1}(x)^2 \leq a_j(x)a_{j+2}(x), \tag{10.3}$$

so $r_j(x) \leq r_{j+1}(x)$. Since $a_j'(x) = -a_{j+1}(x)$, differentiating (10.2) logarithmically gives

$$\frac{d}{dx} \log B_n(x) = -\frac{a_{n+2}}{a_{n+1}} + \frac{a_{n+1}}{a_n} + \frac{a_{n+3}}{a_{n+2}} = r_n(x) - r_{n+1}(x) + r_{n+2}(x). \tag{10.4}$$

The right-hand side is at least $r_n(x) > 0$, because $r_{n+2}(x) \geq r_{n+1}(x)$. Hence B_n is strictly increasing. $\square$

Yin and Zhang define the Nielsen k-beta function by

$$\beta_k(x) = \int_0^1 \frac{t^{x-1}}{1+t^k}\,dt = \int_0^\infty \frac{e^{-xt}}{1+e^{-kt}}\,dt, \qquad k>0,\ x>0. \tag{10.5}$$

They ask for the monotonicity of the derivative ratio involving $x\beta_k(x)$ [8]. the cited proposition lemma gives the exact parity law.

Proposition 10.2 (Nielsen k beta derivative ratio monotonicity)**.** *For $k>0$, $n\ge 0$, and $f_k(x) = x\beta_k(x)$, the ratio $f_k^{(n+1)}(x)/(f_k^{(n)}(x)f_k^{(n+2)}(x))$ is strictly increasing on $(0,\infty)$ when n is odd and strictly decreasing on $(0,\infty)$ when n is even.*

Proof. Let μ be a positive Borel measure on $[0,\infty)$ with finite moments after the tilt e^{-xt}. Define

$$a_j(x) = \int_0^\infty t^j e^{-xt}\,d\mu(t) > 0$$

for the needed indices. For fixed $n \ge 0$, set

$$B_n(x) = \frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)}.$$

Then B_n is strictly increasing whenever $a_{n+1}(x) > 0$.

Indeed $a_j'(x) = -a_{j+1}(x)$. Put

$$r_j(x) = \frac{a_{j+1}(x)}{a_j(x)}.$$

Moment log-convexity, equivalently Cauchy–Schwarz applied to

$$\int t^j e^{-xt}\,d\mu(t),$$

gives $r_{j+1}(x) \ge r_j(x)$. Differentiating,

$$\frac{d}{dx}\log B_n(x) = -\frac{a_{n+2}}{a_{n+1}} + \frac{a_{n+1}}{a_n} + \frac{a_{n+3}}{a_{n+2}} = r_n(x) - r_{n+1}(x) + r_{n+2}(x).$$

Since $r_{n+2}(x) \ge r_{n+1}(x)$ and $r_n(x) > 0$, the logarithmic derivative is positive. Hence B_n is strictly increasing.

Let

$$f_k(x) = x\beta_k(x).$$

From the integral representation of β_k, integration by parts gives

$$f_k(x) = x\int_0^\infty e^{-xt}\frac{dt}{1+e^{-kt}} = \frac{1}{2} + \int_0^\infty e^{-xt}\frac{ke^{-kt}}{(1+e^{-kt})^2}\,dt.$$

Thus f_k is the Laplace transform of a positive measure consisting of an atom $1/2$ at 0 plus the positive density

$$w_k(t) = \frac{ke^{-kt}}{(1+e^{-kt})^2}$$

on $(0,\infty)$. All tilted moments are finite, and $a_j(x) > 0$ for $j \ge 1$.

For $a_j(x) = (-1)^j f_k^{(j)}(x)$, the cited proposition lemma gives that

$$B_{k,n}(x) = \frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)}$$

is strictly increasing for every $k > 0$, $n \geq 0$, and $x > 0$. Since

$$R_{k,n}(x) = (-1)^{n+1} B_{k,n}(x),$$

we get the precise monotonicity law:

if n is odd, $R_{k,n}$ is strictly increasing on $(0, \infty)$; if n is even, $R_{k,n}$ is strictly decreasing on $(0, \infty)$.

This solves the Yin–Zhang Nielsen k-beta derivative-ratio open problem in the parity-refined form. □

Proposition 10.3 (CM Laplace moment ratio monotonicity)**.** *Let $a_j(x) = \int_0^\infty t^j e^{-xt}\, d\mu(t) > 0$ be tilted moments of a positive Laplace measure for the needed indices. Then, for every $n \geq 0$, $B_n(x) = a_{n+1}(x)/(a_n(x)a_{n+2}(x))$ is strictly increasing on its domain.*

Proof. For CM Laplace moment ratio monotonicity, this proof is the corresponding component of Proposition 10.2. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Corollary 10.4 (APP-0010: Nielsen k-beta derivative-ratio monotonicity)**.** *Let $k > 0$, let $n \geq 0$, and put $f_k(x) = x\beta_k(x)$. Then*

$$R_{k,n}(x) = \frac{f_k^{(n+1)}(x)}{f_k^{(n)}(x) f_k^{(n+2)}(x)} \tag{10.6}$$

is strictly increasing on $(0, \infty)$ when n is odd, and strictly decreasing on $(0, \infty)$ when n is even.

Proof. We use Proposition 10.3 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let μ be a positive Borel measure on $[0, \infty)$ with finite moments after the tilt e^{-xt}. Define

$$a_j(x) = \int_0^\infty t^j e^{-xt}\, d\mu(t) > 0$$

for the needed indices. For fixed $n \geq 0$, set

$$B_n(x) = \frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)}.$$

Then B_n is strictly increasing whenever $a_{n+1}(x) > 0$.

Indeed $a_j'(x) = -a_{j+1}(x)$. Put

$$r_j(x) = \frac{a_{j+1}(x)}{a_j(x)}.$$

Moment log-convexity, equivalently Cauchy–Schwarz applied to

$$\int t^j e^{-xt}\, d\mu(t),$$

gives $r_{j+1}(x) \geq r_j(x)$. Differentiating,

$$\frac{d}{dx}\log B_n(x) = -\frac{a_{n+2}}{a_{n+1}} + \frac{a_{n+1}}{a_n} + \frac{a_{n+3}}{a_{n+2}} = r_n(x) - r_{n+1}(x) + r_{n+2}(x).$$

Since $r_{n+2}(x) \geq r_{n+1}(x)$ and $r_n(x) > 0$, the logarithmic derivative is positive. Hence B_n is strictly increasing.

Let

$$f_k(x) = x\beta_k(x).$$

From the integral representation of β_k, integration by parts gives

$$f_k(x) = x\int_0^\infty e^{-xt}\frac{dt}{1+e^{-kt}} = \frac{1}{2} + \int_0^\infty e^{-xt}\frac{ke^{-kt}}{(1+e^{-kt})^2}\,dt.$$

Thus f_k is the Laplace transform of a positive measure consisting of an atom $1/2$ at 0 plus the positive density

$$w_k(t) = \frac{ke^{-kt}}{(1+e^{-kt})^2}$$

on $(0,\infty)$. All tilted moments are finite, and $a_j(x) > 0$ for $j \geq 1$.

For $a_j(x) = (-1)^j f_k^{(j)}(x)$, the cited proposition lemma gives that

$$B_{k,n}(x) = \frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)}$$

is strictly increasing for every $k > 0$, $n \geq 0$, and $x > 0$. Since

$$R_{k,n}(x) = (-1)^{n+1}B_{k,n}(x),$$

we get the precise monotonicity law:

if n is odd, $R_{k,n}$ is strictly increasing on $(0,\infty)$; if n is even, $R_{k,n}$ is strictly decreasing on $(0,\infty)$.

This solves the Yin–Zhang Nielsen k-beta derivative-ratio open problem in the parity-refined form. □

11 Endpoint and parameter applications

The next three applications add one certified endpoint result and two parameter-monotonicity results to the complete-monotonicity layer. Only fully solved source questions are labelled.

11.1 Exact $n = 2$ beta-window endpoint

For $s > 1$, write

$$Z_s(a) = \sum_{m=0}^{\infty}(m+a)^{-s}. \tag{11.1}$$

In Qi–Lim–Nantomah's notation [4], put

$$C_2(x) = \psi''(x) + x\psi^{(3)}(x), \qquad P_2(x) = \psi''(x)\psi''(1/x), \tag{11.2}$$

and define

$$\mathcal{I}_2 = \{\beta \in \mathbb{R} :\ x^\beta C_2(x) - P_2(x) < 0 \text{ for all } x > 0\}. \tag{11.3}$$

The remaining endpoint is controlled on $0 < x < 1$ by

$$R(x) = \frac{P_2(x)}{C_2(x)} = \frac{2Z_3(x)Z_3(1/x)}{3xZ_4(x) - Z_3(x)}, \qquad Q_2(x) = \frac{\log R(x)}{\log x}. \tag{11.4}$$

Definition 11.1 (Rational certificates and finite covers)**.** A rational certificate is a sign or enclosure proof reduced to explicit rational inequalities. A finite cover certificate partitions a compact interval into finitely many rational subintervals and proves the required inequality on each piece, often using an analytic-denominator reduction, a Taylor model, or derivative bounds.

Proposition 11.1 (Q2 analytic denominator reduced assembly terminal certificate)**.** *The terminal endpoint follows from a reduced analytic-denominator assembly: the true covers* $(0, 9/50]$ *and* $[9/10, 1)$*, the endpoint signs on* J*, an analytic-denominator or direct-log finite-middle cover of*

$$\left[\frac{9}{50}, \frac{1409}{5000}\right] \cup \left[\frac{293}{1000}, \frac{9}{10}\right]$$

, and a compact one-crossing/dominance certificate on $[1409/5000, 293/1000] \setminus J$*. This determines* $L_2 = Q_2(\xi)$ *and* $\mathcal{I}_2 = (Q_2(\xi), 3]$.

Proof. The replayable certificate is:

It proves

$$Q_2(x) < q_J$$

on

$$\left[\frac{9}{50}, \frac{1409}{5000}\right] \cup \left[\frac{293}{1000}, \frac{9}{10}\right].$$

The left bridge $[9/50, 221050/1000000]$ uses the already-derived identity

$$3xZ_4(x) - Z_3(x) = 2x^{-3} + \sum_{k=1}^{\infty} \frac{2x - k}{(x+k)^4}.$$

For $x < 1/2$, this gives

$$R(x) > Z_3(1/x).$$

The certificate proves

$$Z_3(1/x) > x^{1157/500}$$

on $[9/50, 221050/1000000]$. Since

$$\frac{1157}{500} < \frac{115727}{50000} < q_J,$$

this gives $Q_2(x) < q_J$ on the cited proposition.

The remaining two subintervals use the direct comparison

$$\log R(x) > \frac{115727}{50000} \log x.$$

Since $\log x < 0$, this implies

$$Q_2(x) < \frac{115727}{50000} < q_J.$$

The script uses power-of-two range reduction for the atanh logarithm enclosure:

$$\log r = \log(2^k r) - k \log 2,$$

so every logarithm is evaluated with an argument in $[1, 2)$.

Together with the earlier true covers $(0, 9/50]$ and $[9/10, 1)$, this covers all outside-compact points below q_J.

The replayable certificate is:

Let

$$H(x) = \Lambda(x) + x\Lambda'(x).$$

Using $Z_s'(x) = -sZ_{s+1}(x)$ and

$$\frac{d}{dx} Z_s(1/x) = sx^{-2}Z_{s+1}(1/x),$$

the script builds a rational interval enclosure for H using $Z_3, \ldots, Z_6$ at x and $Z_3, \ldots, Z_5$ at $1/x$. It proves

$$H(x) > 0 \qquad \left(\frac{1409}{5000} \leq x \leq \frac{293}{1000}\right).$$

Since

$$G'(x) = \log x\, H(x)$$

and $\log x < 0$ on the compact bracket, this gives

$$G'(x) < 0.$$

The existing endpoint certificate proves

$$G(287345/1000000) > 0, \qquad G(287346/1000000) < 0.$$

Therefore G has a unique zero

$$\xi \in J = \left[\frac{287345}{1000000}, \frac{287346}{1000000}\right],$$

with $G > 0$ to the left of J and $G < 0$ to the right of J on the compact bracket.

Because

$$Q_2'(x) = \frac{G(x)}{x(\log x)^2},$$

Q_2 is increasing before ξ and decreasing after ξ on the compact bracket.

The true cover package is now:

$$(0, 9/50] \cup \left[9/50, \frac{1409}{5000}\right] \cup \left[\frac{293}{1000}, 9/10\right] \cup [9/10, 1).$$

All points outside the compact bracket have $Q_2(x) < q_J$, and $q_J < Q_2(\xi)$ by the certified inner witness. On the compact bracket, Q_2 has a unique maximum at the unique zero $\xi \in J$. Thus

$$L_2 = Q_2(\xi).$$

At $\beta = Q_2(\xi)$, the defining inequality has equality at $x = \xi$, so the lower endpoint is excluded. The known upper endpoint remains included. Therefore

$$\mathcal{I}_2 = (Q_2(\xi), 3].$$

This proves the terminal exact endpoint node. □

Proposition 11.2 (Q2 compact taylor monotonicity J terminal certificate)**.** *Using the true endpoint-sign certificate on* $J = [287345/1000000, 287346/1000000]$*, there is a Taylor-model or derivative-bounded rational certificate proving* $G > 0$ *on* $[1409/5000, 287345/1000000]$*,* $G < 0$ *on* $[287346/1000000, 293/1000]$*, and dominance below the witness* q_J*; combined with finite-middle covers outside the compact bracket and the true endpoint outside covers, this determines* $L_2 = Q_2(\xi)$ *and* $\mathcal{I}_2 = (Q_2(\xi), 3]$*.*

Proof. This proof is the component of Proposition 11.1 corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Proposition 11.3 (Q2 remaining finite middle log cover terminal certificate)**.** *Using the true outside covers* $(0, 9/50]$ *and* $[9/10, 1)$*, the true endpoint signs on* J*, and the true enclosure lemmas, there is a finite rational direct-log cover of*

$$\left[\frac{9}{50}, \frac{1409}{5000}\right] \cup \left[\frac{293}{1000}, \frac{9}{10}\right]$$

, proving $Q_2(x) < q_J$ *on every interval, together with a compact one-crossing/dominance certificate on* $[1409/5000, 293/1000] \setminus J$*. This determines* $L_2 = Q_2(\xi)$ *and* $\mathcal{I}_2 = (Q_2(\xi), 3]$*.*

Proof. For Q2 remaining finite middle log cover terminal certificate, this proof is the corresponding component of Proposition 11.2. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Corollary 11.4 (APP-0011: exact $n = 2$ beta-window endpoint)**.** *Let*

$$J = \left[\frac{287345}{1000000}, \frac{287346}{1000000}\right]. \tag{11.5}$$

There is a unique $\xi \in J$ *at which* Q_2 *attains its global supremum on* $(0, 1)$*. Hence, with* $L_2 = Q_2(\xi)$*,*

$$\mathcal{I}_2 = (L_2, 3]. \tag{11.6}$$

Proof. We use Proposition 11.1, Proposition 11.2, and Proposition 11.3 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

This script proves a true auxiliary finite-middle outside-cover node:

The replayable certificate is:

It proves

$$Q_2(x) < q_J$$

on

$$\left[\frac{9}{50}, \frac{1409}{5000}\right] \cup \left[\frac{293}{1000}, \frac{9}{10}\right].$$

The left bridge $[9/50, 221050/1000000]$ uses the already-derived identity

$$3xZ_4(x) - Z_3(x) = 2x^{-3} + \sum_{k=1}^{\infty} \frac{2x-k}{(x+k)^4}.$$

For $x < 1/2$, this gives

$$R(x) > Z_3(1/x).$$

The certificate proves

$$Z_3(1/x) > x^{1157/500}$$

on $[9/50, 221050/1000000]$. Since

$$\frac{1157}{500} < \frac{115727}{50000} < q_J,$$

this gives $Q_2(x) < q_J$ on the cited proposition.

The remaining two subintervals use the direct comparison

$$\log R(x) > \frac{115727}{50000} \log x.$$

Since $\log x < 0$, this implies

$$Q_2(x) < \frac{115727}{50000} < q_J.$$

The script uses power-of-two range reduction for the atanh logarithm enclosure:

$$\log r = \log(2^k r) - k \log 2,$$

so every logarithm is evaluated with an argument in $[1, 2)$.

Together with the earlier true covers $(0, 9/50]$ and $[9/10, 1)$, this covers all outside-compact points below q_J.

This certificate also proves:

the Q2 compact monotonicity J certificate.

The replayable certificate is:

Let

$$H(x) = \Lambda(x) + x\Lambda'(x).$$

Using $Z_s'(x) = -sZ_{s+1}(x)$ and

$$\frac{d}{dx} Z_s(1/x) = sx^{-2} Z_{s+1}(1/x),$$

the script builds a rational interval enclosure for H using $Z_3, \ldots, Z_6$ at x and $Z_3, \ldots, Z_5$ at $1/x$. It proves

$$H(x) > 0 \qquad \left(\frac{1409}{5000} \le x \le \frac{293}{1000}\right).$$

Since

$$G'(x) = \log x \, H(x)$$

and $\log x < 0$ on the compact bracket, this gives

$$G'(x) < 0.$$

The existing endpoint certificate proves

$$G(287345/1000000) > 0, \qquad G(287346/1000000) < 0.$$

Therefore G has a unique zero

$$\xi \in J = \left[\frac{287345}{1000000}, \frac{287346}{1000000}\right],$$

with $G > 0$ to the left of J and $G < 0$ to the right of J on the compact bracket.

Because

$$Q_2'(x) = \frac{G(x)}{x(\log x)^2},$$

Q_2 is increasing before ξ and decreasing after ξ on the compact bracket.

The true cover package is now:

$$(0, 9/50] \cup \left[9/50, \frac{1409}{5000}\right] \cup \left[\frac{293}{1000}, 9/10\right] \cup [9/10, 1).$$

All points outside the compact bracket have $Q_2(x) < q_J$, and $q_J < Q_2(\xi)$ by the certified inner witness. On the compact bracket, Q_2 has a unique maximum at the unique zero $\xi \in J$. Thus

$$L_2 = Q_2(\xi).$$

At $\beta = Q_2(\xi)$, the defining inequality has equality at $x = \xi$, so the lower endpoint is excluded. The known upper endpoint remains included. Therefore

$$\mathcal{I}_2 = (Q_2(\xi), 3].$$

This proves the terminal exact endpoint node. $\square$

11.2 Baricz V_q parameter log-convexity

Baricz records the question whether, for fixed $x > 0$, the function

$$V_q(x) = \frac{2e^{x^2}}{\Gamma(q+1)} \int_x^\infty e^{-t^2}(t^2 - x^2)^q \, dt, \qquad q > -1, \tag{11.7}$$

is log-convex as a function of q [9].

Proposition 11.5 (Baricz Vq strict q logconvexity)**.** *For every fixed $x > 0$, the function $q \mapsto V_q(x)$ is strictly log-convex on $(-1, \infty)$.*

Proof. This proof is the component of Proposition 11.6 corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Proposition 11.6 (Baricz Vq q Laplace normal form)**.** *For every $x > 0$ and $q > -1$, Baricz's function satisfies $V_q(x) = \pi^{-1/2} \int_0^\infty s^{-1/2} e^{-x^2 s}(1+s)^{-(q+1)}\, ds$.*

Proof. Put $u = t^2 - x^2$. Then

$$V_q(x) = \frac{1}{\Gamma(q+1)} \int_0^\infty e^{-u} u^q (u+x^2)^{-1/2}\, du. \tag{11.8}$$

Using

$$r^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-rs}\, ds$$

and Tonelli's theorem, we obtain

$$V_q(x) = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-x^2 s}(1+s)^{-(q+1)}\, ds. \tag{11.9}$$

Let $a = q+1 > 0$. Differentiation under the integral gives

$$(-1)^n \frac{d^n}{da^n} V_{a-1}(x) = \frac{1}{\sqrt{\pi}} \int_0^\infty (\log(1+s))^n s^{-1/2} e^{-x^2 s}(1+s)^{-a}\, ds \geq 0. \tag{11.10}$$

Equivalently, after the change of variables $y = \log(1+s)$, $V_{a-1}(x)$ is a positive Laplace transform in a. The representing measure is not concentrated at a single point, because the density in (11.9) is positive on $s > 0$. Strict Holder inequality for two different exponents therefore gives strict log-convexity in a, and the shift $a = q+1$ gives the stated strict log-convexity in q. $\square$

Corollary 11.7 (APP-0012: Baricz V_q strict q-log-convexity)**.** *For each fixed $x > 0$, the map $q \mapsto V_q(x)$ is strictly log-convex on $(-1, \infty)$.*

Proof. We use Proposition 11.6 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Put $u = t^2 - x^2$. Then $t = (u+x^2)^{1/2}$ and

$$dt = \frac{du}{2(u+x^2)^{1/2}}.$$

Since $e^{x^2} e^{-t^2} = e^{-u}$, the factor 2 in the definition cancels the 1/2 from dt, giving

$$V_q(x) = \frac{1}{\Gamma(q+1)} \int_0^\infty e^{-u} u^q (u+x^2)^{-1/2}\, du.$$

This integral is finite for $q > -1$: near 0 the integrand is $O(u^q)$, and at infinity it is exponentially decaying.

For $r > 0$,

$$r^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-rs}\, ds.$$

With $r = u + x^2$, Tonelli's theorem applies because the integrand is nonnegative. Thus

$$\begin{aligned} V_q(x) &= \frac{1}{\Gamma(q+1)\sqrt{\pi}} \int_0^\infty \int_0^\infty e^{-u} u^q s^{-1/2} e^{-(u+x^2)s}\, ds\, du \\ &= \frac{1}{\Gamma(q+1)\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-x^2 s} \left(\int_0^\infty u^q e^{-(1+s)u}\, du \right) ds \\ &= \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-x^2 s}(1+s)^{-(q+1)}\, ds. \end{aligned}$$

The last step uses

$$\int_0^\infty u^q e^{-(1+s)u}\,du = \frac{\Gamma(q+1)}{(1+s)^{q+1}}.$$

Therefore the Baricz Vq q Laplace normal form is true.

Let $a = q+1 > 0$. The normal form becomes

$$V_{a-1}(x) = \frac{1}{\sqrt{\pi}}\int_0^\infty s^{-1/2}e^{-x^2 s}(1+s)^{-a}\,ds.$$

For each integer $n \geq 0$, differentiation under the integral gives

$$(-1)^n \frac{d^n}{da^n} V_{a-1}(x) = \frac{1}{\sqrt{\pi}}\int_0^\infty (\log(1+s))^n s^{-1/2} e^{-x^2 s}(1+s)^{-a}\,ds \geq 0.$$

The differentiation is justified locally uniformly in $a > 0$: near $s = 0$, $(\log(1+s))^n s^{-1/2} = O(s^{n-1/2})$, and as $s \to \infty$, the factor $e^{-x^2 s}$ dominates every logarithmic and polynomial factor.

Thus $a \mapsto V_{a-1}(x)$ is completely monotone on $(0,\infty)$. Equivalently, after the change of variables $y = \log(1+s)$, it is a positive Laplace transform in a.

Therefore the Baricz Vq q complete monotonicity is true.

For $a > 0$, write

$$F_x(a) = V_{a-1}(x) = \int_{(0,\infty)} e^{-ay}\,d\nu_x(y),$$

where ν_x is the positive pushforward of

$$\frac{1}{\sqrt{\pi}} s^{-1/2} e^{-x^2 s}\,ds$$

under $y = \log(1+s)$. This measure is not concentrated at one point because it has positive density on $s > 0$, hence on $y > 0$.

For $a_1 \neq a_2$ and $0 < \lambda < 1$, strict Holder inequality gives

$$F_x(\lambda a_1 + (1-\lambda)a_2) < F_x(a_1)^\lambda F_x(a_2)^{1-\lambda}.$$

Strictness holds because $e^{-a_1 y}$ and $e^{-a_2 y}$ are not proportional ν_x-almost everywhere unless ν_x is supported at a single y.

Thus $a \mapsto F_x(a)$ is strictly log-convex on $(0,\infty)$. Since $a = q+1$, the shift preserves strict log-convexity, so $q \mapsto V_q(x)$ is strictly log-convex on $(-1,\infty)$ for every fixed $x > 0$.

Therefore the Baricz Vq strict q logconvexity is true, and it implies the source node the Baricz Vq q logconvexity open problem. ☐

11.3 Yin (p,k)-digamma sharp necessity

Yin proved sufficiency of $\alpha \leq 1$ for complete monotonicity of

$$\delta_{p,k,\alpha}(x) = x^\alpha\left[\frac{1}{k}\log\frac{pkx}{x+k(p+1)} - \psi_{p,k}(x)\right], \qquad p \in \mathbb{N},\ k > 0, \tag{11.11}$$

and asked whether this condition is also necessary [10].

Proposition 11.8 (Yin pk digamma alpha necessity)**.** *For $p \in \mathbb{N}$, $k > 0$, if $x^\alpha\left[\frac{1}{k}\log\frac{pkx}{x+k(p+1)} - \psi_{p,k}(x)\right]$ is completely monotone on $(0,\infty)$, then $\alpha \leq 1$.*

Proof. *Setup.* Fix $p \in \mathbb{N}$ and $k > 0$. Define

$$B_{p,k}(x) = \frac{1}{k} \log \frac{pkx}{x + k(p+1)} - \psi_{p,k}(x), \qquad \delta_{p,k,\alpha}(x) = x^{\alpha} B_{p,k}(x).$$

The source gives the finite formula

$$\psi_{p,k}(x) = \frac{1}{k} \log(pk) - \sum_{n=0}^{p} \frac{1}{x + nk}.$$

Therefore

$$\begin{aligned} B_{p,k}(x) &= \frac{1}{k} \log \frac{pkx}{x + k(p+1)} - \frac{1}{k} \log(pk) + \sum_{n=0}^{p} \frac{1}{x + nk} \\ &= \frac{1}{k} \log \frac{x}{x + k(p+1)} + \sum_{n=0}^{p} \frac{1}{x + nk}. \end{aligned}$$

This normal form also gives positivity directly. Put $t = x/k$. Then

$$B_{p,k}(x) = \frac{1}{k} \left[\sum_{n=0}^{p} \frac{1}{t+n} + \log \frac{t}{t+p+1} \right].$$

Since

$$\log \frac{t+p+1}{t} = \int_0^{p+1} \frac{du}{t+u},$$

and $u \mapsto (t+u)^{-1}$ is strictly decreasing, the left Riemann sum dominates:

$$\sum_{n=0}^{p} \frac{1}{t+n} > \int_0^{p+1} \frac{du}{t+u}.$$

Hence $B_{p,k}(x) > 0$ for $x > 0$.

This proves the claim.

From the finite normal form,

$$B_{p,k}(x) = \frac{1}{x} + \frac{1}{k} \log \frac{x}{x + k(p+1)} + \sum_{n=1}^{p} \frac{1}{x + nk}.$$

As $x \to 0^+$, the finite sum over $n \geq 1$ is $O(1)$, and

$$\frac{1}{k} \log \frac{x}{x + k(p+1)} = O(|\log x|).$$

Thus

$$B_{p,k}(x) = \frac{1}{x} + O(|\log x|) \qquad (x \to 0^+).$$

Equivalently,

$$x B_{p,k}(x) \to 1.$$

Therefore

$$\delta_{p,k,\alpha}(x) = x^{\alpha} B_{p,k}(x) = x^{\alpha-1}(x B_{p,k}(x)) \sim x^{\alpha-1}.$$

This proves the claim.

Assume that $\delta_{p,k,\alpha}$ is completely monotone on $(0,\infty)$. Then

$$\delta_{p,k,\alpha}(x) \geq 0, \qquad \delta'_{p,k,\alpha}(x) \leq 0.$$

Since $B_{p,k}(x) > 0$, the function $\delta_{p,k,\alpha}$ is actually positive for every $x > 0$.

Suppose, for contradiction, that $\alpha > 1$. The endpoint asymptotic gives

$$\lim_{x \to 0^+} \delta_{p,k,\alpha}(x) = 0.$$

Fix $x_0 > 0$. Positivity gives $\delta_{p,k,\alpha}(x_0) > 0$. Because $\delta_{p,k,\alpha}(x) \to 0$ as $x \to 0^+$, choose $0 < y < x_0$ with

$$\delta_{p,k,\alpha}(y) < \frac{1}{2}\delta_{p,k,\alpha}(x_0).$$

But $\delta'_{p,k,\alpha} \leq 0$ means the function is nonincreasing as x increases, hence for $0 < y < x_0$,

$$\delta_{p,k,\alpha}(y) \geq \delta_{p,k,\alpha}(x_0),$$

a contradiction.

Therefore $\alpha \leq 1$.

This proves the claim. □

Corollary 11.9 (APP-0013: Yin (p,k)-digamma sharp α-necessity)**.** *If $\delta_{p,k,\alpha}$ is completely monotone on $(0,\infty)$, then $\alpha \leq 1$. Combined with Yin's sufficiency theorem, the sharp condition is $\alpha \leq 1$.*

Proof. We use Proposition 11.8 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Setup. Fix $p \in \mathbb{N}$ and $k > 0$. Define

$$B_{p,k}(x) = \frac{1}{k}\log\frac{pkx}{x+k(p+1)} - \psi_{p,k}(x), \qquad \delta_{p,k,\alpha}(x) = x^\alpha B_{p,k}(x).$$

The source gives the finite formula

$$\psi_{p,k}(x) = \frac{1}{k}\log(pk) - \sum_{n=0}^{p}\frac{1}{x+nk}.$$

Therefore

$$\begin{aligned} B_{p,k}(x) &= \frac{1}{k}\log\frac{pkx}{x+k(p+1)} - \frac{1}{k}\log(pk) + \sum_{n=0}^{p}\frac{1}{x+nk} \\ &= \frac{1}{k}\log\frac{x}{x+k(p+1)} + \sum_{n=0}^{p}\frac{1}{x+nk}. \end{aligned}$$

This normal form also gives positivity directly. Put $t = x/k$. Then

$$B_{p,k}(x) = \frac{1}{k}\left[\sum_{n=0}^{p}\frac{1}{t+n} + \log\frac{t}{t+p+1}\right].$$

Since

$$\log\frac{t+p+1}{t} = \int_0^{p+1}\frac{du}{t+u},$$

and $u \mapsto (t+u)^{-1}$ is strictly decreasing, the left Riemann sum dominates:

$$\sum_{n=0}^{p} \frac{1}{t+n} > \int_0^{p+1} \frac{du}{t+u}.$$

Hence $B_{p,k}(x) > 0$ for $x > 0$.

This yields $B_{p,k}(x) > 0$ for all $x > 0$, i.e. the finite-normal-form positivity claim needed for the endpoint obstruction.

From the finite normal form,

$$B_{p,k}(x) = \frac{1}{x} + \frac{1}{k} \log \frac{x}{x+k(p+1)} + \sum_{n=1}^{p} \frac{1}{x+nk}.$$

As $x \to 0^+$, the finite sum over $n \geq 1$ is $O(1)$, and

$$\frac{1}{k} \log \frac{x}{x+k(p+1)} = O(|\log x|).$$

Thus

$$B_{p,k}(x) = \frac{1}{x} + O(|\log x|) \qquad (x \to 0^+).$$

Equivalently,

$$xB_{p,k}(x) \to 1.$$

Therefore

$$\delta_{p,k,\alpha}(x) = x^{\alpha} B_{p,k}(x) = x^{\alpha-1}(xB_{p,k}(x)) \sim x^{\alpha-1}.$$

This yields the near-zero asymptotic $xB_{p,k}(x) \to 1$, i.e. $B_{p,k}(x) \sim x^{-1}$ as $x \to 0^+$.

Assume that $\delta_{p,k,\alpha}$ is completely monotone on $(0, \infty)$. Then

$$\delta_{p,k,\alpha}(x) \geq 0, \qquad \delta'_{p,k,\alpha}(x) \leq 0.$$

Since $B_{p,k}(x) > 0$, the function $\delta_{p,k,\alpha}$ is actually positive for every $x > 0$.

Suppose, for contradiction, that $\alpha > 1$. The endpoint asymptotic gives

$$\lim_{x \to 0^+} \delta_{p,k,\alpha}(x) = 0.$$

Fix $x_0 > 0$. Positivity gives $\delta_{p,k,\alpha}(x_0) > 0$. Because $\delta_{p,k,\alpha}(x) \to 0$ as $x \to 0^+$, choose $0 < y < x_0$ with

$$\delta_{p,k,\alpha}(y) < \frac{1}{2} \delta_{p,k,\alpha}(x_0).$$

But $\delta'_{p,k,\alpha} \leq 0$ means the function is nonincreasing as x increases, hence for $0 < y < x_0$,

$$\delta_{p,k,\alpha}(y) \geq \delta_{p,k,\alpha}(x_0),$$

a contradiction.

Therefore $\alpha \leq 1$.

This contradiction argument gives $\alpha \leq 1$, and thus the claimed open-problem conclusion follows.

$\square$

12 The Laplace-Transport Application Layer

The applications in this layer are organized by one theme. A problem is first converted into a positive-kernel statement, usually by a logarithmic second difference, a Frullani identity, a moment determinant, or a transform normalization. The conclusion then follows by transporting that kernel through a permanence operation: Stieltjes transformation, Bernstein integration, stable subordination, quotient compression, or triangular coefficient extraction. This section records representative public solves.

Proposition 12.1 (Ramanujan integral Stieltjes). *The Ramanujan integral $I_R(x) = \int_0^\infty e^{-xt}\,dt/[t(\pi^2+\log^2 t)]$ is a Stieltjes function on $(0,\infty)$.*

Proof. For $t > 0$,

$$\frac{1}{t(\pi^2+\log^2 t)} = \frac{1}{\pi(1+t)}\int_0^1 t^{-u}\sin(\pi u)\,du.$$

The right side is a positive superposition of completely monotone powers and is therefore completely monotone as a function of t. The Laplace transform of a completely monotone density is Stieltjes. Integrating the same density against $1-e^{-xt}$ gives a complete Bernstein function by the standard Stieltjes-density characterization of complete Bernstein functions. □

Proposition 12.2 (CM closure product positive mixture). *Completely monotone functions on $(0,\infty)$ are closed under pointwise products, nonnegative finite sums, and nonnegative parameter integrals whenever the integral is finite.*

Proof. Completely monotone functions on $(0,\infty)$ are closed under pointwise products, nonnegative finite sums, and nonnegative parameter integrals whenever the integral is finite.

If ϕ is completely monotone and

$$F(x) = \int_0^\infty e^{-xt}\phi(t)\,dt$$

is finite for every $x > 0$, then F is a Stieltjes function. Indeed, Bernstein's theorem gives

$$\phi(t) = \int_0^\infty e^{-ts}\,d\mu(s),$$

and Tonelli gives

$$F(x) = \int_0^\infty \frac{d\mu(s)}{x+s}.$$

If

$$f(x) = a + bx + \int_0^\infty (1-e^{-xt})m(t)\,dt$$

is a Bernstein representation and m is completely monotone, then f is a complete Bernstein function, subject to the usual Levy integrability condition. □

Corollary 12.3 (APP-0014: Ramanujan integral is Stieltjes). *Let*

$$I_R(x) = \int_0^\infty \frac{e^{-xt}}{t(\pi^2+\log^2 t)}\,dt, \qquad x > 0.$$

Then I_R is a Stieltjes function. Moreover every primitive of the form

$$\widetilde{I}_R(x) = a + \int_0^\infty \frac{1-e^{-xt}}{t(\pi^2+\log^2 t)}\,dt, \qquad a \geq 0,$$

is a complete Bernstein function.

Proof. We use Proposition 12.2 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The node the Ramanujan integral Stieltjes proves that $I_R(x) = \int_0^\infty e^{-xt}\,dt/[t(\pi^2 + \log^2 t)]$ is a Stieltjes function. The node the Ramanujan antiderivative complete Bernstein proves that the primitive $\widetilde{I}_R(x) = a + \int_0^\infty (1 - e^{-xt})\,dt/[t(\pi^2 + \log^2 t)]$ is a complete Bernstein function. These are exactly the two clauses of APP-0014, so the conjunction gives the source-facing application result. □

Proposition 12.4 (Prabhakar Q stable subordination normal form)**.** *For $0 < \alpha < 1$, $\gamma > 0$, $\beta > \alpha\gamma$, and $\lambda > 0$, let $P^\gamma_{\alpha,\beta}$ be Sibisi's transform-normalized Pollard measure satisfying $E^\gamma_{\alpha,\beta}(-u) = \int_0^\infty e^{-ur}\,dP^\gamma_{\alpha,\beta}(r)$. Then the measure $Q^\gamma_{\alpha,\beta}(A \mid \lambda) = \int \Pr((\lambda r)^{1/\alpha} S_\alpha \in A)\,dP^\gamma_{\alpha,\beta}(r)$ satisfies $E^\gamma_{\alpha,\beta}(-\lambda x^\alpha) = \int_0^\infty e^{-xt}\,dQ^\gamma_{\alpha,\beta}(t \mid \lambda)$.*

Proof. Conditioning on r and using the stable transform gives

$$\mathbb{E}e^{-x(\lambda r)^{1/\alpha} S_\alpha} = e^{-\lambda r x^\alpha}.$$

Integration against $P^\eta_{\alpha,\beta}$ gives the Pollard representation of $E^\eta_{\alpha,\beta}(-\lambda x^\alpha)$. The total mass is finite, equal to the value at $x = 0$, and no normalization to a probability measure is needed for the transform identity. □

Corollary 12.5 (APP-0015: Prabhakar Q-measure by stable subordination)**.** *Let $0 < \alpha < 1$, $\eta > 0$, $\beta > \alpha\eta$, and $\lambda > 0$. Let*

$$E^\eta_{\alpha,\beta}(z) = \sum_{m=0}^\infty \frac{(\eta)_m z^m}{m!\,\Gamma(\alpha m + \beta)}$$

be the Prabhakar Mittag-Leffler function. Let $P^\eta_{\alpha,\beta}$ be the transform-normalized Pollard measure characterized in the strict Pollard range by

$$\int_0^\infty e^{-xr}\,dP^\eta_{\alpha,\beta}(r) = E^\eta_{\alpha,\beta}(-x), \qquad x > 0.$$

Let S_α be the positive α-stable variable satisfying $\mathbb{E}e^{-sS_\alpha} = e^{-s^\alpha}$. Define the finite measure

$$Q^\eta_{\alpha,\beta}(A \mid \lambda) = \int \Pr\{(\lambda r)^{1/\alpha} S_\alpha \in A\}\,dP^\eta_{\alpha,\beta}(r).$$

Then

$$\int_0^\infty e^{-xt}\,dQ^\eta_{\alpha,\beta}(t \mid \lambda) = E^\eta_{\alpha,\beta}(-\lambda x^\alpha).$$

Proof. We use Proposition 12.4 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let P be a finite positive measure on $[0, \infty)$, let

$$F(u) = \int_0^\infty e^{-ur}\,dP(r),$$

let $0 < \alpha < 1$, $\lambda > 0$, and let S_α be positive α-stable:

$$\mathbb{E}e^{-xS_\alpha} = e^{-x^\alpha}.$$

Define a finite positive measure Q by

$$Q(A) = \int_{[0,\infty)} \Pr\{(\lambda r)^{1/\alpha} S_\alpha \in A\}\, dP(r).$$

Then Tonelli's theorem gives, for $x > 0$,

$$\int_0^\infty e^{-xt}\, dQ(t) = \int_{[0,\infty)} \mathbb{E}e^{-x(\lambda r)^{1/\alpha} S_\alpha}\, dP(r) = \int_{[0,\infty)} e^{-\lambda r x^\alpha}\, dP(r) = F(\lambda x^\alpha).$$

Apply the lemma to Sibisi's transform-normalized $P = P^\gamma_{\alpha,\beta}$ and

$$F(u) = E^\gamma_{\alpha,\beta}(-u).$$

For $0 < \alpha < 1$, $\gamma > 0$, $\beta > \alpha\gamma$, and $\lambda > 0$, set

$$Q^\gamma_{\alpha,\beta}(A \mid \lambda) = \int_{[0,\infty)} \Pr\{(\lambda r)^{1/\alpha} S_\alpha \in A\}\, dP^\gamma_{\alpha,\beta}(r).$$

Then

$$\int_0^\infty e^{-xt}\, dQ^\gamma_{\alpha,\beta}(t \mid \lambda) = E^\gamma_{\alpha,\beta}(-\lambda x^\alpha).$$

By uniqueness of finite Laplace transforms, this is Sibisi's $Q^\gamma_{\alpha,\beta}(\cdot \mid \lambda)$.

If S_α has density f_α, then the non-atomic density part is

$$q^\gamma_{\alpha,\beta}(t \mid \lambda) = \int_{(0,\infty)} (\lambda r)^{-1/\alpha} f_\alpha\left(\frac{t}{(\lambda r)^{1/\alpha}}\right) dP^\gamma_{\alpha,\beta}(r),$$

with an atom $P^\gamma_{\alpha,\beta}(\{0\})\delta_0$ if $P^\gamma_{\alpha,\beta}$ has an atom at zero. When $P^\gamma_{\alpha,\beta}$ is represented by Sibisi's Pollard density $p^\gamma_{\alpha,\beta}(r)$, this becomes

$$q^\gamma_{\alpha,\beta}(t \mid \lambda) = \int_0^\infty (\lambda r)^{-1/\alpha} f_\alpha\left(\frac{t}{(\lambda r)^{1/\alpha}}\right) p^\gamma_{\alpha,\beta}(r)\, dr.$$

Because

$$E^\gamma_{\alpha,\beta}(0) = \frac{1}{\Gamma(\beta)},$$

the representing measures $P^\gamma_{\alpha,\beta}$ and $Q^\gamma_{\alpha,\beta}(\cdot \mid \lambda)$ have total mass $1/\Gamma(\beta)$. They are finite positive measures, not generally probability measures. The normalized probability version is

$$\widehat{Q}^\gamma_{\alpha,\beta} = \Gamma(\beta) Q^\gamma_{\alpha,\beta}, \qquad \mathcal{L}\widehat{Q}^\gamma_{\alpha,\beta}(x) = \Gamma(\beta) E^\gamma_{\alpha,\beta}(-\lambda x^\alpha).$$

The boundary $\beta = \alpha\gamma$ is not included here because Sibisi's ordinary convolution density uses $1/\Gamma(\beta - \alpha\gamma)$. A limiting treatment is a separate open frontier. □

Proposition 12.6 (From Mills Laplace CM normal form). *For $t > 0$, the standard normal Mills ratio satisfies $r(t) = \int_0^\infty e^{-tu-u^2/2}\, du$. Hence r is completely monotone on $(0, \infty)$.*

Proof. The density defining M_L is a positive moment density. Its order-four Hankel-minor consequence gives

$$M_{L+4} M_L - 4M_{L+3} M_{L+1} + 3M^2_{L+2} > 0.$$

Using the recurrence $M_{n+1} = nM_{n-1} - tM_n$, this determinant inequality reduces to a quadratic inequality for m_L; solving the admissible positive root gives $m_L < U_L$. Iterating the same recurrence expresses M_L in the form

$$M_L = (-1)^L (P_L r - B_L),$$

and substituting the level-L quotient bound gives the displayed alternating upper and lower bounds for r. □

Corollary 12.7 (APP-0016: Mills-ratio bound at every derivative level)**.** *Let*

$$M_L(t) = \int_0^\infty u^L e^{-tu-u^2/2}\, du, \qquad L = 0, 1, 2, \dots.$$

Put $m_L(t) = M_{L+1}(t)/M_L(t)$ *and*

$$U_L(t) = \frac{\sqrt{t^2(t^2+4L+7)^2 + 4(L+1)(t^2+4L+8)(t^2+4L+6)} - t(t^2+4L+7)}{2(t^2+4L+8)}.$$

Then $m_L(t) < U_L(t)$ *for every* $L \geq 0$ *and* $t > 0$*. With*

$$P_0 = 1, \quad P_1 = t, \quad P_{n+1} = tP_n + nP_{n-1},$$

and

$$B_0 = 0, \quad B_1 = 1, \quad B_{n+1} = nB_{n-1} - tB_n,$$

the classical Mills ratio $r(t) = M_0(t)$ *satisfies*

$$r(t) > \frac{B_{L+1} - U_L B_L}{P_{L+1} + U_L P_L} \quad (L \geq 0 \text{ even}),$$

and

$$r(t) < \frac{U_L B_L - B_{L+1}}{P_{L+1} + U_L P_L} \quad (L \geq 1 \text{ odd}).$$

Proof. We use Proposition 12.6 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Set

$$m_L(t) = \frac{M_{L+1}(t)}{M_L(t)}.$$

The recurrence gives

$$\frac{M_{L+2}}{M_L} = L + 1 - tm_L,$$

$$\frac{M_{L+3}}{M_L} = (t^2 + L + 2)m_L - t(L+1),$$

and

$$\frac{M_{L+4}}{M_L} = (L+1)(t^2 + L + 3) - t(t^2 + 2L + 5)m_L.$$

Substitution into the determinant inequality and expansion give

$$(L+1)(t^2+4L+6) - t(t^2+4L+7)m_L - (t^2+4L+8)m_L^2 > 0.$$

Equivalently,

$$(t^2+4L+8)m_L^2 + t(t^2+4L+7)m_L - (L+1)(t^2+4L+6) < 0.$$

Since $m_L > 0$, the positive root yields the uniform strict bound

$$m_L(t) < U_L(t),$$

where

$$U_L(t) = \frac{\sqrt{t^2(t^2+4L+7)^2 + 4(L+1)(t^2+4L+8)(t^2+4L+6)} - t(t^2+4L+7)}{2(t^2+4L+8)}.$$

This is the determinant-to-bound extractor. It is a single theorem for every $L \ge 0$.

Define polynomials $P_n(t)$ by

$$P_0 = 1, \qquad P_1 = t, \qquad P_{n+1} = tP_n + nP_{n-1}.$$

Define $B_n(t)$ by

$$B_0 = 0, \qquad B_1 = 1, \qquad B_{n+1} = nB_{n-1} - tB_n.$$

Then the same recurrence gives

$$M_n(t) = (-1)^n P_n(t) r(t) + B_n(t).$$

The ratio bound $M_{L+1} < U_L M_L$ becomes

$$\left((-1)^{L+1} P_{L+1} - (-1)^L U_L P_L\right) r < U_L B_L - B_{L+1}.$$

Since $P_n(t) > 0$ for $t > 0$ and $P_{2j}(0) > 0$, the coefficient has sign $(-1)^{L+1}$. Thus the bounds alternate:

For even $L \ge 0$,

$$r(t) > \frac{B_{L+1}(t) - U_L(t)B_L(t)}{P_{L+1}(t) + U_L(t)P_L(t)}.$$

For odd $L \ge 1$,

$$r(t) < \frac{U_L(t)B_L(t) - B_{L+1}(t)}{P_{L+1}(t) + U_L(t)P_L(t)}.$$

These formulas are valid at $t = 0$ by direct evaluation of the recurrences and for $t > 0$ by the positivity of M_L. □

Proposition 12.8 (Baricz gamma quotient a2b3 not BF)**.** *For $a = 2$ and $b = 3$, the Baricz Gamma quotient $x \mapsto \Gamma(x)\Gamma(x-a+b)/(\Gamma(x-a)\Gamma(x+b))$ is not a Bernstein function on $(2, \infty)$.*

Proof. Take $a = 2$, $b = 3$, and set $y = x - 2$. The quotient becomes

$$g(y) = \frac{y(y+1)}{(y+3)(y+4)}.$$

A direct differentiation gives

$$g'''(y) = \frac{36(y^4 + 8y^3 + 12y^2 - 40y - 94)}{(y+3)^4(y+4)^4}.$$

Thus $g'''(1) < 0$. If g were Bernstein, then g' would be completely monotone and hence $g''' \ge 0$. The displayed value contradicts that necessary condition. □

Counterexample 12.9 (APP-0017: a gamma quotient need not be Bernstein)**.** The assertion that the gamma quotient

$$x \longmapsto \frac{\Gamma(x+a)\Gamma(x+b)}{\Gamma(x)\Gamma(x+a+b)}$$

is Bernstein for all positive a, b is false.

Proof. We use Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Take $a = 2$, $b = 3$, and set $y = x - 2 > 0$. Then by the Gamma recurrence,

$$\frac{\Gamma(x)\Gamma(x-a+b)}{\Gamma(x-a)\Gamma(x+b)} = \frac{\Gamma(x)\Gamma(x+1)}{\Gamma(x-2)\Gamma(x+3)} = \frac{(x-2)(x-1)}{(x+1)(x+2)} = \frac{y(y+1)}{(y+3)(y+4)}.$$

Let

$$g(y) = \frac{y(y+1)}{(y+3)(y+4)}, \qquad y > 0.$$

If g were a Bernstein function, then g' would be completely monotone. In particular $g''' \geq 0$ would be necessary.

Exact symbolic differentiation gives

$$g'(y) = \frac{6(y^2+4y+2)}{(y+3)^2(y+4)^2},$$

$$g''(y) = -\frac{12(y^3+6y^2+6y-10)}{(y+3)^3(y+4)^3},$$

and

$$g'''(y) = \frac{36(y^4+8y^3+12y^2-40y-94)}{(y+3)^4(y+4)^4}.$$

At $y = 1$,

$$g'''(1) = -\frac{1017}{40000} < 0.$$

Therefore g' is not completely monotone, g is not Bernstein, and the Baricz for-all-a, b Bernstein question has a negative answer. □

Proposition 12.10 (QG tau global supremum)**.** *Let $a_* > 0$ be the unique positive root of $e^{a_*} = 1 + a_* + a_*^2$. Then $\sup_{s\in\mathbb{N},t>0} \tau(s,t) = a_*/(1 + a_* + a_*^2) = 0.298425607525639\ldots$, and the supremum is not attained.*

Proof. Let $a_* > 0$ be the unique positive root of

$$e^{a_*} = 1 + a_* + a_*^2.$$

Then

$$\sup_{s\in\mathbb{N},\ t>0} \tau(s,t) = \frac{a_*}{1 + a_* + a_*^2} = 0.298425607525639\ldots.$$

The supremum is not attained.

For fixed $s \geq 1$, set

$$y = \frac{1}{t+1} \in (0,1).$$

Then
$$\tau_s(y) = \frac{1 - y - (1 + sy)(1 - y)^{s+1}}{sy}.$$
Direct differentiation gives
$$\tau_s'(y) = \frac{(1 - y)^s\{1 + sy + s(s + 1)y^2\} - 1}{sy^2}.$$
Let
$$F_s(y) = (1 - y)^s\{1 + sy + s(s + 1)y^2\}.$$
Then
$$F_s'(y) = \frac{s(s + 1)y(1 - y)^s\big((s + 2)y - 1\big)}{y - 1}.$$
Thus F_s increases on $(0, 1/(s + 2))$ and decreases on $(1/(s + 2), 1)$. Since $F_s(0) = 1$ and $F_s(1) = 0$, the equation $F_s(y) = 1$ has exactly one root $y_s \in (1/(s + 2), 1)$ apart from the endpoint 0, and τ_s has the unique maximum at y_s.

At the maximizing point,
$$(1 - y_s)^s\{1 + sy_s + s(s + 1)y_s^2\} = 1$$
and
$$m_s := \max_{t>0} \tau(s, t) = \frac{(s + 1)y_s(1 - y_s)}{1 + sy_s + s(s + 1)y_s^2}.$$

Treat s as a real parameter. The envelope derivative at the unique maximizer equals the partial derivative in s:
$$\frac{d}{ds}m(s) = \partial_s\tau_s(y_s).$$
Using $F_s(y_s) = 1$, its sign is the opposite of
$$H_s(y_s) = y_s\{1 + (s + 1)y_s\} + (1 + sy_s)\log(1 - y_s).$$
The root satisfies $y_s \geq 1/(s + 1)$. Indeed, for $s = 1$ equality holds, and for $s > 1$
$$F_s\left(\frac{1}{s + 1}\right) = \left(\frac{s}{s + 1}\right)^s \frac{3s + 1}{s + 1} > 1.$$
Equivalently, with $x = 1/s$, this is
$$(1 + x)^{1/x} < \frac{3 + x}{1 + x},$$
which follows because
$$\frac{d}{dx}\left[\log\frac{3 + x}{1 + x} - \frac{\log(1 + x)}{x}\right] < 0$$
on $(0, 1]$ and the expression vanishes at $x = 1$.

For $y \geq 1/(s + 1)$,
$$H_s'(y) = 1 + 2(s + 1)y + s\log(1 - y) - \frac{1 + sy}{1 - y} < \frac{y(1 - (s + 2)y)}{1 - y} < 0,$$
where $\log(1 - y) < -y$. Also
$$H_s\left(\frac{1}{s + 1}\right) = \frac{2}{s + 1} - \frac{2s + 1}{s + 1}\log\left(1 + \frac{1}{s}\right) < 0,$$

using $\log(1+u) > 2u/(2+u)$. Hence $H_s(y_s) < 0$, so $m(s)$ is strictly increasing.

Put $a_s = sy_s$. The critical equation becomes

$$\left(1-\frac{a_s}{s}\right)^s \left(1+a_s+\left(1+\frac{1}{s}\right)a_s^2\right) = 1.$$

The numbers a_s are bounded above by a_*. For $s > a_*$,

$$\log F_s(a_*/s) \le -a_* - \frac{a_*^2}{2s} + a_* + \frac{a_*^2}{se^{a_*}} < 0,$$

so $F_s(a_*/s) < 1$ and $a_s < a_*$; the remaining small s are immediate. Therefore $a_s \to a_*$, because every subsequential limit solves

$$e^{-a}(1+a+a^2) = 1.$$

The equation $e^a = 1+a+a^2$ has exactly one positive root after 0: the function $e^a - 1 - a - a^2$ first decreases and then, after its derivative has its unique positive zero, increases to $+\infty$.

Consequently

$$\lim_{s\to\infty} m_s = \frac{a_*}{1+a_*+a_*^2}.$$

Since $m(s)$ is strictly increasing and s is finite in the source domain, no finite (s,t) attains the limiting value. Thus the source's requested maximum should be read as a supremum.

$$a_* = 1.79328213290076\ldots, \qquad \frac{a_*}{1+a_*+a_*^2} = 0.298425607525639\ldots,$$

which is consistent with the source's observation $\tau_0 > 0.2980$. □

Counterexample 12.11 (APP-0018: exact supremum for the Qi-Guo tau threshold)**.** For $s \in \mathbb{N}$ and $t > 0$, define

$$\tau(s,t) = \frac{1}{s}\left[t - (t+s+1)\left(\frac{t}{t+1}\right)^{s+1}\right].$$

Let $a_* > 0$ be the unique positive solution of

$$e^{a_*} = 1 + a_* + a_*^2.$$

Then

$$\sup_{s\in\mathbb{N},\ t>0} \tau(s,t) = \frac{a_*}{1+a_*+a_*^2} = 0.298425607525639\ldots.$$

The supremum is not attained.

Proof. We use Proposition 12.10 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let $a_* > 0$ be the unique positive root of

$$e^{a_*} = 1 + a_* + a_*^2.$$

Then

$$\sup_{s\in\mathbb{N},\ t>0} \tau(s,t) = \frac{a_*}{1+a_*+a_*^2} = 0.298425607525639\ldots.$$

The supremum is not attained.

For fixed $s \geq 1$, set

$$y = \frac{1}{t+1} \in (0,1).$$

Then

$$\tau_s(y) = \frac{1 - y - (1+sy)(1-y)^{s+1}}{sy}.$$

Direct differentiation gives

$$\tau_s'(y) = \frac{(1-y)^s\{1 + sy + s(s+1)y^2\} - 1}{sy^2}.$$

Let

$$F_s(y) = (1-y)^s\{1 + sy + s(s+1)y^2\}.$$

Then

$$F_s'(y) = \frac{s(s+1)y(1-y)^s\big((s+2)y - 1\big)}{y-1}.$$

Thus F_s increases on $(0, 1/(s+2))$ and decreases on $(1/(s+2), 1)$. Since $F_s(0) = 1$ and $F_s(1) = 0$, the equation $F_s(y) = 1$ has exactly one root $y_s \in (1/(s+2), 1)$ apart from the endpoint 0, and τ_s has the unique maximum at y_s.

At the maximizing point,

$$(1-y_s)^s\{1 + sy_s + s(s+1)y_s^2\} = 1$$

and

$$m_s := \max_{t>0} \tau(s,t) = \frac{(s+1)y_s(1-y_s)}{1 + sy_s + s(s+1)y_s^2}.$$

Treat s as a real parameter. The envelope derivative at the unique maximizer equals the partial derivative in s:

$$\frac{d}{ds}m(s) = \partial_s \tau_s(y_s).$$

Using $F_s(y_s) = 1$, its sign is the opposite of

$$H_s(y_s) = y_s\{1 + (s+1)y_s\} + (1 + sy_s)\log(1-y_s).$$

The root satisfies $y_s \geq 1/(s+1)$. Indeed, for $s = 1$ equality holds, and for $s > 1$

$$F_s\left(\frac{1}{s+1}\right) = \left(\frac{s}{s+1}\right)^s \frac{3s+1}{s+1} > 1.$$

Equivalently, with $x = 1/s$, this is

$$(1+x)^{1/x} < \frac{3+x}{1+x},$$

which follows because

$$\frac{d}{dx}\left[\log\frac{3+x}{1+x} - \frac{\log(1+x)}{x}\right] < 0$$

on $(0,1]$ and the expression vanishes at $x = 1$.

For $y \geq 1/(s+1)$,

$$H_s'(y) = 1 + 2(s+1)y + s\log(1-y) - \frac{1+sy}{1-y} < \frac{y(1-(s+2)y)}{1-y} < 0,$$

where $\log(1-y) < -y$. Also

$$H_s\left(\frac{1}{s+1}\right) = \frac{2}{s+1} - \frac{2s+1}{s+1}\log\left(1+\frac{1}{s}\right) < 0,$$

using $\log(1+u) > 2u/(2+u)$. Hence $H_s(y_s) < 0$, so $m(s)$ is strictly increasing.

Put $a_s = sy_s$. The critical equation becomes

$$\left(1-\frac{a_s}{s}\right)^s \left(1 + a_s + \left(1+\frac{1}{s}\right)a_s^2\right) = 1.$$

The numbers a_s are bounded above by a_*. For $s > a_*$,

$$\log F_s(a_*/s) \le -a_* - \frac{a_*^2}{2s} + a_* + \frac{a_*^2}{se^{a_*}} < 0,$$

so $F_s(a_*/s) < 1$ and $a_s < a_*$; the remaining small s are immediate. Therefore $a_s \to a_*$, because every subsequential limit solves

$$e^{-a}(1+a+a^2) = 1.$$

The equation $e^a = 1+a+a^2$ has exactly one positive root after 0: the function $e^a - 1 - a - a^2$ first decreases and then, after its derivative has its unique positive zero, increases to $+\infty$.

Consequently

$$\lim_{s\to\infty} m_s = \frac{a_*}{1+a_*+a_*^2}.$$

Since $m(s)$ is strictly increasing and s is finite in the source domain, no finite (s,t) attains the limiting value. Thus the source's requested maximum should be read as a supremum.

$$a_* = 1.79328213290076\ldots, \qquad \frac{a_*}{1+a_*+a_*^2} = 0.298425607525639\ldots,$$

which is consistent with the source's observation $\tau_0 > 0.2980$. $\square$

Proposition 12.12 (Wakrim W symbol exact BF range)**.** *For $0 < \alpha < 1$ and $\beta \ge 0$, the Wakrim W-symbol $\Phi_{\alpha,\beta}(s) = s^\alpha(1+(1-\alpha)s^{\alpha-1})^{-\beta}$ is a Bernstein function on $(0,\infty)$ if and only if $0 \le \beta \le 1$.*

Proof. Set $a = 1-\alpha \in (0,1)$. Then

$$\Phi_{\alpha,\beta}(s) = s^{1-a}\left(1+as^{-a}\right)^{-\beta} = \frac{s^{1-a+a\beta}}{(s^a+a)^\beta}.$$

Write

$$p = 1 - a + a\beta = \alpha + (1-\alpha)\beta.$$

Differentiating gives

$$\Phi'_{\alpha,\beta}(s) = s^{p-1}(s^a+a)^{-\beta-1}\left[p(s^a+a) - \beta a s^a\right].$$

Since $p - \beta a = 1 - a = \alpha$,

$$\Phi'_{\alpha,\beta}(s) = s^{-a(1-\beta)}(s^a+a)^{-\beta-1}\left[\alpha s^a + a(\alpha + a\beta)\right].$$

With $y = s^a$, this becomes

$$\Phi'_{\alpha,\beta}(s) = H(s^a),$$

where

$$H(y) = y^{\beta-1}(y+a)^{-\beta}\left(\alpha + \frac{a^2\beta}{y+a}\right).$$

For $0 < \beta \le 1$, each factor in H is completely monotone on $(0,\infty)$: $y^{\beta-1} = y^{-(1-\beta)}$, $(y+a)^{-\beta}$, and the positive sum $\alpha + a^2\beta/(y+a)$. Therefore H is completely monotone by product and positive-sum closure.

The map $s \mapsto s^a$ is Bernstein for $0 < a < 1$. The standard composition theorem says that if f is completely monotone and g is Bernstein, then $f \circ g$ is completely monotone. Hence $\Phi'_{\alpha,\beta}$ is completely monotone, and $\Phi_{\alpha,\beta}$ is Bernstein for $0 < \beta \le 1$. The endpoint $\beta = 0$ is $\Phi_{\alpha,0}(s) = s^\alpha$, also Bernstein.

Combining this local positive range with Wakrim's source proof that $\Phi_{\alpha,\beta}$ is not Bernstein for $\beta > 1$, the exact range is

$$\Phi_{\alpha,\beta} \in BF \iff 0 \le \beta \le 1.$$

□

Counterexample 12.13 (APP-0019: exact Bernstein range for the W-symbol)**.** Let $0 < \alpha < 1$, $\beta \ge 0$, and

$$\Phi_{\alpha,\beta}(s) = s^\alpha \left(1 + (1-\alpha)s^{\alpha-1}\right)^{-\beta}.$$

Then $\Phi_{\alpha,\beta}$ is a Bernstein function if and only if

$$0 \le \beta \le 1.$$

Proof. We use Proposition 12.12 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Set $a = 1-\alpha \in (0,1)$. Then

$$\Phi_{\alpha,\beta}(s) = s^{1-a}\left(1 + as^{-a}\right)^{-\beta} = \frac{s^{1-a+a\beta}}{(s^a+a)^\beta}.$$

Write

$$p = 1 - a + a\beta = \alpha + (1-\alpha)\beta.$$

Differentiating gives

$$\Phi'_{\alpha,\beta}(s) = s^{p-1}(s^a+a)^{-\beta-1}\left[p(s^a+a) - \beta a s^a\right].$$

Since $p - \beta a = 1 - a = \alpha$,

$$\Phi'_{\alpha,\beta}(s) = s^{-a(1-\beta)}(s^a+a)^{-\beta-1}\left[\alpha s^a + a(\alpha + a\beta)\right].$$

With $y = s^a$, this becomes

$$\Phi'_{\alpha,\beta}(s) = H(s^a),$$

where

$$H(y) = y^{\beta-1}(y+a)^{-\beta}\left(\alpha + \frac{a^2\beta}{y+a}\right).$$

For $0 < \beta \le 1$, each factor in H is completely monotone on $(0,\infty)$: $y^{\beta-1} = y^{-(1-\beta)}$, $(y+a)^{-\beta}$, and the positive sum $\alpha + a^2\beta/(y+a)$. Therefore H is completely monotone by product and positive-sum closure.

The map $s \mapsto s^a$ is Bernstein for $0 < a < 1$. The standard composition theorem says that if f is completely monotone and g is Bernstein, then $f \circ g$ is completely monotone. Hence $\Phi'_{\alpha,\beta}$ is completely monotone, and $\Phi_{\alpha,\beta}$ is Bernstein for $0 < \beta \le 1$. The endpoint $\beta = 0$ is $\Phi_{\alpha,0}(s) = s^\alpha$, also Bernstein.

Combining this local positive range with Wakrim's source proof that $\Phi_{\alpha,\beta}$ is not Bernstein for $\beta > 1$, the exact range is

$$\Phi_{\alpha,\beta} \in BF \quad \Longleftrightarrow \quad 0 \le \beta \le 1.$$

□

Proposition 12.14 (Qi hlambda x4 source range not CM)**.** *For $h_\lambda(x) = \Psi(x) - (x^2 + \lambda x + 12)/(12x^4(x+1)^2)$, $x^4 h_\lambda(x)$ is not completely monotone for every $\lambda \le 0$, and $x^4(-h_\mu(x))$ is not completely monotone for every $\mu \ge 4$.*

Proof. Near zero,

$$h_\lambda(x) = -\frac{\lambda}{12x^3} + \left(\frac{\lambda}{6} + \frac{\pi^2}{3} - \frac{37}{12}\right)\frac{1}{x^2} + O(x^{-1}).$$

If $\lambda < 0$, then $x^4 h_\lambda(x) = (-\lambda/12)x + O(x^2)$, whose derivative is positive near zero. If $\lambda = 0$, the coefficient $\pi^2/3 - 37/12$ is positive, again giving a positive derivative for the degree-four transform near zero. Complete monotonicity would require the first derivative to be non-positive. The same small-x test applied to $-x^4 h_\mu$ excludes the claimed μ-range. Hence the degree-four conjecture fails. □

Counterexample 12.15 (APP-0020: Qi h_λ degree-four conjecture is false)**.** Let

$$\Psi(x) = [\psi'(x)]^2 + \psi''(x), \qquad h_\lambda(x) = \Psi(x) - \frac{x^2 + \lambda x + 12}{12x^4(x+1)^2}.$$

The conjecture that $\deg_x^{\mathrm{cm}} h_\lambda = \deg_x^{\mathrm{cm}}(-h_\mu) = 4$ exactly in the stated parameter ranges $\lambda \le 0$, $\mu \ge 4$, is false. Here $\deg_x^{\mathrm{cm}} g$ denotes the source convention: the supremal power r for which $x^r g(x)$ is completely monotone on $(0, \infty)$.

Proof. We use Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

A completely monotone function must be nonincreasing, so its first derivative cannot be positive at any point.

Use the recurrence expansions at $x = 0^+$:

$$\psi'(x) = \frac{1}{x^2} + \zeta(2) - 2\zeta(3)x + 3\zeta(4)x^2 + O(x^3),$$

and

$$\psi''(x) = -\frac{2}{x^3} - 2\zeta(3) + 6\zeta(4)x + O(x^2).$$

Therefore

$$\Psi(x) = \frac{1}{x^4} - \frac{2}{x^3} + \frac{\pi^2}{3x^2} - \frac{4\zeta(3)}{x} + O(1).$$

Also

$$\frac{x^2 + \lambda x + 12}{12x^4(1+x)^2} = \frac{1}{x^4} + \left(\frac{\lambda}{12} - 2\right)\frac{1}{x^3} + \left(\frac{37}{12} - \frac{\lambda}{6}\right)\frac{1}{x^2} + O(x^{-1}).$$

Subtracting gives

$$h_\lambda(x) = -\frac{\lambda}{12x^3} + \left(\frac{\lambda}{6} + \frac{\pi^2}{3} - \frac{37}{12}\right)\frac{1}{x^2} + O(x^{-1}).$$

If $\lambda < 0$, then

$$x^4 h_\lambda(x) = -\frac{\lambda}{12} x + O(x^2),$$

so

$$\frac{d}{dx}\big[x^4 h_\lambda(x)\big] = -\frac{\lambda}{12} + O(x) > 0$$

for all sufficiently small $x > 0$. Hence $x^4 h_\lambda$ is not completely monotone.

If $\lambda = 0$, then

$$x^4 h_0(x) = \left(\frac{\pi^2}{3} - \frac{37}{12}\right) x^2 + O(x^3).$$

Since $4\pi^2 > 37$, the leading coefficient is positive, and

$$\frac{d}{dx}\big[x^4 h_0(x)\big] = 2\left(\frac{\pi^2}{3} - \frac{37}{12}\right) x + O(x^2) > 0$$

for all sufficiently small $x > 0$. Hence $x^4 h_0$ is not completely monotone.

Similarly, for $\mu \geq 4$,

$$x^4(-h_\mu(x)) = \frac{\mu}{12} x + O(x^2),$$

so

$$\frac{d}{dx}\big[x^4(-h_\mu(x))\big] = \frac{\mu}{12} + O(x) > 0$$

for all sufficiently small $x > 0$. Hence $x^4(-h_\mu)$ is not completely monotone for every $\mu \geq 4$.

Thus the degree-four transforms required by Qi's conjecture are not completely monotone on the conjectured source ranges. The conjecture in Remark 7.4 is false. □

Proposition 12.16 (Szabo psi difference alpha0 exact one)**.** *For $0 < d < 1$, the sharp cutoff for complete monotonicity of $y^\alpha[\psi(y+d) - \psi(y) - d/y]$ is $\alpha_0 = 1$.*

Proof. Let $0 < d < 1$, $y > 0$, and

$$H_d(y) = \psi(y+d) - \psi(y) - \frac{d}{y}.$$

Then

$$y^\alpha H_d(y)$$

is strictly completely monotone on $(0, \infty)$ if $\alpha \leq 1$, and is not completely monotone if $\alpha > 1$. Hence the sharp cutoff in Szabo Open Problem 1.5 is

$$\alpha_0 = 1.$$

As $y \downarrow 0$, the digamma Laurent expansion gives

$$\psi(y) = -\frac{1}{y} - \gamma + O(y), \qquad \psi(y+d) = \psi(d) + O(y).$$

Therefore

$$H_d(y) = \psi(y+d) - \psi(y) - \frac{d}{y} = \frac{1-d}{y} + O(1).$$

Since $0 < d < 1$, the leading coefficient is positive. Hence

$$y^\alpha H_d(y) = (1-d) y^{\alpha-1} + O(y^\alpha).$$

If $\alpha > 1$, then

$$\frac{d}{dy}\left(y^{\alpha} H_d(y)\right) = (1-d)(\alpha-1)y^{\alpha-2} + O(y^{\alpha-1}) > 0$$

for all sufficiently small $y > 0$. A completely monotone function must be nonincreasing, so no $\alpha > 1$ can be completely monotone.

Combining this with the source-imported strict complete monotonicity for $\alpha \le 1$ proves $\alpha_0 = 1$.

Szabo's original function is recovered by $y = x + a$ and $d = b - a$, where $a \ge 0$, $b > 0$, and $0 < b - a < 1$. Translation from $(-a, \infty)$ to $(0, \infty)$ preserves complete-monotonicity sign tests. □

Corollary 12.17 (APP-0021: Szabo's cutoff is $\alpha_0 = 1$)**.** *For $0 < d < 1$, define*

$$H_d(y) = \psi(y+d) - \psi(y) - \frac{d}{y}.$$

The exact cutoff for complete monotonicity of $y^{\alpha} H_d(y)$ is

$$\alpha_0 = 1.$$

Proof. We use Proposition 12.16 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let $0 < d < 1$, $y > 0$, and

$$H_d(y) = \psi(y+d) - \psi(y) - \frac{d}{y}.$$

Then

$$y^{\alpha} H_d(y)$$

is strictly completely monotone on $(0, \infty)$ if $\alpha \le 1$, and is not completely monotone if $\alpha > 1$. Hence the sharp cutoff in Szabo Open Problem 1.5 is

$$\alpha_0 = 1.$$

As $y \downarrow 0$, the digamma Laurent expansion gives

$$\psi(y) = -\frac{1}{y} - \gamma + O(y), \qquad \psi(y+d) = \psi(d) + O(y).$$

Therefore

$$H_d(y) = \psi(y+d) - \psi(y) - \frac{d}{y} = \frac{1-d}{y} + O(1).$$

Since $0 < d < 1$, the leading coefficient is positive. Hence

$$y^{\alpha} H_d(y) = (1-d)y^{\alpha-1} + O(y^{\alpha}).$$

If $\alpha > 1$, then

$$\frac{d}{dy}\left(y^{\alpha} H_d(y)\right) = (1-d)(\alpha-1)y^{\alpha-2} + O(y^{\alpha-1}) > 0$$

for all sufficiently small $y > 0$. A completely monotone function must be nonincreasing, so no $\alpha > 1$ can be completely monotone.

Combining this with the source-imported strict complete monotonicity for $\alpha \le 1$ proves $\alpha_0 = 1$.

Szabo's original function is recovered by $y = x + a$ and $d = b - a$, where $a \ge 0$, $b > 0$, and $0 < b - a < 1$. Translation from $(-a, \infty)$ to $(0, \infty)$ preserves complete-monotonicity sign tests. □

Proposition 12.18 (Chen Choi Gurland logF strict CM). *For $F(x) = \Gamma(1/x)\Gamma(3/x)/\Gamma(2/x)^2$, the function $x \mapsto \log F(x)$ is strictly completely monotone on $(0, \infty)$.*

Proof. This proof is the component of Proposition 12.19 corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Proposition 12.19 (Chen Choi Gurland Euler sum normal form). *For $F(x) = \Gamma(1/x)\Gamma(3/x)/\Gamma(2/x)^2$, $\log F(x) = \log(4/3) + \sum_{m=1}^{\infty} A(mx)$, with locally uniform convergence of all derivative series on compact subintervals of $(0, \infty)$.*

Proof. Set

$$A(u) = 2\log(u + 2) - \log(u + 1) - \log(u + 3).$$

The identity

$$A(u) = \int_0^\infty e^{-ut} \frac{e^{-t}(1 - e^{-t})^2}{t} \, dt$$

shows that A is strictly completely monotone. The Weierstrass product for Γ gives

$$\log F(x) = \log \frac{4}{3} + \sum_{m \geq 1} A(mx).$$

Termwise differentiation is justified on compact subintervals of $(0, \infty)$, and each differentiated summand has the required strict alternating sign. This proves logarithmic complete monotonicity. □

Corollary 12.20 (APP-0022: Chen-Choi Gurland ratio is log-completely monotone). *Let*

$$F(x) = \frac{\Gamma(1/x)\Gamma(3/x)}{\Gamma(2/x)^2}, \qquad x > 0.$$

Then F is logarithmically completely monotone; equivalently,

$$(-1)^n (\log F)^{(n)}(x) > 0, \qquad n = 0, 1, 2, \ldots.$$

Proof. We use Proposition 12.19 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$A(u) = 2\log(u + 2) - \log(u + 1) - \log(u + 3).$$

Then $A(u) > 0$ for $u > 0$, since

$$A(u) = \log \frac{(u + 2)^2}{(u + 1)(u + 3)}$$

and the numerator exceeds the denominator by 1. Moreover

$$A'(u) = \frac{2}{u + 2} - \frac{1}{u + 1} - \frac{1}{u + 3} = -\frac{2}{(u + 1)(u + 2)(u + 3)}.$$

Using

$$\int_0^\infty e^{-ut} e^{-at} \, dt = \frac{1}{u + a},$$

we get

$$-A'(u) = \int_0^\infty e^{-ut} e^{-t}(1-e^{-t})^2 \, dt.$$

Since $A(u) \to 0$ as $u \to \infty$, integration from u to infinity gives

$$A(u) = \int_0^\infty e^{-ut} \frac{e^{-t}(1-e^{-t})^2}{t} \, dt.$$

The density is positive, locally integrable at 0, and exponentially decaying at infinity. Hence A is strictly completely monotone.

Weierstrass' product gives

$$\log \Gamma(z) = -\log z - \gamma z - \sum_{m=1}^{\infty} \left(\log(1 + z/m) - z/m\right).$$

Put $z = 1/x$. In

$$\log \Gamma(z) + \log \Gamma(3z) - 2 \log \Gamma(2z),$$

the linear terms cancel. The logarithmic terms give $\log(4/3)$. The summand at index m becomes

$$\log \frac{(1 + 2z/m)^2}{(1 + z/m)(1 + 3z/m)} = \log \frac{(mx + 2)^2}{(mx + 1)(mx + 3)} = A(mx).$$

Thus

$$\log F(x) = \log \frac{4}{3} + \sum_{m=1}^{\infty} A(mx).$$

For every compact $x \in [\varepsilon, M]$ and every $n \geq 0$,

$$\frac{d^n}{dx^n} A(mx) = m^n A^{(n)}(mx) = O_{n,\varepsilon}(m^{-2}),$$

because the three-point second difference in $A^{(n)}$ is $O((mx)^{-n-2})$. Therefore the series may be differentiated termwise locally uniformly.

For $n = 0$, $\log F(x) > 0$ follows from $\log(4/3) > 0$ and $A(mx) > 0$. For $n \geq 1$,

$$(-1)^n (\log F)^{(n)}(x) = \sum_{m=1}^{\infty} m^n (-1)^n A^{(n)}(mx) > 0.$$

Therefore $\log F$ is strictly completely monotone on $(0, \infty)$, proving Chen-Choi Conjecture 1. □

Definition 12.1 (Determinant and triangular compression language)**.** A determinant compression is a reduction of a sign claim to positivity of a smaller determinant, moment matrix, or coefficient system. A triangular nonnegative compression rewrites a family of expressions as an upper-triangular nonnegative linear combination of already controlled quantities. Coefficient extraction notation $[z^k]$ denotes the coefficient of z^k in a formal power series.

Proposition 12.21 (Sokal GS all lambda derivatives positive combination)**.** *For every $\ell \geq 1$, Sokal's generalized-Stieltjes tests satisfy $\partial_\lambda^\ell F_{n,k}^{[\lambda]}(x) = \sum_{r=0}^{k-\ell} (k!/r!)[z^{k-r}](-\log(1-z))^\ell F_{n,r}^{[\lambda]}(x)$, with nonnegative coefficients and empty sum* 0.

Proof. The exponential generating function is

$$\mathcal{F}(z) = (1-z)^{-a} \sum_{j\geq 0} G_j(x) \left(\frac{z}{1-z}\right)^j \frac{1}{j!}.$$

Differentiating ℓ times with respect to λ multiplies $\mathcal{F}$ by $(-\log(1-z))^\ell$. Extracting coefficients gives the displayed triangular formula. Since $-\log(1-z) = \sum_{m\geq 1} z^m/m$ has non-negative coefficients, all compression coefficients are non-negative. $\square$

Proposition 12.22 (Sokal GS lambda derivative generating function)**.** *For fixed n, Sokal's generalized-Stieltjes tests satisfy the formal exponential generating function* $\sum_{k\geq 0} F^{[\lambda]}_{n,k}(x) z^k/k! = (1-z)^{-(n+\lambda)}(-1)^n \sum_{j\geq 0} x^j f^{(}$
$z))^j/j!$.

Proof. Fix $n \geq 0$, $x > 0$, and write $a = n + \lambda$. For this proof f is fixed and only the coefficients depend on λ. Put

$$G_j(x) = (-1)^n x^j f^{(n+j)}(x).$$

Then Sokal's test can be written as

$$F^{[\lambda]}_{n,k}(x) = \sum_{j=0}^{k} \binom{k}{j} (a+j)_{k-j} G_j(x),$$

where $(u)_m = \Gamma(u+m)/\Gamma(u)$.

Consider the exponential generating series in the formal variable z:

$$\mathcal{F}(z) = \sum_{k\geq 0} F^{[\lambda]}_{n,k}(x) \frac{z^k}{k!}.$$

Coefficient extraction is finite in each degree, so this is a formal identity. Interchanging the finite coefficient contributions gives

$$\begin{aligned}
\mathcal{F}(z) &= \sum_{k\geq 0} \sum_{j=0}^{k} \binom{k}{j} (a+j)_{k-j} G_j(x) \frac{z^k}{k!} \\
&= \sum_{j\geq 0} G_j(x) \frac{z^j}{j!} \sum_{m\geq 0} (a+j)_m \frac{z^m}{m!} \\
&= \sum_{j\geq 0} G_j(x) \frac{z^j}{j!} (1-z)^{-a-j} \\
&= (1-z)^{-a} \sum_{j\geq 0} G_j(x) \frac{(z/(1-z))^j}{j!}.
\end{aligned}$$

Since $a = n + \lambda$, differentiating with respect to λ gives

$$\partial_\lambda \mathcal{F}(z) = -\log(1-z) \mathcal{F}(z).$$

But

$$-\log(1-z) = \sum_{m\geq 1} \frac{z^m}{m},$$

so comparison of coefficients of z^k yields

$$\frac{\partial_\lambda F^{[\lambda]}_{n,k}(x)}{k!} = \sum_{r=0}^{k-1} \frac{F^{[\lambda]}_{n,r}(x)}{r!} \frac{1}{k-r}.$$

Multiplying by $k!$ proves the first-derivative formula:

$$\partial_\lambda F^{[\lambda]}_{n,k}(x) = \sum_{r=0}^{k-1} \frac{k!}{r!(k-r)} F^{[\lambda]}_{n,r}(x).$$

The $k = 0$ case gives 0, as expected.

For higher derivatives, the same generating-function identity gives

$$\partial^\ell_\lambda \mathcal{F}(z) = (-\log(1-z))^\ell \mathcal{F}(z).$$

Thus

$$\frac{\partial^\ell_\lambda F^{[\lambda]}_{n,k}(x)}{k!} = \sum_{r=0}^{k} \frac{F^{[\lambda]}_{n,r}(x)}{r!} [z^{k-r}](-\log(1-z))^\ell.$$

Since $-\log(1-z)$ starts with z, the coefficient is zero unless $k - r \geq \ell$, giving the displayed sum over $0 \leq r \leq k - \ell$.

All coefficients are nonnegative because $-\log(1-z) = \sum_{m\geq 1} z^m/m$ has strictly positive Taylor coefficients, and products of power series with nonnegative coefficients have nonnegative coefficients. Therefore every λ-derivative of a Sokal test is a nonnegative linear combination of lower-k tests.

The same generating function gives an integrated order-monotonicity formula. For $\tau \geq 0$,

$$\mathcal{F}_{\lambda+\tau}(z) = (1-z)^{-\tau} \mathcal{F}_\lambda(z),$$

and hence

$$F^{[\lambda+\tau]}_{n,k}(x) = k! \sum_{r=0}^{k} \frac{F^{[\lambda]}_{n,r}(x)}{r!} [z^{k-r}](1-z)^{-\tau}.$$

Since $[z^m](1-z)^{-\tau} = (\tau)_m/m! \geq 0$, nonnegativity of the order-λ tests implies nonnegativity of the order-$\lambda + \tau$ tests by a direct triangular formula. $\square$

Corollary 12.23 (APP-0023: Sokal lambda-derivatives have triangular compression)**.** *Let $n, k \in \mathbb{N}_0$, $\lambda > 0$, $x > 0$, and let f be a C^∞ function on $(0, \infty)$. In Sokal's generalized-Stieltjes application, the quantities below are the Hankel-type expressions associated with a generalized Stieltjes function of order λ. For $j \geq 0$, set*

$$G_j(x) = (-1)^n x^j f^{(n+j)}(x), \qquad a = n + \lambda.$$

For $0 \leq r \leq k$, define

$$F^{[\lambda]}_{n,r}(x) = \sum_{j=0}^{r} \binom{r}{j} (a+j)_{r-j} G_j(x).$$

Then, for every integer $0 \leq \ell \leq k$,

$$\partial^\ell_\lambda F^{[\lambda]}_{n,k}(x) = \sum_{r=0}^{k-\ell} \frac{k!}{r!} [z^{k-r}](-\log(1-z))^\ell \, F^{[\lambda]}_{n,r}(x).$$

For $\ell > k$, $\partial^\ell_\lambda F^{[\lambda]}_{n,k}(x) = 0$. The coefficients in this triangular compression are non-negative.

Proof. We use Proposition 12.22 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Fix $n \geq 0$, $x > 0$, and write $a = n + \lambda$. For this proof f is fixed and only the coefficients depend on λ. Put

$$G_j(x) = (-1)^n x^j f^{(n+j)}(x).$$

Then Sokal's test can be written as

$$F_{n,k}^{[\lambda]}(x) = \sum_{j=0}^{k} \binom{k}{j} (a+j)_{k-j} G_j(x),$$

where $(u)_m = \Gamma(u+m)/\Gamma(u)$.

Consider the exponential generating series in the formal variable z:

$$\mathcal{F}(z) = \sum_{k \geq 0} F_{n,k}^{[\lambda]}(x) \frac{z^k}{k!}.$$

Coefficient extraction is finite in each degree, so this is a formal identity. Interchanging the finite coefficient contributions gives

$$\begin{aligned}
\mathcal{F}(z) &= \sum_{k \geq 0} \sum_{j=0}^{k} \binom{k}{j} (a+j)_{k-j} G_j(x) \frac{z^k}{k!} \\
&= \sum_{j \geq 0} G_j(x) \frac{z^j}{j!} \sum_{m \geq 0} (a+j)_m \frac{z^m}{m!} \\
&= \sum_{j \geq 0} G_j(x) \frac{z^j}{j!} (1-z)^{-a-j} \\
&= (1-z)^{-a} \sum_{j \geq 0} G_j(x) \frac{(z/(1-z))^j}{j!}.
\end{aligned}$$

Since $a = n + \lambda$, differentiating with respect to λ gives

$$\partial_\lambda \mathcal{F}(z) = -\log(1-z)\mathcal{F}(z).$$

But

$$-\log(1-z) = \sum_{m \geq 1} \frac{z^m}{m},$$

so comparison of coefficients of z^k yields

$$\frac{\partial_\lambda F_{n,k}^{[\lambda]}(x)}{k!} = \sum_{r=0}^{k-1} \frac{F_{n,r}^{[\lambda]}(x)}{r!} \frac{1}{k-r}.$$

Multiplying by $k!$ proves the first-derivative formula:

$$\partial_\lambda F_{n,k}^{[\lambda]}(x) = \sum_{r=0}^{k-1} \frac{k!}{r!(k-r)} F_{n,r}^{[\lambda]}(x).$$

The $k = 0$ case gives 0, as expected.

For higher derivatives, the same generating-function identity gives

$$\partial_\lambda^\ell \mathcal{F}(z) = (-\log(1-z))^\ell \mathcal{F}(z).$$

Thus

$$\frac{\partial_\lambda^\ell F_{n,k}^{[\lambda]}(x)}{k!} = \sum_{r=0}^{k} \frac{F_{n,r}^{[\lambda]}(x)}{r!}[z^{k-r}](-\log(1-z))^\ell.$$

Since $-\log(1-z)$ starts with z, the coefficient is zero unless $k-r \geq \ell$, giving the displayed sum over $0 \leq r \leq k-\ell$.

All coefficients are nonnegative because $-\log(1-z) = \sum_{m\geq 1} z^m/m$ has strictly positive Taylor coefficients, and products of power series with nonnegative coefficients have nonnegative coefficients. Therefore every λ-derivative of a Sokal test is a nonnegative linear combination of lower-k tests.

The same generating function gives an integrated order-monotonicity formula. For $\tau \geq 0$,

$$\mathcal{F}_{\lambda+\tau}(z) = (1-z)^{-\tau}\mathcal{F}_\lambda(z),$$

and hence

$$F_{n,k}^{[\lambda+\tau]}(x) = k! \sum_{r=0}^{k} \frac{F_{n,r}^{[\lambda]}(x)}{r!}[z^{k-r}](1-z)^{-\tau}.$$

Since $[z^m](1-z)^{-\tau} = (\tau)_m/m! \geq 0$, nonnegativity of the order-λ tests implies nonnegativity of the order-$\lambda+\tau$ tests by a direct triangular formula. $\square$

Proposition 12.24 (Simon gamma quotient derivative positive laplace density)**.** *For every $0 < \alpha < 1$, the derivative of $F_\alpha(x) = \Gamma(x+\alpha)/(\Gamma(x)x^\alpha)$ is completely monotone, with a positive Laplace-density representation obtained from the decreasing kernel J_α.*

Proof. Put $\beta = 1-\alpha$. The beta integral gives

$$F_\alpha(x) = x^\beta \frac{\Gamma(x+\alpha)}{\Gamma(x+1)} = \frac{x^\beta}{\Gamma(\beta)} \int_0^\infty e^{-xu}(e^u-1)^{-\alpha}\,du.$$

Define

$$J_\alpha(t) = \int_0^t (t-u)^{\alpha-1}(e^u-1)^{-\alpha}\,du.$$

Laplace convolution gives

$$F_\alpha(x) = \frac{x}{\Gamma(\alpha)\Gamma(1-\alpha)} \int_0^\infty e^{-xt} J_\alpha(t)\,dt.$$

After the substitution $u = tv$,

$$J_\alpha(t) = \int_0^1 (1-v)^{\alpha-1}\left(\frac{t}{e^{tv}-1}\right)^\alpha dv.$$

For each $v > 0$, the function $t/(e^{tv}-1)$ is decreasing in t. The integrand is dominated near the endpoints by $(1-v)^{\alpha-1}v^{-\alpha}$, which is integrable because $0 < \alpha < 1$. Hence J_α is decreasing, with

$$J_\alpha(0+) = \Gamma(\alpha)\Gamma(1-\alpha), \qquad J_\alpha(\infty) = 0.$$

Let $\mu_\alpha = -dJ_\alpha$. This is a positive finite measure, and Stieltjes integration by parts yields

$$1 - F_\alpha(x) = \frac{1}{\Gamma(\alpha)\Gamma(1-\alpha)} \int_0^\infty e^{-xt}\,\mu_\alpha(dt).$$

Thus $1 - F_\alpha$ is completely monotone. Differentiating this Laplace representation gives

$$F_\alpha'(x) = \frac{1}{\Gamma(\alpha)\Gamma(1-\alpha)} \int_0^\infty te^{-xt}\,\mu_\alpha(dt),$$

so F_α' is completely monotone. Therefore F_α is a Bernstein function. □

Proposition 12.25 (Bernstein derivative complete monotonicity criterion)**.** *Let f be nonnegative and continuously differentiable on an interval. If f' is completely monotone, then f is a Bernstein function on that interval. Conversely, for a differentiable Bernstein function, f' is completely monotone.*

Proof. Let f be nonnegative and continuously differentiable on an interval (a, ∞). A Bernstein function is exactly a nonnegative function whose derivative is completely monotone. Hence, if f' is completely monotone, the defining criterion is satisfied and f is Bernstein. Conversely, if f is a differentiable Bernstein function, the same defining criterion gives complete monotonicity of f'. This proves the criterion. □

Corollary 12.26 (APP-0024: Simon gamma quotient is Bernstein)**.** *For $0 < \alpha < 1$, define*

$$F_\alpha(x) = \frac{\Gamma(x+\alpha)}{\Gamma(x)x^\alpha}, \qquad x > 0.$$

Then F_α is a Bernstein function. More precisely, $1 - F_\alpha$ is completely monotone.

Proof. We use Proposition 12.25 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For $\alpha, \beta > 0$, $s, t \geq 0$, and $x > 0$, the beta identity gives

$$(x+s)^{-\alpha}(x+t)^{-\beta} = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \frac{u^{\alpha-1}(1-u)^{\beta-1}}{\big(x + us + (1-u)t\big)^{\alpha+\beta}}\,du.$$

If

$$f(x) = \int_{[0,\infty)} (x+s)^{-\alpha}\,d\mu(s), \qquad g(x) = \int_{[0,\infty)} (x+t)^{-\beta}\,d\nu(t),$$

then Tonelli gives

$$f(x)g(x) = \int_{[0,\infty)} (x+r)^{-(\alpha+\beta)}\,d\kappa(r),$$

where κ is the positive pushforward measure defined by

$$\int \varphi(r)\,d\kappa(r) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \iiint_0^1 \varphi\big(us + (1-u)t\big)u^{\alpha-1}(1-u)^{\beta-1}\,du\,d\mu(s)\,d\nu(t).$$

Thus pure generalized Stieltjes transforms satisfy

$$\mathcal{S}_\alpha^0 \mathcal{S}_\beta^0 \subseteq \mathcal{S}_{\alpha+\beta}^0.$$

With the usual nonnegative constant term convention, constants and lower-order terms require the standard order-lift convention already recorded in the source vocabulary; no application should rely on constants being absorbed unless that convention is explicitly cited.

For $0 < \alpha < 1$, set

$$F_\alpha(x) = \frac{\Gamma(x+\alpha)}{\Gamma(x)x^\alpha}, \qquad L_\alpha(x) = \log F_\alpha(x).$$

Then

$$L'_\alpha(x) = \psi(x+\alpha) - \psi(x) - \frac{\alpha}{x} = \int_0^\infty e^{-xt} \left(\frac{1-e^{-\alpha t}}{1-e^{-t}} - \alpha \right) dt.$$

The kernel is positive for $t > 0$, because $u \mapsto 1 - e^{-ut}$ is strictly concave on $[0,1]$, hence

$$1 - e^{-\alpha t} > \alpha(1 - e^{-t}).$$

Therefore L'_α is completely monotone, and $1/F_\alpha$ is logarithmically completely monotone.

This does not prove that F_α is a Bernstein function. Simon's source explicitly states that the Bernstein character of the corresponding $\Phi(\lambda) = \Gamma(1-t+\lambda)/(\lambda^{1-t}\Gamma(\lambda))$ is a puzzling question, and also says the reciprocal LCM property is necessary but not sufficient. $\square$

Proposition 12.27 (Du Wang h3 classification assembly result)**.** *The admitted middle-window increasing theorem for $1/2 \le a \le 1$ and the admitted outer-window nonmonotonicity obstruction for $0 < a < 1/2$ or $1 < a < 2$ assemble to the full Du–Wang h_3 classification on $0 < a < 2$.*

Proof. The middle-window monotonicity theorem on $1/2 \le a \le 1$ and the outer-window nonmonotonicity theorem on $0 < a < 1/2$ and $1 < a < 2$ partition the domain, and together they give exactly the source classification. $\square$

Proposition 12.28 (Du Wang polygamma ratio halfline bound)**.** *For every $t > 1/2$, one has $-2\psi''(t)/\psi'''(t) \ge t - 1/2$.*

Proof. For $t > 1/2$, define

$$S_m(t) = \sum_{n=0}^{\infty} \frac{1}{(n+t)^m}.$$

The polygamma identities give

$$\psi''(t) = -2S_3(t), \qquad \psi'''(t) = 6S_4(t).$$

Thus the desired bound

$$-\frac{2\psi''(t)}{\psi'''(t)} \ge t - \frac{1}{2}$$

is equivalent to

$$2S_3(t) - 3\left(t - \frac{1}{2}\right) S_4(t) \ge 0.$$

Write $y = t - \frac{1}{2} > 0$. Using

$$S_m(t) = \frac{1}{\Gamma(m)} \int_0^\infty \frac{x^{m-1}e^{-tx}}{1-e^{-x}}\, dx$$

and

$$\frac{e^{-tx}}{1-e^{-x}} = e^{-yx} \frac{1}{2\sinh(x/2)},$$

set

$$K(x) = \frac{1}{2\sinh(x/2)}.$$

Then

$$2S_3(t) - 3yS_4(t) = \int_0^\infty e^{-yx} x^2 K(x)\, dx - \frac{y}{2} \int_0^\infty e^{-yx} x^3 K(x)\, dx.$$

Since $x^3 K(x) \sim x^2$ as $x \downarrow 0$ and decays exponentially at infinity, integration by parts gives

$$y \int_0^\infty e^{-yx} x^3 K(x)\, dx = \int_0^\infty e^{-yx} (x^3 K(x))'\, dx.$$

Therefore

$$2S_3(t) - 3yS_4(t) = \int_0^\infty e^{-yx} \left(x^2 K(x) - \frac{1}{2}(x^3 K(x))' \right) dx.$$

Now

$$K'(x) = -\frac{1}{2} K(x) \coth(x/2),$$

so

$$x^2 K(x) - \frac{1}{2}(x^3 K(x))' = \frac{1}{2} x^2 K(x) \left(\frac{x}{2} \coth(x/2) - 1 \right).$$

For $z > 0$, $z \coth z > 1$. The integrand is therefore strictly positive, and

$$2S_3(t) - 3\left(t - \frac{1}{2}\right) S_4(t) > 0.$$

This proves

$$-\frac{2\psi''(t)}{\psi'''(t)} > t - \frac{1}{2}, \qquad t > \frac{1}{2}.$$

This proves the Du Wang polygamma ratio halfline bound.

Let $1/2 \le a \le 1$, $u > 0$, and $t = a + u$. Since $a \ge 1/2$,

$$u = t - a \le t - \frac{1}{2}.$$

The ratio lemma gives

$$t - \frac{1}{2} < -\frac{2\psi''(t)}{\psi'''(t)}.$$

Because $\psi'''(t) > 0$,

$$2\psi''(t) + u\psi'''(t) < 0.$$

Consequently

$$H_a'(u) = -u^2 \{2\psi''(t) + u\psi'''(t)\} > 0.$$

Together with $H_a(0+) = 2\log\Gamma(a) \ge 0$, this gives

$$H_a(u) \ge 0 \qquad (u > 0,\ 1/2 \le a \le 1).$$

This proves the Du Wang h31 middle window u monotonicity.

For $x > 0$, $u = x$, and $t = x + a$,

$$h_3'(x) = \widetilde{h}_3'(x + a) = \frac{h_{31}(x + a)}{x^2} = \frac{H_a(x)}{x^2} \ge 0.$$

The inequality is strict for $x > 0$ except possibly at the endpoint $a = 1, u = 0$, which is outside the open x-interval. Thus h_3 is increasing on $(0, \infty)$ for every

$$\frac{1}{2} \le a \le 1.$$

This monotonicity result combines with the previously proved outer-window nonmonotonicity theorem to complete the open-problem source classification.

The outer-window nonmonotonicity result proves that h_3 is not monotone for

$$0 < a < \frac{1}{2} \qquad \text{or} \qquad 1 < a < 2.$$

The middle-window theorem above proves increasing behavior for

$$\frac{1}{2} \le a \le 1.$$

Therefore Du–Wang Open Problem 2 is classified on $0 < a < 2$:

$$h_3 \text{ is increasing on } (0, \infty) \quad \Longleftrightarrow \quad \frac{1}{2} \le a \le 1,$$

and it is not monotone on the two outer windows.

This proves the Du Wang h3 open problem 2 classification. ☐

Counterexample 12.29 (APP-0025: Du–Wang h_3 monotonicity classification)**.** For $0 < a < 2$, define

$$h_3(x) = \frac{-x^2\psi'(x+a) + 2x\psi(x+a) - 2\log\Gamma(x+a)}{x}, \qquad x > 0.$$

Then h_3 is increasing on $(0, \infty)$ if and only if

$$\frac{1}{2} \le a \le 1.$$

For $0 < a < 1/2$ and for $1 < a < 2$, h_3 is not monotone on $(0, \infty)$.

Proof. We use Proposition 12.28 and Proposition 12.27 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The middle-window monotonicity theorem on $1/2 \le a \le 1$ and the outer-window nonmonotonicity theorem on $0 < a < 1/2$ and $1 < a < 2$ partition the domain, and together they give exactly the source classification. ☐

Proposition 12.30 (Baskakov alpha1 r8 not CM)**.** *The Baskakov $\alpha = 1$, $r = 8$ function $f_1^{[8]}(x) = 1/((1+x)^8 - x^8)$ is not completely monotone on $(0, \infty)$.*

Proof. For even $r = 2m$ and $\alpha = 1$, the source definition reduces to

$$f_1^{[2m]}(x) = \frac{1}{(1+x)^{2m} - x^{2m}}.$$

Let $D_m(x) = (1+x)^{2m} - x^{2m}$. Its roots are

$$\lambda_{k,m} = -\frac{1}{2} - \frac{i}{2}\cot\frac{\pi k}{2m}, \qquad k = 1, \dots, 2m-1,$$

and the corresponding residues are

$$R_{k,m} = \frac{(-1)^{m+k}}{2m}\left(2\sin\frac{\pi k}{2m}\right)^{2m-2}.$$

Thus the inverse-Laplace density of $1/D_m$, if it were nonnegative, would be

$$\rho_m(t) = \frac{e^{-t/2}}{2m}\left[2^{2m-2} + 2\sum_{k=1}^{m-1}(-1)^{m+k}\left(2\sin\frac{\pi k}{2m}\right)^{2m-2}\cos\left(\frac{t}{2}\cot\frac{\pi k}{2m}\right)\right].$$

For $m = 4$, write $h_4(t) = e^{t/2}\rho_4(t)$. The formula simplifies to

$$\begin{aligned} h_4(t) = 8 + 2\cos\frac{t}{2} &- \frac{(2-\sqrt{2})^3}{4}\cos\left(\frac{1+\sqrt{2}}{2}t\right) \\ &- \frac{(2+\sqrt{2})^3}{4}\cos\left(\frac{\sqrt{2}-1}{2}t\right). \end{aligned}$$

At $t = 10\pi$,

$$h_4(10\pi) = 6 + 10\cos(5\pi\sqrt{2}).$$

Since $0 < 5\sqrt{2} - 7 < 1/4$,

$$\cos(5\pi\sqrt{2}) = -\cos(\pi(5\sqrt{2}-7)) < -\frac{\sqrt{2}}{2} < -\frac{3}{5}.$$

Consequently $h_4(10\pi) < 0$, and so $\rho_4(10\pi) < 0$. By uniqueness of Laplace transforms of positive measures, $f_1^{[8]}$ cannot be completely monotone. □

Counterexample 12.31 (APP-0026: Baskakov even-power conjecture is false)**.** For the Abel–Gawronski–Neuschel higher-power Baskakov family

$$f_\alpha^{[r]}(x) = (1+x)^{-r\alpha}\sum_{k=0}^{\infty}\binom{-\alpha}{k}^r\left(\frac{x}{1+x}\right)^{rk},$$

the universal even-power complete-monotonicity conjecture is false. Already at $\alpha = 1$ and $r = 8$,

$$f_1^{[8]}(x) = \frac{1}{(1+x)^8 - x^8}$$

is not completely monotone on $(0,\infty)$.

Proof. We use Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Set

$$D_m(x) = (1+x)^{2m} - x^{2m}.$$

The roots are obtained from

$$\frac{1+x}{x} = \omega_k, \qquad \omega_k = e^{\pi i k/m}, \qquad k = 1,\dots,2m-1.$$

Thus

$$\lambda_{k,m} = \frac{1}{\omega_k - 1} = -\frac{1}{2} - \frac{i}{2}\cot\frac{\pi k}{2m}.$$

At this root,

$$D_m'(\lambda_{k,m}) = 2m\lambda_{k,m}^{2m-1}(\omega_k^{-1} - 1),$$

and direct simplification gives the real residue

$$R_{k,m} = \frac{1}{D_m'(\lambda_{k,m})} = \frac{(-1)^{m+k}}{2m}\left(2\sin\frac{\pi k}{2m}\right)^{2m-2}.$$

Consequently the inverse-Laplace density is

$$\rho_m(t) = \sum_{k=1}^{2m-1} R_{k,m} e^{\lambda_{k,m} t}.$$

Pairing conjugate roots gives

$$\rho_m(t) = \frac{e^{-t/2}}{2m}\left[2^{2m-2} + 2\sum_{k=1}^{m-1}(-1)^{m+k}\left(2\sin\frac{\pi k}{2m}\right)^{2m-2}\cos\left(\frac{t}{2}\cot\frac{\pi k}{2m}\right)\right].$$

This proves the finite trigonometric density formula used by the diagnostic candidate.

For $m = 2$, this gives

$$\rho_2(t) = e^{-t/2}\left(1 - \cos\frac{t}{2}\right) = 2e^{-t/2}\sin^2\frac{t}{4} \geq 0,$$

recovering the previously admitted $r = 4, \alpha = 1$ seed.

For $m = 3$,

$$\rho_3(t) = \frac{e^{-t/2}}{6}\left[16 + 2\cos\left(\frac{\sqrt{3}\,t}{2}\right) - 18\cos\left(\frac{t}{2\sqrt{3}}\right)\right].$$

Set $u = t/(2\sqrt{3})$. Since $\cos(3u) = 4\cos^3 u - 3\cos u$,

$$\rho_3(t) = \frac{4}{3}e^{-t/2}(1-\cos u)^2(2+\cos u) \geq 0.$$

Thus $r = 6, \alpha = 1$ is completely monotone by an explicit positive density.

For $m = 4$, write $h_4(t) = e^{t/2}\rho_4(t)$. The density formula gives

$$h_4(t) = 8 + 2\cos\frac{t}{2} - \frac{(2-\sqrt{2})^3}{4}\cos\left(\frac{1+\sqrt{2}}{2}t\right) - \frac{(2+\sqrt{2})^3}{4}\cos\left(\frac{\sqrt{2}-1}{2}t\right).$$

At $t = 10\pi$,

$$\cos(t/2) = \cos(5\pi) = -1,$$

and both other cosine terms equal $-\cos(5\pi\sqrt{2})$. Since

$$(2-\sqrt{2})^3 + (2+\sqrt{2})^3 = 40,$$

we obtain

$$h_4(10\pi) = 6 + 10\cos(5\pi\sqrt{2}).$$

Now

$$0 < 5\sqrt{2} - 7 < \frac{1}{4}$$

because $7/5 < \sqrt{2} < 29/20$. Hence, with $\delta = 5\sqrt{2} - 7$,

$$\cos(5\pi\sqrt{2}) = \cos(7\pi + \pi\delta) = -\cos(\pi\delta) < -\cos\frac{\pi}{4} = -\frac{\sqrt{2}}{2} < -\frac{3}{5}.$$

Therefore

$$h_4(10\pi) < 6 + 10\left(-\frac{3}{5}\right) = 0,$$

and

$$\rho_4(10\pi) = e^{-5\pi} h_4(10\pi) < 0.$$

Since the inverse-Laplace transform is unique for measures on $[0, \infty)$, $f_1^{[8]}(x) = 1/((1+x)^8 - x^8)$ cannot be completely monotone. $\square$

Definition 12.2 (Log-function derivative-sign regions)**.** For a finite-dimensional class of log-functions $\mathcal{F}$ and an operator $L = -d/dx$, the derivative-sign region D_k consists of functions $f \in \mathcal{F}$ for which $L^k f \geq 0$ on the stated domain. A descending-chain claim asserts $D_{k+1} \subseteq D_k$.

Proposition 12.32 (Ma Weigert Dk chain integration proof)**.** *For $f \in \mathcal{F}_{1,n}$, if $L^{k+1} f \geq 0$ on $(0, \infty)$, then $L^k f \geq 0$ on $(0, \infty)$, because $L^k f(x) = \int_x^\infty L^{k+1} f(t)\, dt$.*

Proof. For $f(x) = p(\log x)/x$, induction gives

$$L^k f(x) = x^{-k-1} p_k(\log x)$$

for a polynomial p_k. Indeed, if the formula holds at level k, then

$$L^{k+1} f(x) = x^{-k-2}\big((k+1)p_k(\log x) - p_k'(\log x)\big).$$

Thus $L^k f(x) \to 0$ as $x \to \infty$. If $f \in D_{k+1}$, put $g = L^k f$. Then $-g' = L^{k+1} f \geq 0$, and hence, for $R > x$,

$$g(x) - g(R) = \int_x^R L^{k+1} f(t)\, dt.$$

Letting $R \to \infty$ gives

$$L^k f(x) = \int_x^\infty L^{k+1} f(t)\, dt \geq 0.$$

Therefore $f \in D_k$. $\square$

Corollary 12.33 (APP-0027: Ma–Weigert derivative regions form a chain)**.** *Let*

$$\mathcal{F}_{1,n} = \left\{ x \mapsto \frac{p(\log x)}{x} : p \in \mathbb{R}[y]_n \right\},$$

and let $L = -d/dx$. If

$$D_k = \{f \in \mathcal{F}_{1,n} : L^k f(x) \geq 0 \text{ for all } x > 0\},$$

then

$$D_{k+1} \subseteq D_k \qquad (k \geq 0,\ n \geq 0).$$

Proof. We use Proposition 12.32 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

We first prove that for each $k \geq 0$,

$$L^k f(x) = x^{-k-1} p_k(\log x)$$

for some polynomial p_k.

For $k = 0$, this is the definition with $p_0 = p$. Suppose

$$L^k f(x) = x^{-k-1} p_k(\log x).$$

Then

$$\begin{aligned} L^{k+1} f(x) &= -\frac{d}{dx}\left(x^{-k-1} p_k(\log x)\right) \\ &= x^{-k-2}\left((k+1)p_k(\log x) - p_k'(\log x)\right). \end{aligned}$$

Thus $p_{k+1}(y) = (k+1)p_k(y) - p_k'(y)$, which is again a polynomial.

Since every polynomial in $\log x$ grows slower than every positive power of x,

$$x^{-k-1} p_k(\log x) \to 0 \qquad (x \to \infty).$$

Therefore

$$\lim_{x\to\infty} L^k f(x) = 0$$

for every $k \geq 0$.

Assume $f \in D_{k+1}$. Then

$$L^{k+1} f(x) \geq 0$$

on $(0, \infty)$. Put

$$g(x) = L^k f(x).$$

By definition of L,

$$-g'(x) = L^{k+1} f(x) \geq 0.$$

Using $g(x) \to 0$ as $x \to \infty$, integrate from x to R:

$$g(x) - g(R) = \int_x^R L^{k+1} f(t)\, dt.$$

Letting $R \to \infty$ gives

$$L^k f(x) = g(x) = \int_x^\infty L^{k+1} f(t)\, dt \geq 0.$$

Hence $f \in D_k$. Therefore

$$D_{k+1} \subseteq D_k$$

for all $k \geq 0$ and all n. □

Proposition 12.34 (QA divisor polygamma even claim refuted)**.** *The Qi–Agarwal/Yin claim that $f_{2\ell}$ is not completely monotone for every $\ell \in \mathbb{N}$ is false, because f_2 is completely monotone.*

Proof. The standard polygamma representation gives, for $k \geq 1$,

$$(-1)^{k+1}\psi^{(k)}(x) = \int_0^\infty \frac{t^k e^{-xt}}{1-e^{-t}}\,dt.$$

Thus $\psi^{(k)}$ is a positive completely monotone function whenever k is odd. If n is odd and $km = n$, both k and m are odd, so each summand $[\psi^{(k)}]^m$ is a finite product of positive completely monotone functions. Finite sums preserve complete monotonicity.

For $n = 2$, the divisor pairs are $(1, 2)$ and $(2, 1)$, hence

$$f_2(x) = [\psi'(x)]^2 + \psi''(x).$$

Qi–Agarwal record the sharp theorem that

$$[\psi'(x)]^2 + \lambda\psi''(x)$$

is completely monotone on $(0, \infty)$ if and only if $\lambda \leq 1$. Taking $\lambda = 1$ proves that f_2 is completely monotone. This refutes the stated even-parity clause at $\ell = 1$. □

Proposition 12.35 (QA divisor polygamma odd CM)**.** *For every odd positive integer n, the divisor-polygamma sum $f_n(x) = \sum_{km=n}[\psi^{(k)}(x)]^m$ is completely monotone on $(0, \infty)$.*

Proof. This proof is the component of Proposition 12.34 corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Counterexample 12.36 (Parity classification assembly principle)**.** Suppose a source parity classification is split into two disjoint parity classes. If the positive class is proved by one theorem and the claimed negative class is refuted by one admissible counterexample, then the corrected parity classification is the conjunction of those two upstream results.

Proof. Let a classification problem split a family $\mathcal{F}$ into two disjoint classes $\mathcal{F}_0$ and $\mathcal{F}_1$. Suppose one theorem proves the asserted positive property for every member of $\mathcal{F}_1$, and another theorem or counterexample determines that the proposed universal assertion on $\mathcal{F}_0$ must be corrected. Then the corrected classification is obtained by taking the conjunction of the two upstream results.

The two classes are disjoint and exhaustive, so every member of $\mathcal{F}$ lies in exactly one class. The first upstream result decides all cases in $\mathcal{F}_1$, and the second decides the corrected status of $\mathcal{F}_0$. Hence their conjunction is a complete classification for $\mathcal{F}$. □

Counterexample 12.37 (APP-0028: divisor-polygamma parity correction)**.** For $n \in \mathbb{N}$, set

$$f_n(x) = \sum_{km=n} \left[\psi^{(k)}(x)\right]^m, \qquad x > 0.$$

Then f_n is completely monotone for every odd n. The asserted opposite parity law that $f_{2\ell}$ is never completely monotone is false:

$$f_2(x) = [\psi'(x)]^2 + \psi''(x)$$

is completely monotone on $(0, \infty)$.

Proof. We use Proposition 12.36 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The node the QA divisor polygamma odd CM proves that every odd divisor-polygamma sum f_n is completely monotone. The node the QA divisor polygamma even claim refuted, itself following from the QA divisor polygamma f2 CM, proves that the source's universal even non-complete-monotonicity clause is false. These two admitted statements exactly resolve the parity claim: the odd assertion holds, while the even negation asserted in the source does not hold in full generality. $\square$

Proposition 12.38 (BZ gamma quotient critical window reduction)**.** *There is a rational interval J around $1.126207061\ldots$ such that $\widetilde{F}' < 0$ on $(-1, \inf J]$, $\widetilde{F}' > 0$ on $[\sup J, \infty)$, and $\widetilde{F}'$ has exactly one zero inside J.*

Proof. The source proves $F' < 0$ on $(-1, 1)$. For $x > 1$, set

$$G(x) = \log \Gamma(x+1), \qquad D(x) = \log \frac{x^2+6}{x+6},$$

and

$$N_0(x) = \psi(x+1)D(x) - G(x)D'(x).$$

Since $F' = N_0/D^2$, the sign of F' on $x > 1$ is the sign of N_0. A direct right-tail estimate gives $N_0(x) > 0$ for $x \geq 8$: with

$$P(x) = x^2 + 12x - 6, \qquad Q(x) = (x^2+6)(x+6),$$

one has $D' = P/Q$; the bounds $\psi(x+1) > \log x$, $\log \Gamma(x+1) \leq x \log x$, and

$$D(x) \geq \frac{xP(x)}{Q(x)} \qquad (x \geq 8)$$

imply $N_0(x) > 0$.

It remains to prove a single crossing on $(1, 8]$. Define

$$r(x) = \frac{\psi(x+1)}{D'(x)} = \psi(x+1)\frac{Q(x)}{P(x)}.$$

Then

$$P(x)^2 r'(x) = S(x),$$

where

$$S(x) = \psi'(x+1)Q(x)P(x) + \psi(x+1)\{Q'(x)P(x) - Q(x)P'(x)\}.$$

Using the elementary bounds, valid for $y \geq 2$,

$$\psi(y) > \log y - \frac{1}{y}, \qquad \psi'(y) > \frac{1}{y} + \frac{1}{2y^2}, \qquad \psi''(y) > -\frac{1}{y^2} - \frac{2}{y^3},$$

one obtains $S'(x) > 0$ on $[1, 8]$. Also

$$S(1) = 343\left(\frac{\pi^2}{6} - 1\right) - 539(1-\gamma) < 0,$$

while $S(8) > 0$. Hence r' has exactly one zero on $[1, 8]$.

Now put $I(x) = D(x)r(x) - G(x)$. Then $N_0(x) = D'(x)I(x)$ and

$$I'(x) = D(x)r'(x).$$

Since $D(x) > 0$ for $x > 1$, I decreases once and then increases once. The removable endpoint has $\lim_{x\downarrow 1} I(x) = 0$, and the quadratic leading coefficient of N_0 at 1 is negative. Finally,

$$N_0(8) = (H_8 - \gamma)\log 5 - \frac{11}{70}\log(40320) > 0.$$

Thus N_0, and hence F', has exactly one zero on $(1, 8]$. Combining this with the source interval $(-1, 1)$ and the right-tail estimate proves the theorem. $\square$

Proposition 12.39 (BZ gamma quotient N single crossing one eight)**.** *Let $N(x)$ be the Bulboaca–Zayed derivative-sign numerator from the Bulboaca–Zayed gamma-quotient derivative normal form. On $[1, 8]$, N has exactly one zero ξ, with $N(x) < 0$ for $1 \le x < \xi$ and $N(x) > 0$ for $\xi < x \le 8$.*

Proof. This proof is the component of Proposition 12.38 corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Corollary 12.40 (APP-0029: Bulboaca–Zayed Gamma quotient one-critical-point theorem)**.** *Let*

$$F(x) = \frac{\log\Gamma(x+1)}{\log(x^2+6) - \log(x+6)}$$

with the removable values filled in on $(-1, \infty)$. Then F' has a unique zero x_m on $(-1, \infty)$. Moreover

$$F'(x) < 0 \quad (-1 < x < x_m), \qquad F'(x) > 0 \quad (x > x_m),$$

and numerically $x_m = 1.126207061051643\ldots$.

Proof. We use Proposition 12.39 and Proposition 12.38 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

proof.

The raw numerator has a removable zero at $x = 1$, since $G(1) = D(1) = 0$. Its quadratic leading coefficient is

$$\lim_{x\downarrow 1}\frac{N_0(x)}{(x-1)^2} = \frac{7(\pi^2/6 - 1) - 11(1-\gamma)}{98} < 0,$$

again by $\pi^2 < 987/100$ and $\gamma < 5773/10000$, which gives $-1347/980000 < 0$. Therefore any normalization dividing out the removable $(x-1)^2$ factor is negative at the left endpoint.

The single-crossing candidate is true:

$$theBulboaca--Zayedgamma-quotientNsinglecrossingoneeight.$$

Together with the source's theorem on $(-1, 1)$ and the previous local right-tail theorem for $x \ge 8$, this proves the critical-window reduction

$$theBulboaca--Zayedgamma-quotientcriticalwindowreduction$$

and then the source problem

$$theBulboacaZayedgammaquotientfullmonotonicity.$$

The finite-envelope candidate is not separately promoted: the proof above is a clean analytic ratio-kernel argument rather than the finite rational cover specified in that candidate. The diagnostic obstruction-map candidate is also not needed after the single-crossing proof. $\square$

Proposition 12.41 (Ramanujan Turan n2 upper window counterfamily)**.** *For the Ramanujan integral Turan gap, there are parameters $\alpha \in (0, 1/2)$ such that $H_2(x; \alpha) < 0$ for some $x > 0$; specifically, with $x = e^{-L}$ and $\alpha_L = 1/2 - 1/(4L)$, this holds for all sufficiently large L.*

Proof. Put

$$A_k(x) = (-1)^k I_R^{(k)}(x) = \int_0^\infty e^{-xt} \frac{t^{k-1}}{\pi^2 + \log^2 t}\, dt.$$

Then

$$H_2(x; \alpha) = A_2(x)^2 - \alpha A_1(x) A_3(x).$$

Let $x = e^{-L}$. After the substitution $y = xt$,

$$A_k(e^{-L}) = e^{kL} \int_0^\infty \frac{e^{-y} y^{k-1}}{\pi^2 + (L + \log y)^2}\, dy.$$

Splitting at $|\log y| \le L^{2/3}$ and expanding the denominator uniformly on the central region gives, for fixed $k = 1, 2, 3$,

$$A_k(e^{-L}) = e^{kL} L^{-2} \Gamma(k) \left(1 - \frac{2\psi(k)}{L} + O(L^{-2})\right).$$

Consequently

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L}) A_3(e^{-L})} = \frac{1}{2} \left(1 + \frac{2(\psi(1) + \psi(3) - 2\psi(2))}{L} + O(L^{-2})\right).$$

Since

$$\psi(1) = -\gamma, \qquad \psi(2) = 1 - \gamma, \qquad \psi(3) = \frac{3}{2} - \gamma,$$

the bracketed correction is $-1/L + O(L^{-2})$, and hence

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L}) A_3(e^{-L})} = \frac{1}{2} - \frac{1}{2L} + O(L^{-2}).$$

Choose $\alpha_L = 1/2 - 1/(4L)$. For all sufficiently large L, $\alpha_L \in (0, 1/2)$ and

$$A_2(e^{-L})^2 - \alpha_L A_1(e^{-L}) A_3(e^{-L}) < 0.$$

Thus $H_2(e^{-L}; \alpha_L) < 0$. Complete monotonicity would imply nonnegativity at order zero, so the source's universal interval statement is false. ☐

Counterexample 12.42 (APP-0030: Ramanujan Turan window has a negative answer)**.** Let

$$I_R(x) = \int_0^\infty e^{-xt} \frac{dt}{t(\pi^2 + \log^2 t)}$$

and, for $n \ge 2$,

$$H_n(x; \alpha) = \left(I_R^{(n)}(x)\right)^2 - \alpha I_R^{(n-1)}(x) I_R^{(n+1)}(x).$$

The proposed complete-monotonicity window

$$\frac{n-2}{n-1} < \alpha < \frac{n-1}{n}$$

is false. It already fails for $n = 2$.

Proof. We use Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Put

$$A_k(x) = (-1)^k I_R^{(k)}(x) = \int_0^\infty e^{-xt} \frac{t^{k-1}}{\pi^2 + \log^2 t}\, dt \qquad (k \geq 1).$$

Then

$$H_2(x; \alpha) = A_2(x)^2 - \alpha A_1(x) A_3(x).$$

We study $x = e^{-L}$ as $L \to \infty$. With $y = xt$,

$$A_k(e^{-L}) = e^{kL} \int_0^\infty \frac{e^{-y} y^{k-1}}{\pi^2 + (L + \log y)^2}\, dy.$$

For fixed $k = 1, 2, 3$,

$$A_k(e^{-L}) = e^{kL} L^{-2} \Gamma(k) \left(1 - \frac{2\psi(k)}{L} + O(L^{-2})\right),$$

where ψ is the digamma function. To justify the expansion, split the y-integral into the central region $|\log y| \leq L^{2/3}$ and its complement. On the central interval, with $z = \log y$, expand uniformly:

$$\frac{1}{\pi^2 + (L + \log y)^2} = L^{-2} \left(1 - \frac{2 \log y}{L} + O\left(\frac{1 + \log^2 y}{L^2}\right)\right),$$

and integrate against $e^{-y} y^{k-1}$. The lower complement $0 < y < e^{-L^{2/3}}$ is $O(e^{-cL^{2/3}})$ for $k = 1, 2, 3$ after the change $y = e^{-u}$, and the upper complement $y > e^{L^{2/3}}$ is super-exponentially small because of e^{-y}. These errors are $o(L^{-N})$ for every fixed N, which is more than needed for the displayed expansion.

Therefore

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L}) A_3(e^{-L})} = \frac{\Gamma(2)^2}{\Gamma(1)\Gamma(3)} \left(1 + \frac{2(\psi(1) + \psi(3) - 2\psi(2))}{L} + O(L^{-2})\right).$$

Using

$$\psi(1) = -\gamma, \qquad \psi(2) = 1 - \gamma, \qquad \psi(3) = \frac{3}{2} - \gamma,$$

we get

$$\psi(1) + \psi(3) - 2\psi(2) = -\frac{1}{2},$$

so

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L}) A_3(e^{-L})} = \frac{1}{2} - \frac{1}{2L} + O(L^{-2}).$$

Choose

$$\alpha_L = \frac{1}{2} - \frac{1}{4L}.$$

For all sufficiently large L, $\alpha_L \in (0, 1/2)$ and

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L}) A_3(e^{-L})} < \alpha_L.$$

Consequently

$$H_2(e^{-L}; \alpha_L) = A_2(e^{-L})^2 - \alpha_L A_1(e^{-L}) A_3(e^{-L}) < 0.$$

Complete monotonicity would imply nonnegativity at order zero. Hence $H_2(\cdot;\alpha_L)$ is not completely monotone for these fixed parameters $\alpha_L \in (0,1/2)$. The source's universal Turan-window statement is false already in the $n=2$ upper window.

the Ramanujan Turan n2 upper window counterfamily the Ramanujan Turan window negative answer

The new mechanism is a logarithmic-density moment-ratio asymptotic obstruction for quadratic Laplace-moment Turan gaps. It is distinct from the public APP-0014 Stieltjes/complete-Bernstein classification of I_R. □

12.1 Wright-density and Mellin-factor transport

Proposition 12.43 (Bazhlekova Wright density topcap bridge normal form)**.** *For the normalized Bazhlekova symbol $h(s) = s^\alpha(1+s^{-p})^{1/2}$, with $0 < \beta = \alpha - p/2 < \alpha < 1$ and the principal branch in the no-cover seed range, define $\mathcal{W}_{\alpha,p}(x) = -\sum_{m\geq 0}\binom{1/2}{m}x^m/\Gamma(pm-\alpha)$. Then $\mathcal{L}^{-1}\{h'\}(t) = t^{-\alpha}\mathcal{W}_{\alpha,p}(t^p)$, and the Wright top-cap limit has the same sign as $\mathcal{W}_{\alpha,p}(\lambda^{-1})$. Thus normalized derivative density positivity and top-cap positivity are the same Wright-function sign problem.*

Proof. Put

$$\alpha = \frac{a}{2}, \qquad p = a-b, \qquad \beta = \frac{b}{2} = \alpha - \frac{p}{2},$$

so that

$$h(s) = s^{b/2}(1+s^{a-b})^{1/2} = s^\alpha(1+s^{-p})^{1/2}.$$

At the two no-cover seeds,

$$(\alpha,p) = \left(\frac{3}{4},\frac{11}{10}\right), \qquad (\alpha,p) = \left(\frac{11}{20},\frac{21}{20}\right),$$

and $0<\beta<\alpha<1$, $1<p<2$.

Define the Wright-normalized series

$$\mathcal{W}_{\alpha,p}(x) = -\sum_{m=0}^{\infty}\binom{1/2}{m}\frac{x^m}{\Gamma(pm-\alpha)}.$$

The coefficient ratio and the gamma denominator show that this series is entire in x.

For $s>1$, the binomial expansion gives

$$h'(s) = \sum_{m=0}^{\infty}\binom{1/2}{m}(\alpha-pm)s^{\alpha-pm-1}.$$

For each m, analytic continuation of the elementary Laplace formula gives

$$\mathcal{L}\left\{\frac{t^{pm-\alpha}}{\Gamma(pm-\alpha)}\right\}(s) = (pm-\alpha)s^{\alpha-pm-1}.$$

Therefore

$$\mathcal{L}^{-1}\{h'\}(t) = -\sum_{m=0}^{\infty}\binom{1/2}{m}\frac{t^{pm-\alpha}}{\Gamma(pm-\alpha)} = t^{-\alpha}\mathcal{W}_{\alpha,p}(t^p).$$

For $1<p<2$ the Wright series has subexponential growth, so the Laplace transform is defined on $s>0$. Equality for $s>1$ extends to $s>0$ by analytic continuation on the principal branch.

Consequently, at the no-cover seeds, complete monotonicity of h' is equivalent to

$$\mathcal{W}_{\alpha,p}(x) \geq 0 \qquad (x > 0),$$

because the inverse Laplace transform is the locally integrable density above.

The previous top-cap audit used the signed polynomials $R_n(y) = (-1)^{n-1}Q_n(y)$. Its formal top-cap function was

$$W^{\mathrm{top}}_{\alpha,p}(\lambda) = 1 - \frac{\Gamma(1-\alpha)}{\alpha} \sum_{m\geq 1} \binom{1/2}{m} \frac{\lambda^{-m}}{\Gamma(pm-\alpha)}.$$

Since

$$-\frac{1}{\Gamma(-\alpha)} = \frac{\alpha}{\Gamma(1-\alpha)},$$

the two normalizations satisfy

$$W^{\mathrm{top}}_{\alpha,p}(\lambda) = \frac{\Gamma(1-\alpha)}{\alpha}\mathcal{W}_{\alpha,p}(\lambda^{-1}).$$

The factor $\Gamma(1-\alpha)/\alpha$ is positive. Thus top-cap positivity and normalized derivative density positivity are the same sign problem after $x = \lambda^{-1}$. □

Proposition 12.44 (Bazhlekova two outside-gap square-root symbols are Bernstein)**.** *Let $g_1(s) = s^{3/2} + s^{2/5}$ and $g_2(s) = s^{11/10} + s^{1/20}$. Then $h_i(s) = \sqrt{g_i(s)}$ is a Bernstein function on $(0,\infty)$ for $i = 1, 2$, even though the exponent gaps are $11/10$ and $21/20$.*

Proof. For

$$h(s) = s^{\alpha}(1 + s^{-p})^{1/2}, \qquad 0 < \alpha - p/2 < \alpha < 1, \quad 1 < p < 2,$$

the the corresponding theorem the Bazhlekova Wright density topcap bridge normal form proves

$$\mathcal{L}^{-1}\{h'\}(t) = t^{-\alpha}\mathcal{W}_{\alpha,p}(t^p),$$

where

$$\mathcal{W}_{\alpha,p}(x) = -\sum_{m\geq 0} \binom{1/2}{m} \frac{x^m}{\Gamma(pm-\alpha)}.$$

Thus $\mathcal{W}_{\alpha,p}(x) > 0$ for all $x > 0$ implies $h' \in CM$, hence $h \in BF$.

For g_1, $(\alpha, p) = (3/4, 11/10)$. For g_2, $(\alpha, p) = (11/20, 21/20)$.

The node the Bazhlekova Wright two seed compact zero ten positive proves

$$\mathcal{W}_{3/4,11/10}(x) > 0, \qquad \mathcal{W}_{11/20,21/20}(x) > 0 \qquad (0 \leq x \leq 10).$$

Its proof artifacts are:

The node the Bazhlekova Wright two seed post ten derivative positive proves

$$\mathcal{W}'_{3/4,11/10}(x) > 0, \qquad \mathcal{W}'_{11/20,21/20}(x) > 0 \qquad (10 \leq x \leq 20),$$

with lower bounds $0.0201343226217\ldots$ and $0.00299977163172\ldots$. Since the functions are positive at 10, this proves positivity on $[10, 20]$. Its proof artifacts are:

The node the Bazhlekova Wright two seed Watson tail from twenty positive proves positivity for $x \geq 20$ by a three-term Watson contour certificate. The replay output at $x = 20$ gives margins

$$0.732374180856\ldots$$

for $(3/4, 11/10)$, and

$$0.100568095684\ldots$$

for $(11/20, 21/20)$. Its proof artifacts are:

Combining the three certified ranges gives the the corresponding theorem the Bazhlekova Wright two seed all x positive:

$$\mathcal{W}_{3/4,11/10}(x) > 0, \qquad \mathcal{W}_{11/20,21/20}(x) > 0 \qquad (x > 0).$$

This proves that the source condition $\alpha - \alpha_m \le 1$ is not necessary for the positivity/subordination package. It does not classify all relaxations and does not claim $\sqrt{g_i} \in CBF$; in fact the off-cut zeros for gap > 1 block the CBF shortcut. □

Counterexample 12.45 (APP-0058: Bazhlekova two-seed Wright/Bernstein gap condition is relaxable)**.** For the two positive-coefficient symbols $g_1(s) = s^{3/2} + s^{2/5}$ and $g_2(s) = s^{11/10} + s^{1/20}$, the Bazhlekova–Bazhlekov propagation positivity and subordination package holds even though $\alpha - \alpha_m > 1$. In particular, for the corresponding propagation functions one has $w \ge 0$, $w_t \ge 0$, $-w_x \ge 0$, and $\phi(t, \tau) = -w_x(\tau, t)$ is a probability-density subordination kernel. Hence the source condition $\alpha - \alpha_m \le 1$ is not necessary.

This resolves the source problem recorded as **APP-0058** [48].

Proof. We use Proposition 12.43 and Proposition 12.44 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For

$$h(s) = s^{\alpha}(1 + s^{-p})^{1/2}, \qquad 0 < \alpha - p/2 < \alpha < 1, \quad 1 < p < 2,$$

the the corresponding theorem the Bazhlekova Wright density topcap bridge normal form proves

$$\mathcal{L}^{-1}\{h'\}(t) = t^{-\alpha}\mathcal{W}_{\alpha,p}(t^p),$$

where

$$\mathcal{W}_{\alpha,p}(x) = -\sum_{m \ge 0} \binom{1/2}{m} \frac{x^m}{\Gamma(pm - \alpha)}.$$

Thus $\mathcal{W}_{\alpha,p}(x) > 0$ for all $x > 0$ implies $h' \in CM$, hence $h \in BF$.

For g_1, $(\alpha, p) = (3/4, 11/10)$. For g_2, $(\alpha, p) = (11/20, 21/20)$.

The node the Bazhlekova Wright two seed compact zero ten positive proves

$$\mathcal{W}_{3/4,11/10}(x) > 0, \qquad \mathcal{W}_{11/20,21/20}(x) > 0 \qquad (0 \le x \le 10).$$

Its proof artifacts are:

The node the Bazhlekova Wright two seed post ten derivative positive proves

$$\mathcal{W}'_{3/4,11/10}(x) > 0, \qquad \mathcal{W}'_{11/20,21/20}(x) > 0 \qquad (10 \le x \le 20),$$

with lower bounds $0.0201343226217\ldots$ and $0.00299977163172\ldots$. Since the functions are positive at 10, this proves positivity on $[10, 20]$. Its proof artifacts are:

The node the Bazhlekova Wright two seed Watson tail from twenty positive proves positivity for $x \ge 20$ by a three-term Watson contour certificate. The replay output at $x = 20$ gives margins

$$0.732374180856\ldots$$

for $(3/4, 11/10)$, and

$$0.100568095684\ldots$$

for $(11/20, 21/20)$. Its proof artifacts are:

Combining the three certified ranges gives the the corresponding theorem the Bazhlekova Wright two seed all x positive:

$$\mathcal{W}_{3/4,11/10}(x) > 0, \qquad \mathcal{W}_{11/20,21/20}(x) > 0 \qquad (x > 0).$$

This proves that the source condition $\alpha - \alpha_m \leq 1$ is not necessary for the positivity/subordination package. It does not classify all relaxations and does not claim $\sqrt{g_i} \in CBF$; in fact the off-cut zeros for gap > 1 block the CBF shortcut. $\square$

Proposition 12.46 (Positive stable negative moment formula)**.** *If S_α is positive α-stable with $0 < \alpha < 1$ and $\mathbb{E}e^{-uS_\alpha} = e^{-u^\alpha}$, then for $\Re r > 0$, $\mathbb{E}S_\alpha^{-r} = \Gamma(r/\alpha)/(\alpha\Gamma(r))$.*

Proof. For Positive stable negative moment formula, this proof is the corresponding component of Proposition 12.47. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Proposition 12.47 (Mellin quotient product-factor criterion)**.** *Let Y, Z, V be positive random variables with V independent of Z. If $\mathbb{E}(VZ)^{it} = \mathbb{E}Y^{it}$ for every real t, then $VZ \overset{d}{=} Y$. In particular, an explicit positive V whose Mellin transform is the quotient of the Mellin transforms of Y and Z on the imaginary line realizes the desired product factorization.*

Proof. For $r > 0$, use the identity

$$x^{-r} = \frac{1}{\Gamma(r)} \int_0^\infty u^{r-1} e^{-ux}\, du, \qquad x > 0.$$

With $x = S_\alpha$ and Fubini,

$$\mathbb{E}S_\alpha^{-r} = \frac{1}{\Gamma(r)} \int_0^\infty u^{r-1} \mathbb{E}e^{-uS_\alpha}\, du = \frac{1}{\Gamma(r)} \int_0^\infty u^{r-1} e^{-u^\alpha}\, du.$$

Substituting $v = u^\alpha$ gives

$$\mathbb{E}S_\alpha^{-r} = \frac{\Gamma(r/\alpha)}{\alpha\Gamma(r)}.$$

The same formula holds for complex r with $\Re r > 0$ by the same absolutely convergent integral.

For $\Re z > 0$,

$$\mathbb{E}W_0^z = \mathbb{E}S_\alpha^{-bz} = \frac{\Gamma(bz/\alpha)}{\alpha\Gamma(bz)} = \frac{\Gamma(az)}{\alpha\Gamma(bz)},$$

because $b/\alpha = a$. Since

$$\mathbb{E}W_0 = \frac{\Gamma(a)}{\alpha\Gamma(b)} < \infty,$$

the size-biased law is well-defined. For $\Re s > -1$,

$$\mathbb{E}W^s = \frac{\mathbb{E}W_0^{s+1}}{\mathbb{E}W_0} = \frac{\Gamma(a(s+1))\Gamma(b)}{\Gamma(b(s+1))\Gamma(a)}.$$

Therefore

$$\mathbb{E}V^s = 2^{(a-b)s}\frac{\Gamma(a(s+1))\Gamma(b)}{\Gamma(b(s+1))\Gamma(a)}.$$

Let $Y_q = |X_q|$, independent of V. For $\Re s > -1$,

$$\mathbb{E}(VY_q)^s = \mathbb{E}V^s\,\mathbb{E}|X_q|^s.$$

Using the source moment formula for X_q,

$$\mathbb{E}|X_q|^s = 2^{bs}\frac{\Gamma(b(s+1))}{\Gamma(b)}.$$

Thus

$$\mathbb{E}(VY_q)^s = 2^{as}\frac{\Gamma(a(s+1))}{\Gamma(a)} = \mathbb{E}|X_p|^s.$$

Taking $s = it$ gives equality of the characteristic functions of

$$\log(V|X_q|) \quad \text{and} \quad \log|X_p|.$$

Hence

$$V|X_q| \overset{d}{=} |X_p|.$$

Both X_q and X_p have continuous symmetric laws about 0, and $V > 0$ is independent of X_q. Multiplication by V preserves the sign symmetry of X_q, and the absolute values agree in distribution. Therefore

$$VX_q \overset{d}{=} X_p.$$

For $p = q$, take $V = 1$. For $p > q$, first-contact verified that the source's Proposition 15 proves the Mellin-ratio candidate is not a characteristic function, so no such positive independent factor exists. This completes the source-aligned classification. □

Definition 12.3 (Generalized Gaussian source normalization)**.** For the Dytso–Bustin–Poor–Shamai source, $X_p \sim N_p(0,1)$ denotes the centered unit-scale generalized Gaussian random variable with density proportional to $\exp(-|x|^p/2)$. Its absolute Mellin transform is $\mathbb{E}|X_p|^s = 2^{s/p}\Gamma((s+1)/p)/\Gamma(1/p)$ for $\Re s > -1$.

Definition 12.4 (Inverse stable power size-biased factor)**.** For $a > b > 0$, set $\alpha = b/a$. If S_α is positive α-stable, $W_0 = S_\alpha^{-b}$, and W is the ordinary W_0-size-biased random variable, then $\mathbb{E}W^s = \Gamma(a(s+1))\Gamma(b)/[\Gamma(b(s+1))\Gamma(a)]$ for $\Re s > -1$.

Corollary 12.48 (APP-0070: Dytso-Bustin-Poor generalized-Gaussian product factorization classification)**.** *For centered unit-scale generalized Gaussian random variables $X_p \sim N_p(0,1)$ and $X_q \sim N_q(0,1)$, there exists a positive random variable V, independent of X_q, such that $X_p \overset{d}{=} VX_q$ if and only if $p \le q$. The unresolved source branch is $0 < p < q$, where one may take $V = 2^{1/p-1/q}W$, with W the ordinary size-biased version of $S_{p/q}^{-1/q}$ and $S_{p/q}$ positive p/q-stable. The $p > q$ nonexistence branch is imported from the source.*

This resolves the source problem recorded as ***APP-0070*** *[59].*

Proof. We use Proposition 7.2, Proposition 12.46, Proposition 12.4, Proposition 12.3, and Proposition 12.47 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For $r > 0$, use the identity

$$x^{-r} = \frac{1}{\Gamma(r)} \int_0^\infty u^{r-1} e^{-ux}\, du, \qquad x > 0.$$

With $x = S_\alpha$ and Fubini,

$$\mathbb{E} S_\alpha^{-r} = \frac{1}{\Gamma(r)} \int_0^\infty u^{r-1} \mathbb{E} e^{-uS_\alpha}\, du = \frac{1}{\Gamma(r)} \int_0^\infty u^{r-1} e^{-u^\alpha}\, du.$$

Substituting $v = u^\alpha$ gives

$$\mathbb{E} S_\alpha^{-r} = \frac{\Gamma(r/\alpha)}{\alpha\Gamma(r)}.$$

The same formula holds for complex r with $\Re r > 0$ by the same absolutely convergent integral.

For $\Re z > 0$,

$$\mathbb{E} W_0^z = \mathbb{E} S_\alpha^{-bz} = \frac{\Gamma(bz/\alpha)}{\alpha\Gamma(bz)} = \frac{\Gamma(az)}{\alpha\Gamma(bz)},$$

because $b/\alpha = a$. Since

$$\mathbb{E} W_0 = \frac{\Gamma(a)}{\alpha\Gamma(b)} < \infty,$$

the size-biased law is well-defined. For $\Re s > -1$,

$$\mathbb{E} W^s = \frac{\mathbb{E} W_0^{s+1}}{\mathbb{E} W_0} = \frac{\Gamma(a(s+1))\Gamma(b)}{\Gamma(b(s+1))\Gamma(a)}.$$

Therefore

$$\mathbb{E} V^s = 2^{(a-b)s} \frac{\Gamma(a(s+1))\Gamma(b)}{\Gamma(b(s+1))\Gamma(a)}.$$

Let $Y_q = |X_q|$, independent of V. For $\Re s > -1$,

$$\mathbb{E}(VY_q)^s = \mathbb{E} V^s\, \mathbb{E}|X_q|^s.$$

Using the source moment formula for X_q,

$$\mathbb{E}|X_q|^s = 2^{bs} \frac{\Gamma(b(s+1))}{\Gamma(b)}.$$

Thus

$$\mathbb{E}(VY_q)^s = 2^{as} \frac{\Gamma(a(s+1))}{\Gamma(a)} = \mathbb{E}|X_p|^s.$$

Taking $s = it$ gives equality of the characteristic functions of

$$\log(V|X_q|) \quad \text{and} \quad \log|X_p|.$$

Hence

$$V|X_q| \stackrel{d}{=} |X_p|.$$

Both X_q and X_p have continuous symmetric laws about 0, and $V > 0$ is independent of X_q. Multiplication by V preserves the sign symmetry of X_q, and the absolute values agree in distribution. Therefore

$$VX_q \stackrel{d}{=} X_p.$$

For $p = q$, take $V = 1$. For $p > q$, first-contact verified that the source's Proposition 15 proves the Mellin-ratio candidate is not a characteristic function, so no such positive independent factor exists. This completes the source-aligned classification. $\square$

13 Moment-ratio and endpoint method applications

The following applications are source-facing consequences of the moment-ratio, Bessel zero, and endpoint-obstruction mechanisms developed earlier.

Counterexample 13.1 (APP-0039: Ramanujan Turan-window complete-monotonicity interval is false)**.** The Mishra–Swaminathan Ramanujan integral Turan-window complete-monotonicity statement has a negative answer: the proposed interval already fails for $n = 2$ and suitable $\alpha \in (0, 1/2)$.

This resolves the source problem recorded as **APP-0039** [13].

Proof. We use Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Put

$$A_k(x) = (-1)^k I_R^{(k)}(x) = \int_0^\infty e^{-xt} \frac{t^{k-1}}{\pi^2 + \log^2 t}\, dt \qquad (k \geq 1).$$

Then

$$H_2(x; \alpha) = A_2(x)^2 - \alpha A_1(x) A_3(x).$$

We study $x = e^{-L}$ as $L \to \infty$. With $y = xt$,

$$A_k(e^{-L}) = e^{kL} \int_0^\infty \frac{e^{-y} y^{k-1}}{\pi^2 + (L + \log y)^2}\, dy.$$

For fixed $k = 1, 2, 3$,

$$A_k(e^{-L}) = e^{kL} L^{-2} \Gamma(k) \left(1 - \frac{2\psi(k)}{L} + O(L^{-2})\right),$$

where ψ is the digamma function. To justify the expansion, split the y-integral into the central region $|\log y| \leq L^{2/3}$ and its complement. On the central interval, with $z = \log y$, expand uniformly:

$$\frac{1}{\pi^2 + (L + \log y)^2} = L^{-2} \left(1 - \frac{2 \log y}{L} + O\left(\frac{1 + \log^2 y}{L^2}\right)\right),$$

and integrate against $e^{-y} y^{k-1}$. The lower complement $0 < y < e^{-L^{2/3}}$ is $O(e^{-cL^{2/3}})$ for $k = 1, 2, 3$ after the change $y = e^{-u}$, and the upper complement $y > e^{L^{2/3}}$ is super-exponentially small because of e^{-y}. These errors are $o(L^{-N})$ for every fixed N, which is more than needed for the displayed expansion.

Therefore

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L}) A_3(e^{-L})} = \frac{\Gamma(2)^2}{\Gamma(1)\Gamma(3)} \left(1 + \frac{2(\psi(1) + \psi(3) - 2\psi(2))}{L} + O(L^{-2})\right).$$

Using

$$\psi(1) = -\gamma, \qquad \psi(2) = 1 - \gamma, \qquad \psi(3) = \frac{3}{2} - \gamma,$$

we get

$$\psi(1) + \psi(3) - 2\psi(2) = -\frac{1}{2},$$

so

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L}) A_3(e^{-L})} = \frac{1}{2} - \frac{1}{2L} + O(L^{-2}).$$

Choose

$$\alpha_L = \frac{1}{2} - \frac{1}{4L}.$$

For all sufficiently large L, $\alpha_L \in (0, 1/2)$ and

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L})A_3(e^{-L})} < \alpha_L.$$

Consequently

$$H_2(e^{-L}; \alpha_L) = A_2(e^{-L})^2 - \alpha_L A_1(e^{-L})A_3(e^{-L}) < 0.$$

Complete monotonicity would imply nonnegativity at order zero. Hence $H_2(\cdot; \alpha_L)$ is not completely monotone for these fixed parameters $\alpha_L \in (0, 1/2)$. The source's universal Turan-window statement is false already in the $n = 2$ upper window.

the Ramanujan Turan n2 upper window counterfamily the Ramanujan Turan window negative answer

The new mechanism is a logarithmic-density moment-ratio asymptotic obstruction for quadratic Laplace-moment Turan gaps. It is distinct from the public APP-0014 Stieltjes/complete-Bernstein classification of I_R. □

Proposition 13.2 (YT Bessel W general zero partial fraction)**.** *For every $\nu > -1$, if $j_{\nu+1,n}$ are the positive zeros of $J_{\nu+1}$, then $G_\nu(s) = W_\nu(\sqrt{s})$ satisfies $G_\nu(s) = 2(\nu+1) + 2\sum_{n\geq 1} s/(s + j_{\nu+1,n}^2)$ and $G_\nu'(s) = 2\sum_{n\geq 1} j_{\nu+1,n}^2/(s + j_{\nu+1,n}^2)^2$. Consequently G_ν is a Bernstein function on $(0, \infty)$.*

Proof. Put $\mu = \nu + 1 > 0$. The canonical product for the modified Bessel function gives, locally uniformly in z,

$$I_\mu(z) = \frac{(z/2)^\mu}{\Gamma(\mu+1)} \prod_{n=1}^{\infty} \left(1 + \frac{z^2}{j_{\mu,n}^2}\right),$$

where $j_{\mu,n}$ are the positive zeros of J_μ. Therefore

$$z\frac{I_\mu'(z)}{I_\mu(z)} = \mu + 2\sum_{n=1}^{\infty} \frac{z^2}{z^2 + j_{\mu,n}^2}.$$

The recurrence

$$I_\mu'(z) = I_{\mu-1}(z) - \frac{\mu}{z} I_\mu(z)$$

then yields

$$W_\nu(z) = z\frac{I_{\mu-1}(z)}{I_\mu(z)} = 2\mu + 2\sum_{n=1}^{\infty} \frac{z^2}{z^2 + j_{\mu,n}^2}.$$

Set $s = z^2$. Then

$$G_\nu(s) := W_\nu(\sqrt{s}) = 2(\nu+1) + 2\sum_{n=1}^{\infty} \frac{s}{s + j_{\nu+1,n}^2}.$$

Termwise differentiation is justified because $j_{\nu+1,n} \sim \pi n$, so the differentiated sums converge locally uniformly on $(0, \infty)$. Thus

$$G_\nu'(s) = 2\sum_{n=1}^{\infty} \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^2}.$$

For every $k \geq 0$,

$$(-1)^k \frac{d^k}{ds^k} \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^2} = (k+1)! \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^{k+2}} \geq 0.$$

The locally uniformly convergent sum preserves these inequalities, hence G'_ν is completely monotone and G_ν is a Bernstein function.

If $0 < \tau \leq 1/2$, then $x^{2\tau}$ is a Bernstein function. By Bernstein-function composition closure,

$$x \mapsto W_\nu(x^\tau) = G_\nu(x^{2\tau})$$

is a Bernstein function on $(0, \infty)$. $\square$

Corollary 13.3 (APP-0040: Yang-Tian Bessel-W power-Bernstein conjecture is true)**.** *For $\tau \in (0, 1/2]$ and $\nu > -1$, $x \mapsto W_\nu(x^\tau)$ is a Bernstein function on $(0, \infty)$.*

This resolves the source problem recorded as ***APP-0040*** *[34].*

Proof. We use Proposition 13.2 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Put $\mu = \nu + 1 > 0$. The canonical product for the modified Bessel function gives, locally uniformly in z,

$$I_\mu(z) = \frac{(z/2)^\mu}{\Gamma(\mu + 1)} \prod_{n=1}^{\infty} \left(1 + \frac{z^2}{j_{\mu,n}^2}\right),$$

where $j_{\mu,n}$ are the positive zeros of J_μ. Therefore

$$z \frac{I'_\mu(z)}{I_\mu(z)} = \mu + 2 \sum_{n=1}^{\infty} \frac{z^2}{z^2 + j_{\mu,n}^2}.$$

The recurrence

$$I'_\mu(z) = I_{\mu-1}(z) - \frac{\mu}{z} I_\mu(z)$$

then yields

$$W_\nu(z) = z \frac{I_{\mu-1}(z)}{I_\mu(z)} = 2\mu + 2 \sum_{n=1}^{\infty} \frac{z^2}{z^2 + j_{\mu,n}^2}.$$

Set $s = z^2$. Then

$$G_\nu(s) := W_\nu(\sqrt{s}) = 2(\nu + 1) + 2 \sum_{n=1}^{\infty} \frac{s}{s + j_{\nu+1,n}^2}.$$

Termwise differentiation is justified because $j_{\nu+1,n} \sim \pi n$, so the differentiated sums converge locally uniformly on $(0, \infty)$. Thus

$$G'_\nu(s) = 2 \sum_{n=1}^{\infty} \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^2}.$$

For every $k \geq 0$,

$$(-1)^k \frac{d^k}{ds^k} \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^2} = (k+1)! \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^{k+2}} \geq 0.$$

The locally uniformly convergent sum preserves these inequalities, hence G'_ν is completely monotone and G_ν is a Bernstein function.

If $0 < \tau \le 1/2$, then $x^{2\tau}$ is a Bernstein function. By Bernstein-function composition closure,

$$x \mapsto W_\nu(x^\tau) = G_\nu(x^{2\tau})$$

is a Bernstein function on $(0, \infty)$. $\square$

Counterexample 13.4 (APP-0041: Qi h_λ degree-four complete-monotonicity conjecture is false)**.** Let

$$\Psi(x) = [\psi'(x)]^2 + \psi''(x), \qquad h_\lambda(x) = \Psi(x) - \frac{x^2 + \lambda x + 12}{12x^4(x+1)^2}.$$

The conjecture that $\deg_x^{\mathrm{cm}} h_\lambda = \deg_x^{\mathrm{cm}}(-h_\mu) = 4$ exactly for $\lambda \le 0$ and $\mu \ge 4$ is false.

This resolves the source problem recorded as **APP-0041** [35].

Proof. We use Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

A completely monotone function must be nonincreasing, so its first derivative cannot be positive at any point.

Use the recurrence expansions at $x = 0^+$:

$$\psi'(x) = \frac{1}{x^2} + \zeta(2) - 2\zeta(3)x + 3\zeta(4)x^2 + O(x^3),$$

and

$$\psi''(x) = -\frac{2}{x^3} - 2\zeta(3) + 6\zeta(4)x + O(x^2).$$

Therefore

$$\Psi(x) = \frac{1}{x^4} - \frac{2}{x^3} + \frac{\pi^2}{3x^2} - \frac{4\zeta(3)}{x} + O(1).$$

Also

$$\frac{x^2 + \lambda x + 12}{12x^4(1+x)^2} = \frac{1}{x^4} + \left(\frac{\lambda}{12} - 2\right)\frac{1}{x^3} + \left(\frac{37}{12} - \frac{\lambda}{6}\right)\frac{1}{x^2} + O(x^{-1}).$$

Subtracting gives

$$h_\lambda(x) = -\frac{\lambda}{12x^3} + \left(\frac{\lambda}{6} + \frac{\pi^2}{3} - \frac{37}{12}\right)\frac{1}{x^2} + O(x^{-1}).$$

If $\lambda < 0$, then

$$x^4 h_\lambda(x) = -\frac{\lambda}{12}x + O(x^2),$$

so

$$\frac{d}{dx}\left[x^4 h_\lambda(x)\right] = -\frac{\lambda}{12} + O(x) > 0$$

for all sufficiently small $x > 0$. Hence $x^4 h_\lambda$ is not completely monotone.

If $\lambda = 0$, then

$$x^4 h_0(x) = \left(\frac{\pi^2}{3} - \frac{37}{12}\right)x^2 + O(x^3).$$

Since $4\pi^2 > 37$, the leading coefficient is positive, and

$$\frac{d}{dx}\left[x^4 h_0(x)\right] = 2\left(\frac{\pi^2}{3} - \frac{37}{12}\right)x + O(x^2) > 0$$

for all sufficiently small $x > 0$. Hence $x^4 h_0$ is not completely monotone.

Similarly, for $\mu \geq 4$,

$$x^4(-h_\mu(x)) = \frac{\mu}{12}x + O(x^2),$$

so

$$\frac{d}{dx}\left[x^4(-h_\mu(x))\right] = \frac{\mu}{12} + O(x) > 0$$

for all sufficiently small $x > 0$. Hence $x^4(-h_\mu)$ is not completely monotone for every $\mu \geq 4$.

Thus the degree-four transforms required by Qi's conjecture are not completely monotone on the conjectured source ranges. The conjecture in Remark 7.4 is false. $\square$

13.1 Derivative-chain closure and Laplace-survival Poissonization

Corollary 13.5 (APP-0059: Guo ASCM/SCM Theorem 45 condition can be waived)**.** *Let I^+ and I_1^+ be open intervals contained in $(0, \infty)$. If $f' \geq 0$, $f' \in ASCM(I_1^+)$, $g' \in SCM(I^+)$, and $R(g) \subset I_1^+$, then $(f \circ g)' \in ASCM(I^+)$. Therefore the condition $2xg'(x) \geq g(x)$ in Guo's Theorem 45 can be waived.*

This resolves the source problem recorded as ***APP-0059*** *[49].*

Proof. We use Proposition 7.2, Proposition 12.2, and Proposition 15.18 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Set $F = f'$, $u = g'$, and

$$H = (f \circ g)' = F(g)u.$$

For $H \in ASCM(I^+)$, it is enough to show that for every $m \geq 2$,

$$B_m(x) := (-1)^{m-1}x^m(f \circ g)^{(m)}(x)$$

is nonnegative and decreasing on I^+. This is the ASCM condition for H, with $m = n + 1$.

Faà di Bruno's formula gives

$$(f \circ g)^{(m)}(x) = \sum_{\Lambda_m} \frac{m!}{i_1! \cdots i_m!} f^{(k)}(g(x)) \prod_{j=1}^{m} \left(\frac{g^{(j)}(x)}{j!}\right)^{i_j},$$

where

$$\Lambda_m = \left\{(i_1, \ldots, i_m) \in \mathbb{N}_0^m : \sum_{j=1}^{m} j i_j = m\right\}, \qquad k = \sum_{j=1}^{m} i_j.$$

For each $j \geq 1$, since $u = g' \in SCM(I^+)$, the function

$$G_j(x) := \frac{(-1)^{j-1}x^j g^{(j)}(x)}{j!} = \frac{(-1)^{j-1}x^j u^{(j-1)}(x)}{j!}$$

is nonnegative and decreasing. In particular, $g' \geq 0$, so g is increasing.

For the f-factor, first take $k = 1$. Then $f^{(1)}(g(x)) = F(g(x))$. The hypothesis $F \geq 0$ gives nonnegativity, and $F \in ASCM$ gives $-y^2F'(y) \geq 0$, hence $F' \leq 0$. Since g is increasing, $F(g(x))$ is nonnegative and decreasing.

For $k \geq 2$, the ASCM condition for F gives

$$A_k(y) := (-1)^{k-1}y^k f^{(k)}(y) = (-1)^{k-1}y^k F^{(k-1)}(y)$$

nonnegative and decreasing on I_1^+. Therefore

$$(-1)^{k-1} f^{(k)}(y) = A_k(y) y^{-k}$$

is also nonnegative and decreasing, because it is the product of two nonnegative decreasing functions. Composing with the increasing function g, the function

$$(-1)^{k-1} f^{(k)}(g(x))$$

is nonnegative and decreasing on I^+.

Now fix a term in the Faà di Bruno sum. The sign exponent satisfies

$$(k-1) + \sum_{j=1}^{m} (j-1) i_j = (k-1) + (m-k) = m-1,$$

and the x-weights satisfy

$$\prod_{j=1}^{m} x^{j i_j} = x^m.$$

Hence

$$(-1)^{m-1} x^m f^{(k)}(g(x)) \prod_{j=1}^{m} \left(\frac{g^{(j)}(x)}{j!} \right)^{i_j}$$

is a positive numerical coefficient times

$$\Big[(-1)^{k-1} f^{(k)}(g(x))\Big] \prod_{j=1}^{m} G_j(x)^{i_j}.$$

Every factor is nonnegative and decreasing. Products and finite sums of nonnegative decreasing functions are nonnegative and decreasing. Thus B_m is nonnegative and decreasing for every $m \geq 2$.

Therefore $H = (f \circ g)' \in ASCM(I^+)$. The source condition $2xg'(x) \geq g(x)$ is unnecessary for Theorem 45. $\square$

Proposition 13.6 (Completely monotone survival functions are Laplace transforms)**.** *A function $S : [0, \infty) \to [0, 1]$ is the Laplace transform of a probability measure on $[0, +\infty)$ exactly when $S(0) = 1$, S is completely monotone on $(0, \infty)$, and the usual right-continuity/normalization conditions hold. Thus a survival function $S = 1 - F_\xi$ has a nonnegative Laplace-transform inverse law precisely under this Bernstein–Widder condition.*

Proof. Since U is uniform on $(0, 1)$,

$$\mathbb{P}(-\log U \leq y) = \mathbb{P}(U \geq e^{-y}) = 1 - e^{-y}, \qquad y \geq 0.$$

Thus $E = -\log U$ has the exponential distribution with rate 1.

Condition on $\zeta = z$. If $z > 0$, then

$$\mathbb{P}\left(\frac{E}{z} \leq x \mid \zeta = z \right) = \mathbb{P}(E \leq xz) = 1 - e^{-xz}.$$

If $z = 0$ in the extended-valued convention, then $\xi = \infty$, so

$$\mathbb{P}(\xi \leq x \mid \zeta = 0) = 0 = 1 - e^{-x \cdot 0}.$$

Taking expectations gives

$$\mathbb{P}(\xi \le x) = \mathbb{E}[1 - e^{-x\zeta}] = 1 - \mathbb{E}e^{-x\zeta} = 1 - L_\zeta(x).$$

This proves the source's forward construction in the finite case $\mathbb{P}(\zeta > 0) = 1$, and proves the corrected extended-valued formula for arbitrary $\zeta \ge 0$.

The atom-at-zero caveat is necessary. By bounded convergence,

$$L_\zeta(x) = \mathbb{E}e^{-x\zeta} \longrightarrow \mathbb{P}(\zeta = 0),$$

so

$$1 - L_\zeta(x) \longrightarrow 1 - \mathbb{P}(\zeta = 0).$$

A finite distribution function on $[0, \infty)$ must tend to 1. Therefore no finite-valued ξ can satisfy the literal formula for all $x \ge 0$ when $\mathbb{P}(\zeta = 0) > 0$.

For the probability-generating-function construction, condition on ζ. If $\kappa_t \mid \zeta \sim \text{Poisson}(t\zeta)$, then

$$\mathbb{E}[x^{\kappa_t} \mid \zeta] = \exp(t\zeta(x-1)) = \exp(-t(1-x)\zeta).$$

Therefore

$$\mathbb{E}[x^{\kappa_t}] = \mathbb{E}e^{-t(1-x)\zeta} = L_\zeta(t(1-x)).$$

This is the source's requested $G_t(x)$.

To realize κ_t as a transform of ζ and uniforms, take iid uniforms (η_i), set $E_i = -\log \eta_i$, $T_n = E_1 + \cdots + E_n$, and

$$N(a) = \max\{n \ge 0 : T_n \le a\}.$$

Then N is a standard Poisson process and $\kappa_t = N(t\zeta)$ supplies the conditional Poisson draw.

The inverse criterion is exactly Bernstein–Widder: normalized completely monotone survival functions are precisely Laplace transforms of probability measures on $[0, \infty)$. It is a bridge for the source inverse problem, not a new standalone theorem. $\square$

Proposition 13.7 (Conditional Poissonization realizes the Laplace PGF)**.** *Let $\zeta \ge 0$ and, for $t > 0$, let $\kappa_t \mid \zeta \sim \text{Poisson}(t\zeta)$. Then κ_t is nonnegative integer-valued and $\mathbb{E}[x^{\kappa_t}] = L_\zeta(t(1-x))$ for every $0 \le x \le 1$. The conditional Poisson draw can be realized from iid uniforms by the usual exponential-interarrival construction of a Poisson process.*

Proof. For Conditional Poissonization realizes the Laplace PGF, this proof is the corresponding component of Proposition 13.6. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Proposition 13.8 (Uniform exponential race gives the Laplace-survival transform)**.** *Let $U \sim \text{Unif}(0,1)$ be independent of a nonnegative random variable ζ, and set $E = -\log U$. If $\mathbb{P}(\zeta > 0) = 1$, then $\xi = E/\zeta$ is finite and satisfies $\mathbb{P}(\xi \le x) = 1 - L_\zeta(x)$ for every $x \ge 0$. For arbitrary $\zeta \ge 0$, the same equality holds for finite x under the extended convention $\xi = \infty$ on $\{\zeta = 0\}$.*

Proof. For Uniform exponential race gives the Laplace-survival transform, this proof is the corresponding component of Proposition 13.6. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Definition 13.1 (Laplace-survival transform)**.** For a nonnegative random variable ζ with Laplace transform $L_\zeta(x) = \mathbb{E}e^{-x\zeta}$, a Laplace-survival transform is a construction of a random variable ξ whose distribution function satisfies $F_\xi(x) = 1 - L_\zeta(x)$. If $\mathbb{P}(\zeta = 0) > 0$, the natural exact construction is extended-valued with $\xi = \infty$ on $\{\zeta = 0\}$.

Corollary 13.9 (APP-0069: Nagel-Weiss/Mecke Laplace-survival and Poissonization constructions)**.** *The Nagel–Weiss/Mecke Laplace-survival construction problem is solved in the finite-valued source scope: if $\mathbb{P}(\zeta > 0) = 1$, then $\xi = (-\log U)/\zeta$ satisfies $F_\xi(x) = 1 - L_\zeta(x)$ for all $x \geq 0$. For arbitrary nonnegative ζ, the same formula holds on the extended half-line by setting $\xi = \infty$ on $\{\zeta = 0\}$, and no finite-valued solution can satisfy the literal formula when $\mathbb{P}(\zeta = 0) > 0$. The source's probability-generating-function construction problem is solved by $\kappa_t \mid \zeta \sim \mathrm{Poisson}(t\zeta)$, which gives $\mathbb{E}[x^{\kappa_t}] = L_\zeta(t(1-x))$.*

*This resolves the source problem recorded as **APP-0069** [58].*

Proof. We use Proposition 7.2, Proposition 13.1, Proposition 13.7, Proposition 13.6, and Proposition 13.8 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Since U is uniform on $(0,1)$,

$$\mathbb{P}(-\log U \leq y) = \mathbb{P}(U \geq e^{-y}) = 1 - e^{-y}, \qquad y \geq 0.$$

Thus $E = -\log U$ has the exponential distribution with rate 1.

Condition on $\zeta = z$. If $z > 0$, then

$$\mathbb{P}\left(\frac{E}{z} \leq x \mid \zeta = z\right) = \mathbb{P}(E \leq xz) = 1 - e^{-xz}.$$

If $z = 0$ in the extended-valued convention, then $\xi = \infty$, so

$$\mathbb{P}(\xi \leq x \mid \zeta = 0) = 0 = 1 - e^{-x \cdot 0}.$$

Taking expectations gives

$$\mathbb{P}(\xi \leq x) = \mathbb{E}[1 - e^{-x\zeta}] = 1 - \mathbb{E}e^{-x\zeta} = 1 - L_\zeta(x).$$

This proves the source's forward construction in the finite case $\mathbb{P}(\zeta > 0) = 1$, and proves the corrected extended-valued formula for arbitrary $\zeta \geq 0$.

The atom-at-zero caveat is necessary. By bounded convergence,

$$L_\zeta(x) = \mathbb{E}e^{-x\zeta} \longrightarrow \mathbb{P}(\zeta = 0),$$

so

$$1 - L_\zeta(x) \longrightarrow 1 - \mathbb{P}(\zeta = 0).$$

A finite distribution function on $[0, \infty)$ must tend to 1. Therefore no finite-valued ξ can satisfy the literal formula for all $x \geq 0$ when $\mathbb{P}(\zeta = 0) > 0$.

For the probability-generating-function construction, condition on ζ. If $\kappa_t \mid \zeta \sim \mathrm{Poisson}(t\zeta)$, then

$$\mathbb{E}[x^{\kappa_t} \mid \zeta] = \exp(t\zeta(x-1)) = \exp(-t(1-x)\zeta).$$

Therefore

$$\mathbb{E}[x^{\kappa_t}] = \mathbb{E}e^{-t(1-x)\zeta} = L_\zeta(t(1-x)).$$

This is the source's requested $G_t(x)$.

To realize κ_t as a transform of ζ and uniforms, take iid uniforms (η_i), set $E_i = -\log \eta_i$, $T_n = E_1 + \cdots + E_n$, and

$$N(a) = \max\{n \geq 0 : T_n \leq a\}.$$

Then N is a standard Poisson process and $\kappa_t = N(t\zeta)$ supplies the conditional Poisson draw.

The inverse criterion is exactly Bernstein–Widder: normalized completely monotone survival functions are precisely Laplace transforms of probability measures on $[0, \infty)$. It is a bridge for the source inverse problem, not a new standalone theorem. $\square$

14 Bessel endpoint obstruction

The next group belongs to the same endpoint-obstruction mechanism used throughout the paper: a universal positivity or concavity statement is reduced to a one-point Riccati certificate, and exact rational bounds then decide the sign. The same certificate resolves the square-root log-concavity problem, the Riccati form of that problem, a related quadratic ratio bound, and the proposed three-regime certificate route.

Counterexample 14.1 (Not Bessel I sqrt log concavity nu ge 0)**.** not(For every $\nu \geq 0$, the function $u \mapsto \sqrt{u}\, I_\nu(u)$ is strictly log-concave on $(0, \infty)$.)

Proof. Let

$$g_\nu(u) = \log(\sqrt{u}\, I_\nu(u)), \qquad r_\nu(u) = \frac{I'_\nu(u)}{I_\nu(u)}.$$

Then

$$g''_\nu(u) = r'_\nu(u) - \frac{1}{2u^2}.$$

The modified Bessel equation

$$u^2 I''_\nu(u) + u I'_\nu(u) - (u^2 + \nu^2) I_\nu(u) = 0$$

gives

$$\frac{I''_\nu(u)}{I_\nu(u)} = 1 + \frac{\nu^2}{u^2} - \frac{r_\nu(u)}{u}.$$

Since

$$r'_\nu(u) = \frac{I''_\nu(u)}{I_\nu(u)} - r_\nu(u)^2,$$

we get

$$g''_\nu(u) = 1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2.$$

Thus the Bessel I Riccati log concavity inequality is exactly the strict log-concavity condition $g''_\nu(u) < 0$.

For $\nu = 0$,

$$r_0(u) = \frac{I'_0(u)}{I_0(u)} = \frac{I_1(u)}{I_0(u)}.$$

At $u = 10$, the Riccati expression is

$$g''_0(10) = \frac{199}{200} - \frac{r_0(10)}{10} - r_0(10)^2.$$

It is enough to prove

$$r_0(10) < \frac{9487}{10000},$$

because

$$\frac{199}{200} - \frac{1}{10}\frac{9487}{10000} - \left(\frac{9487}{10000}\right)^2 = \frac{9831}{100000000} > 0.$$

$$I_0(10) = \sum_{k=0}^{\infty} \frac{25^k}{(k!)^2}, \qquad I_1(10) = \sum_{k=0}^{\infty} \frac{5 \cdot 25^k}{k!(k+1)!}.$$

With $N = 20$, set

$$L_0 = \sum_{k=0}^{20} \frac{25^k}{(k!)^2} < I_0(10).$$

For the I_1 tail after $N = 20$, the term ratio is bounded by

$$q = \frac{25}{22 \cdot 23} < 1,$$

so the script computes an exact rational upper bound $U_1 > I_1(10)$ and verifies

$$10000\, U_1 < 9487\, L_0.$$

Therefore $I_1(10)/I_0(10) < 9487/10000$, and hence

$$g_0''(10) > 0.$$

This proves the Bessel-I log-concavity counterexample to T-I sqrt-log concavity for $\nu > 0$, and the counterexample to T-I Riccati log-concavity inequality. □

Counterexample 14.2 (APP-0043: Riccati log-concavity inequality for I_ν is false)**.** The inequality

$$1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2 < 0, \qquad r_\nu(u) = \frac{I_\nu'(u)}{I_\nu(u)},$$

does not hold for every $\nu \geq 0$ and $u > 0$.

This resolves the source problem recorded as **APP-0043** [36].

Proof. We use Proposition 9.1 and Proposition 14.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$g_\nu(u) = \log(\sqrt{u}\, I_\nu(u)), \qquad r_\nu(u) = \frac{I_\nu'(u)}{I_\nu(u)}.$$

Then

$$g_\nu''(u) = r_\nu'(u) - \frac{1}{2u^2}.$$

The modified Bessel equation

$$u^2 I_\nu''(u) + u I_\nu'(u) - (u^2 + \nu^2) I_\nu(u) = 0$$

gives

$$\frac{I_\nu''(u)}{I_\nu(u)} = 1 + \frac{\nu^2}{u^2} - \frac{r_\nu(u)}{u}.$$

Since

$$r_\nu'(u) = \frac{I_\nu''(u)}{I_\nu(u)} - r_\nu(u)^2,$$

we get

$$g_\nu''(u) = 1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2.$$

Thus the Bessel I Riccati log concavity inequality is exactly the strict log-concavity condition $g_\nu''(u) < 0$.

For $\nu = 0$,

$$r_0(u) = \frac{I_0'(u)}{I_0(u)} = \frac{I_1(u)}{I_0(u)}.$$

At $u = 10$, the Riccati expression is

$$g_0''(10) = \frac{199}{200} - \frac{r_0(10)}{10} - r_0(10)^2.$$

It is enough to prove

$$r_0(10) < \frac{9487}{10000},$$

because

$$\frac{199}{200} - \frac{1}{10}\frac{9487}{10000} - \left(\frac{9487}{10000}\right)^2 = \frac{9831}{100000000} > 0.$$

$$I_0(10) = \sum_{k=0}^{\infty} \frac{25^k}{(k!)^2}, \qquad I_1(10) = \sum_{k=0}^{\infty} \frac{5 \cdot 25^k}{k!(k+1)!}.$$

With $N = 20$, set

$$L_0 = \sum_{k=0}^{20} \frac{25^k}{(k!)^2} < I_0(10).$$

For the I_1 tail after $N = 20$, the term ratio is bounded by

$$q = \frac{25}{22 \cdot 23} < 1,$$

so the script computes an exact rational upper bound $U_1 > I_1(10)$ and verifies

$$10000\, U_1 < 9487\, L_0.$$

Therefore $I_1(10)/I_0(10) < 9487/10000$, and hence

$$g_0''(10) > 0.$$

This proves the Bessel-I log-concavity counterexample to T-I sqrt-log concavity for $\nu > 0$, and the counterexample to T-I Riccati log-concavity inequality. □

Counterexample 14.3 (APP-0044: Bessel ratio quadratic lower bound is false)**.** The claimed inequality

$$q_\nu(u)^2 + \frac{(2\nu+1)q_\nu(u)}{u} + \frac{\nu + \frac{1}{2}}{u^2} > 1, \qquad q_\nu(u) = \frac{I_{\nu+1}(u)}{I_\nu(u)},$$

is false for all $\nu \geq 0$ and $u > 0$.

This resolves the source problem recorded as **APP-0044** [36].

Proof. We use Proposition 9.1 and Proposition 14.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let
$$g_\nu(u) = \log(\sqrt{u}\, I_\nu(u)), \qquad r_\nu(u) = \frac{I'_\nu(u)}{I_\nu(u)}.$$
Then
$$g''_\nu(u) = r'_\nu(u) - \frac{1}{2u^2}.$$
The modified Bessel equation
$$u^2 I''_\nu(u) + uI'_\nu(u) - (u^2 + \nu^2)I_\nu(u) = 0$$
gives
$$\frac{I''_\nu(u)}{I_\nu(u)} = 1 + \frac{\nu^2}{u^2} - \frac{r_\nu(u)}{u}.$$
Since
$$r'_\nu(u) = \frac{I''_\nu(u)}{I_\nu(u)} - r_\nu(u)^2,$$
we get
$$g''_\nu(u) = 1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2.$$
Thus the Bessel I Riccati log concavity inequality is exactly the strict log-concavity condition $g''_\nu(u) < 0$.

For $\nu = 0$,
$$r_0(u) = \frac{I'_0(u)}{I_0(u)} = \frac{I_1(u)}{I_0(u)}.$$
At $u = 10$, the Riccati expression is
$$g''_0(10) = \frac{199}{200} - \frac{r_0(10)}{10} - r_0(10)^2.$$
It is enough to prove
$$r_0(10) < \frac{9487}{10000},$$
because
$$\frac{199}{200} - \frac{1}{10}\frac{9487}{10000} - \left(\frac{9487}{10000}\right)^2 = \frac{9831}{100000000} > 0.$$
$$I_0(10) = \sum_{k=0}^{\infty} \frac{25^k}{(k!)^2}, \qquad I_1(10) = \sum_{k=0}^{\infty} \frac{5 \cdot 25^k}{k!(k+1)!}.$$
With $N = 20$, set
$$L_0 = \sum_{k=0}^{20} \frac{25^k}{(k!)^2} < I_0(10).$$
For the I_1 tail after $N = 20$, the term ratio is bounded by
$$q = \frac{25}{22 \cdot 23} < 1,$$

so the script computes an exact rational upper bound $U_1 > I_1(10)$ and verifies

$$10000\, U_1 < 9487\, L_0.$$

Therefore $I_1(10)/I_0(10) < 9487/10000$, and hence

$$g_0''(10) > 0.$$

This proves the Bessel-I log-concavity counterexample to T-I sqrt-log concavity for $\nu > 0$, and the counterexample to T-I Riccati log-concavity inequality. □

Counterexample 14.4 (APP-0045: Three-regime Bessel log-concavity certificate route is refuted)**.** No valid three-regime certificate can prove the universal Riccati log-concavity inequality for I_ν, because the universal inequality itself is false.

This resolves the source problem recorded as **APP-0045** [36].

Proof. We use Proposition 9.1 and Proposition 14.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$g_\nu(u) = \log(\sqrt{u}\, I_\nu(u)), \qquad r_\nu(u) = \frac{I_\nu'(u)}{I_\nu(u)}.$$

Then

$$g_\nu''(u) = r_\nu'(u) - \frac{1}{2u^2}.$$

The modified Bessel equation

$$u^2 I_\nu''(u) + u I_\nu'(u) - (u^2 + \nu^2) I_\nu(u) = 0$$

gives

$$\frac{I_\nu''(u)}{I_\nu(u)} = 1 + \frac{\nu^2}{u^2} - \frac{r_\nu(u)}{u}.$$

Since

$$r_\nu'(u) = \frac{I_\nu''(u)}{I_\nu(u)} - r_\nu(u)^2,$$

we get

$$g_\nu''(u) = 1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2.$$

Thus the Bessel I Riccati log concavity inequality is exactly the strict log-concavity condition $g_\nu''(u) < 0$.

For $\nu = 0$,

$$r_0(u) = \frac{I_0'(u)}{I_0(u)} = \frac{I_1(u)}{I_0(u)}.$$

At $u = 10$, the Riccati expression is

$$g_0''(10) = \frac{199}{200} - \frac{r_0(10)}{10} - r_0(10)^2.$$

It is enough to prove

$$r_0(10) < \frac{9487}{10000},$$

because

$$\frac{199}{200} - \frac{1}{10}\frac{9487}{10000} - \left(\frac{9487}{10000}\right)^2 = \frac{9831}{100000000} > 0.$$

$$I_0(10) = \sum_{k=0}^{\infty} \frac{25^k}{(k!)^2}, \qquad I_1(10) = \sum_{k=0}^{\infty} \frac{5 \cdot 25^k}{k!(k+1)!}.$$

With $N = 20$, set

$$L_0 = \sum_{k=0}^{20} \frac{25^k}{(k!)^2} < I_0(10).$$

For the I_1 tail after $N = 20$, the term ratio is bounded by

$$q = \frac{25}{22 \cdot 23} < 1,$$

so the script computes an exact rational upper bound $U_1 > I_1(10)$ and verifies

$$10000\, U_1 < 9487\, L_0.$$

Therefore $I_1(10)/I_0(10) < 9487/10000$, and hence

$$g_0''(10) > 0.$$

This proves the Bessel-I log-concavity counterexample to T-I sqrt-log concavity for $\nu > 0$, and the counterexample to T-I Riccati log-concavity inequality. □

14.1 Bessel K endpoint quotient monotonicity

Proposition 14.5 (BPV Bessel K open range is nonmonotone)**.** *For every $|\nu| < 1$, the function $g_\nu(u) = K_\nu'(u)/K_\nu(u)^2$ is not monotone on $(0, \infty)$. More precisely, $g_\nu'(u) > 0$ for all sufficiently small $u > 0$, while $g_\nu'(u) < 0$ for all sufficiently large u.*

Proof. Since $K_{-\nu} = K_\nu$, it is enough to treat $0 \le \nu < 1$. Put

$$y_\nu(u) = \frac{uK_\nu'(u)}{K_\nu(u)}.$$

The modified Bessel equation gives

$$K_\nu''(u) = \left(1 + \frac{\nu^2}{u^2}\right) K_\nu(u) - \frac{1}{u} K_\nu'(u).$$

Differentiating $g_\nu = K_\nu'/K_\nu^2$ yields

$$g_\nu'(u) = \frac{K_\nu K_\nu'' - 2(K_\nu')^2}{K_\nu^3} = \frac{F_\nu(u)}{u^2 K_\nu(u)},$$

where

$$F_\nu(u) = u^2 + \nu^2 - y_\nu(u) - 2y_\nu(u)^2.$$

Since $K_\nu(u) > 0$ on $(0, \infty)$, the sign of g_ν' is the sign of F_ν.

First assume $0 < \nu < 1$. The standard small-argument expansion

$$K_\nu(u) \sim 2^{\nu-1}\Gamma(\nu)u^{-\nu}$$

gives $y_\nu(u) \to -\nu$ as $u \downarrow 0$. Hence

$$F_\nu(u) \to \nu^2 + \nu - 2\nu^2 = \nu(1-\nu) > 0.$$

Thus $g_\nu'(u) > 0$ for all sufficiently small $u > 0$.

For $\nu = 0$, the standard expansions

$$K_0(u) \sim -\log(u/2) - \gamma, \qquad K_0'(u) = -K_1(u) \sim -\frac{1}{u}$$

give

$$y_0(u) \sim -\frac{1}{-\log(u/2)-\gamma} \to 0^-.$$

Writing $a(u) = -y_0(u) > 0$, for all sufficiently small u we have $a(u) < 1/2$, and therefore

$$F_0(u) = u^2 - y_0(u) - 2y_0(u)^2 = u^2 + a(u)(1-2a(u)) > 0.$$

So $g_0'(u) > 0$ near zero.

For large u, the standard expansion

$$\frac{K_\nu'(u)}{K_\nu(u)} = -1 - \frac{1}{2u} + O(u^{-2})$$

gives

$$y_\nu(u) = -u - \frac{1}{2} + O(u^{-1}).$$

Substituting into F_ν,

$$F_\nu(u) = u^2 + \nu^2 - y_\nu(u) - 2y_\nu(u)^2 = -u^2 - u + \nu^2 + O(1) < 0$$

for all sufficiently large u. Thus $g_\nu'(u) < 0$ eventually.

The derivative has opposite signs near the two endpoints. Therefore g_ν is neither increasing nor decreasing on $(0,\infty)$ for every $|\nu| < 1$. $\square$

Proposition 14.6 (BPV BesselK halforder logderivative square unimodal)**.** *For $\nu = 1/2$, the BPV quotient $g(u) = K_{1/2}'(u)/K_{1/2}(u)^2$ satisfies $g'(u) > 0$ on $(0, (\sqrt{2}-1)/2)$ and $g'(u) < 0$ on $((\sqrt{2}-1)/2, \infty)$. Equivalently, it has a unique critical point at $(\sqrt{2}-1)/2$, a strict global maximum.*

Proof. Put $C = \sqrt{\pi/2}$. The half-order identity gives

$$K_{1/2}(u) = Cu^{-1/2}e^{-u}.$$

Hence

$$\frac{K_{1/2}'(u)}{K_{1/2}(u)} = \frac{d}{du}\log K_{1/2}(u) = -1 - \frac{1}{2u}.$$

Therefore

$$g(u) = \frac{K_{1/2}'(u)/K_{1/2}(u)}{K_{1/2}(u)} = -C^{-1}e^u\left(u^{1/2} + \frac{1}{2}u^{-1/2}\right).$$

Differentiating,

$$g'(u) = -C^{-1}e^u \left(u^{1/2} + u^{-1/2} - \frac{1}{4}u^{-3/2}\right) = -C^{-1}e^u u^{-3/2} \left(u^2 + u - \frac{1}{4}\right).$$

The quadratic $u^2 + u - \frac{1}{4}$ has the single positive zero

$$r = \frac{-1 + \sqrt{2}}{2}.$$

It is negative on $(0, r)$ and positive on (r, ∞). Since the remaining prefactor in g' is strictly negative, $g'(u) > 0$ on $(0, r)$ and $g'(u) < 0$ on (r, ∞).

Thus the half-order quotient increases first and then decreases. It is not decreasing on $(0, \infty)$. $\square$

Corollary 14.7 (APP-0056: BPV Bessel K logarithmic-derivative quotient monotonicity is classified)**.** *The function $u \mapsto K_\nu'(u)/K_\nu(u)^2$ is strictly decreasing on $(0, \infty)$ for $|\nu| \geq 1$, by BPV Theorem 2(a), and is nonmonotone on $(0, \infty)$ for $|\nu| < 1$, by the endpoint-sign theorem.*

*This resolves the source problem recorded as **APP-0056** [47].*

Proof. We use Proposition 14.6 and Proposition 14.5 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Since $K_{-\nu} = K_\nu$, it is enough to treat $0 \leq \nu < 1$. Put

$$y_\nu(u) = \frac{uK_\nu'(u)}{K_\nu(u)}.$$

The modified Bessel equation gives

$$K_\nu''(u) = \left(1 + \frac{\nu^2}{u^2}\right) K_\nu(u) - \frac{1}{u}K_\nu'(u).$$

Differentiating $g_\nu = K_\nu'/K_\nu^2$ yields

$$g_\nu'(u) = \frac{K_\nu K_\nu'' - 2(K_\nu')^2}{K_\nu^3} = \frac{F_\nu(u)}{u^2 K_\nu(u)},$$

where

$$F_\nu(u) = u^2 + \nu^2 - y_\nu(u) - 2y_\nu(u)^2.$$

Since $K_\nu(u) > 0$ on $(0, \infty)$, the sign of g_ν' is the sign of F_ν.

First assume $0 < \nu < 1$. The standard small-argument expansion

$$K_\nu(u) \sim 2^{\nu-1}\Gamma(\nu)u^{-\nu}$$

gives $y_\nu(u) \to -\nu$ as $u \downarrow 0$. Hence

$$F_\nu(u) \to \nu^2 + \nu - 2\nu^2 = \nu(1 - \nu) > 0.$$

Thus $g_\nu'(u) > 0$ for all sufficiently small $u > 0$.

For $\nu = 0$, the standard expansions

$$K_0(u) \sim -\log(u/2) - \gamma, \qquad K_0'(u) = -K_1(u) \sim -\frac{1}{u}$$

give

$$y_0(u) \sim -\frac{1}{-\log(u/2)-\gamma} \to 0^-.$$

Writing $a(u) = -y_0(u) > 0$, for all sufficiently small u we have $a(u) < 1/2$, and therefore

$$F_0(u) = u^2 - y_0(u) - 2y_0(u)^2 = u^2 + a(u)(1-2a(u)) > 0.$$

So $g_0'(u) > 0$ near zero.

For large u, the standard expansion

$$\frac{K_\nu'(u)}{K_\nu(u)} = -1 - \frac{1}{2u} + O(u^{-2})$$

gives

$$y_\nu(u) = -u - \frac{1}{2} + O(u^{-1}).$$

Substituting into F_ν,

$$F_\nu(u) = u^2 + \nu^2 - y_\nu(u) - 2y_\nu(u)^2 = -u^2 - u + \nu^2 + O(1) < 0$$

for all sufficiently large u. Thus $g_\nu'(u) < 0$ eventually.

The derivative has opposite signs near the two endpoints. Therefore g_ν is neither increasing nor decreasing on $(0,\infty)$ for every $|\nu| < 1$. □

Proposition 14.8 (Half-order modified Bessel K closed form)**.** *For $u > 0$, $K_{1/2}(u) = \sqrt{\pi/(2u)}e^{-u}$.*

Proof. At the admissible endpoint $\nu = 1/2$,

$$K_{1/2}(u) = \sqrt{\frac{\pi}{2u}}e^{-u}.$$

With $c = \sqrt{\pi/2}$,

$$u^2K_{1/2}'(u) = -ce^{-u}\left(u^{3/2} + \frac{1}{2}u^{1/2}\right).$$

Differentiating gives

$$\frac{d}{du}\{u^2K_{1/2}'(u)\} = ce^{-u}u^{-1/2}\left(u^2 - u - \frac{1}{4}\right).$$

The quadratic has positive root $(1+\sqrt{2})/2$. Hence the derivative is positive on $((1+\sqrt{2})/2, 2)$, so $u \mapsto u^2K_{1/2}'(u)$ is increasing on a subinterval of $(0,2)$ and cannot be strictly decreasing there. □

Proposition 14.9 (Half-order Bessel K endpoint derivative has a sign root)**.** *Let $F(u) = u^2K_{1/2}'(u)$. Then $F'(u) = \sqrt{\pi/2}e^{-u}u^{-1/2}(u^2 - u - 1/4)$, so $F'(u) > 0$ for $u > (1+\sqrt{2})/2$, in particular on a nonempty subinterval of $(0,2)$.*

Proof. For Half-order Bessel K endpoint derivative has a sign root, this proof is the corresponding component of Proposition 14.8. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Definition 14.1 (BPV Bessel K endpoint monotonicity question)**.** Baricz–Ponnusamy–Vuorinen Question 7 asks whether $u \mapsto u^2 K_\nu'(u)$ is strictly decreasing on $(0, 2)$ for every $|\nu| \le 1/2$, after proving the corresponding decreasing statement on $(0, 1)$.

Counterexample 14.10 (APP-0067: BPV Bessel K Question 7 has a half-order endpoint counterexample)**.** Baricz–Ponnusamy–Vuorinen Question 7 has a negative answer: $u \mapsto u^2 K_\nu'(u)$ is not strictly decreasing on $(0, 2)$ for all $|\nu| \le 1/2$. The endpoint $\nu = 1/2$ has $(u^2 K_{1/2}'(u))' > 0$ on $((1 + \sqrt{2})/2, 2)$.

This resolves the source problem recorded as **APP-0067** [47].

Proof. We use Proposition 9.1, Proposition 14.8, Proposition 14.9, and Proposition 14.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

At the admissible endpoint $\nu = 1/2$,

$$K_{1/2}(u) = \sqrt{\frac{\pi}{2u}} e^{-u}.$$

With $c = \sqrt{\pi/2}$,

$$u^2 K_{1/2}'(u) = -ce^{-u} \left(u^{3/2} + \frac{1}{2} u^{1/2} \right).$$

Differentiating gives

$$\frac{d}{du}\{u^2 K_{1/2}'(u)\} = ce^{-u} u^{-1/2} \left(u^2 - u - \frac{1}{4} \right).$$

The quadratic has positive root $(1 + \sqrt{2})/2$. Hence the derivative is positive on $((1 + \sqrt{2})/2, 2)$, so $u \mapsto u^2 K_{1/2}'(u)$ is increasing on a subinterval of $(0, 2)$ and cannot be strictly decreasing there. ☐

15 Endpoint obstruction certificates

This section collects source problems whose solutions are obtained by reducing the proposed universal statement to an explicit obstruction. Each result is placed with the certificate or counterexample that supplies its proof, so the source question appears as a consequence of the surrounding method.

Counterexample 15.1 (APP-0031: Baricz gamma-quotient Bernstein test is false)**.** Baricz's question asking whether $x \mapsto \Gamma(x)\Gamma(x - a + b)/(\Gamma(x - a)\Gamma(x + b))$ is a Bernstein function on (a, ∞) for every $a, b > 0$ has a negative answer.

This resolves the source problem recorded as **APP-0031** [16].

Proof. We use Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Take $a = 2$, $b = 3$, and set $y = x - 2 > 0$. Then by the Gamma recurrence,

$$\frac{\Gamma(x)\Gamma(x - a + b)}{\Gamma(x - a)\Gamma(x + b)} = \frac{\Gamma(x)\Gamma(x + 1)}{\Gamma(x - 2)\Gamma(x + 3)} = \frac{(x - 2)(x - 1)}{(x + 1)(x + 2)} = \frac{y(y + 1)}{(y + 3)(y + 4)}.$$

Let

$$g(y) = \frac{y(y + 1)}{(y + 3)(y + 4)}, \qquad y > 0.$$

If g were a Bernstein function, then g' would be completely monotone. In particular $g''' \ge 0$ would be necessary.

Exact symbolic differentiation gives

$$g'(y) = \frac{6(y^2+4y+2)}{(y+3)^2(y+4)^2},$$

$$g''(y) = -\frac{12(y^3+6y^2+6y-10)}{(y+3)^3(y+4)^3},$$

and

$$g'''(y) = \frac{36(y^4+8y^3+12y^2-40y-94)}{(y+3)^4(y+4)^4}.$$

At $y=1$,

$$g'''(1) = -\frac{1017}{40000} < 0.$$

Therefore g' is not completely monotone, g is not Bernstein, and the Baricz for-all-a, b Bernstein question has a negative answer. $\square$

Proposition 15.2 (From Mills all L moment ratio quadratic bound)**.** *Let $M_L(t) = (-1)^L r^{(L)}(t) = \int_0^\infty u^L e^{-tu-u^2/2}\,du$ for the standard normal Mills ratio. For every $L \geq 0$ and $t \geq 0$, with $m_L = M_{L+1}/M_L$, one has $m_L(t) < U_L(t)$, where $U_L(t)$ is the positive root* $\frac{\sqrt{t^2(t^2+4L+7)^2+4(L+1)(t^2+4L+8)(t^2+4L+6)}-t(t^2+4L+7}{2(t^2+4L+8)}$

Proof. Set

$$m_L(t) = \frac{M_{L+1}(t)}{M_L(t)}.$$

The recurrence gives

$$\frac{M_{L+2}}{M_L} = L+1-tm_L,$$

$$\frac{M_{L+3}}{M_L} = (t^2+L+2)m_L - t(L+1),$$

and

$$\frac{M_{L+4}}{M_L} = (L+1)(t^2+L+3) - t(t^2+2L+5)m_L.$$

Substitution into the determinant inequality and expansion give

$$(L+1)(t^2+4L+6) - t(t^2+4L+7)m_L - (t^2+4L+8)m_L^2 > 0.$$

Equivalently,

$$(t^2+4L+8)m_L^2 + t(t^2+4L+7)m_L - (L+1)(t^2+4L+6) < 0.$$

Since $m_L > 0$, the positive root yields the uniform strict bound

$$m_L(t) < U_L(t),$$

where

$$U_L(t) = \frac{\sqrt{t^2(t^2+4L+7)^2+4(L+1)(t^2+4L+8)(t^2+4L+6)} - t(t^2+4L+7)}{2(t^2+4L+8)}.$$

This is the determinant-to-bound extractor. It is a single theorem for every $L \geq 0$.

Define polynomials $P_n(t)$ by

$$P_0 = 1, \qquad P_1 = t, \qquad P_{n+1} = tP_n + nP_{n-1}.$$

Define $B_n(t)$ by

$$B_0 = 0, \qquad B_1 = 1, \qquad B_{n+1} = nB_{n-1} - tB_n.$$

Then the same recurrence gives

$$M_n(t) = (-1)^n P_n(t) r(t) + B_n(t).$$

The ratio bound $M_{L+1} < U_L M_L$ becomes

$$\big((-1)^{L+1} P_{L+1} - (-1)^L U_L P_L\big)\, r < U_L B_L - B_{L+1}.$$

Since $P_n(t) > 0$ for $t > 0$ and $P_{2j}(0) > 0$, the coefficient has sign $(-1)^{L+1}$. Thus the bounds alternate:

For even $L \geq 0$,

$$r(t) > \frac{B_{L+1}(t) - U_L(t)B_L(t)}{P_{L+1}(t) + U_L(t)P_L(t)}.$$

For odd $L \geq 1$,

$$r(t) < \frac{U_L(t)B_L(t) - B_{L+1}(t)}{P_{L+1}(t) + U_L(t)P_L(t)}.$$

These formulas are valid at $t = 0$ by direct evaluation of the recurrences and for $t > 0$ by the positivity of M_L. □

Corollary 15.3 (APP-0032: From's all-L Mills-ratio bound family is explicit)**.** *Let $P_0 = 1$, $P_1 = t$, $P_{n+1} = tP_n + nP_{n-1}$, and $B_0 = 0$, $B_1 = 1$, $B_{n+1} = nB_{n-1} - tB_n$. With $U_L(t)$ as in the all-L moment-ratio bound, the standard normal Mills ratio satisfies, for even $L \geq 0$, $r(t) > [B_{L+1} - U_L B_L]/[P_{L+1} + U_L P_L]$, and for odd $L \geq 1$, $r(t) < [U_L B_L - B_{L+1}]/[P_{L+1} + U_L P_L]$, for all $t \geq 0$.*

*This resolves the source problem recorded as **APP-0032** [15].*

Proof. We use Proposition 12.6 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Set

$$m_L(t) = \frac{M_{L+1}(t)}{M_L(t)}.$$

The recurrence gives

$$\frac{M_{L+2}}{M_L} = L + 1 - tm_L,$$

$$\frac{M_{L+3}}{M_L} = (t^2 + L + 2)m_L - t(L+1),$$

and

$$\frac{M_{L+4}}{M_L} = (L+1)(t^2 + L + 3) - t(t^2 + 2L + 5)m_L.$$

Substitution into the determinant inequality and expansion give

$$(L+1)(t^2 + 4L + 6) - t(t^2 + 4L + 7)m_L - (t^2 + 4L + 8)m_L^2 > 0.$$

Equivalently,

$$(t^2 + 4L + 8)m_L^2 + t(t^2 + 4L + 7)m_L - (L + 1)(t^2 + 4L + 6) < 0.$$

Since $m_L > 0$, the positive root yields the uniform strict bound

$$m_L(t) < U_L(t),$$

where

$$U_L(t) = \frac{\sqrt{t^2(t^2 + 4L + 7)^2 + 4(L + 1)(t^2 + 4L + 8)(t^2 + 4L + 6)} - t(t^2 + 4L + 7)}{2(t^2 + 4L + 8)}.$$

This is the determinant-to-bound extractor. It is a single theorem for every $L \geq 0$.

Define polynomials $P_n(t)$ by

$$P_0 = 1, \qquad P_1 = t, \qquad P_{n+1} = tP_n + nP_{n-1}.$$

Define $B_n(t)$ by

$$B_0 = 0, \qquad B_1 = 1, \qquad B_{n+1} = nB_{n-1} - tB_n.$$

Then the same recurrence gives

$$M_n(t) = (-1)^n P_n(t)r(t) + B_n(t).$$

The ratio bound $M_{L+1} < U_L M_L$ becomes

$$\left((-1)^{L+1}P_{L+1} - (-1)^L U_L P_L\right) r < U_L B_L - B_{L+1}.$$

Since $P_n(t) > 0$ for $t > 0$ and $P_{2j}(0) > 0$, the coefficient has sign $(-1)^{L+1}$. Thus the bounds alternate:

For even $L \geq 0$,

$$r(t) > \frac{B_{L+1}(t) - U_L(t)B_L(t)}{P_{L+1}(t) + U_L(t)P_L(t)}.$$

For odd $L \geq 1$,

$$r(t) < \frac{U_L(t)B_L(t) - B_{L+1}(t)}{P_{L+1}(t) + U_L(t)P_L(t)}.$$

These formulas are valid at $t = 0$ by direct evaluation of the recurrences and for $t > 0$ by the positivity of M_L. □

Proposition 15.4 (Ramanujan density logsquare CM)**.** *The function $\phi_0(t) = 1/[t(\pi^2 + \log^2 t)]$ is completely monotone on $(0, \infty)$.*

Proof. The density

$$\phi_0(t) = \frac{1}{t(\pi^2 + \log^2 t)}$$

is completely monotone on $(0, \infty)$. Consequently I_R is a Stieltjes function. Moreover, the antiderivative

$$\widetilde{I}_R(x) = a + \int_0^\infty (1 - e^{-xt})\phi_0(t)\, dt$$

is a complete Bernstein function.

For $u = \log t$,

$$\int_0^1 e^{-au} \sin(\pi a)\, da = \frac{\pi(1+e^{-u})}{u^2+\pi^2}.$$

Substituting $u = \log t$ gives

$$\int_0^1 t^{-a} \sin(\pi a)\, da = \frac{\pi(1+t^{-1})}{\pi^2+\log^2 t}.$$

Therefore

$$\phi_0(t) = \frac{1}{t(\pi^2+\log^2 t)} = \frac{1}{\pi(1+t)} \int_0^1 t^{-a} \sin(\pi a)\, da.$$

For $0 < a < 1$, t^{-a} is completely monotone because

$$t^{-a} = \frac{1}{\Gamma(a)} \int_0^\infty e^{-ts} s^{a-1}\, ds.$$

Also $t \mapsto (1+t)^{-1}$ is completely monotone. Complete monotonicity is closed under products and positive mixtures, and $\sin(\pi a) > 0$ on $(0,1)$. Hence ϕ_0 is completely monotone.

If ϕ is completely monotone and

$$F(x) = \int_0^\infty e^{-xt} \phi(t)\, dt$$

is finite for $x > 0$, then F is Stieltjes: write $\phi(t) = \int_0^\infty e^{-ts}\, d\mu(s)$ and use Tonelli to obtain

$$F(x) = \int_0^\infty \frac{d\mu(s)}{x+s}.$$

Applying this to ϕ_0 proves that I_R is Stieltjes.

Finally, the source already proves that

$$\widetilde{I}_R(x) = a + \int_0^\infty (1-e^{-xt})\phi_0(t)\, dt$$

is a Bernstein function. The Levy density ϕ_0 is completely monotone, and it satisfies the Bernstein integrability condition

$$\int_0^\infty (1 \wedge t)\phi_0(t)\, dt < \infty.$$

The standard complete-Bernstein criterion for Levy densities therefore gives that $\widetilde{I}_R$ is a complete Bernstein function. □

Corollary 15.5 (APP-0033: Ramanujan antiderivative is complete Bernstein)**.** *The antiderivative $\widetilde{I}_R(x) = a + \int_0^\infty (1-e^{-xt})\, dt/[t(\pi^2+\log^2 t)]$ of the Ramanujan integral is a complete Bernstein function.*

This resolves the source problem recorded as ***APP-0033*** *[13].*

Proof. We use Proposition 12.2 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For $u = \log t$,

$$\int_0^1 e^{-au} \sin(\pi a)\, da = \frac{\pi(1+e^{-u})}{u^2+\pi^2}.$$

Substituting $u = \log t$ gives

$$\int_0^1 t^{-a} \sin(\pi a)\, da = \frac{\pi(1+t^{-1})}{\pi^2 + \log^2 t}.$$

Therefore

$$\phi_0(t) = \frac{1}{t(\pi^2 + \log^2 t)} = \frac{1}{\pi(1+t)} \int_0^1 t^{-a} \sin(\pi a)\, da.$$

For $0 < a < 1$, t^{-a} is completely monotone because

$$t^{-a} = \frac{1}{\Gamma(a)} \int_0^\infty e^{-ts} s^{a-1}\, ds.$$

Also $t \mapsto (1+t)^{-1}$ is completely monotone. Complete monotonicity is closed under products and positive mixtures, and $\sin(\pi a) > 0$ on $(0,1)$. Hence ϕ_0 is completely monotone.

If ϕ is completely monotone and

$$F(x) = \int_0^\infty e^{-xt} \phi(t)\, dt$$

is finite for $x > 0$, then F is Stieltjes: write $\phi(t) = \int_0^\infty e^{-ts}\, d\mu(s)$ and use Tonelli to obtain

$$F(x) = \int_0^\infty \frac{d\mu(s)}{x+s}.$$

Applying this to ϕ_0 proves that I_R is Stieltjes.

Finally, the source already proves that

$$\widetilde{I}_R(x) = a + \int_0^\infty (1 - e^{-xt}) \phi_0(t)\, dt$$

is a Bernstein function. The Levy density ϕ_0 is completely monotone, and it satisfies the Bernstein integrability condition

$$\int_0^\infty (1 \wedge t) \phi_0(t)\, dt < \infty.$$

The standard complete-Bernstein criterion for Levy densities therefore gives that $\widetilde{I}_R$ is a complete Bernstein function.

the CM closure product positive mixture the Stieltjes density criterion S1 the CBF Levy density CM criterion the Ramanujan density logsquare CM the Ramanujan integral Stieltjes the Ramanujan antiderivative complete Bernstein

the Ramanujan Turan window CM open problem ☐

Corollary 15.6 (APP-0034: Bulboaca-Zayed gamma-quotient monotonicity solved)**.** *For the continuous extension $\widetilde{F}$ of $F(x) = \log \Gamma(x+1)/(\log(x^2+6) - \log(x+6))$ on $(-1, \infty)$, prove analytically that $\widetilde{F}'$ has a unique zero $x_m \simeq 1.126207061\ldots$, with $\widetilde{F}' < 0$ on $(-1, x_m)$ and $\widetilde{F}' > 0$ on (x_m, ∞).*

This resolves the source problem recorded as ***APP-0034*** *[5].*

Proof. We use Proposition 12.39 and Proposition 12.38 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

proof.

The raw numerator has a removable zero at $x = 1$, since $G(1) = D(1) = 0$. Its quadratic leading coefficient is

$$\lim_{x \downarrow 1} \frac{N_0(x)}{(x-1)^2} = \frac{7(\pi^2/6 - 1) - 11(1-\gamma)}{98} < 0,$$

again by $\pi^2 < 987/100$ and $\gamma < 5773/10000$, which gives $-1347/980000 < 0$. Therefore any normalization dividing out the removable $(x-1)^2$ factor is negative at the left endpoint.

The single-crossing candidate is true:

$$theBulboaca--Zayedgamma-quotientNsinglecrossingoneeight.$$

Together with the source's theorem on $(-1,1)$ and the previous local right-tail theorem for $x \geq 8$, this proves the critical-window reduction

$$theBulboaca--Zayedgamma-quotientcriticalwindowreduction$$

and then the source problem

$$theBulboacaZayedgammaquotientfullmonotonicity.$$

The finite-envelope candidate is not separately promoted: the proof above is a clean analytic ratio-kernel argument rather than the finite rational cover specified in that candidate. The diagnostic obstruction-map candidate is also not needed after the single-crossing proof. ☐

Proposition 15.7 (Keady self bijection inverse CM negative example)**.** *The function $f(x) = x^{-1} + 100e^{-x}$ is a completely monotone decreasing bijection $(0,\infty) \to (0,\infty)$, but f^{-1} is not completely monotone.*

Proof. This proof is the component of Proposition 15.8 corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. ☐

Proposition 15.8 (Keady inverse third derivative sign certificate)**.** *For $f(x) = x^{-1} + 100e^{-x}$ and $g = f^{-1}$, the inverse-derivative certificate gives $g'''(f(1/8)) > 0$.*

Proof. There exists a completely monotone decreasing bijection

$$f : (0,\infty) \to (0,\infty)$$

whose inverse is not completely monotone. One explicit example is

$$f(x) = \frac{1}{x} + 100e^{-x}.$$

For every $n \geq 0$,

$$(-1)^n f^{(n)}(x) = \frac{n!}{x^{n+1}} + 100e^{-x} > 0.$$

Thus f is strictly completely monotone. Also

$$f'(x) = -x^{-2} - 100e^{-x} < 0,$$

so f is strictly decreasing. Finally,

$$\lim_{x\downarrow 0} f(x) = \infty, \qquad \lim_{x\to\infty} f(x) = 0.$$

Therefore f is a decreasing bijection from $(0,\infty)$ onto $(0,\infty)$.

Let $g = f^{-1}$ and write $y = f(x)$. Inverse differentiation gives

$$g'''(f(x)) = \frac{3(f''(x))^2 - f'''(x)f'(x)}{(f'(x))^5}.$$

For $f_a(x) = x^{-1} + ae^{-x}$, define

$$N_a(x) = 3(f_a''(x))^2 - f_a'''(x)f_a'(x).$$

A direct calculation gives

$$N_a(x) = \frac{6}{x^6} + ae^{-x}\left(-\frac{6}{x^4} + \frac{12}{x^3} - \frac{1}{x^2}\right) + 2a^2e^{-2x}.$$

At $a = 100$ and $x_0 = 1/8$,

$$N_{100}(1/8) = 1{,}572{,}864 - 1{,}849{,}600e^{-1/8} + 20{,}000e^{-1/4}.$$

Using $e^{-1/8} > 7/8$ and $e^{-1/4} < 1$,

$$N_{100}(1/8) < 1{,}572{,}864 - 1{,}849{,}600 \cdot \frac{7}{8} + 20{,}000 = -25{,}536 < 0.$$

Since $f'(1/8) < 0$, the denominator $(f'(1/8))^5$ is negative. Hence

$$g'''(f(1/8)) = \frac{N_{100}(1/8)}{(f'(1/8))^5} > 0.$$

A completely monotone function must satisfy $(-1)^3 g''' \geq 0$, i.e. $g''' \leq 0$. Therefore $g = f^{-1}$ is not completely monotone. □

Counterexample 15.9 (APP-0035: Keady self-bijection inverse-CM question is negative)**.** Keady Question 3 has a negative answer for the self-range case: there is a completely monotone decreasing bijection $(0, \infty) \to (0, \infty)$ whose inverse is not completely monotone.

This resolves the source problem recorded as **APP-0035** [33].

Proof. We use Proposition 15.8 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

There exists a completely monotone decreasing bijection

$$f : (0, \infty) \to (0, \infty)$$

whose inverse is not completely monotone. One explicit example is

$$f(x) = \frac{1}{x} + 100e^{-x}.$$

For every $n \geq 0$,

$$(-1)^n f^{(n)}(x) = \frac{n!}{x^{n+1}} + 100e^{-x} > 0.$$

Thus f is strictly completely monotone. Also

$$f'(x) = -x^{-2} - 100e^{-x} < 0,$$

so f is strictly decreasing. Finally,

$$\lim_{x\downarrow 0} f(x) = \infty, \qquad \lim_{x\to\infty} f(x) = 0.$$

Therefore f is a decreasing bijection from $(0,\infty)$ onto $(0,\infty)$.

Let $g = f^{-1}$ and write $y = f(x)$. Inverse differentiation gives

$$g'''(f(x)) = \frac{3(f''(x))^2 - f'''(x)f'(x)}{(f'(x))^5}.$$

For $f_a(x) = x^{-1} + ae^{-x}$, define

$$N_a(x) = 3(f_a''(x))^2 - f_a'''(x)f_a'(x).$$

A direct calculation gives

$$N_a(x) = \frac{6}{x^6} + ae^{-x}\left(-\frac{6}{x^4} + \frac{12}{x^3} - \frac{1}{x^2}\right) + 2a^2e^{-2x}.$$

At $a = 100$ and $x_0 = 1/8$,

$$N_{100}(1/8) = 1{,}572{,}864 - 1{,}849{,}600e^{-1/8} + 20{,}000e^{-1/4}.$$

Using $e^{-1/8} > 7/8$ and $e^{-1/4} < 1$,

$$N_{100}(1/8) < 1{,}572{,}864 - 1{,}849{,}600 \cdot \frac{7}{8} + 20{,}000 = -25{,}536 < 0.$$

Since $f'(1/8) < 0$, the denominator $(f'(1/8))^5$ is negative. Hence

$$g'''(f(1/8)) = \frac{N_{100}(1/8)}{(f'(1/8))^5} > 0.$$

A completely monotone function must satisfy $(-1)^3g''' \ge 0$, i.e. $g''' \le 0$. Therefore $g = f^{-1}$ is not completely monotone. $\square$

Counterexample 15.10 (APP-0036: Baskakov even-line complete-monotonicity fails at $r = 8$)**.** Abel–Gawronski–Neuschel's even-power Baskakov complete-monotonicity conjecture is false; it fails at $\alpha = 1$, $r = 8$.

This resolves the source problem recorded as **APP-0036** [25].

Proof. We use Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Set

$$D_m(x) = (1+x)^{2m} - x^{2m}.$$

The roots are obtained from

$$\frac{1+x}{x} = \omega_k, \qquad \omega_k = e^{\pi ik/m}, \qquad k = 1,\dots,2m-1.$$

Thus

$$\lambda_{k,m} = \frac{1}{\omega_k - 1} = -\frac{1}{2} - \frac{i}{2}\cot\frac{\pi k}{2m}.$$

At this root,

$$D'_m(\lambda_{k,m}) = 2m\lambda_{k,m}^{2m-1}(\omega_k^{-1} - 1),$$

and direct simplification gives the real residue

$$R_{k,m} = \frac{1}{D'_m(\lambda_{k,m})} = \frac{(-1)^{m+k}}{2m}\left(2\sin\frac{\pi k}{2m}\right)^{2m-2}.$$

Consequently the inverse-Laplace density is

$$\rho_m(t) = \sum_{k=1}^{2m-1} R_{k,m}e^{\lambda_{k,m}t}.$$

Pairing conjugate roots gives

$$\rho_m(t) = \frac{e^{-t/2}}{2m}\left[2^{2m-2} + 2\sum_{k=1}^{m-1}(-1)^{m+k}\left(2\sin\frac{\pi k}{2m}\right)^{2m-2}\cos\left(\frac{t}{2}\cot\frac{\pi k}{2m}\right)\right].$$

This proves the finite trigonometric density formula used by the diagnostic candidate.

For $m = 2$, this gives

$$\rho_2(t) = e^{-t/2}\left(1 - \cos\frac{t}{2}\right) = 2e^{-t/2}\sin^2\frac{t}{4} \geq 0,$$

recovering the previously admitted $r = 4, \alpha = 1$ seed.

For $m = 3$,

$$\rho_3(t) = \frac{e^{-t/2}}{6}\left[16 + 2\cos\left(\frac{\sqrt{3}\,t}{2}\right) - 18\cos\left(\frac{t}{2\sqrt{3}}\right)\right].$$

Set $u = t/(2\sqrt{3})$. Since $\cos(3u) = 4\cos^3 u - 3\cos u$,

$$\rho_3(t) = \frac{4}{3}e^{-t/2}(1-\cos u)^2(2+\cos u) \geq 0.$$

Thus $r = 6, \alpha = 1$ is completely monotone by an explicit positive density.

For $m = 4$, write $h_4(t) = e^{t/2}\rho_4(t)$. The density formula gives

$$h_4(t) = 8 + 2\cos\frac{t}{2} - \frac{(2-\sqrt{2})^3}{4}\cos\left(\frac{1+\sqrt{2}}{2}t\right) - \frac{(2+\sqrt{2})^3}{4}\cos\left(\frac{\sqrt{2}-1}{2}t\right).$$

At $t = 10\pi$,

$$\cos(t/2) = \cos(5\pi) = -1,$$

and both other cosine terms equal $-\cos(5\pi\sqrt{2})$. Since

$$(2-\sqrt{2})^3 + (2+\sqrt{2})^3 = 40,$$

we obtain

$$h_4(10\pi) = 6 + 10\cos(5\pi\sqrt{2}).$$

Now

$$0 < 5\sqrt{2} - 7 < \frac{1}{4}$$

because $7/5 < \sqrt{2} < 29/20$. Hence, with $\delta = 5\sqrt{2} - 7$,

$$\cos(5\pi\sqrt{2}) = \cos(7\pi + \pi\delta) = -\cos(\pi\delta) < -\cos\frac{\pi}{4} = -\frac{\sqrt{2}}{2} < -\frac{3}{5}.$$

Therefore

$$h_4(10\pi) < 6 + 10\left(-\frac{3}{5}\right) = 0,$$

and

$$\rho_4(10\pi) = e^{-5\pi} h_4(10\pi) < 0.$$

Since the inverse-Laplace transform is unique for measures on $[0, \infty)$, $f_1^{[8]}(x) = 1/((1+x)^8 - x^8)$ cannot be completely monotone. ☐

Counterexample 15.11 (APP-0048: BMR tau-Gauss ordinary concavity is false)**.** The Bansal–Mehrez–Raina tau-Gauss source function $a \mapsto {}_2\phi_1^{\tau}(a, c-a; c; z)$ is not concave in general on $(0, c)$. In the subcase $c = 2$, $\tau = 1$, and $z = 1 - e^{-4}$, one has $F''(a, z) > 0$ for all sufficiently small $a > 0$.

This resolves the source problem recorded as **APP-0048** [39].

Proof. We use Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For $0 < z < 1$, define probability weights

$$w_k(a, z) = \frac{A_k(a) z^k / k!}{F(a, z)}.$$

Let

$$g_k(a) = \partial_a \log A_k(a), \qquad h_k(a) = \partial_a^2 \log A_k(a).$$

Then direct differentiation of the log-sum gives

$$\partial_a^2 \log F(a, z) = \sum_{k \geq 0} w_k(a, z) h_k(a) + \mathrm{Var}_{w(a,z)}(g_k(a)).$$

Thus global log-concavity is equivalent to the variance-domination inequality

$$\mathrm{Var}_{w(a,z)}(g_k(a)) \leq \sum_{k \geq 0} w_k(a, z)(-h_k(a))$$

for all $0 < a < c$, $c, \tau > 0$, and $0 < z < 1$. This is the sharp remaining gate; coefficientwise log-concavity alone is insufficient.

Take the classical subcase

$$c = 2, \qquad \tau = 1, \qquad z = 1 - e^{-4}.$$

Then

$$F(a, z) = \sum_{k=0}^{\infty} \frac{(a)_k (2-a)_k}{(2)_k} \frac{z^k}{k!}.$$

For $k \geq 1$, as $a \downarrow 0$,

$$\frac{(a)_k (2-a)_k}{(2)_k \, k!} = \frac{a}{k} + a^2 \left(\frac{1}{k} - \frac{1}{k^2} - \frac{1}{k(k+1)}\right) + O(a^3),$$

with locally uniform summability for fixed $0 < z < 1$. Hence

$$F(a,z) = 1 + aL(z) + a^2M(z) + O(a^3),$$

where

$$L(z) = \sum_{k\geq 1} \frac{z^k}{k} = -\log(1-z)$$

and

$$M(z) = \sum_{k\geq 1} z^k \left(\frac{1}{k} - \frac{1}{k^2} - \frac{1}{k(k+1)}\right) = \frac{L(z)}{z} - \mathrm{Li}_2(z) - 1.$$

For $z = 1 - e^{-4}$, $L(z) = 4$, $L(z)/z > 4$, and $\mathrm{Li}_2(z) < \pi^2/6 < 2$. Therefore

$$M(z) > 4 - 2 - 1 = 1 > 0.$$

It follows that

$$F''(a,z) \to 2M(z) > 0 \qquad (a \downarrow 0),$$

so $F''(a,z) > 0$ for all sufficiently small $a > 0$. Therefore the source function is not concave in a in general. □

Corollary 15.12 (APP-0049: Baricz arithmetic/arithmetic zero-balanced hypergeometric threshold)**.** *For $c > 0$, define $F_a(r) = {}_2F_1(a, c-a; c; r)$ on $0 < a < c$, $0 < r < 1$. The uniform arithmetic/arithmetic inequality $(F_{a_1}(r) + F_{a_2}(r))/2 \leq F_{(a_1+a_2)/2}(r)$ for all $a_1, a_2 \in (0,c)$ and all $r \in (0,1)$ holds if and only if $0 < c \leq 1$.*

*This resolves the source problem recorded as **APP-0049** [40].*

Proof. We use Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For $c > 0$, set

$$F_a(r) = {}_2F_1(a, c-a; c; r), \qquad 0 < a < c, \quad 0 < r < 1.$$

The Baricz arithmetic/arithmetic source slice asks when

$$\frac{F_{a_1}(r) + F_{a_2}(r)}{2} \leq F_{(a_1+a_2)/2}(r)$$

holds for all $a_1, a_2 \in (0,c)$ and all $r \in (0,1)$.

The answer is exactly $0 < c \leq 1$.

The existing local node the Hypergeometric Fa geometric mean logconvex slice imports Baricz's theorem:

$$\sqrt{F_{a_1}(r)F_{a_2}(r)} \leq \frac{F_{a_1}(r) + F_{a_2}(r)}{2} \leq F_{(a_1+a_2)/2}(r)$$

for $0 < c \leq 1$, $a_1, a_2 \in (0,c)$, and $0 < r < 1$. This proves the A/A inequality on that range.

Fix $c > 1$. For $0 < r < 1$,

$$F_a(r) = \sum_{k=0}^{\infty} \frac{(a)_k(c-a)_k}{(c)_k} \frac{r^k}{k!}.$$

For $k \geq 1$, as $a \to 0$,

$$(a)_k = a(k-1)!\left(1 + aH_{k-1} + O(a^2)\right)$$

and

$$\frac{(c-a)_k}{(c)_k} = 1 - a\sum_{j=0}^{k-1}\frac{1}{c+j} + O(a^2).$$

Hence

$$\frac{(a)_k(c-a)_k}{(c)_k k!} = \frac{a}{k} + \frac{a^2}{k}\left(H_{k-1} - \sum_{j=0}^{k-1}\frac{1}{c+j}\right) + O(a^3).$$

For fixed $r < 1$, normal convergence of the differentiated hypergeometric series justifies termwise expansion, so

$$F_a(r) = 1 + aL(r) + a^2 M_c(r) + O(a^3),$$

where

$$L(r) = \sum_{k\geq 1}\frac{r^k}{k} = -\log(1-r)$$

and

$$M_c(r) = \sum_{k\geq 1}\left(H_{k-1} - \sum_{j=0}^{k-1}\frac{1}{c+j}\right)\frac{r^k}{k}.$$

The bracket is

$$B_{c,k} = \psi(k) + \gamma - \psi(c+k) + \psi(c),$$

so

$$B_{c,k} \to \gamma + \psi(c).$$

Because $c > 1$, $\psi(c) > \psi(1) = -\gamma$, and therefore $\gamma + \psi(c) > 0$. Thus $B_{c,k} \geq \eta > 0$ for all sufficiently large k, and

$$M_c(r) \to +\infty \qquad (r \uparrow 1).$$

Choose $r_0 \in (0,1)$ with $M_c(r_0) > 0$. Then

$$\partial_a^2 F_a(r_0) \to 2M_c(r_0) > 0 \qquad (a \downarrow 0).$$

For some sufficiently small interior $a_0 \in (0,c)$, $\partial_a^2 F_{a_0}(r_0) > 0$. By continuity, $\partial_a^2 F_a(r_0) > 0$ on a small interval around a_0. Taking $h > 0$ small enough that $a_0 \pm h \in (0,c)$, Taylor's theorem gives

$$\frac{F_{a_0-h}(r_0) + F_{a_0+h}(r_0)}{2} > F_{a_0}(r_0).$$

Since $a_0 = ((a_0-h)+(a_0+h))/2$, the A/A inequality fails.

Therefore the source A/A inequality holds uniformly if and only if $0 < c \leq 1$. $\square$

Proposition 15.13 (Stolarsky power mean pgt1 asymptotic obstruction)**.** *For $p > 1$, $d \neq 0$, and $y > |d|$, the shifted power mean $G(y) = H_p(y+d, y-d)$ satisfies $G''(y) > 0$. In particular, $G(y) = y + \frac{(p-1)d^2}{2y} + O(y^{-3})$ as $y \to \infty$, so G' is eventually increasing.*

Proof. Let

$$c = \frac{a+b}{2}, \qquad d = \frac{a-b}{2}.$$

Since $a, b > 0$ and $a \neq b$, we have $c > |d| > 0$. Put $y = x + c$. Then on the natural half-line $y > |d|$,

$$F(x) = G(y), \qquad G(y) = \left(\frac{(y+d)^p + (y-d)^p}{2}\right)^{1/p}.$$

Translation does not affect the sign pattern of derivatives, so it is enough to show that G is not Bernstein.

For $|t| < 1$, define

$$\phi_p(t) = H_p(1+t, 1-t) = 2^{-1/p}\left((1+t)^p + (1-t)^p\right)^{1/p}.$$

By homogeneity,

$$G(y) = y\phi_p(d/y).$$

Let

$$A = 1+t, \qquad B = 1-t, \qquad N = A^p + B^p.$$

Then $\phi_p(t) = 2^{-1/p}N^{1/p}$. Differentiating twice gives

$$\phi_p''(t) = 2^{-1/p}(p-1)N^{1/p-2}\left(N(A^{p-2}+B^{p-2}) - (A^{p-1}-B^{p-1})^2\right).$$

The bracket simplifies to

$$A^{p-2}B^{p-2}(A+B)^2,$$

because

$$(A^p+B^p)(A^{p-2}+B^{p-2}) - (A^{p-1}-B^{p-1})^2 = A^{p-2}B^{p-2}(A^2+2AB+B^2).$$

For $p > 1$ and $|t| < 1$, all factors are positive, so

$$\phi_p''(t) > 0.$$

Since $t = d/y$,

$$G'(y) = \phi_p(t) - t\phi_p'(t)$$

and

$$G''(y) = \frac{t^2}{y}\phi_p''(t) = \frac{d^2}{y^3}\phi_p''(d/y) > 0$$

for every $y > |d|$.

For $|u| < 1$, define

$$A(u) = \frac{(1+u)^p + (1-u)^p}{2}.$$

The odd terms cancel in the binomial expansion, giving

$$A(u) = 1 + \frac{p(p-1)}{2}u^2 + O(u^4) \qquad (u \to 0).$$

Since $z \mapsto z^{1/p}$ is analytic near $z = 1$,

$$A(u)^{1/p} = 1 + \frac{1}{p}\cdot\frac{p(p-1)}{2}u^2 + O(u^4) = 1 + \frac{p-1}{2}u^2 + O(u^4).$$

With $u = d/y$, this yields

$$G(y) = yA(d/y)^{1/p} = y + \frac{(p-1)d^2}{2y} + O(y^{-3}) \qquad (y \to \infty).$$

Differentiating the same analytic expansion gives

$$G'(y) = 1 - \frac{(p-1)d^2}{2y^2} + O(y^{-4}),$$

and hence

$$G''(y) = \frac{(p-1)d^2}{y^3} + O(y^{-5}).$$

Because $p > 1$ and $d \neq 0$, the leading coefficient is positive. This recovers the tail obstruction from the exact curvature certificate above.

If G were a Bernstein function, then G' would be completely monotone. In particular,

$$(G')'(y) = G''(y) \leq 0$$

throughout the interval. This contradicts the positivity of $G''(y)$ proved above.

Thus G, and equivalently F, is not a Bernstein function. $\square$

Counterexample 15.14 (APP-0050: Stolarsky shifted power means with $p > 1$ are not Bernstein)**.** not(For some $p > 1$, the shifted power mean $x \mapsto H_p(x+a, x+b)$, where $H_p(a,b) = ((a^p+b^p)/2)^{1/p}$, is a Bernstein function for every positive pair of shifts a, b.)

This resolves the source problem recorded as **APP-0050** [41].

Proof. We use Proposition 15.13 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$c = \frac{a+b}{2}, \qquad d = \frac{a-b}{2}.$$

Since $a, b > 0$ and $a \neq b$, we have $c > |d| > 0$. Put $y = x + c$. Then on the natural half-line $y > |d|$,

$$F(x) = G(y), \qquad G(y) = \left(\frac{(y+d)^p + (y-d)^p}{2}\right)^{1/p}.$$

Translation does not affect the sign pattern of derivatives, so it is enough to show that G is not Bernstein.

For $|t| < 1$, define

$$\phi_p(t) = H_p(1+t, 1-t) = 2^{-1/p}\left((1+t)^p + (1-t)^p\right)^{1/p}.$$

By homogeneity,

$$G(y) = y\phi_p(d/y).$$

Let

$$A = 1+t, \qquad B = 1-t, \qquad N = A^p + B^p.$$

Then $\phi_p(t) = 2^{-1/p}N^{1/p}$. Differentiating twice gives

$$\phi_p''(t) = 2^{-1/p}(p-1)N^{1/p-2}\left(N(A^{p-2}+B^{p-2}) - (A^{p-1}-B^{p-1})^2\right).$$

The bracket simplifies to

$$A^{p-2}B^{p-2}(A+B)^2,$$

because

$$(A^p+B^p)(A^{p-2}+B^{p-2}) - (A^{p-1}-B^{p-1})^2 = A^{p-2}B^{p-2}(A^2+2AB+B^2).$$

For $p > 1$ and $|t| < 1$, all factors are positive, so

$$\phi_p''(t) > 0.$$

Since $t = d/y$,

$$G'(y) = \phi_p(t) - t\phi_p'(t)$$

and

$$G''(y) = \frac{t^2}{y}\phi_p''(t) = \frac{d^2}{y^3}\phi_p''(d/y) > 0$$

for every $y > |d|$.

For $|u| < 1$, define

$$A(u) = \frac{(1+u)^p + (1-u)^p}{2}.$$

The odd terms cancel in the binomial expansion, giving

$$A(u) = 1 + \frac{p(p-1)}{2}u^2 + O(u^4) \qquad (u \to 0).$$

Since $z \mapsto z^{1/p}$ is analytic near $z = 1$,

$$A(u)^{1/p} = 1 + \frac{1}{p} \cdot \frac{p(p-1)}{2}u^2 + O(u^4) = 1 + \frac{p-1}{2}u^2 + O(u^4).$$

With $u = d/y$, this yields

$$G(y) = yA(d/y)^{1/p} = y + \frac{(p-1)d^2}{2y} + O(y^{-3}) \qquad (y \to \infty).$$

Differentiating the same analytic expansion gives

$$G'(y) = 1 - \frac{(p-1)d^2}{2y^2} + O(y^{-4}),$$

and hence

$$G''(y) = \frac{(p-1)d^2}{y^3} + O(y^{-5}).$$

Because $p > 1$ and $d \neq 0$, the leading coefficient is positive. This recovers the tail obstruction from the exact curvature certificate above.

If G were a Bernstein function, then G' would be completely monotone. In particular,

$$(G')'(y) = G''(y) \leq 0$$

throughout the interval. This contradicts the positivity of $G''(y)$ proved above.

Thus G, and equivalently F, is not a Bernstein function. □

15.1 Poisson-root and local complete-monotonicity obstructions

Proposition 15.15 (Townes finite-interval BLT root-preservation counterexample)**.** *There exists a real-valued infinitely divisible probability law with bilateral Laplace transform L finite near $[0, 1]$, such that L is completely monotone on $[0, 1]$ but $L^{1/n}$ is not completely monotone on $[0, 1]$ for some integer $n \geq 2$.*

Proof. Let
$$\eta = 10^{-4}$$
and define
$$L(s) = \exp\left(-10\eta s + 10^{-4}\left(e^{-10\eta s} - 1\right) + (e^{\eta s} - 1)\right).$$
Then L is the bilateral Laplace transform of
$$X = \eta(10 + 10P - Q),$$
where $P \sim \mathrm{Pois}(10^{-4})$, $Q \sim \mathrm{Pois}(1)$, and P, Q are independent. Thus X is a real-valued infinitely divisible law. Its support is unbounded below because of the negative jump $-\eta Q$, so this does not use the already-depleted nonnegative-support theorem.

We prove:
$$L \text{ is completely monotone on } [0,1],$$
but
$$L^{1/100} \text{ is not completely monotone on } [0,1].$$

For $s \in [0,1]$, the Esscher-normalized law has unscaled variable
$$Z_{a,b} = 10 + 10P_a - Q_b,$$
where
$$a = 10^{-4}e^{-10\eta s}, \qquad b = e^{\eta s}.$$
Hence
$$a \in [a_0, A], \qquad b \in [1, B],$$
with
$$a_0 = 10^{-4}e^{-10^{-3}} > 9.99 \cdot 10^{-5}, \qquad A = 10^{-4}, \qquad B = e^{10^{-4}} < 1.001.$$

Since
$$(-1)^k L^{(k)}(s) = L(s)\eta^k \mathbb{E}[Z_{a,b}^k],$$
it is enough to prove $\mathbb{E}[Z_{a,b}^k] \geq 0$ for every $k \geq 0$ and every (a,b) in the rectangle above.

Even moments are nonnegative. For odd moments, it suffices to prove the tail dominance
$$\mathbb{P}(Z_{a,b} > r) \geq \mathbb{P}(Z_{a,b} < -r) \qquad (r \geq 0).$$
Indeed, for $j \geq 0$,
$$\mathbb{E}[Z_{a,b}^{2j+1}] = (2j+1)\int_0^\infty r^{2j}\left(\mathbb{P}(Z_{a,b} > r) - \mathbb{P}(Z_{a,b} < -r)\right) dr.$$

The variable $Z_{a,b}$ is stochastically increasing in a and decreasing in b, so the tail-dominance difference is smallest at $a = a_0$, $b = B$.

Because $Z_{a,b}$ is integer-valued, it is enough to check integer $r \geq 0$. For $0 \leq r \leq 9$,
$$\mathbb{P}(Z_{a,b} > r) \geq \mathbb{P}(P_a = 0, Q_b = 0) = e^{-a-b} > e^{-A-B},$$
whereas
$$\mathbb{P}(Z_{a,b} < -r) \leq \mathbb{P}(Q_B \geq 11).$$

The elementary Poisson tail bound

$$\mathbb{P}(Q_B \geq N) \leq e^{-B} \frac{B^N}{N!} \frac{N}{N-B} \qquad (N > B)$$

gives

$$\mathbb{P}(Q_B \geq 11) < 3 \cdot 10^{-8} < e^{-A-B}.$$

For $r \geq 10$, put $m = \lfloor r/10 \rfloor$. Then $m \geq 1$, and the event $P_a = m,\ Q_b = 0$ gives

$$\mathbb{P}(Z_{a,b} > r) \geq e^{-a-b} \frac{a^m}{m!} \geq e^{-A-B} \frac{a_0^m}{m!}.$$

On the other hand,

$$Z_{a,b} < -r \quad \Longrightarrow \quad Q_b \geq r + 11,$$

so

$$\mathbb{P}(Z_{a,b} < -r) \leq \mathbb{P}(Q_B \geq r + 11).$$

For $10m \leq r \leq 10m + 9$, this is bounded by

$$e^{-B} \frac{B^{10m+11}}{(10m+11)!} \frac{10m+11}{10m+11-B}.$$

Therefore

$$\frac{\mathbb{P}(Z_{a,b} < -r)}{\mathbb{P}(Z_{a,b} > r)} \leq 6 \, \frac{1.001^{10m+11} m!}{(9.99 \cdot 10^{-5})^m (10m+11)!}.$$

At $m = 1$ the right-hand side is $< 10^{-14}$. The ratio of the bound at $m+1$ to the bound at m is at most

$$\frac{1.001^{10}(m+1)}{(9.99 \cdot 10^{-5})(10m+12)(10m+13) \cdots (10m+21)} < 1$$

for every $m \geq 1$. Thus the tail dominance holds for every $r \geq 10$.

This proves L is completely monotone on $[0, 1]$.

Let

$$R(s) = L(s)^{1/100}.$$

For the unscaled infinitely divisible seed

$$Z = 10 + 10P - Q$$

at $s = 0$, the first three cumulants are

$$c_1 = 9.001, \qquad c_2 = 1.01, \qquad c_3 = -0.9.$$

At time $t = 1/100$, the third raw moment is the Bell-polynomial expression

$$m_3(t) = tc_3 + 3t^2 c_1 c_2 + t^3 c_1^3.$$

Thus

$$m_3(1/100) = -0.009 + 0.002727303 + 0.000729243027001 = -0.005543453972999 < 0.$$

Consequently

$$(-1)^3 R'''(0) = \eta^3 m_3(1/100) < 0.$$

This violates complete monotonicity. Therefore $L^{1/100}$ is not completely monotone on $[0,1]$.

The source PGF is

$$G(z) = L(1-z).$$

If the corresponding Poisson mixture were 100-divisible, then the PGF root H would satisfy $H(z)^{100} = G(z)$ and $H(1) = 1$. Since $G(z) > 0$ on $[0,1]$, this root is the principal root

$$H(z) = L(1-z)^{1/100} = R(1-z).$$

But R is not completely monotone at $s = 0$, equivalently H is not absolutely monotone at $z = 1$. A PGF cannot have a negative factorial-moment derivative. Hence this mixed Poisson law is not DID. ☐

Proposition 15.16 (Townes finite-interval BLT root preservation is false)**.** *The Townes finite-interval BLT root-preservation statement is false: there exists a real-valued infinitely divisible law with bilateral Laplace transform L completely monotone on* $[0,1]$*, but* $L^{1/100}$ *is not completely monotone on* $[0,1]$.

Proof. For Townes finite-interval BLT root preservation is false, this proof is the corresponding component of Proposition 15.15. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. ☐

Counterexample 15.17 (APP-0051: Townes real-valued Poisson-mixture ID-to-DID conjecture is false)**.** There exists a real-valued infinitely divisible distribution whose bilateral Laplace transform is completely monotone on $[0,1]$, so it is valid for Poisson mixing, but whose corresponding mixed Poisson distribution is not discretely infinitely divisible.

This resolves the source problem recorded as **APP-0051** [42].

Proof. We use Proposition 11.1, Proposition 15.15, and Proposition 15.16 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$\eta = 10^{-4}$$

and define

$$L(s) = \exp\left(-10\eta s + 10^{-4}\left(e^{-10\eta s} - 1\right) + \left(e^{\eta s} - 1\right)\right).$$

Then L is the bilateral Laplace transform of

$$X = \eta(10 + 10P - Q),$$

where $P \sim \mathrm{Pois}(10^{-4})$, $Q \sim \mathrm{Pois}(1)$, and P, Q are independent. Thus X is a real-valued infinitely divisible law. Its support is unbounded below because of the negative jump $-\eta Q$, so this does not use the already-depleted nonnegative-support theorem.

We prove:

$$L \text{ is completely monotone on } [0,1],$$

but

$$L^{1/100} \text{ is not completely monotone on } [0,1].$$

For $s \in [0,1]$, the Esscher-normalized law has unscaled variable

$$Z_{a,b} = 10 + 10P_a - Q_b,$$

where

$$a = 10^{-4}e^{-10\eta s}, \qquad b = e^{\eta s}.$$

Hence

$$a \in [a_0, A], \qquad b \in [1, B],$$

with

$$a_0 = 10^{-4}e^{-10^{-3}} > 9.99 \cdot 10^{-5}, \qquad A = 10^{-4}, \qquad B = e^{10^{-4}} < 1.001.$$

Since

$$(-1)^k L^{(k)}(s) = L(s)\eta^k \mathbb{E}[Z_{a,b}^k],$$

it is enough to prove $\mathbb{E}[Z_{a,b}^k] \geq 0$ for every $k \geq 0$ and every (a, b) in the rectangle above.

Even moments are nonnegative. For odd moments, it suffices to prove the tail dominance

$$\mathbb{P}(Z_{a,b} > r) \geq \mathbb{P}(Z_{a,b} < -r) \qquad (r \geq 0).$$

Indeed, for $j \geq 0$,

$$\mathbb{E}[Z_{a,b}^{2j+1}] = (2j+1)\int_0^\infty r^{2j}\left(\mathbb{P}(Z_{a,b} > r) - \mathbb{P}(Z_{a,b} < -r)\right) dr.$$

The variable $Z_{a,b}$ is stochastically increasing in a and decreasing in b, so the tail-dominance difference is smallest at $a = a_0$, $b = B$.

Because $Z_{a,b}$ is integer-valued, it is enough to check integer $r \geq 0$. For $0 \leq r \leq 9$,

$$\mathbb{P}(Z_{a,b} > r) \geq \mathbb{P}(P_a = 0, Q_b = 0) = e^{-a-b} > e^{-A-B},$$

whereas

$$\mathbb{P}(Z_{a,b} < -r) \leq \mathbb{P}(Q_B \geq 11).$$

The elementary Poisson tail bound

$$\mathbb{P}(Q_B \geq N) \leq e^{-B}\frac{B^N}{N!}\frac{N}{N-B} \qquad (N > B)$$

gives

$$\mathbb{P}(Q_B \geq 11) < 3 \cdot 10^{-8} < e^{-A-B}.$$

For $r \geq 10$, put $m = \lfloor r/10 \rfloor$. Then $m \geq 1$, and the event $P_a = m,\ Q_b = 0$ gives

$$\mathbb{P}(Z_{a,b} > r) \geq e^{-a-b}\frac{a^m}{m!} \geq e^{-A-B}\frac{a_0^m}{m!}.$$

On the other hand,

$$Z_{a,b} < -r \quad \Longrightarrow \quad Q_b \geq r + 11,$$

so

$$\mathbb{P}(Z_{a,b} < -r) \leq \mathbb{P}(Q_B \geq r + 11).$$

For $10m \leq r \leq 10m + 9$, this is bounded by

$$e^{-B}\frac{B^{10m+11}}{(10m+11)!}\frac{10m+11}{10m+11-B}.$$

Therefore
$$\frac{\mathbb{P}(Z_{a,b} < -r)}{\mathbb{P}(Z_{a,b} > r)} \le 6\,\frac{1.001^{10m+11} m!}{(9.99 \cdot 10^{-5})^m (10m+11)!}.$$
At $m = 1$ the right-hand side is $< 10^{-14}$. The ratio of the bound at $m+1$ to the bound at m is at most
$$\frac{1.001^{10}(m+1)}{(9.99 \cdot 10^{-5})(10m+12)(10m+13)\cdots(10m+21)} < 1$$
for every $m \ge 1$. Thus the tail dominance holds for every $r \ge 10$.

This proves L is completely monotone on $[0, 1]$.

Let
$$R(s) = L(s)^{1/100}.$$
For the unscaled infinitely divisible seed
$$Z = 10 + 10P - Q$$
at $s = 0$, the first three cumulants are
$$c_1 = 9.001, \qquad c_2 = 1.01, \qquad c_3 = -0.9.$$
At time $t = 1/100$, the third raw moment is the Bell-polynomial expression
$$m_3(t) = tc_3 + 3t^2 c_1 c_2 + t^3 c_1^3.$$
Thus
$$m_3(1/100) = -0.009 + 0.002727303 + 0.000729243027001 = -0.005543453972999 < 0.$$
Consequently
$$(-1)^3 R'''(0) = \eta^3 m_3(1/100) < 0.$$
This violates complete monotonicity. Therefore $L^{1/100}$ is not completely monotone on $[0, 1]$.

The source PGF is
$$G(z) = L(1 - z).$$
If the corresponding Poisson mixture were 100-divisible, then the PGF root H would satisfy $H(z)^{100} = G(z)$ and $H(1) = 1$. Since $G(z) > 0$ on $[0, 1]$, this root is the principal root
$$H(z) = L(1 - z)^{1/100} = R(1 - z).$$
But R is not completely monotone at $s = 0$, equivalently H is not absolutely monotone at $z = 1$. A PGF cannot have a negative factorial-moment derivative. Hence this mixed Poisson law is not DID. $\square$

Proposition 15.18 (Complete-monotonicity closure calculus principle)**.** *Complete monotonicity, logarithmic complete monotonicity, Stieltjes, Bernstein, and complete-Bernstein conclusions are transported by their standard closure operations, reciprocal transforms, derivative criteria, positive mixtures, and composition rules.*

Proof. A completely monotone function has a positive Laplace representation

$$f(x) = \int_{[0,\infty)} e^{-xt}\, d\mu(t),$$

so

$$(-1)^n f^{(n)}(x) = \int_{[0,\infty)} t^n e^{-xt}\, d\mu(t) \geq 0.$$

Positive sums and mixtures add the representing measures. The Stieltjes and Bernstein derivative criteria reduce to the same positivity of representing measures or of f'. Thus once an expression is reduced by these standard closure operations to positive measures, reciprocal transforms, derivative criteria, or Bernstein composition, the required complete-monotonicity or Bernstein conclusion follows. □

Proposition 15.19 (ArsinhSquare log derivative completely monotone)**.** *For $\varphi(x) = \operatorname{arsinh}^2 \sqrt{x}$, the logarithmic derivative $\varphi'(x)/\varphi(x)$ is completely monotone on $(0, \infty)$.*

Proof. A Stieltjes function

$$F(z) = \frac{a}{z} + b + \int_0^\infty \frac{1}{z+s}\, d\mu(s), \qquad a, b \geq 0, \quad \mu \geq 0,$$

maps the upper half-plane into the lower half-plane:

$$\operatorname{Im} F(z) \leq 0, \qquad \operatorname{Im} z > 0.$$

Let $z = -t + i0$ with $t > 1$. On principal branches,

$$\sqrt{z} = i\sqrt{t}, \qquad \sqrt{1+z} = i\sqrt{t-1}.$$

Also

$$\operatorname{arsinh}(i\sqrt{t}) = \log\left(i\sqrt{t} + i\sqrt{t-1}\right) = A(t) + \frac{\pi i}{2},$$

where

$$A(t) = \operatorname{arcosh}\sqrt{t} = \log(\sqrt{t} + \sqrt{t-1}) > 0.$$

Therefore

$$h(-t+i0) = -\frac{1}{\sqrt{t(t-1)}\left(A(t) + \frac{\pi i}{2}\right)} = -\frac{A(t) - \frac{\pi i}{2}}{\sqrt{t(t-1)}\left(A(t)^2 + \frac{\pi^2}{4}\right)}.$$

Thus

$$\operatorname{Im} h(-t+i0) = \frac{\pi/2}{\sqrt{t(t-1)}\left(A(t)^2 + \frac{\pi^2}{4}\right)} > 0$$

for every $t > 1$. This contradicts the Stieltjes half-plane sign condition. Hence h is not Stieltjes.

Equivalently, the forced Stieltjes inversion density on $(1, \infty)$ would be

$$\rho_h(t) = -\frac{1}{\pi} \operatorname{Im} h(-t+i0) = -\frac{1}{2\sqrt{t(t-1)}\left(A(t)^2 + \frac{\pi^2}{4}\right)} < 0.$$

Although h is not Stieltjes, it is completely monotone on $(0, \infty)$.

Put
$$a(x) = \operatorname{arsinh}\sqrt{x}.$$
Then
$$a'(x) = \frac{1}{2\sqrt{x}\sqrt{1+x}},$$
a product of completely monotone factors. Hence a is a Bernstein function. The reciprocal of a positive Bernstein function is completely monotone, so $1/a$ is completely monotone. Therefore
$$h(x) = 2a'(x)\frac{1}{a(x)}$$
is a product of completely monotone functions, and is completely monotone. □

Counterexample 15.20 (APP-0053: arsinh-square logarithmic derivative is not Stieltjes)**.** For $\varphi(x) = \operatorname{arsinh}^2\sqrt{x}$, the logarithmic derivative $\varphi'(x)/\varphi(x) = 1/(\sqrt{x}\sqrt{1+x}\operatorname{arsinh}\sqrt{x})$ is not a Stieltjes function; its upper boundary value on $(-\infty,-1)$ has strictly positive imaginary part.

This resolves the source problem recorded as **APP-0053** [44].

Proof. We use Proposition 15.18, Proposition 15.19, and Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

A Stieltjes function
$$F(z) = \frac{a}{z} + b + \int_0^\infty \frac{1}{z+s}\,d\mu(s), \qquad a,b \geq 0, \quad \mu \geq 0,$$
maps the upper half-plane into the lower half-plane:
$$\operatorname{Im} F(z) \leq 0, \qquad \operatorname{Im} z > 0.$$

Let $z = -t + i0$ with $t > 1$. On principal branches,
$$\sqrt{z} = i\sqrt{t}, \qquad \sqrt{1+z} = i\sqrt{t-1}.$$
Also
$$\operatorname{arsinh}(i\sqrt{t}) = \log\left(i\sqrt{t} + i\sqrt{t-1}\right) = A(t) + \frac{\pi i}{2},$$
where
$$A(t) = \operatorname{arcosh}\sqrt{t} = \log(\sqrt{t} + \sqrt{t-1}) > 0.$$
Therefore
$$h(-t+i0) = -\frac{1}{\sqrt{t(t-1)}\left(A(t) + \frac{\pi i}{2}\right)} = -\frac{A(t) - \frac{\pi i}{2}}{\sqrt{t(t-1)}\left(A(t)^2 + \frac{\pi^2}{4}\right)}.$$
Thus
$$\operatorname{Im} h(-t+i0) = \frac{\pi/2}{\sqrt{t(t-1)}\left(A(t)^2 + \frac{\pi^2}{4}\right)} > 0$$
for every $t > 1$. This contradicts the Stieltjes half-plane sign condition. Hence h is not Stieltjes.

Equivalently, the forced Stieltjes inversion density on $(1,\infty)$ would be
$$\rho_h(t) = -\frac{1}{\pi}\operatorname{Im} h(-t+i0) = -\frac{1}{2\sqrt{t(t-1)}\left(A(t)^2 + \frac{\pi^2}{4}\right)} < 0.$$

Although h is not Stieltjes, it is completely monotone on $(0,\infty)$.

Put

$$a(x) = \operatorname{arsinh}\sqrt{x}.$$

Then

$$a'(x) = \frac{1}{2\sqrt{x}\sqrt{1+x}},$$

a product of completely monotone factors. Hence a is a Bernstein function. The reciprocal of a positive Bernstein function is completely monotone, so $1/a$ is completely monotone. Therefore

$$h(x) = 2a'(x)\frac{1}{a(x)}$$

is a product of completely monotone functions, and is completely monotone. □

Proposition 15.21 (Signed Atomic Hausdorff Negativity Obstruction)**.** *If a sequence has a finite signed Hausdorff representing measure on* $[0,1]$ *with a negative atom at some point* $r>0$*, then the sequence has no positive Hausdorff representing measure on* $[0,1]$.

Proof. A Bernstein function has the representation

$$F(x) = c + bx + \int_{(0,\infty)} (1-e^{-xt})\, d\mu(t),$$

where $b \geq 0$, $\mu \geq 0$, and $\int (1\wedge t)\, d\mu(t) < \infty$. Hence

$$d_n = F(n+1) - F(n) = b + \int_{(0,\infty)} e^{-nt}(1-e^{-t})\, d\mu(t).$$

Pushing forward $(1-e^{-t})\, d\mu(t)$ under $s = e^{-t}$, and adding the atom $b\delta_1$, gives a finite positive measure ρ on $[0,1]$ such that

$$d_n = \int_{[0,1]} s^n \, d\rho(s).$$

Finiteness follows from $1-e^{-t} \asymp 1\wedge t$.

Let a sequence (d_n) have a finite signed representing measure σ on $[0,1]$. If σ has a negative atom at some point $r \in (0,1]$, then (d_n) has no positive representing measure on $[0,1]$.

The Hausdorff moment problem on the compact interval $[0,1]$ is determinate for finite signed measures. Therefore any finite positive representing measure would have to equal σ, contradicting the negative mass at r. If moments are indexed only for $n \geq 1$, the possible ambiguity is an atom at 0, which does not affect a negative atom at $r > 0$.

Set $q = p^{-1} \in (0,1)$. Then

$$a_n = \frac{p^n + N}{p^n + D} = \frac{1+Nq^n}{1+Dq^n}.$$

The first differences are

$$\begin{aligned} d_n &= a_{n+1} - a_n \\ &= \frac{1+Nq^{n+1}}{1+Dq^{n+1}} - \frac{1+Nq^n}{1+Dq^n} \\ &= (D-N)(1-q)\frac{q^n}{(1+Dq^n)(1+Dq^{n+1})}. \end{aligned}$$

For $u = q^n$,

$$\frac{(1-q)u}{(1+Du)(1+Dqu)} = \frac{1}{D}\left(\frac{1}{1+Dqu} - \frac{1}{1+Du}\right).$$

Since $0 < D \le -N < 1$, the geometric expansions converge absolutely, and

$$d_n = (D-N)\sum_{m\ge 1}(1-q^m)(-D)^{m-1}q^{mn}.$$

Thus (d_n) has the finite signed Hausdorff representation

$$\sigma = (D-N)\sum_{m\ge 1}(1-q^m)(-D)^{m-1}\delta_{q^m}.$$

The total variation is finite because

$$\sum_{m\ge 1}(1-q^m)D^{m-1} \le \frac{1}{1-D}.$$

The atom at q^2 has mass

$$\sigma(\{q^2\}) = -(D-N)D(1-q^2) < 0.$$

By the signed-atom obstruction, (d_n) is not a Hausdorff moment sequence of a positive measure.

For $p > 1$, $-1 < N < 0$, and $0 < D \le -N$, the sequence

$$a_n = \frac{p^n + N}{p^n + D}$$

is not interpolated by any Bernstein function.

If such a Bernstein function F existed, the first bridge lemma would force $d_n = a_{n+1} - a_n$ to be a Hausdorff moment sequence for a finite positive measure on $[0,1]$. The explicit GRWS expansion gives the unique finite signed representing measure for (d_n), and that measure has negative mass at q^2. This contradicts positivity.

The boundary $D = -N$ is included: then $D - N = 2D > 0$, $0 < D < 1$, and the negative atom remains $-2D^2(1-q^2)$. The boundary $D = 0$ is excluded: there the negative atom disappears and

$$d_n = (-N)(1-q)q^n$$

has a positive one-atom representation. $\square$

Proposition 15.22 (BF Integer Increments Are Hausdorff CM)**.** *If F is a Bernstein function and $a_n = F(n)$, then the first differences $d_n = a_{n+1} - a_n$ form a Hausdorff moment sequence on $[0,1]$.*

Proof. For BF Integer Increments Are Hausdorff CM, this proof is the corresponding component of Proposition 15.21. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Definition 15.1 (GRWS weight-squared sequence)**.** For a geometrically regular weighted shift with parameters $p > 1$, $-1 < N, D < 1$, the weights are $\alpha_n = \sqrt{(p^n+N)/(p^n+D)}$, and the weights-squared interpolation sequence is $a_n = \alpha_n^2 = (p^n+N)/(p^n+D)$.

Proposition 15.23 (GRWS SectorII Difference Atomic Expansion)**.** *For $p > 1$, $q = p^{-1}$, $-1 < N < 0$, and $0 < D \le -N$, the first differences of $a_n = (1 + Nq^n)/(1 + Dq^n)$ satisfy $a_{n+1} - a_n = (D-N)\sum_{m\ge1}(1-q^m)(-D)^{m-1}q^{mn}$. This is the moment sequence of a finite signed atomic measure on $[0, 1]$ whose atom at q^2 is negative.*

Proof. For GRWS SectorII Difference Atomic Expansion, this proof is the corresponding component of Proposition 15.21. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Counterexample 15.24 (APP-0054: GRWS Sector II weights-squared Bernstein interpolation fails)**.** For $p > 1$, $-1 < N < 0$, and $0 < D \le -N$, the GRWS weights-squared sequence $a_n = (p^n + N)/(p^n + D)$ is not interpolated by any Bernstein function.

This resolves the source problem recorded as **APP-0054** [45].

Proof. We use Proposition 9.1, Proposition 15.21, Proposition 15.22, and Proposition 15.23 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

A Bernstein function has the representation

$$F(x) = c + bx + \int_{(0,\infty)} (1 - e^{-xt})\, d\mu(t),$$

where $b \ge 0$, $\mu \ge 0$, and $\int(1 \wedge t)\, d\mu(t) < \infty$. Hence

$$d_n = F(n + 1) - F(n) = b + \int_{(0,\infty)} e^{-nt}(1 - e^{-t})\, d\mu(t).$$

Pushing forward $(1 - e^{-t})\, d\mu(t)$ under $s = e^{-t}$, and adding the atom $b\delta_1$, gives a finite positive measure ρ on $[0, 1]$ such that

$$d_n = \int_{[0,1]} s^n\, d\rho(s).$$

Finiteness follows from $1 - e^{-t} \asymp 1 \wedge t$.

Let a sequence (d_n) have a finite signed representing measure σ on $[0, 1]$. If σ has a negative atom at some point $r \in (0, 1]$, then (d_n) has no positive representing measure on $[0, 1]$.

The Hausdorff moment problem on the compact interval $[0, 1]$ is determinate for finite signed measures. Therefore any finite positive representing measure would have to equal σ, contradicting the negative mass at r. If moments are indexed only for $n \ge 1$, the possible ambiguity is an atom at 0, which does not affect a negative atom at $r > 0$.

Set $q = p^{-1} \in (0, 1)$. Then

$$a_n = \frac{p^n + N}{p^n + D} = \frac{1 + Nq^n}{1 + Dq^n}.$$

The first differences are

$$\begin{aligned} d_n &= a_{n+1} - a_n \\ &= \frac{1 + Nq^{n+1}}{1 + Dq^{n+1}} - \frac{1 + Nq^n}{1 + Dq^n} \\ &= (D - N)(1 - q)\frac{q^n}{(1 + Dq^n)(1 + Dq^{n+1})}. \end{aligned}$$

For $u = q^n$,

$$\frac{(1-q)u}{(1+Du)(1+Dqu)} = \frac{1}{D}\left(\frac{1}{1+Dqu} - \frac{1}{1+Du}\right).$$

Since $0 < D \le -N < 1$, the geometric expansions converge absolutely, and

$$d_n = (D-N)\sum_{m\ge 1}(1-q^m)(-D)^{m-1}q^{mn}.$$

Thus (d_n) has the finite signed Hausdorff representation

$$\sigma = (D-N)\sum_{m\ge 1}(1-q^m)(-D)^{m-1}\delta_{q^m}.$$

The total variation is finite because

$$\sum_{m\ge 1}(1-q^m)D^{m-1} \le \frac{1}{1-D}.$$

The atom at q^2 has mass

$$\sigma(\{q^2\}) = -(D-N)D(1-q^2) < 0.$$

By the signed-atom obstruction, (d_n) is not a Hausdorff moment sequence of a positive measure.

For $p > 1$, $-1 < N < 0$, and $0 < D \le -N$, the sequence

$$a_n = \frac{p^n + N}{p^n + D}$$

is not interpolated by any Bernstein function.

If such a Bernstein function F existed, the first bridge lemma would force $d_n = a_{n+1} - a_n$ to be a Hausdorff moment sequence for a finite positive measure on $[0,1]$. The explicit GRWS expansion gives the unique finite signed representing measure for (d_n), and that measure has negative mass at q^2. This contradicts positivity.

The boundary $D = -N$ is included: then $D - N = 2D > 0$, $0 < D < 1$, and the negative atom remains $-2D^2(1-q^2)$. The boundary $D = 0$ is excluded: there the negative atom disappears and

$$d_n = (-N)(1-q)q^n$$

has a positive one-atom representation. □

15.2 Coefficient-ratio, reciprocal-JCM, and product-kernel obstructions

Proposition 15.25 (Second-coefficient obstruction for analytic quotient CM transfer)**.** *Let $0 < t < 1$ and $g(x) = 1+px+qx^2+O(x^3)$, and set $f(x) = g(tx)$. Then $[x^2]f(x)/g(x) = (1-t)\{p^2 - q(1+t)\}$. In particular, if this coefficient is negative, then $(f/g)''(0) < 0$, so f/g is not completely monotone on any interval $(0,\varepsilon)$.*

Proof. Define

$$g(x) = 1 + \frac{x}{4} + x^2 + \sum_{n\ge 3}\frac{x^n}{n!}, \qquad f(x) = g(x/2).$$

Then g and f are entire. The coefficients of g are

$$b_0 = 1, \qquad b_1 = \frac{1}{4}, \qquad b_2 = 1, \qquad b_n = \frac{1}{n!} \quad (n \ge 3),$$

so $b_n > 0$ for every n. Since $f(x) = g(x/2)$, its coefficients satisfy

$$a_n = b_n 2^{-n}.$$

Thus

$$c_n = \frac{a_n}{b_n} = 2^{-n}.$$

For every $n, k \geq 0$,

$$\Delta^k c_n = (1 - \tfrac{1}{2})^k 2^{-n} = 2^{-n-k} > 0.$$

Hence $\{a_n/b_n\}_{n\geq 0}$ is strictly completely monotone, and therefore completely monotone, in the source sense.

Now write the quotient near the left endpoint. More generally, if

$$g(x) = 1 + px + qx^2 + O(x^3), \qquad 0 < t < 1, \qquad f(x) = g(tx),$$

then

$$\frac{f(x)}{g(x)} = 1 + p(t-1)x + (1-t)\{p^2 - q(1+t)\}x^2 + O(x^3).$$

In the present example $p = 1/4$, $q = 1$, and $t = 1/2$. Therefore

$$[x^2]\frac{f(x)}{g(x)} = \frac{1}{2}\left(\frac{1}{16} - \frac{3}{2}\right) = -\frac{23}{32},$$

so

$$\left(\frac{f}{g}\right)''(0) = -\frac{23}{16} < 0.$$

By analyticity, $(f/g)''(x) < 0$ for all sufficiently small $x > 0$. A completely monotone function on $(0, \varepsilon)$ must satisfy $F''(x) \geq 0$, so f/g is not completely monotone on any interval $(0, \varepsilon)$. It is therefore not strictly completely monotone either. □

Proposition 15.26 (Geometric sequences are strictly completely monotone)**.** *For $0 < t < 1$, the sequence $c_n = t^n$ is strictly completely monotone under the forward-difference convention $\Delta c_n = c_n - c_{n+1}$: for every $n, k \geq 0$, $\Delta^k c_n = (1-t)^k t^n > 0$.*

Proof. For Geometric sequences are strictly completely monotone, this proof is the corresponding component of Proposition 15.25. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Counterexample 15.27 (APP-0060: Baricz coefficient-ratio complete-monotonicity transfer is false)**.** Baricz Problem 4.3(c) has a negative answer. There exist entire power series $f(x) = \sum a_n x^n$ and $g(x) = \sum b_n x^n$, with $b_n > 0$, such that $\{a_n/b_n\}_{n\geq 0}$ is strictly completely monotone, but f/g is not completely monotone on any interval $(0, \varepsilon)$. One example is $g(x) = 1 + x/4 + x^2 + \sum_{n\geq 3} x^n/n!$ and $f(x) = g(x/2)$.

This resolves the source problem recorded as **APP-0060** [50].

Proof. We use Proposition 9.1, Proposition 15.25, Proposition 7.2, Proposition 9.1, and Proposition 15.26 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Define

$$g(x) = 1 + \frac{x}{4} + x^2 + \sum_{n\ge 3} \frac{x^n}{n!}, \qquad f(x) = g(x/2).$$

Then g and f are entire. The coefficients of g are

$$b_0 = 1, \qquad b_1 = \frac{1}{4}, \qquad b_2 = 1, \qquad b_n = \frac{1}{n!} \quad (n \ge 3),$$

so $b_n > 0$ for every n. Since $f(x) = g(x/2)$, its coefficients satisfy

$$a_n = b_n 2^{-n}.$$

Thus

$$c_n = \frac{a_n}{b_n} = 2^{-n}.$$

For every $n, k \ge 0$,

$$\Delta^k c_n = (1 - \tfrac{1}{2})^k 2^{-n} = 2^{-n-k} > 0.$$

Hence $\{a_n/b_n\}_{n\ge 0}$ is strictly completely monotone, and therefore completely monotone, in the source sense.

Now write the quotient near the left endpoint. More generally, if

$$g(x) = 1 + px + qx^2 + O(x^3), \qquad 0 < t < 1, \qquad f(x) = g(tx),$$

then

$$\frac{f(x)}{g(x)} = 1 + p(t-1)x + (1-t)\{p^2 - q(1+t)\}x^2 + O(x^3).$$

In the present example $p = 1/4$, $q = 1$, and $t = 1/2$. Therefore

$$[x^2]\frac{f(x)}{g(x)} = \frac{1}{2}\left(\frac{1}{16} - \frac{3}{2}\right) = -\frac{23}{32},$$

so

$$\left(\frac{f}{g}\right)''(0) = -\frac{23}{16} < 0.$$

By analyticity, $(f/g)''(x) < 0$ for all sufficiently small $x > 0$. A completely monotone function on $(0, \varepsilon)$ must satisfy $F''(x) \ge 0$, so f/g is not completely monotone on any interval $(0, \varepsilon)$. It is therefore not strictly completely monotone either. $\square$

Proposition 15.28 (Slice endpoint concavity obstruction for reciprocal JCM nets)**.** *Let $A_m, B_m > 0$ and $\beta_{m,n} = 1/(B_m + A_m n)$ for $m, n \in \mathbb{Z}_+$. If $\{\beta_{m,n}\}$ is jointly completely monotone, then $r_m = B_m/A_m$ is discretely concave: $r_{m+1} - 2r_m + r_{m-1} \le 0$ for every $m \ge 1$.*

Proof. For $a_0, b_0 > 0$ and $0 < b_1 < a_1 < b_2 < b_3$, let

$$\beta_{m,n} = \frac{1}{b_0(m+b_1)(m+b_2)(m+b_3) + a_0(m+a_1)n}, \qquad m, n \in \mathbb{Z}_+.$$

The target is to prove that this net is not jointly completely monotone, giving a negative answer to Khasnis-Sholapurkar Question 4.6.

Let $A_m, B_m > 0$ and

$$\beta_{m,n} = \frac{1}{B_m + A_m n}.$$

Set $r_m = B_m/A_m$. If $\{\beta_{m,n}\}$ is jointly completely monotone, then

$$r_{m+1} - 2r_m + r_{m-1} \le 0$$

for every $m \ge 1$.

Indeed, joint complete monotonicity is equivalent to the Hausdorff representation

$$\beta_{m,n} = \int_{[0,1]^2} x^m t^n \, d\mu(x,t)$$

for a positive finite measure μ. For a Borel set $E \subset [0,1]$, define

$$\nu_m(E) = \int_{[0,1]\times E} x^m \, d\mu(x,t).$$

Cauchy-Schwarz gives the log-convexity inequality

$$\nu_m(E)^2 \le \nu_{m-1}(E)\nu_{m+1}(E).$$

For fixed m, the n-slice has the unique Hausdorff representing density

$$d\nu_m(t) = \phi_m(t)\, dt, \qquad \phi_m(t) = \frac{1}{A_m} t^{r_m - 1},$$

because

$$\int_0^1 t^n \phi_m(t)\, dt = \frac{1}{A_m(n + r_m)} = \frac{1}{B_m + A_m n}.$$

Taking $E = (0, \varepsilon)$ gives

$$\nu_m(E) = \frac{\varepsilon^{r_m}}{A_m r_m} = \frac{\varepsilon^{r_m}}{B_m}.$$

Thus

$$\left(\frac{\varepsilon^{r_m}}{B_m}\right)^2 \le \frac{\varepsilon^{r_{m-1}}}{B_{m-1}} \frac{\varepsilon^{r_{m+1}}}{B_{m+1}},$$

or

$$\frac{B_{m-1}B_{m+1}}{B_m^2} \varepsilon^{-\{r_{m+1} - 2r_m + r_{m-1}\}} \le 1.$$

If $r_{m+1} - 2r_m + r_{m-1} > 0$, the left side tends to $+\infty$ as $\varepsilon \downarrow 0$, contradiction. This proves the lemma. $\square$

Proposition 15.29 (Degree-gap obstruction for reciprocal polynomial JCM nets)**.** *Let A, B be positive polynomial sequences on $\mathbb{Z}_+$ with positive leading coefficients and* $\deg B - \deg A \ge 2$*. Then the two-parameter net $\beta_{m,n} = 1/(B(m) + A(m)n)$ is not jointly completely monotone.*

Proof. For Degree-gap obstruction for reciprocal polynomial JCM nets, this proof is the corresponding component of Proposition 15.28. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. $\square$

Counterexample 15.30 (APP-0061: Khasnis-Sholapurkar Question 4.6 has a negative answer)**.** For every $a_0, b_0 > 0$ and $0 < b_1 < a_1 < b_2 < b_3$, the net $\beta_{m,n} = 1/[b_0(m+b_1)(m+b_2)(m+b_3) + a_0(m+a_1)n]$ on $\mathbb{Z}_+^2$ is not jointly completely monotone. Thus Khasnis-Sholapurkar Question 4.6 is answered negatively.

This resolves the source problem recorded as **APP-0061** [51].

Proof. We use Proposition 9.1, Proposition 15.28, and Proposition 15.29 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For $a_0, b_0 > 0$ and $0 < b_1 < a_1 < b_2 < b_3$, let

$$\beta_{m,n} = \frac{1}{b_0(m+b_1)(m+b_2)(m+b_3) + a_0(m+a_1)n}, \qquad m, n \in \mathbb{Z}_+.$$

The target is to prove that this net is not jointly completely monotone, giving a negative answer to Khasnis-Sholapurkar Question 4.6.

Let $A_m, B_m > 0$ and

$$\beta_{m,n} = \frac{1}{B_m + A_m n}.$$

Set $r_m = B_m/A_m$. If $\{\beta_{m,n}\}$ is jointly completely monotone, then

$$r_{m+1} - 2r_m + r_{m-1} \le 0$$

for every $m \ge 1$.

Indeed, joint complete monotonicity is equivalent to the Hausdorff representation

$$\beta_{m,n} = \int_{[0,1]^2} x^m t^n \, d\mu(x,t)$$

for a positive finite measure μ. For a Borel set $E \subset [0,1]$, define

$$\nu_m(E) = \int_{[0,1]\times E} x^m \, d\mu(x,t).$$

Cauchy-Schwarz gives the log-convexity inequality

$$\nu_m(E)^2 \le \nu_{m-1}(E)\nu_{m+1}(E).$$

For fixed m, the n-slice has the unique Hausdorff representing density

$$d\nu_m(t) = \phi_m(t)\, dt, \qquad \phi_m(t) = \frac{1}{A_m} t^{r_m - 1},$$

because

$$\int_0^1 t^n \phi_m(t)\, dt = \frac{1}{A_m(n+r_m)} = \frac{1}{B_m + A_m n}.$$

Taking $E = (0, \varepsilon)$ gives

$$\nu_m(E) = \frac{\varepsilon^{r_m}}{A_m r_m} = \frac{\varepsilon^{r_m}}{B_m}.$$

Thus

$$\left(\frac{\varepsilon^{r_m}}{B_m}\right)^2 \le \frac{\varepsilon^{r_{m-1}}}{B_{m-1}} \frac{\varepsilon^{r_{m+1}}}{B_{m+1}},$$

or

$$\frac{B_{m-1}B_{m+1}}{B_m^2} \varepsilon^{-\{r_{m+1} - 2r_m + r_{m-1}\}} \le 1.$$

If $r_{m+1} - 2r_m + r_{m-1} > 0$, the left side tends to $+\infty$ as $\varepsilon \downarrow 0$, contradiction. This proves the lemma. $\square$

Proposition 15.31 (Additive product-kernel rectangle annihilator)**.** *If X and W each contain at least two points, then any finite-valued kernel on $X \times W$ of the form $K((x,w),(x',w')) = AK_X(x,x') + BK_W(w,w')$ is not strictly positive definite on $X \times W$. A four-point checkerboard coefficient vector on a nontrivial product rectangle gives a zero quadratic form.*

Proof. Let X and W be sets with at least two points. Let $K_X\colon X \times X \to \mathbb{R}$ and $K_W\colon W \times W \to \mathbb{R}$ be finite-valued kernels, and let $A, B \in \mathbb{R}$. Define

$$K((x,w),(x',w')) = AK_X(x,x') + BK_W(w,w').$$

Then K is not strictly positive definite on $X \times W$.

Choose $x_0 \neq x_1$ in X and $w_0 \neq w_1$ in W. The four points

$$p_{ij} = (x_i, w_j), \qquad i,j \in \{0,1\},$$

are pairwise distinct. Put $a = (1,-1)$, $b = (1,-1)$, and $c_{ij} = a_i b_j$. Thus the coefficient vector is

$$(c_{00}, c_{01}, c_{10}, c_{11}) = (1,-1,-1,1),$$

which is nonzero.

The quadratic form is

$$\begin{aligned} Q &= \sum_{i,j,k,\ell} c_{ij} c_{k\ell} \left(AK_X(x_i,x_k) + BK_W(w_j,w_\ell)\right) \\ &= A\left(\sum_j b_j\right)^2 \sum_{i,k} a_i a_k K_X(x_i,x_k) + B\left(\sum_i a_i\right)^2 \sum_{j,\ell} b_j b_\ell K_W(w_j,w_\ell). \end{aligned}$$

Since $\sum_i a_i = \sum_j b_j = 0$, we have $Q = 0$. This is a zero quadratic form for a nonzero coefficient vector on distinct points, contradicting strict positive definiteness.

This proof does not address singleton X, singleton $Y \times Z$, or other one-factor degeneracies. The four-point rectangle cannot be formed there, and strict positive definiteness depends on the remaining one-factor kernel and constants.

This proof also does not address the adjacent source cases $r > \lambda$, $\mu_f \neq 0$, $D_f = 0$, complete-Bernstein analogues, or any non-additive variant. ☐

Counterexample 15.32 (APP-0062: Barbosa-Menegatto endpoint additive kernel is not strictly positive definite)**.** In the Barbosa–Menegatto generalized-Stieltjes endpoint $D_f > 0$, $r = \lambda$, $\mu_f = 0$, $C_f D_f > 0$, the kernel $G_\lambda(t,u,v) = C_f h(u,v)^{-\lambda} + D_f g(t)^{-\lambda}$ is not strictly positive definite on $X \times Y \times Z$ whenever X is nontrivial and $Y \times Z$ is nontrivial. Thus the exact nontrivial product-factor endpoint has a negative answer.

This resolves the source problem recorded as **APP-0062** [52].

Proof. We use Proposition 18.3 and Proposition 15.31 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let X and W be sets with at least two points. Let $K_X\colon X \times X \to \mathbb{R}$ and $K_W\colon W \times W \to \mathbb{R}$ be finite-valued kernels, and let $A, B \in \mathbb{R}$. Define

$$K((x,w),(x',w')) = AK_X(x,x') + BK_W(w,w').$$

Then K is not strictly positive definite on $X \times W$.

Choose $x_0 \neq x_1$ in X and $w_0 \neq w_1$ in W. The four points

$$p_{ij} = (x_i, w_j), \qquad i, j \in \{0, 1\},$$

are pairwise distinct. Put $a = (1, -1)$, $b = (1, -1)$, and $c_{ij} = a_i b_j$. Thus the coefficient vector is

$$(c_{00}, c_{01}, c_{10}, c_{11}) = (1, -1, -1, 1),$$

which is nonzero.

The quadratic form is

$$\begin{aligned} Q &= \sum_{i,j,k,\ell} c_{ij} c_{k\ell} \left(A K_X(x_i, x_k) + B K_W(w_j, w_\ell) \right) \\ &= A \left(\sum_j b_j \right)^2 \sum_{i,k} a_i a_k K_X(x_i, x_k) + B \left(\sum_i a_i \right)^2 \sum_{j,\ell} b_j b_\ell K_W(w_j, w_\ell). \end{aligned}$$

Since $\sum_i a_i = \sum_j b_j = 0$, we have $Q = 0$. This is a zero quadratic form for a nonzero coefficient vector on distinct points, contradicting strict positive definiteness.

This proof does not address singleton X, singleton $Y \times Z$, or other one-factor degeneracies. The four-point rectangle cannot be formed there, and strict positive definiteness depends on the remaining one-factor kernel and constants.

This proof also does not address the adjacent source cases $r > \lambda$, $\mu_f \neq 0$, $D_f = 0$, complete-Bernstein analogues, or any non-additive variant. □

15.3 Scalar reductions for operator and matrix inequalities

Proposition 15.33 (Scalar subcase refutes universal operator inequality). *If an operator inequality is universally quantified over all Hilbert spaces and all tuples of positive invertible operators satisfying given hypotheses, then a one-dimensional scalar tuple satisfying those hypotheses and violating the resulting scalar inequality refutes the universal operator inequality.*

Proof. For Scalar subcase refutes universal operator inequality, this proof is the corresponding component of Proposition 15.34. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Proposition 15.34 (Fractional-linear scalar GKE certificate). *For $g_0(x) = 2(x-1)/(x+1)$, equal weights $(1/2, 1/2)$, and scalar data $(A_1, A_2) = (4, 1)$, the scalar generalized Karcher equation has unique positive solution $X = 2$, while $g_0^{-1}((g_0(4) + g_0(1))/2) = 13/7$.*

Proof. Let

$$g_0(x) = \frac{2(x-1)}{x+1} = 2 - \frac{4}{x+1}.$$

Yamazaki lists this fractional-linear example among the normalized operator-monotone functions in L. Independently, $x \mapsto (x+1)^{-1}$ is operator-monotone decreasing on $(0, \infty)$, so g_0 is operator-monotone increasing. Also

$$g_0(1) = 0, \qquad g_0'(x) = \frac{4}{(x+1)^2}, \qquad g_0'(1) = 1.$$

Hence $g_0 \in L$.

Take $H = \mathbb{C}$, equal weights $\omega = (1/2, 1/2)$, and scalar operators

$$A_1 = 4, \qquad A_2 = 1.$$

For a positive scalar X, the generalized Karcher equation becomes

$$\frac{1}{2} g_0(4/X) + \frac{1}{2} g_0(1/X) = 0.$$

Since

$$g_0(a/X) = \frac{2(a - X)}{a + X},$$

this is equivalent to

$$\frac{4 - X}{4 + X} + \frac{1 - X}{1 + X} = 0.$$

The numerator is

$$(4 - X)(1 + X) + (1 - X)(4 + X) = 8 - 2X^2.$$

The unique positive solution is $X = 2$. Therefore

$$\sigma_{g_0}\big((1/2, 1/2); 4, 1\big) = 2.$$

On the right-hand side of the conjectured inequality,

$$g_0(4) = \frac{6}{5}, \qquad g_0(1) = 0,$$

so

$$\frac{1}{2} g_0(4) + \frac{1}{2} g_0(1) = \frac{3}{5}.$$

Solving $u = 2(x - 1)/(x + 1)$ gives

$$g_0^{-1}(u) = \frac{2 + u}{2 - u}.$$

Hence

$$g_0^{-1}\left(\frac{3}{5}\right) = \frac{2 + 3/5}{2 - 3/5} = \frac{13}{7}.$$

Since the scalar operator norm is absolute value and all quantities here are positive, the conjectured inequality would require

$$2 \le \frac{13}{7},$$

which is false.

Thus Yamazaki's displayed operator-norm conjecture is false.

This proof only refutes the universal conjecture by a scalar counterexample. It does not classify all $g \in L$, all scalar pairs, all weights, or any positive validity region. It does not make claims about adjacent Yamazaki norm inequalities or unitarily invariant norm statements not covered by the displayed source conjecture. □

Counterexample 15.35 (APP-0063: Yamazaki generalized-Karcher operator-norm conjecture is false)**.** Yamazaki's generalized-Karcher operator-norm conjecture is false. In the one-dimensional scalar case, with $g_0(x) = 2(x - 1)/(x + 1)$, $\omega = (1/2, 1/2)$, $A_1 = 4$, and $A_2 = 1$, the generalized Karcher solution is $\sigma_{g_0} = 2$, while the conjectured right-hand side is $g_0^{-1}((g_0(4) + g_0(1))/2) = 13/7$. Thus the conjectured inequality would require $2 \le 13/7$, a contradiction.

This resolves the source problem recorded as **APP-0063** [53].

Proof. We use Proposition 9.1, Proposition 15.33, and Proposition 15.34 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$g_0(x) = \frac{2(x-1)}{x+1} = 2 - \frac{4}{x+1}.$$

Yamazaki lists this fractional-linear example among the normalized operator-monotone functions in L. Independently, $x \mapsto (x+1)^{-1}$ is operator-monotone decreasing on $(0, \infty)$, so g_0 is operator-monotone increasing. Also

$$g_0(1) = 0, \qquad g_0'(x) = \frac{4}{(x+1)^2}, \qquad g_0'(1) = 1.$$

Hence $g_0 \in L$.

Take $H = \mathbb{C}$, equal weights $\omega = (1/2, 1/2)$, and scalar operators

$$A_1 = 4, \qquad A_2 = 1.$$

For a positive scalar X, the generalized Karcher equation becomes

$$\frac{1}{2} g_0(4/X) + \frac{1}{2} g_0(1/X) = 0.$$

Since

$$g_0(a/X) = \frac{2(a-X)}{a+X},$$

this is equivalent to

$$\frac{4-X}{4+X} + \frac{1-X}{1+X} = 0.$$

The numerator is

$$(4-X)(1+X) + (1-X)(4+X) = 8 - 2X^2.$$

The unique positive solution is $X = 2$. Therefore

$$\sigma_{g_0}\big((1/2, 1/2); 4, 1\big) = 2.$$

On the right-hand side of the conjectured inequality,

$$g_0(4) = \frac{6}{5}, \qquad g_0(1) = 0,$$

so

$$\frac{1}{2} g_0(4) + \frac{1}{2} g_0(1) = \frac{3}{5}.$$

Solving $u = 2(x-1)/(x+1)$ gives

$$g_0^{-1}(u) = \frac{2+u}{2-u}.$$

Hence

$$g_0^{-1}\left(\frac{3}{5}\right) = \frac{2 + 3/5}{2 - 3/5} = \frac{13}{7}.$$

Since the scalar operator norm is absolute value and all quantities here are positive, the conjectured inequality would require

$$2 \le \frac{13}{7},$$

which is false.

Thus Yamazaki's displayed operator-norm conjecture is false.

This proof only refutes the universal conjecture by a scalar counterexample. It does not classify all $g \in L$, all scalar pairs, all weights, or any positive validity region. It does not make claims about adjacent Yamazaki norm inequalities or unitarily invariant norm statements not covered by the displayed source conjecture. □

Proposition 15.36 (Scalar case refutes universal matrix existence)**.** *If a matrix existence assertion is universally quantified over all positive matrix sizes and all data satisfying its hypotheses, then an admissible one-dimensional scalar datum satisfying the same hypotheses and admitting no scalar solution refutes the universal matrix assertion.*

Proof. Take the one-dimensional scalar case

$$p = 1, \qquad q = 2, \qquad X = 1, \qquad Y = 2.$$

The data satisfy $0 < X < Y$. If positive scalars a, b solved the source system, then

$$\frac{a+b}{2} = 1, \qquad \sqrt{\frac{a^2+b^2}{2}} = 2.$$

Thus $a + b = 2$ and $a^2 + b^2 = 8$. But for $a, b \geq 0$,

$$a^2 + b^2 \leq a^2 + 2ab + b^2 = (a+b)^2 = 4,$$

which contradicts $a^2 + b^2 = 8$. Hence no positive scalar solution exists. The contradiction is a fortiori valid for strictly positive scalars.

For $0 < p < q$ and $a, b \geq 0$, not both zero, define

$$M_r(a, b) = \left(\frac{a^r + b^r}{2}\right)^{1/r}.$$

Using ℓ^r-norm monotonicity in $\mathbb{R}^2$,

$$\frac{M_q(a,b)}{M_p(a,b)} = 2^{1/p-1/q} \frac{\|(a,b)\|_q}{\|(a,b)\|_p} \leq 2^{1/p-1/q}.$$

Thus scalar inverse data with $X > 0$ must satisfy

$$1 \leq \frac{Y}{X} \leq 2^{1/p-1/q},$$

with strict upper inequality when $a, b > 0$ and both are nonzero. In the certificate above, $Y/X = 2 > \sqrt{2} = 2^{1-1/2}$.

This proves a negative answer to the literal unrestricted source question. It does not classify the corrected inverse problem obtained by adding an upper scalar or matrix compatibility condition. □

Definition 15.2 (Two-point scalar power means)**.** For $a, b \geq 0$, not both zero, and $r > 0$, the two-point scalar power mean is $M_r(a,b) = ((a^r + b^r)/2)^{1/r}$.

Proposition 15.37 (Scalar power-mean ratio endpoint)**.** *For $0 < p < q$ and $a, b \geq 0$, not both zero, the two-point scalar power means satisfy $1 \leq M_q(a,b)/M_p(a,b) \leq 2^{1/p-1/q}$. In particular, any positive scalar solution of the two-equation (p,q)-power-mean inverse problem with scalar data $X, Y > 0$ must satisfy $Y/X \leq 2^{1/p-1/q}$.*

Proof. For Scalar power-mean ratio endpoint, this proof is the corresponding component of Proposition 15.36. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. ☐

Counterexample 15.38 (APP-0064: Dinh-Le-Nguyen-Vo matrix power-mean inverse question is false)**.** The literal Dinh–Le–Nguyen–Vo matrix power-mean inverse question is false. In the one-dimensional scalar case, the admissible data $p = 1$, $q = 2$, $X = 1$, and $Y = 2$ satisfy $0 < X < Y$, but the system $(a + b)/2 = 1$, $((a^2 + b^2)/2)^{1/2} = 2$ has no nonnegative, hence no positive, scalar solution.

This resolves the source problem recorded as **APP-0064** [54].

Proof. We use Proposition 9.1, Proposition 15.2, Proposition 15.37, and Proposition 15.36 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Take scalar data

$$p = 1, \quad q = 2, \quad X = 1, \quad Y = 2.$$

If positive scalars a, b solved the two equations, then

$$\frac{a + b}{2} = 1, \qquad \left(\frac{a^2 + b^2}{2}\right)^{1/2} = 2.$$

Thus $a + b = 2$ and $a^2 + b^2 = 8$. But for $a, b \geq 0$,

$$a^2 + b^2 \leq (a + b)^2 = 4,$$

contradicting $a^2 + b^2 = 8$. Therefore the asserted universal inverse existence statement fails already in the one-dimensional scalar case. ☐

15.4 Kummer-series modulus obstruction

Proposition 15.39 (Real-series imaginary-axis modulus second-coefficient obstruction)**.** *Let* $F(z) = 1 + Az + Bz^2 + O(z^3)$ *near* 0*, with* $A, B \in \mathbb{R}$*. Then* $|F(it)|^2 = 1 + (A^2 - 2B)t^2 + O(t^4)$*. In particular, if* $A^2 - 2B < 0$*, then* $|F(it)| < F(0) = 1$ *for all sufficiently small nonzero real* t*.*

Proof. For $c \notin \{0, -1, -2, \ldots\}$, the standard Kummer series is

$$\mathrm{M}(a, c, z) = {}_1F_1(a; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n}{(c)_n} \frac{z^n}{n!}.$$

Hence

$$\mathrm{M}(a, c, z) = 1 + \frac{a}{c} z + \frac{a(a + 1)}{2c(c + 1)} z^2 + O(z^3).$$

Let $F(z) = 1 + Az + Bz^2 + O(z^3)$ with real A, B. Then

$$F(it) = 1 + iAt - Bt^2 + O(t^3), \qquad F(-it) = 1 - iAt - Bt^2 + O(t^3),$$

so

$$|F(it)|^2 = F(it)F(-it) = 1 + (A^2 - 2B)t^2 + O(t^4).$$

Consequently, if $A^2 - 2B < 0$, then $|F(it)| < 1 = F(0)$ for all sufficiently small nonzero real t.

Take

$$\alpha = 1, \qquad \beta = \frac{5}{2}, \qquad c = \alpha - \beta = -\frac{3}{2}.$$

This is admissible because $c \notin \{0, -1, -2, \ldots\}$. For

$$F(z) = \mathrm{M}(1, -3/2, z),$$

the coefficients are

$$A = \frac{1}{-3/2} = -\frac{2}{3}, \qquad B = \frac{1 \cdot 2}{2(-3/2)(-1/2)} = \frac{4}{3}.$$

Thus

$$A^2 - 2B = \frac{4}{9} - \frac{8}{3} = -\frac{20}{9}.$$

Therefore

$$|\mathrm{M}(1, -3/2, it)|^2 = 1 - \frac{20}{9}t^2 + O(t^4) < 1$$

for all sufficiently small nonzero real t. Since $\mathrm{Re}(it) = 0$ and $\mathrm{M}(1, -3/2, 0) = 1$, this gives

$$|\mathrm{M}(1, -3/2, it)| < 1 = \mathrm{M}(1, -3/2, \mathrm{Re}(it)).$$

This contradicts the source's proposed universal inequality.

Conclusion: the Garrappa–Gerhold–Popolizio–Simon Remark 6 Kummer M global modulus question has a negative answer.

Scope caveat: this does not prove or disprove the separate GGPS Conjecture 25 for the two-parameter Mittag-Leffler function $E_{\alpha,\beta}$ and reciprocal complete monotonicity.

Verification:

Local symbolic check: $c = -3/2$, $A = -2/3$, $B = 4/3$, $A^2 - 2B = -20/9$, and ${}_1F_1(1; -3/2; z) = 1 - 2z/3 + 4z^2/3 + O(z^3)$. □

Definition 15.3 (GGPS Remark 6 Kummer M global modulus question)**.** Garrappa–Gerhold–Popolizio–Simon Remark 6 asks whether, for all $\alpha, \beta > 0$, $|\mathrm{M}(\alpha, \alpha-\beta, z)| \geq \mathrm{M}(\alpha, \alpha-\beta, \mathrm{Re}\, z)$ holds for every $z \in \mathbb{C}$, and states that this global question seems still open.

Definition 15.4 (Kummer M local series)**.** For $c \notin \{0, -1, -2, \ldots\}$, the Kummer confluent hypergeometric function has the local expansion $\mathrm{M}(a, c, z) = {}_1F_1(a; c; z) = 1 + (a/c)z + a(a+1)z^2/[2c(c+1)] + O(z^3)$.

Counterexample 15.40 (APP-0068: Kummer M global modulus inequality fails at $(\alpha, \beta) = (1, 5/2)$)**.** The Garrappa–Gerhold–Popolizio–Simon Remark 6 universal Kummer inequality is false. For $\alpha = 1$, $\beta = 5/2$, and all sufficiently small nonzero real t, $|\mathrm{M}(1, -3/2, it)| < 1 = \mathrm{M}(1, -3/2, \mathrm{Re}(it))$.

This resolves the source problem recorded as **APP-0068** [57].

Proof. We use Proposition 9.1, Proposition 15.39, Proposition 15.3, and Proposition 15.4 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For $c \notin \{0, -1, -2, \ldots\}$, the standard Kummer series is

$$\mathrm{M}(a, c, z) = {}_1F_1(a; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n}{(c)_n} \frac{z^n}{n!}.$$

Hence

$$\mathrm{M}(a, c, z) = 1 + \frac{a}{c}z + \frac{a(a+1)}{2c(c+1)}z^2 + O(z^3).$$

Let $F(z) = 1 + Az + Bz^2 + O(z^3)$ with real A, B. Then

$$F(it) = 1 + iAt - Bt^2 + O(t^3), \qquad F(-it) = 1 - iAt - Bt^2 + O(t^3),$$

so

$$|F(it)|^2 = F(it)F(-it) = 1 + (A^2 - 2B)t^2 + O(t^4).$$

Consequently, if $A^2 - 2B < 0$, then $|F(it)| < 1 = F(0)$ for all sufficiently small nonzero real t.

Take

$$\alpha = 1, \qquad \beta = \frac{5}{2}, \qquad c = \alpha - \beta = -\frac{3}{2}.$$

This is admissible because $c \notin \{0, -1, -2, \ldots\}$. For

$$F(z) = \mathrm{M}(1, -3/2, z),$$

the coefficients are

$$A = \frac{1}{-3/2} = -\frac{2}{3}, \qquad B = \frac{1 \cdot 2}{2(-3/2)(-1/2)} = \frac{4}{3}.$$

Thus

$$A^2 - 2B = \frac{4}{9} - \frac{8}{3} = -\frac{20}{9}.$$

Therefore

$$|\mathrm{M}(1, -3/2, it)|^2 = 1 - \frac{20}{9}t^2 + O(t^4) < 1$$

for all sufficiently small nonzero real t. Since $\mathrm{Re}(it) = 0$ and $\mathrm{M}(1, -3/2, 0) = 1$, this gives

$$|\mathrm{M}(1, -3/2, it)| < 1 = \mathrm{M}(1, -3/2, \mathrm{Re}(it)).$$

This contradicts the source's proposed universal inequality.

Conclusion: the Garrappa–Gerhold–Popolizio–Simon Remark 6 Kummer M global modulus question has a negative answer.

Scope caveat: this does not prove or disprove the separate GGPS Conjecture 25 for the two-parameter Mittag-Leffler function $E_{\alpha,\beta}$ and reciprocal complete monotonicity.

Verification:

Local symbolic check: $c = -3/2$, $A = -2/3$, $B = 4/3$, $A^2 - 2B = -20/9$, and ${}_1F_1(1; -3/2; z) = 1 - 2z/3 + 4z^2/3 + O(z^3)$. □

16 Bessel-I square-root endpoint obstruction

This section collects applications in which a universal positivity, convexity, or monotonicity claim is decided by reducing it to a local obstruction. The common method is to isolate a boundary, endpoint, or explicit witness where the proposed sign pattern cannot hold.

Counterexample 16.1 (APP-0042: Modified Bessel square-root log-concavity fails)**.** not(For every $\nu \geq 0$, the function $u \mapsto \sqrt{u}\, I_\nu(u)$ is strictly log-concave on $(0, \infty)$.)

This resolves the source problem recorded as **APP-0042** [36].

Proof. We use Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$g_\nu(u) = \log(\sqrt{u}\, I_\nu(u)), \qquad r_\nu(u) = \frac{I'_\nu(u)}{I_\nu(u)}.$$

Then

$$g''_\nu(u) = r'_\nu(u) - \frac{1}{2u^2}.$$

The modified Bessel equation

$$u^2 I_\nu''(u) + u I_\nu'(u) - (u^2 + \nu^2) I_\nu(u) = 0$$

gives

$$\frac{I_\nu''(u)}{I_\nu(u)} = 1 + \frac{\nu^2}{u^2} - \frac{r_\nu(u)}{u}.$$

Since

$$r_\nu'(u) = \frac{I_\nu''(u)}{I_\nu(u)} - r_\nu(u)^2,$$

we get

$$g_\nu''(u) = 1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2.$$

Thus the Bessel I Riccati log concavity inequality is exactly the strict log-concavity condition $g_\nu''(u) < 0$.

For $\nu = 0$,

$$r_0(u) = \frac{I_0'(u)}{I_0(u)} = \frac{I_1(u)}{I_0(u)}.$$

At $u = 10$, the Riccati expression is

$$g_0''(10) = \frac{199}{200} - \frac{r_0(10)}{10} - r_0(10)^2.$$

It is enough to prove

$$r_0(10) < \frac{9487}{10000},$$

because

$$\frac{199}{200} - \frac{1}{10}\frac{9487}{10000} - \left(\frac{9487}{10000}\right)^2 = \frac{9831}{100000000} > 0.$$

$$I_0(10) = \sum_{k=0}^{\infty} \frac{25^k}{(k!)^2}, \qquad I_1(10) = \sum_{k=0}^{\infty} \frac{5 \cdot 25^k}{k!(k+1)!}.$$

With $N = 20$, set

$$L_0 = \sum_{k=0}^{20} \frac{25^k}{(k!)^2} < I_0(10).$$

For the I_1 tail after $N = 20$, the term ratio is bounded by

$$q = \frac{25}{22 \cdot 23} < 1,$$

so the script computes an exact rational upper bound $U_1 > I_1(10)$ and verifies

$$10000\, U_1 < 9487\, L_0.$$

Therefore $I_1(10)/I_0(10) < 9487/10000$, and hence

$$g_0''(10) > 0.$$

This proves the Bessel-I log-concavity counterexample to T-I sqrt-log concavity for $\nu > 0$, and the counterexample to T-I Riccati log-concavity inequality. $\square$

17 Determinant and moment-ratio applications

This section develops applications governed by determinant compression and moment-ratio comparisons. The method turns recursive derivative information into a finite or structured positivity problem, so bounds and counterexamples can be read from a smaller normal form.

Corollary 17.1 (APP-0038: Ma-Weigert derivative-sign regions form a chain)**.** *In the Ma–Weigert log-function class $\mathcal{F}_{1,n}$, the derivative-sign regions $D_k = \{f : L^k f \geq 0\}$, with $L = -d/dx$, form a descending chain: $D_{k+1} \subseteq D_k$ for every k, n.*

*This resolves the source problem recorded as **APP-0038** [11].*

Proof. We use Proposition 12.32 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

We first prove that for each $k \geq 0$,

$$L^k f(x) = x^{-k-1} p_k(\log x)$$

for some polynomial p_k.

For $k = 0$, this is the definition with $p_0 = p$. Suppose

$$L^k f(x) = x^{-k-1} p_k(\log x).$$

Then

$$\begin{aligned} L^{k+1} f(x) &= -\frac{d}{dx}\left(x^{-k-1} p_k(\log x)\right) \\ &= x^{-k-2}\left((k+1)p_k(\log x) - p_k'(\log x)\right). \end{aligned}$$

Thus $p_{k+1}(y) = (k+1)p_k(y) - p_k'(y)$, which is again a polynomial.

Since every polynomial in $\log x$ grows slower than every positive power of x,

$$x^{-k-1} p_k(\log x) \to 0 \qquad (x \to \infty).$$

Therefore

$$\lim_{x\to\infty} L^k f(x) = 0$$

for every $k \geq 0$.

Assume $f \in D_{k+1}$. Then

$$L^{k+1} f(x) \geq 0$$

on $(0, \infty)$. Put

$$g(x) = L^k f(x).$$

By definition of L,

$$-g'(x) = L^{k+1} f(x) \geq 0.$$

Using $g(x) \to 0$ as $x \to \infty$, integrate from x to R:

$$g(x) - g(R) = \int_x^R L^{k+1} f(t)\, dt.$$

Letting $R \to \infty$ gives

$$L^k f(x) = g(x) = \int_x^\infty L^{k+1} f(t)\, dt \geq 0.$$

Hence $f \in D_k$. Therefore

$$D_{k+1} \subseteq D_k$$

for all $k \geq 0$ and all n. $\square$

18 Dual-certificate and algebraic extremal applications

This section contains applications in which an algebraic extremum is identified by a dual certificate. The structural principle supplies the extremal norm or moment identity, and the source-specific computation verifies that the candidate object attains the certified value.

Definition 18.1 (Finite dimensional l1 dual certificate language)**.** Let V be a real vector space and let $\mathcal{A} \subset V$ be a finite family. The elements of $\mathcal{A}$ are the coordinate atoms. A primal representation of $v \in V$ is a finite expansion $v = \sum_{a\in\mathcal{A}} c_a a$, and its ℓ^1-norm is $\sum_{a\in\mathcal{A}} |c_a|$. A dual certificate for the lower bound M is a linear functional $\Lambda : V \to \mathbb{R}$ such that $\Lambda(v) = M$ and $|\Lambda(a)| \le 1$ for every $a \in \mathcal{A}$.

Proposition 18.1 (Finite-dimensional dual certificate for exact l1 representation norm)**.** *In the finite-dimensional ℓ^1 dual-certificate setting, if v has a primal representation of norm M and a dual certificate Λ with $\Lambda(v) = M$, then the exact minimal ℓ^1-representation norm of v by the coordinate atoms is M.*

Proof. Let V be a real vector space, let $\mathcal{A} \subset V$ be a finite coordinate family, and let $v \in V$. Suppose there is a representation

$$v = \sum_{a\in\mathcal{A}} c_a a, \qquad \sum_{a\in\mathcal{A}} |c_a| = M,$$

and a linear functional $\Lambda : V \to \mathbb{R}$ such that

$$\Lambda(v) = M, \qquad |\Lambda(a)| \le 1 \quad (a \in \mathcal{A}).$$

Then the least ℓ^1-coefficient norm of a representation of v by elements of $\mathcal{A}$ is exactly M.

For any other representation $v = \sum_{a\in\mathcal{A}} b_a a$, linearity and the dual bound give

$$M = \Lambda(v) = \sum_{a\in\mathcal{A}} b_a \Lambda(a) \le \sum_{a\in\mathcal{A}} |b_a|\,|\Lambda(a)| \le \sum_{a\in\mathcal{A}} |b_a|.$$

Thus every representation has norm at least M. The displayed primal representation has norm M, so the minimum is M. ☐

Definition 18.2 (Weyl algebra balanced words and coefficient norm)**.** The real Weyl algebra A_1 is generated by X and D with relation $DX - XD = 1$. A balanced word of degree (n, n) is a word containing exactly n copies of D and n copies of X. The coefficient ℓ^1-norm of a finite word expansion $\sum_w a_w w$ is $\sum_w |a_w|$.

Corollary 18.2 (APP-0046: Tao–Sawin finite Weyl l1 minimum is exact)**.** *In the Weyl algebra DX-XD equals one, the least l1 coefficient norm of a balanced degree (n,n) word combination equal to one is two to the n divided by n factorial.*

$$L_n = \min\{\sum_w |a_w| : \sum_w a_w w = 1,\ w \text{ has } n \text{ copies of } D \text{ and } n \text{ copies of } X\} = \frac{2^n}{n!}.$$

*This resolves the source problem recorded as **APP-0046** [37].*

Proof. We use Proposition 18.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let A_1 be the Weyl algebra over a field of characteristic 0, generated by X, D with

$$DX - XD = 1.$$

Let W_n be the set of words containing exactly n copies of D and n copies of X. For

$$P = \sum_{w \in W_n} a_w w$$

define

$$\|P\|_1 = \sum_{w \in W_n} |a_w|.$$

Let

$$L_n = \min \left\{ \|P\|_1 : \sum_{w \in W_n} a_w w = 1 \text{ in } A_1 \right\}.$$

Then

$$L_n = \frac{2^n}{n!}.$$

Set

$$P_n = \frac{1}{n!} \operatorname{ad}_D^n(X^n).$$

Since $\operatorname{ad}_D(X) = 1$, repeated derivation gives

$$\operatorname{ad}_D^n(X^n) = n!.$$

Also

$$\operatorname{ad}_D^n(X^n) = \sum_{a=0}^{n} (-1)^a \binom{n}{a} D^{n-a} X^n D^a.$$

Thus $P_n = 1$ in A_1, and

$$\|P_n\|_1 = \frac{1}{n!} \sum_{a=0}^{n} \binom{n}{a} = \frac{2^n}{n!}.$$

Therefore $L_n \leq 2^n/n!$.

For $q \in \mathbb{C}$, define Φ_q on normal-ordered monomials by

$$\Phi_q(X^r D^s) = \begin{cases} q^r r!, & r = s, \\ 0, & r \neq s. \end{cases}$$

The functional relevant to the lower bound is

$$\Lambda_n = \Phi_{-1/2},$$

so

$$\Lambda_n(X^i D^i) = (-1)^i 2^{-i} i!.$$

In particular, $\Lambda_n(1) = 1$.

For letters $a_i \in \{X, D\}$,

$$\Phi_q(a_1 \cdots a_m) = \sum_{\pi} \prod_{\{i,j\} \in \pi,\ i<j} C_q(a_i, a_j),$$

where π runs over pairings pairing each X with a D, and

$$C_q(X,D) = q, \qquad C_q(D,X) = q+1, \qquad C_q(X,X) = C_q(D,D) = 0.$$

To prove this, use formal exponentials. Since

$$e^{uD}e^{vX} = e^{uv}e^{vX}e^{uD},$$

normal ordering gives

$$e^{t_1a_1}\cdots e^{t_ma_m} = \exp\left(\sum_{\substack{i<j \\ a_i=D,\ a_j=X}} t_it_j\right) e^{UX}e^{VD},$$

where

$$U = \sum_{i:a_i=X} t_i, \qquad V = \sum_{i:a_i=D} t_i.$$

Applying Φ_q gives

$$\Phi_q(e^{UX}e^{VD}) = e^{qUV}.$$

Hence

$$\Phi_q(e^{t_1a_1}\cdots e^{t_ma_m}) = \exp\left(\sum_{i<j} C_q(a_i,a_j)t_it_j\right).$$

Extracting the coefficient of $t_1\cdots t_m$ gives the Wick pairing formula.

Take $q=-1/2$. Then

$$C_{-1/2}(X,D) = -\frac{1}{2}, \qquad C_{-1/2}(D,X) = \frac{1}{2}.$$

Every nonzero contraction has absolute value $1/2$. If $w \in W_n$, then pairings of the n X's with the n D's are in bijection with permutations, so there are exactly $n!$ possible pairings. Therefore

$$|\Lambda_n(w)| = |\Phi_{-1/2}(w)| \le n!\left(\frac{1}{2}\right)^n = \frac{n!}{2^n}.$$

Now let

$$P = \sum_{w\in W_n} a_w w$$

and suppose $P=1$ in A_1. Applying Λ_n,

$$1 = |\Lambda_n(1)| = |\Lambda_n(P)| \le \sum_{w\in W_n} |a_w|\,|\Lambda_n(w)| \le \frac{n!}{2^n}\sum_{w\in W_n}|a_w|.$$

Thus

$$\|P\|_1 \ge \frac{2^n}{n!}.$$

Combining this with the upper bound proves

$$L_n = \frac{2^n}{n!}.$$

This solves the finite homogeneous Weyl algebra ℓ^1 coefficient model problem from the Tao-Sawin comment thread. It does not solve the full operator-norm problem for commutators close to the identity. $\square$

18.1 Finite certificate and Chebyshev-corridor obstruction

Proposition 18.3 (Exact finite certificate verification principle)**.** *A universal analytic or algebraic assertion may be reduced to finitely many exact rational inequalities when the reduction supplies rigorous enclosures on each cell and a certified tail or endpoint cover.*

Proof. Suppose the domain is covered by finitely many certified cells $C_1, \ldots, C_N$ and a tail region T:

$$D = C_1 \cup \cdots \cup C_N \cup T.$$

On each cell, rational outward-rounded interval arithmetic gives an enclosure I_j for the target expression with $I_j \subset [0, \infty)$. The tail certificate gives the same sign on T. Therefore the target expression is nonnegative at every point of D. If one cell gives an enclosure contained in $(-\infty, 0)$, the same finite certificate refutes a universal nonnegativity claim. □

Definition 18.3 (Snake polynomials and Chebyshev-expansion language)**.** A nonnegative continuous majorant μ on $[-1, 1]$ defines a degree-n snake-polynomial problem by the corridor condition $|p(x)| \leq \mu(x)$. A degree-n snake polynomial is the extremal polynomial, unique up to orientation in the source convention, with $n+1$ alternating contact points. Chebyshev expansion means expansion in the first-kind Chebyshev basis $T_0, T_1, T_2, \ldots$.

Proposition 18.4 (Snake-polynomial corridor alternation certificate)**.** *In the source convention for degree n snake polynomials, a polynomial $p \in \mathcal{P}_n$ with $|p(x)| \leq \mu(x)$ on $[-1, 1]$ and $n+1$ alternating contact points $p(\tau_i) = \pm\mu(\tau_i)$ is the associated snake polynomial, up to orientation.*

Proof. Let $\mu(x) = 48/223 + (175/223)|x|$ on $[-1, 1]$, and define

$$\omega(x) = \frac{6875}{1338}x^4 - \frac{5825}{1338}x^2 + \frac{48}{223}.$$

For $t = |x|$, exact factorization gives

$$\mu(t) - \omega(t) = \frac{25t(1-t)(275t^2 + 275t + 42)}{1338} \geq 0$$

and

$$\mu(t) + \omega(t) = \frac{(5t-3)^2(275t^2 + 330t + 64)}{1338} \geq 0.$$

Thus $|\omega(x)| \leq \mu(x)$. The five points $-1, -3/5, 0, 3/5, 1$ are alternating contact points, since $\omega(\pm 1) = \mu(\pm 1)$, $\omega(\pm 3/5) = -\mu(\pm 3/5)$, and $\omega(0) = \mu(0)$. This verifies the degree-four corridor/alternation certificate. Finally, using $x^2 = (T_2 + T_0)/2$ and $x^4 = (T_4 + 4T_2 + 3T_0)/8$,

$$\omega = -\frac{371}{10704}T_0 + \frac{175}{446}T_2 + \frac{6875}{10704}T_4,$$

so the Chebyshev expansion has a negative coefficient. □

Counterexample 18.5 (APP-0055: Snake-polynomial Chebyshev positivity conjecture is false)**.** Conjecture 7.1 for snake polynomials is false as written. For the continuous nonnegative even convex majorant $\mu(x) = 48/223 + (175/223)|x|$, the associated degree-four snake polynomial is $\omega(x) = (6875/1338)x^4 - (5825/1338)x^2 + 48/223$. Its Chebyshev expansion is $\omega = -(371/10704)T_0 + (175/446)T_2 + (6875/10704)T_4$, which has a negative coefficient.

This resolves the source problem recorded as **APP-0055** [46].

Proof. We use Proposition 18.3, Proposition 18.3, and Proposition 18.4 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$\mu(x) = \frac{48}{223} + \frac{175}{223}|x|, \qquad -1 \leq x \leq 1.$$

This majorant is nonnegative, continuous, even, and convex on $[-1, 1]$.

Define the degree-four polynomial

$$\omega(x) = \frac{6875}{1338}x^4 - \frac{5825}{1338}x^2 + \frac{48}{223}.$$

For $t = |x| \in [0, 1]$, exact algebra gives

$$\mu(t) - \omega(t) = \frac{25t(1-t)(275t^2 + 275t + 42)}{1338} \geq 0$$

and

$$\mu(t) + \omega(t) = \frac{(5t-3)^2(275t^2 + 330t + 64)}{1338} \geq 0.$$

Hence $|\omega(x)| \leq \mu(x)$ on $[-1, 1]$.

The five contact points $-1, -3/5, 0, 3/5, 1$ have alternating signs:

$$\omega(\pm 1) = 1 = \mu(\pm 1),$$

$$\omega\left(\pm\frac{3}{5}\right) = -\frac{153}{223} = -\mu\left(\pm\frac{3}{5}\right),$$

and

$$\omega(0) = \frac{48}{223} = \mu(0).$$

Thus ω satisfies the corridor bound and the five-point alternation certificate required for the degree-four snake polynomial associated with μ, up to the source's harmless orientation convention.

Using

$$x^2 = \frac{T_2(x) + T_0(x)}{2}, \qquad x^4 = \frac{T_4(x) + 4T_2(x) + 3T_0(x)}{8},$$

we obtain

$$\omega(x) = -\frac{371}{10704}T_0(x) + \frac{175}{446}T_2(x) + \frac{6875}{10704}T_4(x).$$

The T_0-coefficient is strictly negative. Reversing the orientation replaces ω by $-\omega$, which makes the T_2 and T_4 coefficients negative.

Therefore Conjecture 7.1 is false as written. □

18.2 Total-positivity and inverse-polynomial certificates

Definition 18.4 (Finite-order total nonnegativity for kernels)**.** A kernel K on an ordered set is TN_p if, for every $r \leq p$ and every increasing row tuple $x_1 < \cdots < x_r$ and increasing column tuple $y_1 < \cdots < y_r$, the determinant $\det(K(x_i, y_j))_{i,j=1}^r$ is nonnegative.

Proposition 18.6 (Negative symmetric minor obstructs PSD and finite-order total nonnegativity)**.** *Let K be a symmetric kernel on an ordered set, and let $x_1 < \cdots < x_p$. If the determinant of the symmetric evaluation matrix $(K(x_i, x_j))_{i,j=1}^p$ is negative, then that matrix is not positive semidefinite and K is not TN_p.*

Proof. Take

$$x = y = (-2, -1, 0, 1, 2).$$

This tuple is strictly increasing. The matrix of $1 + x_i x_j$ is

$$\begin{pmatrix} 5 & 3 & 1 & -1 & -3 \\ 3 & 2 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 & 3 \\ -3 & -1 & 1 & 3 & 5 \end{pmatrix}.$$

Applying $t \mapsto \max(t, 0)^2$ entrywise gives

$$A = \left(K_{JKS}(x_i, x_j)^2\right)_{i,j=1}^5 = \begin{pmatrix} 25 & 9 & 1 & 0 & 0 \\ 9 & 4 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 4 & 9 \\ 0 & 0 & 1 & 9 & 25 \end{pmatrix}.$$

Write

$$B = \begin{pmatrix} 25 & 9 \\ 9 & 4 \end{pmatrix}, \qquad C = \begin{pmatrix} 4 & 9 \\ 9 & 25 \end{pmatrix}, \qquad u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Then

$$A = \begin{pmatrix} B & u & 0 \\ u^T & 1 & u^T \\ 0 & u & C \end{pmatrix}.$$

Both B and C have determinant 19. The Schur complement gives

$$\det A = \det(B)\det(C)\left(1 - u^T B^{-1} u - u^T C^{-1} u\right).$$

Since

$$u^T B^{-1} u = u^T C^{-1} u = \frac{11}{19},$$

we get

$$\det A = 19^2 \left(1 - \frac{22}{19}\right) = -57.$$

A positive semidefinite matrix has nonnegative principal determinants. Since $\det A = -57 < 0$, the symmetric evaluation matrix is not positive semidefinite. Thus K_{JKS}^2 fails the stronger symmetric PSD witness target from the

Also, TN_5 requires every minor of order at most 5, formed from increasing row and column nodes, to be nonnegative. The same determinant is an admissible 5×5 minor with increasing row and column tuple $(-2, -1, 0, 1, 2)$. Its negativity proves

$$K_{JKS}^2 \notin TN_5(\mathbb{R} \times \mathbb{R}).$$

This proves only the $\alpha = 2$ branch of the source question. It does not solve the full integer family and makes no claim about $\alpha \geq 3$. $\square$

Definition 18.5 (Jain-Karlin-Schoenberg kernel)**.** The Jain–Karlin–Schoenberg kernel is $K_{JKS}(x, y) = \max(1 + xy, 0)$ on $\mathbb{R} \times \mathbb{R}$. Its α-power is the entrywise kernel $K_{JKS}(x, y)^\alpha$.

Counterexample 18.7 (APP-0065: JKS alpha-two kernel is not TN5 by a symmetric determinant witness)**.** For the Jain–Karlin–Schoenberg kernel $K_{JKS}(x, y) = \max(1 + xy, 0)$, the square kernel K_{JKS}^2 is not $TN_5(\mathbb{R} \times \mathbb{R})$. More strongly, at $x = y = (-2, -1, 0, 1, 2)$, the symmetric evaluation matrix $(K_{JKS}(x_i, x_j)^2)_{i,j=1}^5$ has determinant -57, so it is not positive semidefinite.

This resolves the source problem recorded as **APP-0065** [55].

Proof. We use Proposition 12.1, Proposition 18.4, Proposition 18.6, and Proposition 18.5 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Take

$$x = y = (-2, -1, 0, 1, 2).$$

This tuple is strictly increasing. The matrix of $1 + x_i x_j$ is

$$\begin{pmatrix} 5 & 3 & 1 & -1 & -3 \\ 3 & 2 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 & 3 \\ -3 & -1 & 1 & 3 & 5 \end{pmatrix}.$$

Applying $t \mapsto \max(t, 0)^2$ entrywise gives

$$A = \left(K_{JKS}(x_i, x_j)^2\right)_{i,j=1}^5 = \begin{pmatrix} 25 & 9 & 1 & 0 & 0 \\ 9 & 4 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 4 & 9 \\ 0 & 0 & 1 & 9 & 25 \end{pmatrix}.$$

Write

$$B = \begin{pmatrix} 25 & 9 \\ 9 & 4 \end{pmatrix}, \qquad C = \begin{pmatrix} 4 & 9 \\ 9 & 25 \end{pmatrix}, \qquad u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Then

$$A = \begin{pmatrix} B & u & 0 \\ u^T & 1 & u^T \\ 0 & u & C \end{pmatrix}.$$

Both B and C have determinant 19. The Schur complement gives

$$\det A = \det(B) \det(C) \left(1 - u^T B^{-1} u - u^T C^{-1} u\right).$$

Since

$$u^T B^{-1} u = u^T C^{-1} u = \frac{11}{19},$$

we get

$$\det A = 19^2 \left(1 - \frac{22}{19}\right) = -57.$$

A positive semidefinite matrix has nonnegative principal determinants. Since $\det A = -57 < 0$, the symmetric evaluation matrix is not positive semidefinite. Thus K_{JKS}^2 fails the stronger symmetric PSD witness target from the

Also, TN_5 requires every minor of order at most 5, formed from increasing row and column nodes, to be nonnegative. The same determinant is an admissible 5×5 minor with increasing row and column tuple $(-2, -1, 0, 1, 2)$. Its negativity proves

$$K_{JKS}^2 \notin TN_5(\mathbb{R} \times \mathbb{R}).$$

This proves only the $\alpha = 2$ branch of the source question. It does not solve the full integer family and makes no claim about $\alpha \geq 3$. $\square$

Proposition 18.8 (Analytic Bernstein functions have a Pringsheim negative-axis obstruction)**.** *Let $f(z) = \sum_{n\geq 1} c_n z^n$ be analytic at 0 with finite radius of convergence R. If f is a Bernstein function on $(0, \infty)$, then $(-1)^{n-1}c_n \geq 0$. Consequently $-f(-z)$ has nonnegative Taylor coefficients and, by Pringsheim's theorem, f has a singularity at the negative real point $-R$.*

Proof. Assume $\Delta = a^2 - 3b \geq 0$. Since $b \geq 0$, one has $\sqrt{\Delta} \leq a$. Put

$$r_- = \frac{a - \sqrt{\Delta}}{3}, \qquad r_+ = \frac{a + \sqrt{\Delta}}{3}.$$

Then $0 \leq r_- \leq r_+$, and

$$P'(x) = 3x^2 + 2ax + b = 3(x + r_-)(x + r_+).$$

Hence

$$g(x) = \frac{1}{P'(x)}$$

is completely monotone on $(0, \infty)$. Indeed, $x \mapsto (x + r)^{-1}$ is completely monotone for $r \geq 0$, and products of completely monotone functions are completely monotone. The repeated-root and $r = 0$ endpoint cases follow by the same formula or by limits; for example $1/(3x^2)$ is completely monotone.

The inverse satisfies

$$\varphi'(\lambda) = g(\varphi(\lambda)).$$

Li's Lemma 2.4 gives the inverse-ODE criterion: if g is completely monotone and a positive solution satisfies $y' = g(y)$, then y is Bernstein. For completeness, the sign induction is short. The base $y' > 0$ is clear. If $(-1)^{j-1}y^{(j)} \geq 0$ for $1 \leq j \leq n$, Faà di Bruno applied to $y^{(n+1)} = (g \circ y)^{(n)}$ writes the derivative as a sum of Bell-polynomial terms. A term with k derivatives on g has sign

$$(-1)^k(-1)^{n-k} = (-1)^n,$$

because $g^{(k)}$ has sign $(-1)^k$, and the product of $y^{(j)}$-terms has total sign $(-1)^{n-k}$. Thus $(-1)^n y^{(n+1)} \geq 0$. Applying this to $y = \varphi$ proves that φ' is completely monotone, so φ is Bernstein.

Assume $a^2 < 3b$. Then $b > 0$, and

$$P'(x) = 3x^2 + 2ax + b > 0$$

for every real x. Therefore $P : \mathbb{R} \to \mathbb{R}$ is strictly increasing and onto, and its real inverse is real analytic in a neighborhood of every real λ. In particular, the branch φ continued from 0 along the real axis has no singularity at any negative real point.

Because $b > 0$, φ is analytic at 0:

$$\varphi(\lambda) = \sum_{n\geq 1} c_n \lambda^n.$$

The radius of this Taylor series is finite. If it were infinite, φ would be entire and would satisfy $P(\varphi(\lambda)) = \lambda$ for every λ. For large $|w|$, $|P(w)| \geq |w|^3/2$, so this identity would imply

$$|\varphi(\lambda)| \leq C(1 + |\lambda|)^{1/3}.$$

Cauchy's estimate would then force $\varphi' \equiv 0$, contradicting $P'(\varphi)\varphi' = 1$. Let $R < \infty$ be the radius of convergence.

Suppose for contradiction that φ is Bernstein. Since φ' is completely monotone and φ is analytic at 0,

$$(-1)^{n-1} c_n \geq 0, \qquad n \geq 1.$$

Define

$$F(z) = -\varphi(-z) = \sum_{n\geq 1} (-1)^{n-1} c_n z^n.$$

Then F has nonnegative Taylor coefficients and the same finite radius R. By Pringsheim's theorem, F has a singularity at $z = R$. Equivalently, φ has a singularity at $\lambda = -R$.

This contradicts the real-axis analyticity proved above, because $-R < 0$. Therefore φ is not Bernstein when $a^2 < 3b$.

Combining the two directions gives the exact cubic criterion.

D-InversePolynomialBFProblem: Li's inverse-polynomial coefficient-condition vocabulary. the QuadraticDenominator CM NonpositiveRoots: reciprocal quadratic with nonpositive real zeros is completely monotone. the InverseODE CM to BF: Li Lemma 2.4 / Faà di Bruno inverse-ODE criterion. the AnalyticBF Pringsheim NegativeAxisObstruction: analytic Bernstein functions with finite radius have the Pringsheim negative-axis obstruction for their Taylor branch. □

Proposition 18.9 (Completely monotone inverse ODE implies Bernstein inverse)**.** *Let g be completely monotone on $(0, \infty)$, and let $y > 0$ solve $y' = g(y)$ on $(0, \infty)$. Then y' is completely monotone, and hence y is a Bernstein function, under the usual smoothness hypotheses of Li's inverse-ODE criterion.*

Proof. For Completely monotone inverse ODE implies Bernstein inverse, this proof is the corresponding component of Proposition 18.8. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Proposition 18.10 (Quadratic denominator with nonpositive roots is completely monotone)**.** *If $q(x) = c(x + r_1)(x + r_2)$ with $c > 0$ and $0 \leq r_1 \leq r_2$, then $1/q(x)$ is completely monotone on $(0, \infty)$, including repeated-root and $r_1 = 0$ endpoint cases.*

Proof. For Quadratic denominator with nonpositive roots is completely monotone, this proof is the corresponding component of Proposition 18.8. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Definition 18.6 (Inverse-polynomial Bernstein coefficient problem)**.** For $n \geq 2$ and coefficients $a_j \geq 0$, the inverse-polynomial Bernstein coefficient problem asks for conditions under which the positive inverse $\varphi(\lambda)$ of $\lambda = \varphi^n + a_1\varphi^{n-1} + \cdots + a_{n-1}\varphi$ is a Bernstein function.

Corollary 18.11 (APP-0066: Li cubic inverse-polynomial Bernstein branch has discriminant criterion)**.** *Let $a, b \geq 0$, and let φ be the positive inverse of $\lambda = \varphi^3 + a\varphi^2 + b\varphi$. Then φ is a Bernstein function on $(0, \infty)$ if and only if $a^2 \geq 3b$.*

*This resolves the source problem recorded as **APP-0066** [56].*

Proof. We use Proposition 9.1, Proposition 18.8, Proposition 18.6, Proposition 18.9, and Proposition 18.10 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Assume $\Delta = a^2 - 3b \geq 0$. Since $b \geq 0$, one has $\sqrt{\Delta} \leq a$. Put

$$r_- = \frac{a - \sqrt{\Delta}}{3}, \qquad r_+ = \frac{a + \sqrt{\Delta}}{3}.$$

Then $0 \leq r_- \leq r_+$, and

$$P'(x) = 3x^2 + 2ax + b = 3(x + r_-)(x + r_+).$$

Hence

$$g(x) = \frac{1}{P'(x)}$$

is completely monotone on $(0, \infty)$. Indeed, $x \mapsto (x + r)^{-1}$ is completely monotone for $r \geq 0$, and products of completely monotone functions are completely monotone. The repeated-root and $r = 0$ endpoint cases follow by the same formula or by limits; for example $1/(3x^2)$ is completely monotone.

The inverse satisfies

$$\varphi'(\lambda) = g(\varphi(\lambda)).$$

Li's Lemma 2.4 gives the inverse-ODE criterion: if g is completely monotone and a positive solution satisfies $y' = g(y)$, then y is Bernstein. For completeness, the sign induction is short. The base $y' > 0$ is clear. If $(-1)^{j-1} y^{(j)} \geq 0$ for $1 \leq j \leq n$, Faà di Bruno applied to $y^{(n+1)} = (g \circ y)^{(n)}$ writes the derivative as a sum of Bell-polynomial terms. A term with k derivatives on g has sign

$$(-1)^k (-1)^{n-k} = (-1)^n,$$

because $g^{(k)}$ has sign $(-1)^k$, and the product of $y^{(j)}$-terms has total sign $(-1)^{n-k}$. Thus $(-1)^n y^{(n+1)} \geq 0$. Applying this to $y = \varphi$ proves that φ' is completely monotone, so φ is Bernstein.

Assume $a^2 < 3b$. Then $b > 0$, and

$$P'(x) = 3x^2 + 2ax + b > 0$$

for every real x. Therefore $P : \mathbb{R} \to \mathbb{R}$ is strictly increasing and onto, and its real inverse is real analytic in a neighborhood of every real λ. In particular, the branch φ continued from 0 along the real axis has no singularity at any negative real point.

Because $b > 0$, φ is analytic at 0:

$$\varphi(\lambda) = \sum_{n \geq 1} c_n \lambda^n.$$

The radius of this Taylor series is finite. If it were infinite, φ would be entire and would satisfy $P(\varphi(\lambda)) = \lambda$ for every λ. For large $|w|$, $|P(w)| \geq |w|^3/2$, so this identity would imply

$$|\varphi(\lambda)| \leq C(1 + |\lambda|)^{1/3}.$$

Cauchy's estimate would then force $\varphi' \equiv 0$, contradicting $P'(\varphi)\varphi' = 1$. Let $R < \infty$ be the radius of convergence.

Suppose for contradiction that φ is Bernstein. Since φ' is completely monotone and φ is analytic at 0,

$$(-1)^{n-1} c_n \geq 0, \qquad n \geq 1.$$

Define

$$F(z) = -\varphi(-z) = \sum_{n\geq 1}(-1)^{n-1}c_n z^n.$$

Then F has nonnegative Taylor coefficients and the same finite radius R. By Pringsheim's theorem, F has a singularity at $z = R$. Equivalently, φ has a singularity at $\lambda = -R$.

This contradicts the real-axis analyticity proved above, because $-R < 0$. Therefore φ is not Bernstein when $a^2 < 3b$.

Combining the two directions gives the exact cubic criterion.

D-InversePolynomialBFProblem: Li's inverse-polynomial coefficient-condition vocabulary. the QuadraticDenominator CM NonpositiveRoots: reciprocal quadratic with nonpositive real zeros is completely monotone. the InverseODE CM to BF: Li Lemma 2.4 / Faà di Bruno inverse-ODE criterion. the AnalyticBF Pringsheim NegativeAxisObstruction: analytic Bernstein functions with finite radius have the Pringsheim negative-axis obstruction for their Taylor branch. □

19 Gamma and polygamma applications

This section contains applications whose main mechanism is gamma-function curvature, polygamma sign control, or beta-window reduction. The recurring strategy is to rewrite parameter inequalities as logarithmic-derivative or endpoint statements and then propagate the resulting sign information.

Counterexample 19.1 (APP-0037: Du-Wang h_3 is increasing exactly on $\frac{1}{2} \leq a \leq 1$)**.** For $0 < a < 2$, Du–Wang's function h_3 is increasing on $(0, \infty)$ exactly for $1/2 \leq a \leq 1$, and is not monotone for $0 < a < 1/2$ or $1 < a < 2$.

This resolves the source problem recorded as **APP-0037** [24].

Proof. We use Proposition 12.28 and Proposition 12.27 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The middle-window monotonicity theorem on $1/2 \leq a \leq 1$ and the outer-window nonmonotonicity theorem on $0 < a < 1/2$ and $1 < a < 2$ partition the domain, and together they give exactly the source classification. □

19.1 Multiple-gamma normal forms and Bernstein obstruction

Proposition 19.2 (Special-function normal-form calculus principle)**.** *A special-function inequality or monotonicity claim may be transported to an explicit derivative, recurrence, integral, product, or partial-fraction normal form, after which sign, convexity, or monotonicity is decided in that normal form.*

Proof. Let the target expression be represented on its domain by an identity

$$F(x) = N(x),$$

where N is written using recurrence relations, logarithmic derivatives, an integral formula, a product, or a partial fraction expansion. If the normal form proves the required sign pattern, for example

$$(-1)^k N^{(k)}(x) \geq 0 \qquad (x \in I),$$

then the identity gives $(-1)^k F^{(k)}(x) \geq 0$ on the same domain. The same substitution transfers monotonicity and convexity inequalities, since differentiating equal normal forms preserves equality of the derivatives used in the sign test. □

Proposition 19.3 (DasSwaminathan logGn unit sign)**.** *For the Barnes multiple-gamma functions G_n, if $F_n(x) = \log G_n(x)$, then $(-1)^{n+1}F_n(x) > 0$ for $0 < x < 1$ and $\operatorname{sgn} F_n'(1) = (-1)^n$.*

Proof. The value $G_n(1) = 1$ is part of the normalization. For $k \le n+1$, the recurrence gives

$$G_n(k+1) = G_n(k)G_{n-1}(k).$$

Inducting on n, $G_{n-1}(k) = 1$ for $1 \le k \le n$, so all displayed values are 1.

Thus $F_n(j) = 0$ for $j = 1, \ldots, n+1$.

For $0 < x < 1$,

$$(-1)^{n+1}F_n(x) > 0.$$

Also

$$\operatorname{sgn} F_n'(1) = (-1)^n.$$

The divided difference on the nodes $x, 1, 2, \ldots, n+1$ satisfies

$$F_n[x, 1, 2, \ldots, n+1] = \frac{F_n^{(n+1)}(\xi)}{(n+1)!} > 0$$

for some $\xi \in (x, n+1)$. Since $F_n(1) = \cdots = F_n(n+1) = 0$,

$$F_n[x, 1, 2, \ldots, n+1] = \frac{F_n(x)}{\prod_{j=1}^{n+1}(x-j)}.$$

The product has sign $(-1)^{n+1}$, proving the unit-interval sign.

Letting $x \to 1^-$ gives the repeated-node divided difference

$$F_n[1, 1, 2, \ldots, n+1] = \frac{F_n'(1)}{\prod_{j=2}^{n+1}(1-j)} > 0.$$

The product has sign $(-1)^n$, so $\operatorname{sgn} F_n'(1) = (-1)^n$. $\square$

Counterexample 19.4 (APP-0052: Das-Swaminathan multiple-gamma Bernstein question is false)**.** The Das-Swaminathan source question asking whether $f_n(x) = \log G_n(x+1)/(x^n \log x)$ is a Bernstein function has a negative answer: $f_4(x) < 0$ for all sufficiently small $x > 0$, so f_4 is not Bernstein.

This resolves the source problem recorded as **APP-0052** [43].

Proof. We use Proposition 19.2, Proposition 19.3, and Proposition 9.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The value $G_n(1) = 1$ is part of the normalization. For $k \le n+1$, the recurrence gives

$$G_n(k+1) = G_n(k)G_{n-1}(k).$$

Inducting on n, $G_{n-1}(k) = 1$ for $1 \le k \le n$, so all displayed values are 1.

Thus $F_n(j) = 0$ for $j = 1, \ldots, n+1$.

For $0 < x < 1$,

$$(-1)^{n+1}F_n(x) > 0.$$

Also

$$\operatorname{sgn} F_n'(1) = (-1)^n.$$

The divided difference on the nodes $x, 1, 2, \ldots, n+1$ satisfies

$$F_n[x, 1, 2, \ldots, n+1] = \frac{F_n^{(n+1)}(\xi)}{(n+1)!} > 0$$

for some $\xi \in (x, n+1)$. Since $F_n(1) = \cdots = F_n(n+1) = 0$,

$$F_n[x, 1, 2, \ldots, n+1] = \frac{F_n(x)}{\prod_{j=1}^{n+1}(x-j)}.$$

The product has sign $(-1)^{n+1}$, proving the unit-interval sign.

Letting $x \to 1^-$ gives the repeated-node divided difference

$$F_n[1, 1, 2, \ldots, n+1] = \frac{F_n'(1)}{\prod_{j=2}^{n+1}(1-j)} > 0.$$

The product has sign $(-1)^n$, so $\operatorname{sgn} F_n'(1) = (-1)^n$. $\square$

20 HCM endpoint obstruction certificates

This section collects applications in which a universal positivity, convexity, or monotonicity claim is decided by reducing it to a local obstruction. The common method is to isolate a boundary, endpoint, or explicit witness where the proposed sign pattern cannot hold.

Definition 20.1 (Hyperbolically completely monotone density)**.** A positive density f on $(0, \infty)$ is hyperbolically completely monotone if, for each fixed $u > 0$, the function $v \mapsto f(uv)f(u/v)$ is completely monotone as a function of $v + v^{-1}$, in the usual hyperbolic variable.

Proposition 20.1 (BPV noncentral chi square HCM leading Bell obstruction chain)**.** *For the noncentral chi-square HCM problem, the leading small-u HCM sign constraints are generated by the Bell recurrence*

$$B_0 = 1, \qquad B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_{n-k} c_{k+1}.$$

With

$$c_1 = \frac{\lambda/\mu - 1}{2}, \qquad c_j = j!\, b_j(\mu)\lambda^j \quad (j \geq 2),$$

and

$$\log \sum_{j \geq 0} \frac{y^j}{4^j j! (\mu/2)_j} = \sum_{j \geq 1} b_j(\mu) y^j,$$

the HCM necessary condition is $(-1)^n B_n \geq 0$ for every $n \geq 1$.

Proof. The source asks for the optimal parameter range in which

$$\chi_{\mu,\lambda}(x) = \frac{1}{2} e^{-(x+\lambda)/2} \left(\frac{x}{\lambda}\right)^{\mu/4 - 1/2} I_{\mu/2-1}(\sqrt{\lambda x})$$

is hyperbolically completely monotone.

The local small-u hyperbolic-product expansion is

$$\log\{\chi_{\mu,\lambda}(uv)\chi_{\mu,\lambda}(u/v)\} = C_u + \frac{u}{2}\left(\frac{\lambda}{\mu} - 1\right) w - \frac{\lambda^2 u^2}{4\mu^2(\mu+2)}(w^2 - 2) + O(u^3), \qquad w = v + v^{-1}.$$

Consequences:

$$\lambda > \mu \quad \Longrightarrow \quad \chi_{\mu,\lambda} \notin HCM,$$

and

$$(\lambda-\mu)^2 < \frac{2\lambda^2}{\mu+2} \quad \Longrightarrow \quad \chi_{\mu,\lambda} \notin HCM.$$

Equivalently, inside $0 < \lambda \le \mu$, the second obstruction excludes

$$\frac{\lambda}{\mu} > \frac{1}{1+\sqrt{2/(\mu+2)}}.$$

Remaining frontier:

$$0 < \lambda \le \mu, \qquad (\lambda-\mu)^2 \ge \frac{2\lambda^2}{\mu+2}.$$

This note is partial theory growth only. Do not stage as a solved application unless the full optimal HCM range is proved or the source problem is explicitly narrowed. □

Counterexample 20.2 (APP-0047: BPV noncentral chi-square HCM range has no positive noncentrality)**.** For every $\mu > 0$ and every positive noncentrality $\lambda > 0$, the noncentral chi-square density $\chi_{\mu,\lambda}$ is not hyperbolically completely monotone. Equivalently, the BPV optimal HCM range for positive noncentrality is empty; if the central case $\lambda = 0$ is adjoined, the remaining HCM range is exactly the central gamma line.

This resolves the source problem recorded as **APP-0047** [38].

Proof. We use Proposition 20.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let $a = \mu/2 > 0$. The complete Bell generating function is

$$\sum_{n\ge 0} \frac{B_n z^n}{n!} = \exp\left(\sum_{j\ge 1} \frac{c_j z^j}{j!}\right) = e^{-z/2} S_\mu(\lambda z).$$

Indeed $b_1(\mu) = 1/(2\mu)$, so the $j=1$ term of $\log S_\mu(\lambda z)$ contributes $\lambda z/(2\mu)$, and the exponential factor $e^{-z/2}$ supplies the remaining $-z/2$.

Define

$$A_n = \frac{(-1)^n B_n}{n!}.$$

Then

$$\sum_{n\ge 0} A_n t^n = e^{t/2} S_\mu(-\lambda t) = e^{t/2}\, {}_0F_1\left(;a;-\frac{\lambda t}{4}\right).$$

Equivalently,

$$A_n = \frac{2^{-n}}{(a)_n} L_n^{a-1}\left(\frac{\lambda}{2}\right),$$

so the leading HCM signs are exactly the fixed-argument Laguerre signs.

Use the classical identity

$$ {}_0F_1\left(;a;-\frac{x^2}{4}\right) = \Gamma(a)\left(\frac{x}{2}\right)^{1-a} J_{a-1}(x).$$

Since $a > 0$, the Bessel function J_{a-1} has a positive real zero $j_{a-1,1} > 0$. For $\lambda > 0$, set

$$t_0 = \frac{j_{a-1,1}^2}{\lambda} > 0.$$

Then

$$e^{t_0/2} {}_0F_1\left(;a;-\frac{\lambda t_0}{4}\right) = 0.$$

If every leading HCM sign were nonnegative, then all coefficients A_n would be nonnegative and $A_0 = 1$, forcing

$$\sum_{n\geq 0} A_n t_0^n > 0,$$

contradicting the displayed zero. Therefore, for every $\mu > 0$ and $\lambda > 0$, there is an integer $n \geq 1$ such that

$$(-1)^n B_n < 0.$$

The leading expansion gives, for any fixed compact w-window,

$$\frac{1}{H_u(w)} \frac{d^n}{dw^n} H_u(w) = B_n u^n + O(u^{n+1}) \qquad (u \downarrow 0).$$

Choosing $u > 0$ sufficiently small yields

$$(-1)^n \frac{d^n}{dw^n} H_u(w) < 0,$$

which violates complete monotonicity of $w \mapsto H_u(w)$.

Conclusion. For every $\mu > 0$ and every positive noncentrality $\lambda > 0$,

$$\chi_{\mu,\lambda} \notin HCM.$$

Thus the BPV optimal HCM range for positive noncentrality is empty. If one adjoins the central case $\lambda = 0$, the remaining density is a gamma density and is HCM; hence the extended closed range is exactly the central line $\lambda = 0$.

Strict APP status: candidate for the next APP slot after APP-0046 because the source explicitly asks the optimal range, the local theorem answers it, and the result is not among public APP-0001–APP-0045. □

21 Gamma and polygamma applications

This section contains applications whose main mechanism is gamma-function curvature, polygamma sign control, or beta-window reduction. The recurring strategy is to rewrite parameter inequalities as logarithmic-derivative or endpoint statements and then propagate the resulting sign information.

Proposition 21.1 (Bendikov-Cygan renewal generating function from SBF potential density)**.** *For a Bendikov–Cygan source-normalized special Bernstein function ψ, the renewal generating function $F(z) = \sum_{k\geq 0} C_\psi(k) z^k$ satisfies $F(z) = 1/\psi(1-z)$. Using the special-Bernstein potential representation $1/\psi(\lambda) = b + \int_0^\infty e^{-\lambda t} u(t)\, dt$ with $b \geq 0$ and nonincreasing u, one has $C_\psi(0) = b + \int_0^\infty e^{-t} u(t)\, dt$ and $C_\psi(k) = k!^{-1} \int_0^\infty t^k e^{-t} u(t)\, dt$ for $k \geq 1$.*

Proof. Let

$$K(z) = \sum_{n\geq 1} c(\psi, n) z^n.$$

The Bendikov–Cygan discrete subordination coefficient calculation gives

$$K(z) = 1 - \psi(1-z),$$

under the source normalization $\psi(0) = 0$, $\psi(1) = 1$. Since $C(0) = 1$ and

$$C(n) = \sum_{j=1}^{n} c(\psi, j) C(n-j),$$

the generating function $F(z) = \sum_{n\geq 0} C(n) z^n$ satisfies

$$F(z) = \frac{1}{1-K(z)} = \frac{1}{\psi(1-z)}.$$

For $\psi \in \mathrm{SBF}$, the source-cited potential representation gives

$$\frac{1}{\psi(\lambda)} = b + \int_0^\infty e^{-\lambda t} u(t)\, dt,$$

where $b \geq 0$ and u is nonincreasing. Substituting $\lambda = 1 - z$,

$$F(z) = b + \int_0^\infty e^{-t} e^{zt} u(t)\, dt = b + \sum_{k\geq 0} \frac{z^k}{k!} \int_0^\infty t^k e^{-t} u(t)\, dt.$$

Therefore

$$C(0) = b + \int_0^\infty e^{-t} u(t)\, dt = 1$$

and, for $k \geq 1$,

$$C(k) = \frac{1}{k!} \int_0^\infty t^k e^{-t} u(t)\, dt.$$

For $k \geq 1$,

$$C(k+1) - C(k) = \frac{1}{(k+1)!} \int_0^\infty t^k e^{-t} (t - (k+1)) u(t)\, dt.$$

If $T \sim \Gamma(k+1, 1)$, this is

$$C(k+1) - C(k) = \frac{1}{k+1} \operatorname{Cov}(T, u(T)).$$

The covariance is nonpositive because $t \mapsto t$ is increasing and u is nonincreasing:

$$\operatorname{Cov}(T, u(T)) = \frac{1}{2} \iint (t-s)(u(t) - u(s))\, d\mu_k(t) d\mu_k(s) \leq 0.$$

Thus $C(k+1) \leq C(k)$ for $k \geq 1$.

For $k = 0$, the atom b must be kept:

$$C(1) - C(0) = \int_0^\infty (t-1) e^{-t} u(t)\, dt - b = \operatorname{Cov}_{\mathrm{Exp}(1)}(T, u(T)) - b \leq 0.$$

Hence $C(1) \leq C(0)$, completing the monotonicity proof. $\square$

Proposition 21.2 (Gamma moments of a decreasing density form a decreasing normalized sequence)**.** *Let $u : (0, \infty) \to [0, \infty)$ be nonincreasing and locally integrable with finite gamma-weighted moments. Define $M_k = k!^{-1} \int_0^\infty t^k e^{-t} u(t)\, dt$ for $k \geq 1$. Then $M_{k+1} \leq M_k$ for every $k \geq 1$. If additionally $M_0 = b + \int_0^\infty e^{-t} u(t)\, dt$ with $b \geq 0$, then $M_1 \leq M_0$.*

Proof. For Gamma moments of a decreasing density form a decreasing normalized sequence, this proof is the corresponding component of Proposition 21.1. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Corollary 21.3 (APP-0071: Bendikov-Cygan SBF renewal monotonicity)**.** *For every Bendikov–Cygan source-normalized special Bernstein function ψ, the discrete renewal sequence $C_\psi(k)$ is nonincreasing: $C_\psi(k+1) \leq C_\psi(k)$ for every $k \geq 0$. Consequently the Bendikov–Cygan open question asking whether special Bernstein functions have decreasing discrete renewal sequences is answered affirmatively, with decreasing interpreted as nonincreasing.*

This resolves the source problem recorded as ***APP-0071*** *[60].*

Proof. We use Proposition 7.2, Proposition 21.1, and Proposition 21.2 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$K(z) = \sum_{n \geq 1} c(\psi, n) z^n.$$

The Bendikov–Cygan discrete subordination coefficient calculation gives

$$K(z) = 1 - \psi(1 - z),$$

under the source normalization $\psi(0) = 0$, $\psi(1) = 1$. Since $C(0) = 1$ and

$$C(n) = \sum_{j=1}^{n} c(\psi, j) C(n - j),$$

the generating function $F(z) = \sum_{n \geq 0} C(n) z^n$ satisfies

$$F(z) = \frac{1}{1 - K(z)} = \frac{1}{\psi(1 - z)}.$$

For $\psi \in \mathrm{SBF}$, the source-cited potential representation gives

$$\frac{1}{\psi(\lambda)} = b + \int_0^\infty e^{-\lambda t} u(t)\, dt,$$

where $b \geq 0$ and u is nonincreasing. Substituting $\lambda = 1 - z$,

$$F(z) = b + \int_0^\infty e^{-t} e^{zt} u(t)\, dt = b + \sum_{k \geq 0} \frac{z^k}{k!} \int_0^\infty t^k e^{-t} u(t)\, dt.$$

Therefore

$$C(0) = b + \int_0^\infty e^{-t} u(t)\, dt = 1$$

and, for $k \geq 1$,

$$C(k) = \frac{1}{k!} \int_0^\infty t^k e^{-t} u(t)\, dt.$$

For $k \geq 1$,

$$C(k+1) - C(k) = \frac{1}{(k+1)!} \int_0^\infty t^k e^{-t}(t - (k+1))u(t)\, dt.$$

If $T \sim \Gamma(k+1, 1)$, this is

$$C(k+1) - C(k) = \frac{1}{k+1} \operatorname{Cov}(T, u(T)).$$

The covariance is nonpositive because $t \mapsto t$ is increasing and u is nonincreasing:

$$\operatorname{Cov}(T, u(T)) = \frac{1}{2} \iint (t-s)(u(t) - u(s))\, d\mu_k(t) d\mu_k(s) \leq 0.$$

Thus $C(k+1) \leq C(k)$ for $k \geq 1$.

For $k = 0$, the atom b must be kept:

$$C(1) - C(0) = \int_0^\infty (t-1)e^{-t} u(t)\, dt - b = \operatorname{Cov}_{\operatorname{Exp}(1)}(T, u(T)) - b \leq 0.$$

Hence $C(1) \leq C(0)$, completing the monotonicity proof. □

Proposition 21.4 (Conditional star kernels are Gaussian Laplace transforms)**.** *For points $u_1, u_2, u_3 \in \mathbb{R}^2$, positive a_i, $A = \sum_i a_i$, and $p_a = (a_1a_2/A, a_1a_3/A, a_2a_3/A)$, the function $A^{-1}\exp(-p_a \cdot r)$, with $r_{ij} = |u_i - u_j|^2$, is a positive Laplace transform in (a_1, a_2, a_3). Consequently $A^{-\beta}\exp(-p_a \cdot r)$ is completely monotone for every $\beta > 1$.*

Proof. Let $G = K_5 - e$, realized as two nonadjacent apices joined to the terminal triangle $\{2, 3, 4\}$. Write

$$a = (a_1, a_2, a_3) = (x_{02}, x_{03}, x_{04}), \qquad b = (b_1, b_2, b_3) = (x_{12}, x_{13}, x_{14}),$$

$$c = (c_{12}, c_{13}, c_{23}) = (x_{23}, x_{24}, x_{34}), \qquad A = a_1 + a_2 + a_3, \quad B = b_1 + b_2 + b_3.$$

For a triangle edge-vector $w = (w_1, w_2, w_3)$, put

$$E(w) = w_1w_2 + w_1w_3 + w_2w_3.$$

The claim is that $T_G^{-\beta}$ is completely monotone in all nine positive edge variables for every $\beta > 1$.

Lemma 1: star-mesh factorization of T_{K_5-e}.

For $u = (u_1, u_2, u_3)$, define

$$p_u = \left(\frac{u_1u_2}{u_1 + u_2 + u_3}, \frac{u_1u_3}{u_1 + u_2 + u_3}, \frac{u_2u_3}{u_1 + u_2 + u_3} \right).$$

Then

$$T_{K_5-e}(a, b, c) = AB\, E(c + p_a + p_b).$$

Indeed, if a network H on three terminals has spanning-tree polynomial T_H, adjoining a new apex with conductances u_1, u_2, u_3 to the terminals multiplies the reduced Laplacian determinant by $U = u_1 + u_2 + u_3$ and replaces H by the star-mesh transformed network with added terminal edge conductances p_u. This is the Schur complement of the apex diagonal block U in the weighted Laplacian:

$$L_H + \operatorname{diag}(u) - \frac{uu^T}{U}.$$

The last matrix is the terminal Laplacian of H plus a triangle with edge conductances $u_i u_j/U$. Applying this twice, once for each apex, leaves the terminal triangle with edge vector $c + p_a + p_b$, whose spanning-tree polynomial is E.

For $w_i > 0$ and $\beta > 1/2$,

$$E(w)^{-\beta} = C_\beta \int_{Y>0} \exp\{-w_1(Y_{11} + Y_{22} - 2Y_{12}) - w_2 Y_{11} - w_3 Y_{22}\}(\det Y)^{\beta-3/2}\, dY$$

with $C_\beta > 0$, where Y ranges over positive definite 2×2 symmetric matrices.

This is the standard 2×2 Riesz/Wishart integral applied to

$$M(w) = \begin{pmatrix} w_1 + w_2 & -w_1 \\ -w_1 & w_1 + w_3 \end{pmatrix}, \qquad \det M(w) = E(w).$$

The exponent is $-\operatorname{tr}(M(w)Y)$, and the Wishart integral is valid for $\beta > (2-1)/2 = 1/2$.

For $Y > 0$, choose vectors $u, v \in \mathbb{R}^2$ with Gram matrix Y. Then

$$r_{12} = Y_{11} + Y_{22} - 2Y_{12} = |u - v|^2, \quad r_{13} = Y_{11} = |u|^2, \quad r_{23} = Y_{22} = |v|^2.$$

For every $r_{ij} = |u_i - u_j|^2$ with $u_1, u_2, u_3 \in \mathbb{R}^2$,

$$A^{-1} \exp(-p_a \cdot r) = \pi^{-1} \int_{\mathbb{R}^2} \exp\left(-\sum_{i=1}^{3} a_i |z - u_i|^2\right) dz.$$

This is the elementary Gaussian completion identity

$$\sum_i a_i |z - u_i|^2 = A \left| z - \frac{\sum_i a_i u_i}{A} \right|^2 \frac{1}{A} \sum_{i<j} a_i a_j |u_i - u_j|^2.$$

Thus $A^{-1} e^{-p_a \cdot r}$ is a positive Laplace transform in the variables a_i. For $\beta > 1$,

$$A^{-\beta} e^{-p_a \cdot r} = A^{-(\beta-1)} \big(A^{-1} e^{-p_a \cdot r}\big)$$

is again completely monotone, because

$$A^{-(\beta-1)} = \Gamma(\beta - 1)^{-1} \int_0^\infty t^{\beta-2} e^{-tA}\, dt.$$

The same statement holds for $B^{-\beta} e^{-p_b \cdot r}$.

From Lemma 1,

$$T_G^{-\beta} = A^{-\beta} B^{-\beta} E(c + p_a + p_b)^{-\beta}.$$

Insert Lemma 2 with $w = c + p_a + p_b$. For each positive Y, the factor involving c is

$$\exp(-c \cdot r(Y)),$$

a positive Laplace kernel in the triangle-edge variables. The factors involving a and b are

$$A^{-\beta} \exp(-p_a \cdot r(Y)), \qquad B^{-\beta} \exp(-p_b \cdot r(Y)),$$

which are completely monotone by Lemma 3 for every $\beta > 1$. The product is a completely monotone function in disjoint variable blocks, and integration against the positive Riesz measure preserves complete monotonicity by Tonelli.

Therefore $T_{K_5 - e}^{-\beta}$ is completely monotone on $(0, \infty)^9$ for every $\beta > 1$. In Scott–Sokal notation, $K_5 - e \in G_\beta$ for every $1 < \beta < 3/2$. $\square$

Proposition 21.5 (Star-mesh apex elimination for spanning-tree polynomials)**.** *Let a graph network on three terminals have spanning-tree polynomial T_H. Adjoining one apex with positive conductances u_1, u_2, u_3 to the terminals multiplies the spanning-tree polynomial by $U = u_1 + u_2 + u_3$ and adds terminal triangle conductances $u_i u_j / U$.*

Proof. For Star-mesh apex elimination for spanning-tree polynomials, this proof is the corresponding component of Proposition 21.4. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. ☐

Proposition 21.6 (Triangle Kirchhoff polynomial has a two-dimensional Riesz representation)**.** *For $E(w) = w_1w_2 + w_1w_3 + w_2w_3$ and every $\beta > 1/2$, the inverse power $E(w)^{-\beta}$ is a positive Laplace transform on $(0,\infty)^3$, via the 2×2 Riesz/Wishart integral.*

Proof. For Triangle Kirchhoff polynomial has a two-dimensional Riesz representation, this proof is the corresponding component of Proposition 21.4. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. ☐

Corollary 21.7 (APP-0074: Scott-Sokal K5-e spanning-tree polynomial complete monotonicity)**.** *Let $G = K_5 - e$. For every $\beta > 1$, the inverse power $T_G^{-\beta}$ of the spanning-tree polynomial is completely monotone on the positive edge cone. In particular, $K_5 - e \in G_\beta$ for every $1 < \beta < 3/2$, answering the Scott–Sokal source question for this graph affirmatively.*

*This resolves the source problem recorded as **APP-0074** [63].*

Proof. We use Proposition 7.2, Proposition 21.4, Proposition 21.5, and Proposition 21.6 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let $G = K_5 - e$, realized as two nonadjacent apices joined to the terminal triangle $\{2, 3, 4\}$. Write

$$a = (a_1, a_2, a_3) = (x_{02}, x_{03}, x_{04}), \qquad b = (b_1, b_2, b_3) = (x_{12}, x_{13}, x_{14}),$$
$$c = (c_{12}, c_{13}, c_{23}) = (x_{23}, x_{24}, x_{34}), \qquad A = a_1 + a_2 + a_3, \quad B = b_1 + b_2 + b_3.$$

For a triangle edge-vector $w = (w_1, w_2, w_3)$, put

$$E(w) = w_1w_2 + w_1w_3 + w_2w_3.$$

The claim is that $T_G^{-\beta}$ is completely monotone in all nine positive edge variables for every $\beta > 1$.

Lemma 1: star-mesh factorization of T_{K_5-e}.

For $u = (u_1, u_2, u_3)$, define

$$p_u = \left(\frac{u_1u_2}{u_1 + u_2 + u_3}, \frac{u_1u_3}{u_1 + u_2 + u_3}, \frac{u_2u_3}{u_1 + u_2 + u_3}\right).$$

Then

$$T_{K_5-e}(a, b, c) = AB\, E(c + p_a + p_b).$$

Indeed, if a network H on three terminals has spanning-tree polynomial T_H, adjoining a new apex with conductances u_1, u_2, u_3 to the terminals multiplies the reduced Laplacian determinant by $U = u_1 + u_2 + u_3$ and replaces H by the star-mesh transformed network with added terminal edge conductances p_u. This is the Schur complement of the apex diagonal block U in the weighted Laplacian:

$$L_H + \operatorname{diag}(u) - \frac{uu^T}{U}.$$

The last matrix is the terminal Laplacian of H plus a triangle with edge conductances $u_i u_j/U$. Applying this twice, once for each apex, leaves the terminal triangle with edge vector $c + p_a + p_b$, whose spanning-tree polynomial is E.

For $w_i > 0$ and $\beta > 1/2$,

$$E(w)^{-\beta} = C_\beta \int_{Y>0} \exp\{-w_1(Y_{11} + Y_{22} - 2Y_{12}) - w_2 Y_{11} - w_3 Y_{22}\}(\det Y)^{\beta-3/2}\, dY$$

with $C_\beta > 0$, where Y ranges over positive definite 2×2 symmetric matrices.

This is the standard 2×2 Riesz/Wishart integral applied to

$$M(w) = \begin{pmatrix} w_1 + w_2 & -w_1 \\ -w_1 & w_1 + w_3 \end{pmatrix}, \qquad \det M(w) = E(w).$$

The exponent is $-\operatorname{tr}(M(w)Y)$, and the Wishart integral is valid for $\beta > (2-1)/2 = 1/2$.

For $Y > 0$, choose vectors $u, v \in \mathbb{R}^2$ with Gram matrix Y. Then

$$r_{12} = Y_{11} + Y_{22} - 2Y_{12} = |u - v|^2, \quad r_{13} = Y_{11} = |u|^2, \quad r_{23} = Y_{22} = |v|^2.$$

For every $r_{ij} = |u_i - u_j|^2$ with $u_1, u_2, u_3 \in \mathbb{R}^2$,

$$A^{-1} \exp(-p_a \cdot r) = \pi^{-1} \int_{\mathbb{R}^2} \exp\left(-\sum_{i=1}^{3} a_i |z - u_i|^2\right) dz.$$

This is the elementary Gaussian completion identity

$$\sum_i a_i |z - u_i|^2 = A \left| z - \frac{\sum_i a_i u_i}{A} \right|^2 \frac{1}{A} \sum_{i<j} a_i a_j |u_i - u_j|^2.$$

Thus $A^{-1} e^{-p_a \cdot r}$ is a positive Laplace transform in the variables a_i. For $\beta > 1$,

$$A^{-\beta} e^{-p_a \cdot r} = A^{-(\beta-1)} \big(A^{-1} e^{-p_a \cdot r}\big)$$

is again completely monotone, because

$$A^{-(\beta-1)} = \Gamma(\beta - 1)^{-1} \int_0^\infty t^{\beta-2} e^{-tA}\, dt.$$

The same statement holds for $B^{-\beta} e^{-p_b \cdot r}$.

From Lemma 1,

$$T_G^{-\beta} = A^{-\beta} B^{-\beta} E(c + p_a + p_b)^{-\beta}.$$

Insert Lemma 2 with $w = c + p_a + p_b$. For each positive Y, the factor involving c is

$$\exp(-c \cdot r(Y)),$$

a positive Laplace kernel in the triangle-edge variables. The factors involving a and b are

$$A^{-\beta} \exp(-p_a \cdot r(Y)), \qquad B^{-\beta} \exp(-p_b \cdot r(Y)),$$

which are completely monotone by Lemma 3 for every $\beta > 1$. The product is a completely monotone function in disjoint variable blocks, and integration against the positive Riesz measure preserves complete monotonicity by Tonelli.

Therefore $T_{K_5-e}^{-\beta}$ is completely monotone on $(0, \infty)^9$ for every $\beta > 1$. In Scott–Sokal notation, $K_5 - e \in G_\beta$ for every $1 < \beta < 3/2$. □

Proposition 21.8 (K4 Kirchhoff polynomial has a rank-three Riesz representation)**.** *For the complete graph K_4, the spanning-tree polynomial is a rank-three determinant $T_{K_4}(y) = \det \sum_e y_e b_e b_e^T$. For every $\beta > 1$, $T_{K_4}(y)^{-\beta}$ is a positive Laplace transform in the six edge variables via the 3×3 Riesz/Wishart integral; the three edge variables on any triangle correspond to squared distances of three Euclidean points in the Riesz support.*

Proof. Root K_4 at vertex 3. For the remaining vertices $0, 2, 4$, use edge vectors

$$b_{03} = e_0, \quad b_{23} = e_2, \quad b_{43} = e_4, \quad b_{02} = e_0 - e_2, \quad b_{04} = e_0 - e_4, \quad b_{24} = e_2 - e_4.$$

Then

$$T_{K_4}(y) = \det M(y), \qquad M(y) = \sum_{ij} y_{ij} b_{ij} b_{ij}^T.$$

For $\beta > 1$, the 3×3 Riesz/Wishart integral gives

$$\det M(y)^{-\beta} = C_\beta \int_{Y>0} \exp(-\operatorname{tr}(M(y)Y))\,(\det Y)^{\beta-2}\,dY,$$

with $C_\beta > 0$. Writing

$$U_{ij} = b_{ij}^T Y b_{ij},$$

the exponent is $-\sum y_{ij} U_{ij}$. If $Y = LL^T$, then $U_{ij} = |L^T b_{ij}|^2$. In particular the triple

$$(U_{02}, U_{04}, U_{24})$$

is the squared-distance triple of the three points $L^T e_0, L^T e_4, L^T e_2$, hence can be represented in $\mathbb{R}^2$.

For fixed such a squared-distance triple, choose points $\xi_0, \xi_2, \xi_4 \in \mathbb{R}^2$ with

$$U_{02} = |\xi_0 - \xi_2|^2, \quad U_{04} = |\xi_0 - \xi_4|^2, \quad U_{24} = |\xi_2 - \xi_4|^2.$$

The Gaussian completion identity gives

$$S^{-1} \exp\left(-\frac{paU_{02} + pdU_{04} + adU_{24}}{S}\right) = \pi^{-1} \int_{\mathbb{R}^2} \exp\left(-p|z-\xi_0|^2 - a|z-\xi_2|^2 - d|z-\xi_4|^2\right)\,dz.$$

Multiplying by the gamma representation

$$S^{-(\beta-1)} = \Gamma(\beta-1)^{-1} \int_0^\infty t^{\beta-2} e^{-tS}\,dt$$

shows that, for every $\beta > 1$,

$$S^{-\beta} \exp\left(-\frac{paU_{02} + pdU_{04} + adU_{24}}{S}\right)$$

is a positive Laplace transform in the variables p, a, d.

From the star-mesh identity and the Riesz integral,

$$T_{W_4}^{-\beta} = S^{-\beta} T_{K_4}(y)^{-\beta}$$

is an integral of the product

$$\exp(-qU_{02}) \exp(-sU_{04}) \exp(-rU_{03}) \exp(-bU_{23}) \exp(-cU_{34})$$

with the three-arm star factor above. Each factor is a positive Laplace kernel in its edge-variable block, and the Riesz and Gaussian/gamma measures are positive. Tonelli's theorem therefore gives a positive Laplace representation for $T_{W_4}^{-\beta}$ on $(0, \infty)^8$.

Hence $T_{W_4}^{-\beta}$ is completely monotone for every $\beta > 1$. In Scott–Sokal notation, $W_4 \in G_\beta$ throughout the open in the cited source interval $1 < \beta < 3/2$. ☐

Corollary 21.9 (APP-0075: Scott-Sokal W4 spanning-tree polynomial complete monotonicity)**.** *Let W_4 be the wheel with four spokes. For every $\beta > 1$, the inverse power $T_{W_4}^{-\beta}$ of the spanning-tree polynomial is completely monotone on the positive edge cone. In particular, $W_4 \in G_\beta$ for every $1 < \beta < 3/2$, answering the Scott–Sokal source question for this graph affirmatively.*

*This resolves the source problem recorded as **APP-0075** [63].*

Proof. We use Proposition 7.2, Proposition 21.4, Proposition 21.5, and Proposition 21.8 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Root K_4 at vertex 3. For the remaining vertices $0, 2, 4$, use edge vectors

$$b_{03} = e_0, \quad b_{23} = e_2, \quad b_{43} = e_4, \quad b_{02} = e_0 - e_2, \quad b_{04} = e_0 - e_4, \quad b_{24} = e_2 - e_4.$$

Then

$$T_{K_4}(y) = \det M(y), \qquad M(y) = \sum_{ij} y_{ij} b_{ij} b_{ij}^T.$$

For $\beta > 1$, the 3×3 Riesz/Wishart integral gives

$$\det M(y)^{-\beta} = C_\beta \int_{Y>0} \exp(-\operatorname{tr}(M(y)Y)) \, (\det Y)^{\beta-2} \, dY,$$

with $C_\beta > 0$. Writing

$$U_{ij} = b_{ij}^T Y b_{ij},$$

the exponent is $-\sum y_{ij} U_{ij}$. If $Y = LL^T$, then $U_{ij} = |L^T b_{ij}|^2$. In particular the triple

$$(U_{02}, U_{04}, U_{24})$$

is the squared-distance triple of the three points $L^T e_0, L^T e_4, L^T e_2$, hence can be represented in $\mathbb{R}^2$.

For fixed such a squared-distance triple, choose points $\xi_0, \xi_2, \xi_4 \in \mathbb{R}^2$ with

$$U_{02} = |\xi_0 - \xi_2|^2, \quad U_{04} = |\xi_0 - \xi_4|^2, \quad U_{24} = |\xi_2 - \xi_4|^2.$$

The Gaussian completion identity gives

$$S^{-1} \exp\left(-\frac{paU_{02} + pdU_{04} + adU_{24}}{S}\right) = \pi^{-1} \int_{\mathbb{R}^2} \exp\left(-p|z - \xi_0|^2 - a|z - \xi_2|^2 - d|z - \xi_4|^2\right) \, dz.$$

Multiplying by the gamma representation

$$S^{-(\beta-1)} = \Gamma(\beta - 1)^{-1} \int_0^\infty t^{\beta-2} e^{-tS} \, dt$$

shows that, for every $\beta > 1$,

$$S^{-\beta} \exp\left(-\frac{paU_{02} + pdU_{04} + adU_{24}}{S}\right)$$

is a positive Laplace transform in the variables p, a, d.

From the star-mesh identity and the Riesz integral,

$$T_{W_4}^{-\beta} = S^{-\beta} T_{K_4}(y)^{-\beta}$$

is an integral of the product

$$\exp(-qU_{02}) \exp(-sU_{04}) \exp(-rU_{03}) \exp(-bU_{23}) \exp(-cU_{34})$$

with the three-arm star factor above. Each factor is a positive Laplace kernel in its edge-variable block, and the Riesz and Gaussian/gamma measures are positive. Tonelli's theorem therefore gives a positive Laplace representation for $T_{W_4}^{-\beta}$ on $(0, \infty)^8$.

Hence $T_{W_4}^{-\beta}$ is completely monotone for every $\beta > 1$. In Scott–Sokal notation, $W_4 \in G_\beta$ throughout the open in the cited source interval $1 < \beta < 3/2$. □

22 Determinant and moment-ratio applications

This section develops applications governed by determinant compression and moment-ratio comparisons. The method turns recursive derivative information into a finite or structured positivity problem, so bounds and counterexamples can be read from a smaller normal form.

Proposition 22.1 (Sturm-Darboux positive spectral families have positive initial Wronskians)**.** *Let $f_1, \dots, f_m$ be positive solutions on $\mathbb{R}$ of $f_j'' = (q + \lambda_j) f_j$, with $\lambda_1 < \cdots < \lambda_m$, and suppose the left endpoint asymptotics are ordered so that every two-function Wronskian tends to zero at $-\infty$ and each Darboux transform preserves positive ordered left asymptotics. Then every initial Wronskian $W(f_1, \dots, f_k)$ is strictly positive on $\mathbb{R}$, for $1 \le k \le m$.*

Proof. For every $m \ge 1$, every

$$0 < x_1 < \cdots < x_m$$

and every

$$0 \le s_1 < \cdots < s_m,$$

one has

$$\det \left[I_{s_j}(x_i) \right]_{i,j=1}^m > 0.$$

Thus the modified-Bessel kernel $K(x, s) = I_s(x)$ is strictly totally positive on $(0, \infty) \times [0, \infty)$. This answers the Buchstaber–Glutsyuk real-order open question affirmatively.

Set $x = e^y$ and

$$f_s(y) = I_s(e^y), \qquad y \in \mathbb{R},\ s \ge 0.$$

The modified Bessel equation

$$x^2 I_s''(x) + x I_s'(x) - (x^2 + s^2) I_s(x) = 0$$

becomes

$$f_s''(y) = \left(e^{2y} + s^2 \right) f_s(y).$$

Also

$$I_s(x) = \frac{(x/2)^s}{\Gamma(s+1)} (1 + O(x^2))$$

as $x \downarrow 0$, so for $s > 0$

$$f_s(y) = 2^{-s} \Gamma(s+1)^{-1} e^{sy} (1 + O(e^{2y}))$$

as $y \to -\infty$, while

$$f_0(y) = 1 + O(e^{2y}).$$

First take $0 \le a < b$. Since $f_a, f_b > 0$ and both solve the same scalar equation with spectral parameters a^2, b^2,

$$W(f_a, f_b)' = (b^2 - a^2) f_a f_b > 0.$$

The left endpoint asymptotics give $W(f_a, f_b)(y) \to 0$ as $y \to -\infty$, including the endpoint case $a = 0$. Hence

$$W(f_a, f_b)(y) > 0 \qquad (y \in \mathbb{R}).$$

Now prove all initial Wronskians by Darboux induction. Let

$$0 \le s_1 < \cdots < s_m$$

and put $p = f'_{s_1}/f_{s_1}$, $A = D - p$. For $j \geq 2$,

$$u_j = Af_{s_j} = \frac{W(f_{s_1}, f_{s_j})}{f_{s_1}} > 0.$$

The Darboux transform gives

$$u_j'' = \big(q_1(y) + s_j^2\big)u_j, \qquad q_1 = e^{2y} - 2(\log f_{s_1})''.$$

Moreover $u_j(y) \sim (s_j - s_1)c_j e^{s_j y}$ if $s_1 > 0$, and $u_j(y) \sim s_j c_j e^{s_j y}$ if $s_1 = 0$, with $c_j = 2^{-s_j}\Gamma(s_j+1)^{-1} > 0$. Thus the transformed family again satisfies the same ordered positive spectral-family hypotheses.

The Wronskian identity

$$W(f_{s_1}, \dots, f_{s_k}) = f_{s_1}\, W(u_2, \dots, u_k)$$

follows by subtracting (f_{s_j}/f_{s_1}) multiples of the first column in the Wronskian matrix, or equivalently from $Af = f_{s_1}(f/f_{s_1})'$. Since $f_{s_1} > 0$, induction on k gives

$$W(f_{s_1}, \dots, f_{s_k})(y) > 0$$

for every $1 \leq k \leq m$ and every $y \in \mathbb{R}$.

Positive initial Wronskians imply a strict extended complete Chebyshev system. For completeness, the local argument is as follows. If the determinant

$$\det[f_{s_j}(y_i)]_{i,j=1}^m$$

vanished at $y_1 < \cdots < y_m$, there would be a nontrivial linear combination $\sum c_j f_{s_j}$ with m zeros. Repeated generalized Rolle reduction using the positive Wronskians gives a contradiction, since an m-term ECT family permits at most $m-1$ zeros counted with multiplicity. The sign is determined by the confluent limit:

$$\det[f_{s_j}(y_i)]_{i,j=1}^m \sim \frac{W(f_{s_1}, \dots, f_{s_m})(y)}{\prod_{r=0}^{m-1} r!} \prod_{i<j}(y_j - y_i) > 0.$$

Therefore the determinant is positive for all $y_1 < \cdots < y_m$. Returning to $x_i = e^{y_i}$ proves the claimed strict total positivity. ☐

Proposition 22.2 (Positive initial Wronskians imply strict ECT evaluation determinants)**.** *If $f_1, \dots, f_m$ are C^{m-1} functions on an interval and $W(f_1, \dots, f_k) > 0$ for every $1 \leq k \leq m$, then for every $y_1 < \cdots < y_m$ in the interval, $\det[f_j(y_i)]_{i,j=1}^m > 0$.*

Proof. For Positive initial Wronskians imply strict ECT evaluation determinants, this proof is the corresponding component of Proposition 22.1. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. ☐

Corollary 22.3 (APP-0072: Buchstaber-Glutsyuk real-order Bessel I strict total positivity)**.** *For every $m \geq 1$, every $0 < x_1 < \cdots < x_m$, and every $0 \leq s_1 < \cdots < s_m$, $\det[I_{s_j}(x_i)]_{i,j=1}^m > 0$. Consequently the Buchstaber–Glutsyuk open question is answered affirmatively: the modified-Bessel kernel $K(x,s) = I_s(x)$ is strictly totally positive on $(0,\infty) \times [0,\infty)$.*

*This resolves the source problem recorded as **APP-0072** [61].*

Proof. We use Proposition 7.3, Proposition 22.1, and Proposition 22.2 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For every $m \geq 1$, every

$$0 < x_1 < \cdots < x_m$$

and every

$$0 \leq s_1 < \cdots < s_m,$$

one has

$$\det \left[I_{s_j}(x_i) \right]_{i,j=1}^{m} > 0.$$

Thus the modified-Bessel kernel $K(x, s) = I_s(x)$ is strictly totally positive on $(0, \infty) \times [0, \infty)$. This answers the Buchstaber–Glutsyuk real-order open question affirmatively.

Set $x = e^y$ and

$$f_s(y) = I_s(e^y), \qquad y \in \mathbb{R}, \ s \geq 0.$$

The modified Bessel equation

$$x^2 I_s''(x) + x I_s'(x) - (x^2 + s^2) I_s(x) = 0$$

becomes

$$f_s''(y) = \left(e^{2y} + s^2 \right) f_s(y).$$

Also

$$I_s(x) = \frac{(x/2)^s}{\Gamma(s+1)} (1 + O(x^2))$$

as $x \downarrow 0$, so for $s > 0$

$$f_s(y) = 2^{-s} \Gamma(s+1)^{-1} e^{sy} (1 + O(e^{2y}))$$

as $y \to -\infty$, while

$$f_0(y) = 1 + O(e^{2y}).$$

First take $0 \leq a < b$. Since $f_a, f_b > 0$ and both solve the same scalar equation with spectral parameters a^2, b^2,

$$W(f_a, f_b)' = (b^2 - a^2) f_a f_b > 0.$$

The left endpoint asymptotics give $W(f_a, f_b)(y) \to 0$ as $y \to -\infty$, including the endpoint case $a = 0$. Hence

$$W(f_a, f_b)(y) > 0 \qquad (y \in \mathbb{R}).$$

Now prove all initial Wronskians by Darboux induction. Let

$$0 \leq s_1 < \cdots < s_m$$

and put $p = f_{s_1}'/f_{s_1}$, $A = D - p$. For $j \geq 2$,

$$u_j = A f_{s_j} = \frac{W(f_{s_1}, f_{s_j})}{f_{s_1}} > 0.$$

The Darboux transform gives

$$u_j'' = \left(q_1(y) + s_j^2 \right) u_j, \qquad q_1 = e^{2y} - 2(\log f_{s_1})''.$$

Moreover $u_j(y) \sim (s_j - s_1) c_j e^{s_j y}$ if $s_1 > 0$, and $u_j(y) \sim s_j c_j e^{s_j y}$ if $s_1 = 0$, with $c_j = 2^{-s_j} \Gamma(s_j + 1)^{-1} > 0$. Thus the transformed family again satisfies the same ordered positive spectral-family hypotheses.

The Wronskian identity

$$W(f_{s_1},\dots,f_{s_k}) = f_{s_1}\,W(u_2,\dots,u_k)$$

follows by subtracting (f_{s_j}/f_{s_1}) multiples of the first column in the Wronskian matrix, or equivalently from $Af = f_{s_1}(f/f_{s_1})'$. Since $f_{s_1} > 0$, induction on k gives

$$W(f_{s_1},\dots,f_{s_k})(y) > 0$$

for every $1 \le k \le m$ and every $y \in \mathbb{R}$.

Positive initial Wronskians imply a strict extended complete Chebyshev system. For completeness, the local argument is as follows. If the determinant

$$\det[f_{s_j}(y_i)]_{i,j=1}^m$$

vanished at $y_1 < \cdots < y_m$, there would be a nontrivial linear combination $\sum c_j f_{s_j}$ with m zeros. Repeated generalized Rolle reduction using the positive Wronskians gives a contradiction, since an m-term ECT family permits at most $m-1$ zeros counted with multiplicity. The sign is determined by the confluent limit:

$$\det[f_{s_j}(y_i)]_{i,j=1}^m \sim \frac{W(f_{s_1},\dots,f_{s_m})(y)}{\prod_{r=0}^{m-1} r!}\prod_{i<j}(y_j - y_i) > 0.$$

Therefore the determinant is positive for all $y_1 < \cdots < y_m$. Returning to $x_i = e^{y_i}$ proves the claimed strict total positivity. □

Proposition 22.4 (Discrete Gaussian Toeplitz kernels are strictly totally positive)**.** *For every $0 < q < 1$, the bi-infinite sequence $g_q(n) = q^{n^2}$ has a strictly totally positive Toeplitz kernel $T_{g_q}(i,j) = g_q(i-j)$ on $\mathbb{Z}\times\mathbb{Z}$.*

Proof. Belton–Guillot–Khare–Putinar Question 12.2 asks whether totally positive Polya-frequency sequences are dense in all Polya-frequency sequences. I prove the pointwise/product-topology version:

$$\forall a \in PF, \quad \exists b^{(r)} \in TP \cap PF \quad \text{such that} \quad b_n^{(r)} \to a_n \quad (n \in \mathbb{Z}).$$

Here PF means the Toeplitz kernel $T_a(i,j) = a_{i-j}$ is totally nonnegative on $\mathbb{Z}\times\mathbb{Z}$, and TP means all ordered minors are strictly positive.

Fix $0 < q < 1$ and set

$$g_q(n) = q^{n^2}.$$

For strictly increasing integer grids $i_1 < \cdots < i_m$ and $j_1 < \cdots < j_m$,

$$\det[g_q(i_r - j_s)]_{r,s=1}^m = q^{\sum_r i_r^2 + \sum_s j_s^2} \det[(q^{-2i_r})^{j_s}]_{r,s=1}^m.$$

The bases $x_r = q^{-2i_r}$ are strictly increasing positive numbers. After factoring the positive powers $x_r^{j_1}$, the remaining determinant is the generalized Vandermonde determinant with strictly increasing nonnegative integer exponents $j_s - j_1$. It is positive. Hence g_q is a strictly totally positive Polya-frequency sequence.

Let $a \neq 0$ be a PF sequence that is not a bilateral geometric sequence $a_n = C\rho^n$. Then every finite family of distinct integer translates $n \mapsto a_{n-j_s}$ is linearly independent.

First, 1×1 minors give $a_n \geq 0$, and 2×2 Toeplitz minors give log-concavity on the support:

$$a_n^2 \geq a_{n-1}a_{n+1}.$$

Thus the support is an interval and, on the positive part of the support, the ratios

$$r_n = \frac{a_n}{a_{n-1}}$$

are nonincreasing.

If the support is finite or one-sided, the Laurent series $A(z) = \sum_n a_n z^n$ is nonzero and converges on a nonempty annulus. If the support is all of $\mathbb{Z}$, the monotone ratio limits

$$L_+ = \lim_{n\to+\infty} r_n, \qquad L_- = \lim_{n\to-\infty} r_n$$

exist in $[0, \infty]$ with $L_- \geq L_+$. If $L_- = L_+$, monotonicity forces all ratios to be equal, so $a_n = C\rho^n$, contrary to the non-geometric assumption. Hence $L_- > L_+$, and $A(z)$ converges on the nonempty annulus

$$\frac{1}{L_-} < |z| < \frac{1}{L_+},$$

with the usual interpretations when a limit is 0 or ∞.

Suppose now that a nontrivial finite translate dependence holds:

$$\sum_{s=1}^{m} c_s a_{n-j_s} = 0 \qquad (n \in \mathbb{Z}).$$

Multiplying the Laurent series by the nonzero Laurent polynomial $P(z) = \sum_s c_s z^{j_s}$ gives $P(z)A(z) = 0$ throughout a nonempty annulus. Since A is not identically zero and P is not the zero polynomial, this is impossible. The finite translate family is therefore independent.

Let a be a nonzero non-geometric PF sequence and put

$$b_n^{(q)} = (g_q * a)_n = \sum_{k\in\mathbb{Z}} q^{k^2} a_{n-k}.$$

The exponential bounds from Lemma 2 make this absolutely convergent, and $b_n^{(q)} \to a_n$ for each fixed n as $q \downarrow 0$.

The Toeplitz kernels satisfy $T_{b^{(q)}} = T_{g_q} T_a$. For finite increasing grids I, J of size m, the absolutely convergent Cauchy–Binet expansion is

$$\det T_{b^{(q)}}[I, J] = \sum_{K\in\binom{\mathbb{Z}}{m}} \det T_{g_q}[I, K] \det T_a[K, J].$$

Every summand is nonnegative, because T_{g_q} is TP and T_a is TN. Lemma 2 says the J-columns of T_a are independent, so some coordinate set K has $\det T_a[K, J] \neq 0$. Since T_a is TN, that determinant is positive. The corresponding T_{g_q}-minor is strictly positive, so the whole sum is strictly positive. Therefore $b^{(q)}$ is TP.

If $a_n = C\rho^n$ with $C > 0$ and $\rho > 0$, set

$$b_n^{(\varepsilon)} = C\rho^n e^{-\varepsilon n^2}.$$

Then

$$b_{i-j}^{(\varepsilon)} = C\rho^i \rho^{-j} e^{-\varepsilon i^2} e^{-\varepsilon j^2} e^{2\varepsilon ij}.$$

Every Toeplitz minor is a positive row/column rescaling of the generalized Vandermonde determinant

$$\det[(e^{2\varepsilon i_r})^{j_s}]_{r,s=1}^m,$$

which is positive for $i_1 < \cdots < i_m$ and $j_1 < \cdots < j_m$. Hence $b^{(\varepsilon)}$ is TP and $b_n^{(\varepsilon)} \to C\rho^n$ as $\varepsilon \downarrow 0$.

If $a = 0$, then $\eta g_q \to 0$ pointwise as $\eta \downarrow 0$, and ηg_q is TP for each $\eta > 0$. □

Proposition 22.5 (Non-geometric Polya-frequency sequences have independent translates)**.** *Let $a \neq 0$ be a Polya-frequency sequence on $\mathbb{Z}$. If a is not a bilateral geometric sequence $a_n = C\rho^n$, then every finite family of distinct translates $n \mapsto a_{n-j}$ is linearly independent. Equivalently, every finite set of Toeplitz columns of T_a is linearly independent.*

Proof. For Non-geometric Polya-frequency sequences have independent translates, this proof is the corresponding component of Proposition 22.4. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Corollary 22.6 (APP-0073: BGKP PF-sequence total-positivity density)**.** *Every Polya-frequency sequence $a = (a_n)_{n\in\mathbb{Z}}$ is the pointwise limit of totally positive Polya-frequency sequences. Equivalently, totally positive Polya-frequency sequences are dense in all Polya-frequency sequences for the product topology on $\mathbb{R}^{\mathbb{Z}}$. This answers Belton–Guillot–Khare–Putinar Question 12.2 affirmatively under the source-consistent pointwise topology.*

*This resolves the source problem recorded as **APP-0073** [62].*

Proof. We use Proposition 12.1, Proposition 18.4, Proposition 22.4, and Proposition 22.5 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Belton–Guillot–Khare–Putinar Question 12.2 asks whether totally positive Polya-frequency sequences are dense in all Polya-frequency sequences. I prove the pointwise/product-topology version:

$$\forall a \in PF, \quad \exists b^{(r)} \in TP \cap PF \quad \text{such that} \quad b_n^{(r)} \to a_n \quad (n \in \mathbb{Z}).$$

Here PF means the Toeplitz kernel $T_a(i,j) = a_{i-j}$ is totally nonnegative on $\mathbb{Z} \times \mathbb{Z}$, and TP means all ordered minors are strictly positive.

Fix $0 < q < 1$ and set

$$g_q(n) = q^{n^2}.$$

For strictly increasing integer grids $i_1 < \cdots < i_m$ and $j_1 < \cdots < j_m$,

$$\det[g_q(i_r - j_s)]_{r,s=1}^m = q^{\sum_r i_r^2 + \sum_s j_s^2} \det[(q^{-2i_r})^{j_s}]_{r,s=1}^m.$$

The bases $x_r = q^{-2i_r}$ are strictly increasing positive numbers. After factoring the positive powers $x_r^{j_1}$, the remaining determinant is the generalized Vandermonde determinant with strictly increasing nonnegative integer exponents $j_s - j_1$. It is positive. Hence g_q is a strictly totally positive Polya-frequency sequence.

Let $a \neq 0$ be a PF sequence that is not a bilateral geometric sequence $a_n = C\rho^n$. Then every finite family of distinct integer translates $n \mapsto a_{n-j_s}$ is linearly independent.

First, 1×1 minors give $a_n \geq 0$, and 2×2 Toeplitz minors give log-concavity on the support:

$$a_n^2 \geq a_{n-1}a_{n+1}.$$

Thus the support is an interval and, on the positive part of the support, the ratios

$$r_n = \frac{a_n}{a_{n-1}}$$

are nonincreasing.

If the support is finite or one-sided, the Laurent series $A(z) = \sum_n a_n z^n$ is nonzero and converges on a nonempty annulus. If the support is all of $\mathbb{Z}$, the monotone ratio limits

$$L_+ = \lim_{n\to+\infty} r_n, \qquad L_- = \lim_{n\to-\infty} r_n$$

exist in $[0,\infty]$ with $L_- \ge L_+$. If $L_- = L_+$, monotonicity forces all ratios to be equal, so $a_n = C\rho^n$, contrary to the non-geometric assumption. Hence $L_- > L_+$, and $A(z)$ converges on the nonempty annulus

$$\frac{1}{L_-} < |z| < \frac{1}{L_+},$$

with the usual interpretations when a limit is 0 or ∞.

Suppose now that a nontrivial finite translate dependence holds:

$$\sum_{s=1}^{m} c_s a_{n-j_s} = 0 \qquad (n \in \mathbb{Z}).$$

Multiplying the Laurent series by the nonzero Laurent polynomial $P(z) = \sum_s c_s z^{j_s}$ gives $P(z)A(z) = 0$ throughout a nonempty annulus. Since A is not identically zero and P is not the zero polynomial, this is impossible. The finite translate family is therefore independent.

Let a be a nonzero non-geometric PF sequence and put

$$b_n^{(q)} = (g_q * a)_n = \sum_{k\in\mathbb{Z}} q^{k^2} a_{n-k}.$$

The exponential bounds from Lemma 2 make this absolutely convergent, and $b_n^{(q)} \to a_n$ for each fixed n as $q \downarrow 0$.

The Toeplitz kernels satisfy $T_{b^{(q)}} = T_{g_q} T_a$. For finite increasing grids I, J of size m, the absolutely convergent Cauchy–Binet expansion is

$$\det T_{b^{(q)}}[I,J] = \sum_{K\in\binom{\mathbb{Z}}{m}} \det T_{g_q}[I,K] \det T_a[K,J].$$

Every summand is nonnegative, because T_{g_q} is TP and T_a is TN. Lemma 2 says the J-columns of T_a are independent, so some coordinate set K has $\det T_a[K,J] \ne 0$. Since T_a is TN, that determinant is positive. The corresponding T_{g_q}-minor is strictly positive, so the whole sum is strictly positive. Therefore $b^{(q)}$ is TP.

If $a_n = C\rho^n$ with $C > 0$ and $\rho > 0$, set

$$b_n^{(\varepsilon)} = C\rho^n e^{-\varepsilon n^2}.$$

Then

$$b_{i-j}^{(\varepsilon)} = C\rho^i \rho^{-j} e^{-\varepsilon i^2} e^{-\varepsilon j^2} e^{2\varepsilon ij}.$$

Every Toeplitz minor is a positive row/column rescaling of the generalized Vandermonde determinant

$$\det[(e^{2\varepsilon i_r})^{j_s}]_{r,s=1}^m,$$

which is positive for $i_1 < \cdots < i_m$ and $j_1 < \cdots < j_m$. Hence $b^{(\varepsilon)}$ is TP and $b_n^{(\varepsilon)} \to C\rho^n$ as $\varepsilon \downarrow 0$.

If $a = 0$, then $\eta g_q \to 0$ pointwise as $\eta \downarrow 0$, and ηg_q is TP for each $\eta > 0$. $\square$

Proposition 22.7 (Square-root Laplace gamma tilt gives strict log-convexity)**.** *For $x > 0$, let $k_x(s) = x(2\pi)^{-1/2}s^{-3/2}\exp(-x^2/(2s))$ on $(0,\infty)$. Then k_x is a positive probability density and, for every $a > 0$, $\Gamma(a)^{-1}\int_0^\infty y^{a-1}e^{-y}e^{-x\sqrt{2y}}\,dy = \int_0^\infty (1+s)^{-a}k_x(s)\,ds$. Consequently the logarithmic second derivative in a of this gamma-normalized transform equals the tilted variance of $\log(1+S_x)$, and is strictly positive.*

Proof. Yang's source normalization is

$$R_p(x) = \int_0^\infty t^p \exp(-t^2/2 - xt)\,dt, \qquad p > -1, \quad x > 0.$$

Put $a = (p+1)/2$, so $p = 2a-1$. With $y = t^2/2$, one has

$$t^{2a-1}\,dt = 2^{a-1}y^{a-1}\,dy$$

and therefore

$$R_{2a-1}(x) = 2^{a-1}\int_0^\infty y^{a-1}e^{-y}e^{-x\sqrt{2y}}\,dy.$$

For $x > 0$, use the square-root Laplace identity

$$e^{-x\sqrt{2y}} = \frac{x}{\sqrt{2\pi}}\int_0^\infty s^{-3/2}\exp\left(-\frac{x^2}{2s}\right)e^{-sy}\,ds.$$

This is the standard Levy-density representation of the function $y \mapsto e^{-x\sqrt{2y}}$. The kernel

$$k_x(s) = \frac{x}{\sqrt{2\pi}}s^{-3/2}\exp\left(-\frac{x^2}{2s}\right), \qquad s > 0,$$

is positive and integrates to 1, as is seen by evaluating the identity at $y = 0$.

Since all terms are nonnegative, Tonelli's theorem gives

$$\begin{aligned} R_{2a-1}(x) &= 2^{a-1}\int_0^\infty y^{a-1}e^{-y}\left(\int_0^\infty e^{-sy}k_x(s)\,ds\right)dy \\ &= 2^{a-1}\int_0^\infty k_x(s)\left(\int_0^\infty y^{a-1}e^{-(1+s)y}\,dy\right)ds \\ &= 2^{a-1}\Gamma(a)\int_0^\infty (1+s)^{-a}k_x(s)\,ds. \end{aligned}$$

Thus

$$\frac{R_{2a-1}(x)}{\Gamma(a)} = 2^{a-1}\int_0^\infty (1+s)^{-a}k_x(s)\,ds.$$

Let S_x be distributed with density k_x, and set $Z_x = \log(1+S_x)$. Then

$$\frac{R_{2a-1}(x)}{\Gamma(a)} = 2^{a-1}\mathbb{E}e^{-aZ_x}.$$

For the tilted law

$$d\mathbb{P}_{a,x} = \frac{e^{-aZ_x}}{\mathbb{E}e^{-aZ_x}}\,d\mathbb{P}_{Z_x},$$

ordinary differentiation of the finite Laplace transform gives

$$\frac{d^2}{da^2}\log\frac{R_{2a-1}(x)}{\Gamma(a)} = \mathrm{Var}_{a,x}(Z_x) \geq 0.$$

The added affine term $(a-1)\log 2$ contributes no second derivative.

Because $a = (p+1)/2$, the p-curvature is

$$\frac{d^2}{dp^2} \log \frac{R_p(x)}{\Gamma((p+1)/2)} = \frac{1}{4} \operatorname{Var}_{a,x}\bigl(\log(1+S_x)\bigr) \geq 0.$$

For $x > 0$, k_x is positive on all of $(0, \infty)$, so $\log(1+S_x)$ is nonconstant under every tilted law. The variance is therefore strictly positive, proving strict log-convexity. □

Definition 22.1 (Yang order-p Mills moment kernel)**.** For $x > 0$ and $p > -1$, Yang's order-p Mills moment is $R_p(x) = \int_0^\infty t^p \exp(-t^2/2 - xt)\, dt$. The half-gamma normalized Yang Mills ratio is $R_p(x)/\Gamma((p+1)/2)$.

Corollary 22.8 (APP-0076: Yang Mills-ratio half-gamma log-convexity)**.** *Let $R_p(x) = \int_0^\infty t^p \exp(-t^2/2 - xt)\, dt$ for $x > 0$ and $p > -1$. For every fixed $x > 0$, the function $p \mapsto R_p(x)/\Gamma((p+1)/2)$ is strictly log-convex on $(-1, \infty)$. This proves Yang's Remark 11 conjectural log-convexity under the source normalization.*

This resolves the source problem recorded as ***APP-0076*** *[64].*

Proof. We use Proposition 7.3, Proposition 22.7, Proposition 22.1, and Proposition 12.6 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Yang's source normalization is

$$R_p(x) = \int_0^\infty t^p \exp(-t^2/2 - xt)\, dt, \qquad p > -1, \quad x > 0.$$

Put $a = (p+1)/2$, so $p = 2a - 1$. With $y = t^2/2$, one has

$$t^{2a-1}\, dt = 2^{a-1} y^{a-1}\, dy$$

and therefore

$$R_{2a-1}(x) = 2^{a-1} \int_0^\infty y^{a-1} e^{-y} e^{-x\sqrt{2y}}\, dy.$$

For $x > 0$, use the square-root Laplace identity

$$e^{-x\sqrt{2y}} = \frac{x}{\sqrt{2\pi}} \int_0^\infty s^{-3/2} \exp\left(-\frac{x^2}{2s}\right) e^{-sy}\, ds.$$

This is the standard Levy-density representation of the function $y \mapsto e^{-x\sqrt{2y}}$. The kernel

$$k_x(s) = \frac{x}{\sqrt{2\pi}} s^{-3/2} \exp\left(-\frac{x^2}{2s}\right), \qquad s > 0,$$

is positive and integrates to 1, as is seen by evaluating the identity at $y = 0$.

Since all terms are nonnegative, Tonelli's theorem gives

$$\begin{aligned} R_{2a-1}(x) &= 2^{a-1} \int_0^\infty y^{a-1} e^{-y} \left(\int_0^\infty e^{-sy} k_x(s)\, ds \right) dy \\ &= 2^{a-1} \int_0^\infty k_x(s) \left(\int_0^\infty y^{a-1} e^{-(1+s)y}\, dy \right) ds \\ &= 2^{a-1} \Gamma(a) \int_0^\infty (1+s)^{-a} k_x(s)\, ds. \end{aligned}$$

Thus

$$\frac{R_{2a-1}(x)}{\Gamma(a)} = 2^{a-1}\int_0^\infty (1+s)^{-a} k_x(s)\,ds.$$

Let S_x be distributed with density k_x, and set $Z_x = \log(1+S_x)$. Then

$$\frac{R_{2a-1}(x)}{\Gamma(a)} = 2^{a-1}\mathbb{E}e^{-aZ_x}.$$

For the tilted law

$$d\mathbb{P}_{a,x} = \frac{e^{-aZ_x}}{\mathbb{E}e^{-aZ_x}}\,d\mathbb{P}_{Z_x},$$

ordinary differentiation of the finite Laplace transform gives

$$\frac{d^2}{da^2}\log\frac{R_{2a-1}(x)}{\Gamma(a)} = \mathrm{Var}_{a,x}(Z_x) \geq 0.$$

The added affine term $(a-1)\log 2$ contributes no second derivative.

Because $a = (p+1)/2$, the p-curvature is

$$\frac{d^2}{dp^2}\log\frac{R_p(x)}{\Gamma((p+1)/2)} = \frac{1}{4}\,\mathrm{Var}_{a,x}\big(\log(1+S_x)\big) \geq 0.$$

For $x > 0$, k_x is positive on all of $(0,\infty)$, so $\log(1+S_x)$ is nonconstant under every tilted law. The variance is therefore strictly positive, proving strict log-convexity. □

Proposition 22.9 (Finite reciprocal characteristic-function identity certificate)**.** *Let φ be a characteristic function with reciprocal $q = 1/\varphi$ on a neighborhood of the origin, let $\mu_1,\ldots,\mu_n$ be distinct nonzero real numbers, and let $\theta_k = \prod_{j\neq k}\mu_k/(\mu_k-\mu_j)$. If the finite identity $\sum_{k=1}^n \theta_k \prod_{j\neq k} q(\mu_j t) = 1$ holds for all real t, then φ satisfies the Rastegar–Roitershtein functional equation $\prod_j \varphi(\mu_j t) = \sum_k \theta_k \varphi(\mu_k t)$.*

Proof. The local Sympy audit verified:

$(\theta_1,\theta_2,\theta_3) = (-3/5, 3/2, 1/10)$; $\sum_k \theta_k = 1$; $\sum_k \theta_k \prod_{j\neq k}(1+a\mu_j^2 t^2) = 1$ identically; the original functional-equation difference is identically 0; $h_2(\mu) - p_2(\mu) = 0$.

This is a finite symbolic characteristic-function certificate, not a numerical Taylor match.

The reusable bridge is the finite reciprocal-characteristic certificate: if a nonzero real vector μ, the source weights θ_k, and a positive-definite reciprocal polynomial $q(t) = 1 + at^2$ satisfy

$$\sum_{k=1}^n \theta_k \prod_{j\neq k} q(\mu_j t) = 1$$

identically, then $\varphi = 1/q$ gives an exact source functional-equation solution. When this X is not in the source conclusion class, it is a source-level counterexample.

The present $n = 3$ data provide such a certificate. □

Counterexample 22.10 (APP-0078: Rastegar-Roitershtein condition-(3) removal conjecture is false)**.** The Rastegar–Roitershtein conjecture that condition (1.3), numbered condition (3) in the preprint, is unnecessary in Theorem 1.1 for $n \geq 3$ is false. For $n = 3$, $\mu = (1, 2, -2/3)$, the source weights are $\theta = (-3/5, 3/2, 1/10)$, and the centered Laplace characteristic function $\varphi(t) = 1/(1+at^2)$, $a > 0$, satisfies the source functional equation identically, while the law is nondegenerate, has mean zero, and is not one-sided exponential.

This resolves the source problem recorded as **APP-0078** [66].

Proof. We use Proposition 18.3 and Proposition 22.9 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The local Sympy audit verified:

$(\theta_1, \theta_2, \theta_3) = (-3/5, 3/2, 1/10)$; $\sum_k \theta_k = 1$; $\sum_k \theta_k \prod_{j \neq k}(1 + a\mu_j^2 t^2) = 1$ identically; the original functional-equation difference is identically 0; $h_2(\mu) - p_2(\mu) = 0$.

This is a finite symbolic characteristic-function certificate, not a numerical Taylor match.

The reusable bridge is the finite reciprocal-characteristic certificate: if a nonzero real vector μ, the source weights θ_k, and a positive-definite reciprocal polynomial $q(t) = 1 + at^2$ satisfy

$$\sum_{k=1}^{n} \theta_k \prod_{j \neq k} q(\mu_j t) = 1$$

identically, then $\varphi = 1/q$ gives an exact source functional-equation solution. When this X is not in the source conclusion class, it is a source-level counterexample.

The present $n = 3$ data provide such a certificate. $\square$

Proposition 22.11 (Two-by-two determinantal polynomial sign-conjugacy blindness)**.** *For a real symmetric matrix* $M = \begin{pmatrix} a\&b \\ b\&c \end{pmatrix}$ *and* $Z = \operatorname{diag}(z_1, z_2)$, $\det(I + MZ) = 1 + az_1 + cz_2 + (ac - b^2)z_1 z_2$. *Hence the determinantal polynomial records the diagonal entries and determinant but not the sign of the off-diagonal entry.*

Proof. Let

$$A = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 2 \end{pmatrix}, \qquad Z = \operatorname{diag}(z_1, z_2).$$

Both A and B are positive definite, since their leading principal minors are positive and

$$\det A = \det B = 2 - \frac{1}{4} = \frac{7}{4} > 0.$$

For a real symmetric 2×2 matrix

$$M = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

one has

$$\det(I + MZ) = 1 + az_1 + cz_2 + (ac - b^2)z_1 z_2.$$

Therefore the sign of b is invisible to the determinantal stable polynomial. In the present example,

$$\det(I + AZ) = \det(I + BZ) = 1 + z_1 + 2z_2 + \frac{7}{4} z_1 z_2,$$

and also

$$\det(I + A) = \det(I + B) = \frac{23}{4}.$$

Hence the normalized polynomials are identical:

$$\frac{\det(I + AZ)}{\det(I + A)} = \frac{\det(I + BZ)}{\det(I + B)}.$$

The BBL order hypothesis holds immediately by equality. Equivalently, if one checks the stronger proper-position relation directly, $p_A = p_B$ gives

$$p_B + ip_A = (1+i)p_A,$$

which is stable because p_A is a determinantal stable polynomial and multiplication by a nonzero constant preserves stability.

However,

$$B - A = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

has eigenvalues 1 and -1. Thus $B - A$ is indefinite and $A \nleq B$ in Loewner order. Similarly $A - B$ is indefinite, so the matrices are not Loewner-comparable.

For completeness, the example is not a commuting or diagonal case:

$$AB = \begin{pmatrix} 3/4 & 1/2 \\ -1/2 & 15/4 \end{pmatrix}, \qquad BA = \begin{pmatrix} 3/4 & -1/2 \\ 1/2 & 15/4 \end{pmatrix},$$

so $AB \neq BA$.

The obstruction is the non-injectivity of $M \mapsto \det(I + MZ)$. In dimension two, the polynomial records a, c, and $ac - b^2$, but not the sign of b. Diagonal sign conjugation $M \mapsto DMD$, with $D = \operatorname{diag}(1, -1)$, preserves all principal-minor data and hence the normalized determinantal stable polynomial, while literal Loewner comparison between M and DMD can fail.

This refutes the literal source question. It does not refute possible repaired quotient formulations modulo diagonal sign conjugacy. $\square$

Counterexample 22.12 (APP-0079: BBL determinantal stable Loewner converse is false)**.** Borcea–Branden–Liggett Question 4.1 has a negative answer. For $A = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 2 \end{pmatrix}$, both matrices are positive definite and the normalized determinantal stable polynomials $\det(I + AZ)/\det(I + A)$ and $\det(I + BZ)/\det(I + B)$ are identical, so the BBL order hypothesis holds. But $B - A = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ is indefinite, hence $A \nleq B$ in Loewner order.

This resolves the source problem recorded as **APP-0079** [67].

Proof. We use Proposition 18.3, Proposition 22.11, and Proposition 12.1 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

Let

$$A = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 2 \end{pmatrix}, \qquad Z = \operatorname{diag}(z_1, z_2).$$

Both A and B are positive definite, since their leading principal minors are positive and

$$\det A = \det B = 2 - \frac{1}{4} = \frac{7}{4} > 0.$$

For a real symmetric 2×2 matrix

$$M = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

one has

$$\det(I + MZ) = 1 + az_1 + cz_2 + (ac - b^2)z_1 z_2.$$

Therefore the sign of b is invisible to the determinantal stable polynomial. In the present example,

$$\det(I + AZ) = \det(I + BZ) = 1 + z_1 + 2z_2 + \frac{7}{4}z_1z_2,$$

and also

$$\det(I + A) = \det(I + B) = \frac{23}{4}.$$

Hence the normalized polynomials are identical:

$$\frac{\det(I + AZ)}{\det(I + A)} = \frac{\det(I + BZ)}{\det(I + B)}.$$

The BBL order hypothesis holds immediately by equality. Equivalently, if one checks the stronger proper-position relation directly, $p_A = p_B$ gives

$$p_B + ip_A = (1 + i)p_A,$$

which is stable because p_A is a determinantal stable polynomial and multiplication by a nonzero constant preserves stability.

However,

$$B - A = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

has eigenvalues 1 and -1. Thus $B - A$ is indefinite and $A \npreceq B$ in Loewner order. Similarly $A - B$ is indefinite, so the matrices are not Loewner-comparable.

For completeness, the example is not a commuting or diagonal case:

$$AB = \begin{pmatrix} 3/4 & 1/2 \\ -1/2 & 15/4 \end{pmatrix}, \qquad BA = \begin{pmatrix} 3/4 & -1/2 \\ 1/2 & 15/4 \end{pmatrix},$$

so $AB \neq BA$.

The obstruction is the non-injectivity of $M \mapsto \det(I + MZ)$. In dimension two, the polynomial records a, c, and $ac - b^2$, but not the sign of b. Diagonal sign conjugation $M \mapsto DMD$, with $D = \operatorname{diag}(1, -1)$, preserves all principal-minor data and hence the normalized determinantal stable polynomial, while literal Loewner comparison between M and DMD can fail.

This refutes the literal source question. It does not refute possible repaired quotient formulations modulo diagonal sign conjugacy. ☐

Proposition 22.13 (Generalized Vandermonde determinants have strict Chebyshev sign)**.** *If* $0 < b_1 < \cdots < b_m$ *and* $s_1 < \cdots < s_m$*, then* $\det[b_j^{s_i}]_{i,j=1}^m > 0$*. Equivalently,* $u \mapsto e^{s_i u}$ *is an extended complete Chebyshev system on real intervals.*

Proof. For every fixed $0 < q < 1$, prove that

$$K_q(x, y) = \Gamma_q(x + y)$$

is strictly totally positive of all orders on $(0, \infty)^2$.

Let μ be a positive Borel measure on $(0, \infty)$ with at least m distinct support points. Suppose the Mellin moments

$$M(s) = \int_0^\infty t^s \, d\mu(t)$$

are finite for all $s = x_i + y_j$, where

$$0 < x_1 < \cdots < x_m, \qquad 0 < y_1 < \cdots < y_m.$$

Then

$$\det[M(x_i + y_j)]_{i,j=1}^m > 0.$$

If μ has infinite support and the required moments are finite on the domain, then $K(x, y) = M(x+y)$ is STP_∞.

By Andreief's identity, first for finite-support measures and then by approximation/monotone convergence of the nonnegative Cauchy-Binet expansion,

$$\det[M(x_i + y_j)]_{i,j=1}^m = \frac{1}{m!} \int_{(0,\infty)^m} \det[t_k^{x_i}]_{i,k=1}^m \det[t_k^{y_j}]_{j,k=1}^m \prod_{k=1}^m d\mu(t_k).$$

The integrand vanishes on diagonals. On the chamber $0 < t_1 < \cdots < t_m$, both determinants are positive. Indeed, after writing $t_k = e^{u_k}$, the functions $u \mapsto e^{x_i u}$ form an extended complete Chebyshev system because their Wronskian is

$$\det \left[\frac{d^{r-1}}{du^{r-1}} e^{x_i u} \right]_{i,r=1}^m = e^{(x_1+\cdots+x_m)u} \prod_{1 \le i < j \le m} (x_j - x_i) > 0.$$

The same argument applies to the y_j. Since μ has at least m support points, the ordered chamber has positive μ^m-mass on sets where the integrand is strictly positive. Thus the determinant is strictly positive.

The moment hypotheses justify the determinant integral: expanding the product of determinants bounds the absolute integrand by a finite sum of products of the finite moments $M(x_i + y_j)$.

For $0 < q < 1$, the Jackson q-gamma function used by the source is

$$\Gamma_q(z) = (1 - q)^{1-z} \frac{(q; q)_\infty}{(q^z; q)_\infty}, \qquad z > 0.$$

The q-binomial theorem gives

$$\frac{1}{(q^z; q)_\infty} = \sum_{n=0}^\infty \frac{q^{nz}}{(q; q)_n}.$$

Therefore

$$\Gamma_q(z) = (1 - q)(q; q)_\infty \sum_{n=0}^\infty \frac{(q^n/(1 - q))^z}{(q; q)_n}.$$

Equivalently,

$$\Gamma_q(z) = \int_0^\infty t^z \, d\mu_q(t),$$

where

$$\mu_q = (1 - q)(q; q)_\infty \sum_{n=0}^\infty \frac{1}{(q; q)_n} \delta_{q^n/(1-q)}.$$

This is a positive atomic measure with infinitely many distinct support points $q^n/(1 - q)$. The moments are finite for all $z > 0$, since the defining series converges for $q^z \in (0, 1)$.

Applying the Mellin moment kernel lemma to $M(z) = \Gamma_q(z)$ gives, for every $m \ge 1$ and every strictly ordered $x_i, y_j > 0$,

$$\det[\Gamma_q(x_i + y_j)]_{i,j=1}^m > 0.$$

Thus $K_q(x, y) = \Gamma_q(x + y)$ is STP_∞ on $(0, \infty)^2$. □

Proposition 22.14 (Positive Mellin moment kernels with infinite support are strictly totally positive)**.** *Let μ be a positive Borel measure on $(0,\infty)$ with infinite support and finite Mellin moments on the relevant domain. Then $K(x,y)=\int_0^\infty t^{x+y}\,d\mu(t)$ is STP_∞ on that domain.*

Proof. For Positive Mellin moment kernels with infinite support are strictly totally positive, this proof is the corresponding component of Proposition 22.13. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Definition 22.2 (q-gamma and q-Pochhammer notation on the positive axis)**.** For $0<q<1$, the q-Pochhammer symbols are $(a;q)_n=\prod_{j=0}^{n-1}(1-aq^j)$ and $(a;q)_\infty=\prod_{j=0}^{\infty}(1-aq^j)$. The Jackson q-gamma function on $(0,\infty)$ is $\Gamma_q(z)=(1-q)^{1-z}(q;q)_\infty/(q^z;q)_\infty$.

Proposition 22.15 (q-binomial atomic Mellin representation for q-gamma)**.** *For $0<q<1$ and $z>0$, $\Gamma_q(z)=(1-q)(q;q)_\infty\sum_{n=0}^{\infty}(q^n/(1-q))^z/(q;q)_n$. Equivalently, Γ_q is the Mellin transform of the positive atomic measure $(1-q)(q;q)_\infty\sum_{n\geq 0}(q;q)_n^{-1}\delta_{q^n/(1-q)}$.*

Proof. For q-binomial atomic Mellin representation for q-gamma, this proof is the corresponding component of Proposition 22.13. corresponding to the displayed statement. The cited argument establishes the shared certificate or normal form; applying that component to the present notation gives the claim. □

Corollary 22.16 (APP-0080: q-gamma Hankel kernel is strictly totally positive)**.** *For every fixed $0<q<1$, the kernel $K_q(x,y)=\Gamma_q(x+y)$ is STP_∞ on $(0,\infty)^2$. Equivalently, for every $m\geq 1$ and every $0<x_1<\cdots<x_m$, $0<y_1<\cdots<y_m$, one has $\det[\Gamma_q(x_i+y_j)]_{i,j=1}^m>0$.*

*This resolves the source problem recorded as **APP-0080** [68].*

Proof. We use Proposition 12.1, Proposition 22.13, Proposition 22.14, and Proposition 22.15 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

For every fixed $0<q<1$, prove that

$$K_q(x,y)=\Gamma_q(x+y)$$

is strictly totally positive of all orders on $(0,\infty)^2$.

Let μ be a positive Borel measure on $(0,\infty)$ with at least m distinct support points. Suppose the Mellin moments

$$M(s)=\int_0^\infty t^s\,d\mu(t)$$

are finite for all $s=x_i+y_j$, where

$$0<x_1<\cdots<x_m,\qquad 0<y_1<\cdots<y_m.$$

Then

$$\det[M(x_i+y_j)]_{i,j=1}^m>0.$$

If μ has infinite support and the required moments are finite on the domain, then $K(x,y)=M(x+y)$ is STP_∞.

By Andreief's identity, first for finite-support measures and then by approximation/monotone convergence of the nonnegative Cauchy-Binet expansion,

$$\det[M(x_i+y_j)]_{i,j=1}^m=\frac{1}{m!}\int_{(0,\infty)^m}\det[t_k^{x_i}]_{i,k=1}^m\det[t_k^{y_j}]_{j,k=1}^m\prod_{k=1}^m d\mu(t_k).$$

The integrand vanishes on diagonals. On the chamber $0 < t_1 < \cdots < t_m$, both determinants are positive. Indeed, after writing $t_k = e^{u_k}$, the functions $u \mapsto e^{x_i u}$ form an extended complete Chebyshev system because their Wronskian is

$$\det\left[\frac{d^{r-1}}{du^{r-1}} e^{x_i u}\right]_{i,r=1}^{m} = e^{(x_1+\cdots+x_m)u} \prod_{1\le i<j\le m} (x_j - x_i) > 0.$$

The same argument applies to the y_j. Since μ has at least m support points, the ordered chamber has positive μ^m-mass on sets where the integrand is strictly positive. Thus the determinant is strictly positive.

The moment hypotheses justify the determinant integral: expanding the product of determinants bounds the absolute integrand by a finite sum of products of the finite moments $M(x_i + y_j)$.

For $0 < q < 1$, the Jackson q-gamma function used by the source is

$$\Gamma_q(z) = (1-q)^{1-z} \frac{(q;q)_\infty}{(q^z;q)_\infty}, \qquad z > 0.$$

The q-binomial theorem gives

$$\frac{1}{(q^z;q)_\infty} = \sum_{n=0}^{\infty} \frac{q^{nz}}{(q;q)_n}.$$

Therefore

$$\Gamma_q(z) = (1-q)(q;q)_\infty \sum_{n=0}^{\infty} \frac{(q^n/(1-q))^z}{(q;q)_n}.$$

Equivalently,

$$\Gamma_q(z) = \int_0^\infty t^z \, d\mu_q(t),$$

where

$$\mu_q = (1-q)(q;q)_\infty \sum_{n=0}^{\infty} \frac{1}{(q;q)_n} \delta_{q^n/(1-q)}.$$

This is a positive atomic measure with infinitely many distinct support points $q^n/(1-q)$. The moments are finite for all $z > 0$, since the defining series converges for $q^z \in (0,1)$.

Applying the Mellin moment kernel lemma to $M(z) = \Gamma_q(z)$ gives, for every $m \ge 1$ and every strictly ordered $x_i, y_j > 0$,

$$\det[\Gamma_q(x_i + y_j)]_{i,j=1}^m > 0.$$

Thus $K_q(x,y) = \Gamma_q(x+y)$ is STP_∞ on $(0,\infty)^2$. □

23 Endpoint obstruction certificates

This section collects applications in which a universal positivity, convexity, or monotonicity claim is decided by reducing it to a local obstruction. The common method is to isolate a boundary, endpoint, or explicit witness where the proposed sign pattern cannot hold.

Definition 23.1 (Tricomi Psi standard Laplace integral representation)**.** For $a > 0$ and $z > 0$, the Tricomi confluent hypergeometric function satisfies $\Psi(a,c,z) = \Gamma(a)^{-1} \int_0^\infty e^{-zt} t^{a-1} (1+t)^{c-a-1} \, dt$ whenever the displayed integral is interpreted in the source's standard admissible range. In particular, for the integer parameters used in the counterexample, the integral is an elementary positive Laplace integral.

Proposition 23.1 (Tricomi Psi integer polynomial complete-monotonicity counterfamily)**.** *For every integer $m \geq 1$, the quotient $z \mapsto \Psi(m, 2m+1, z)^2/\Psi(2m, 2m+1, z)$ is a polynomial in z^{-1} with nonnegative coefficients, hence is completely monotone on $(0, \infty)$.*

Proof. The same calculation gives a reusable family. For every integer $m \geq 1$, set $a = m$ and $c = 2m + 1$. Then

$$\Psi(m, 2m+1, z) = \frac{1}{\Gamma(m)} \int_0^\infty e^{-zt} t^{m-1} (1+t)^m \, dt.$$

Expanding $(1+t)^m$ gives a finite sum of positive multiples of z^{-m-k}. Meanwhile

$$\Psi(2m, 2m+1, z) = \frac{1}{\Gamma(2m)} \int_0^\infty e^{-zt} t^{2m-1} \, dt = z^{-2m}.$$

Hence the quotient is the square of a polynomial in z^{-1} with nonnegative coefficients, and is completely monotone. These are all outside the proposed window because $c = 2m + 1 > 1$. ☐

Counterexample 23.2 (APP-0077: Ferreira-Simon Tricomi Psi quotient complete-monotonicity window is false)**.** Ferreira–Simon Conjecture 2, read literally as the assertion that $z \mapsto \Psi(a, c, z)^2/\Psi(2a, c, z)$ is completely monotone on $(0, \infty)$ if and only if $c \in [0, 1]$, is false. At $a = 1$ and $c = 3$, the quotient equals $(1 + z^{-1})^2$, which is completely monotone although $3 \notin [0, 1]$.

This resolves the source problem recorded as **APP-0077** [65].

Proof. We use Proposition 7.3, Proposition 23.1, Proposition 23.1, and Proposition 7.2 as the upstream structural result. The source-specific objects in the statement are then verified as follows.

The same calculation gives a reusable family. For every integer $m \geq 1$, set $a = m$ and $c = 2m+1$. Then

$$\Psi(m, 2m+1, z) = \frac{1}{\Gamma(m)} \int_0^\infty e^{-zt} t^{m-1} (1+t)^m \, dt.$$

Expanding $(1+t)^m$ gives a finite sum of positive multiples of z^{-m-k}. Meanwhile

$$\Psi(2m, 2m+1, z) = \frac{1}{\Gamma(2m)} \int_0^\infty e^{-zt} t^{2m-1} \, dt = z^{-2m}.$$

Hence the quotient is the square of a polynomial in z^{-1} with nonnegative coefficients, and is completely monotone. These are all outside the proposed window because $c = 2m + 1 > 1$. ☐

24 Conclusion

The theory developed here is organized around a single principle: positivity should be transported only through mechanisms that preserve the relevant analytic structure. Zeta-law compression turns Dirichlet sums into probabilistic identities, special-function curvature is tested through Laplace kernels, and application-level claims are resolved by moving between Stieltjes, Bernstein, determinant, ratio, and endpoint-obstruction forms.

This perspective gives the paper its hierarchy. Definitions fix the common objects, lemmas and propositions isolate reusable transformations, theorems state the main transport mechanisms, corollaries and counterexamples record their consequences, and the application ledger points each source problem to the result that resolves it. The result is a connected theory of positivity transport rather than a list of unrelated solved examples.

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